

Draft Environmental Assessment for Proposed Amendment to the Fishery Management Plan for the Bering Sea and Aleutian Islands King and Tanner Crabs

Rebuilding Plan for Eastern Bering Sea Snow Crab Final Action Analysis

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Abstract: This draft environmental assessment analyzes an amendment to the BSAI Crab FMP to rebuild the Eastern Bering Sea (EBS) snow crab (*Chionoecetes opilio*) stock in compliance with section 304(e)(3) of the Magnuson-Stevens Fishery Conservation and Management Act (MSA). A range of alternative rebuilding time frames is considered based on whether or not directed fishing is permitted during stock recovery. The impacts of the alternatives considered on crab resources, fishery participants, habitat, and other components of the human environment are discussed in the analysis.

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List of Acronyms and Abbreviations

Acronym or Abbreviation	Meaning	Acronym or Abbreviation	Meaning
AAC	Alaska Administrative Code	MFMT	Maximum fishing mortality threshold
ABC	Acceptable biological catch	MMB	Mature male biomass
ACL	Annual Catch Limit	MSA	Magnuson Stevens Fishery Conservation
ADF&G	Alaska Department of Fish and Game	MSST	Minimum stock size threshold
AFSC	Alaska Fisheries Science Center	MSY	Maximum sustainable yield
AIGKC	Aleutian Is. golden king crab	t	Ton, or metric ton
AKFIN	Alaska Fisheries Information Network	NAICS	North American Industry Classification
APICDA	Aleutian Pribilof Island Community	NAO	NOAA Administrative Order
BBEDC	Bristol Bay Economic Development	NEPA	National Environmental Policy Act
BBRKC	Bristol Bay red king crab	NMFS	National Marine Fishery Service
BCAR	Bottom Contact Area ratio	NOAA	National Oceanic and Atmospheric
B _{MSY}	Biomass corresponding to MSY	NPFMC	North Pacific Fishery Management
BSAI	Bering Sea and Aleutian Islands	NPPSD	North Pacific Pelagic Seabird Database
CAS	Catch Accounting System	NS1	National Standard 1
CBSFA	Central Bering Sea Fishermen's	NSEDC	North Sound Economic Development
CEQ	Council on Environmental Quality	NSRKC	Norton Sound red king crab
CFEC	Commercial Fisheries Entry Commission	Observer	North Pacific Groundfish and Halibut
CFR	Code of Federal Regulations	OFL	Overfishing level
CL	Carapace length	OMB	Office of Management and Budget
CMIP6	Coupled Model Intercomparison Project,	OY	Optimum yield
Council	North Pacific Fishery Management	PBR	Potential biological removal
CP	Catcher/processor	PIBKC	Pribilof Is. blue king crab
CPUE	Catch per unit effort	PIGKC	Pribilof Is. golden king crab
CPT	Crab Plan Team	PIRKC	Pribilof Is. red king crab
CSA	Catch survey analysis	PPA	Preliminary Preferred Alternative
CV	Catcher vessel	PSC	Prohibited species catch
CVRF	Coastal Villages Region Fund	PRA	Paperwork Reduction Act
DPS	Distinct population segment	PSEIS	Programmatic Supplemental
E.O.	Executive Order	RFA	Regulatory Flexibility Act
EA	Environmental Assessment	RFFA	Reasonably foreseeable future action
EBS	Eastern Bering Sea	RIR	Regulatory Impact Review
EEZ	Exclusive Economic Zone	RPA	Reasonable and prudent alternative
EFH	Essential fish habitat	S-R	Stock-recruitment
EIS	Environmental Impact Statement	SAFE	Stock Assessment and Fishery Evaluation
ESA	Endangered Species Act	SAR	Swept area ratio
ESB	Estimated Spawning Biomass	SBA	Small Business Act
ESP	Ecosystem and Socioeconomic Profile	Secretary	Secretary of Commerce
ESU	Endangered species unit	SHS	State Harvest Strategy
F	Fishing mortality rate	SMBKC	Saint Matthew Island blue king crab
FAR	Fraction of Attributable Risk	SMIHCA	Saint Matthew Island Habitat
FAST	Fisheries, Aquatic Sciences & Technology	SSC	Scientific and Statistical Committee
FE	Fishing Effects	SST	Sea surface temperature
FMA	Fisheries Monitoring and Analysis	TAC	Total allowable catch
FMP	Fishery management plan	T _{min}	Minimum time to rebuild
FONSI	Finding of No Significant Impact	T _{max}	Maximum time to rebuild
FR	Federal Register	U.S.	United States
FRFA	Final Regulatory Flexibility Analysis	USCG	United States Coast Guard
ft	Foot or feet	USFWS	United States Fish and Wildlife Service
GHL	Guideline harvest level	WAIRKC	Western Aleutian Is. red king crab
GOA	Gulf of Alaska	YDFDA	Yukon Delta Fisheries Development
IRFA	Initial Regulatory Flexibility Analysis		
IPA	Incentive Plan Agreement		
lb(s)	Pound(s)		
LLP	License limitation program		
LOA	Length overall		
m	Meter or meters		
M	Natural mortality rate		

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Executive Summary

This draft Environmental Assessment (EA) analyzes an amendment to the BSAI Crab FMP to rebuild the Eastern Bering Sea (EBS) snow crab (*Chionoecetes opilio*) stock in compliance with section 304(e)(3) of the Magnuson-Stevens Fishery Conservation and Management Act (MSA). A range of alternative rebuilding time frames is considered based on whether or not directed fishing is permitted during stock recovery. The impacts of the alternatives considered on crab resources, fishery participants, habitat, and other components of the human environment are discussed in the analysis.

Purpose and Need

Pursuant to the MSA section 304(e)(4)(A) and the National Standard Guidelines, the purpose of this proposed action is to develop a rebuilding plan to prevent overfishing and to rebuild the EBS snow crab stock. Rebuilding should take place in as short a time as possible, taking into account the status and biology of any overfished stocks of fish, the needs of fishing communities, and the interaction of the overfished stock of fish within the marine ecosystem. This action is necessary to facilitate compliance with the requirements of the MSA to end and prevent overfishing, rebuild overfished stocks, and achieve optimum yield. The Council approved the following purpose and need statement in June 2022.

The Eastern Bering Sea snow crab stock was determined to be overfished on October 19, 2021, because the estimated mature male biomass is below the minimum stock size threshold specified in the crab fishery management plan (FMP). To comply with provisions of the Magnuson-Stevens Fishery Conservation and Management Act (MSA), a rebuilding plan must be implemented prior to the start of the 2023/2024 fishing season. This action is necessary to facilitate compliance with MSA requirements to rebuild overfished stocks and achieve optimum yield.

Alternatives

In June 2022, the Council provided EBS snow crab rebuilding alternatives for analysis. At the December 2022 meeting, the Council selected Alternative 2, option 2 as its preliminary preferred alternative (PPA), which would allow for a directed fishery to open based on the state harvest strategy while the stock is rebuilding.

Alternative 1: No Action

Alternative 2 (PPA): Adopt a rebuilding plan and specify a target rebuilding time not to exceed T_{MAX} , as recommended by the SSC. The stock will be considered “rebuilt” once it reaches B_{MSY} .

Option 1: No directed fishing until the stock is rebuilt, allow bycatch removals only

Option 2 (PPA): Allow bycatch removals and a directed snow crab fishery under the current State of Alaska harvest strategy

Environmental Assessment

The proposed action is to establish a rebuilding plan for EBS Snow Crab. The environmental assessment provided in this analysis is a comprehensive review of five resource components that may be affected by the proposed alternatives for implementing a rebuilding plan for EBS snow crab. In accordance with National Standard 1 (NS1) guidelines, continued monitoring of the progress made during rebuilding will be conducted on a biennial basis. Additionally, the status of the EBS snow crab stock will continue to be monitored through annual stock assessments.

Impacts on the snow crab population

Alternative 2 is projected to allow EBS snow crab to rebuild within ten years, as required under the MSA. The analysis relies on a projection model that takes into account what is known about the EBS snow crab stock and environmental conditions affecting its recovery. The model used to assess snow crab rebuilding was reviewed and approved by the SSC in October 2022. The minimum estimated time for rebuilding (T_{MIN}), taking into account the biology of the species and current environmental conditions, is 6 years, which under the National Standard 1 guidelines dictates a maximum time for rebuilding (T_{MAX}) of ten years. The main driver in speed of rebuilding for this stock is not fishing mortality, rather it is likely related to recruitment and the conditions that allow for increased recruitment into the population, such as the Arctic Oscillation and physical indicators including, but not limited to temperature, sea ice extent, resource availability, and predator prey relationships. Ecosystem conditions (section 3.4.1) may improve, and improvements would result in reduced natural mortality and increased production and will be monitored during rebuilding through ecosystem indicators. The allowance, in the projections, for recruitment to eventually increase and contribute to stock growth assumes that existing ecosystem conditions or other constraints on production will not continue indefinitely. However, if recruitment remains at low levels, the population may take substantially longer to show rebuilding progress and may never reach B_{MSY} .

Alternative 2 includes two options, which either prohibit a directed snow crab fishery, with bycatch removals only (option 1) or allow it to occur, subject to the current State of Alaska harvest strategy (option 2). Because fishing mortality is not the primary driver of the current snow crab population status, either option does not substantively change the projection of T_{MIN} , and under both options it is assumed that the stock would rebuild within ten years.

Impacts of snow crab bycatch in other fisheries on rebuilding

Non-target bycatch of snow crab in other crab and groundfish fisheries will not substantially affect rebuilding time, based on model projections. In projections that apply average bycatch levels in all non-target fisheries (crab and groundfish) during rebuilding, the median time for stock recovery was not differentiable from the no fishing scenario. To account for unobserved mortality, the rebuilding projections also simulated scenarios with 5x and 100x the level of observed bycatch. Even in these projections, there was a minimal difference in median rebuilding time under each scenario. Therefore, analysts conclude that recovery of the EBS snow crab stock is likely to not be affected by current or predicted bycatch levels, based on average historical bycatch.

Under either option of Alternative 2, bycatch removals may continue, and all of the existing measures in other directed crab fisheries or groundfish fisheries that minimize non-target impacts on EBS snow crab, including a prohibited species catch (PSC) limit and closure area for the non-pelagic and pelagic trawl groundfish fisheries, are expected to be maintained throughout the rebuilding period. As a result, no measures to modify EBS snow crab bycatch management in the groundfish fisheries are included in this rebuilding analysis.

Impacts on other environmental components

Under either option of the rebuilding plan alternative, directed fishing for EBS snow crab will either continue at a similar or at a reduced level compared to status quo. When a fishery is allowed, it will be prosecuted in space and time within the footprint that the fishery has established over recent years. As a result, there are likely no negative effects on habitat as a direct result of implementing a rebuilding plan.

Minimal impacts on marine mammals would be observed as a result of implementing a rebuilding plan. However, there are potential implications for bearded seals as a result of the snow crab stock decline which may include: varying food web interactions and potential increased resource partitioning

interspecifically. The larger trophic level effects for bearded seals and snow crab likely require future work especially given the unprecedented state of snow crab, and the current climate mediated changes in the phenology of arctic sea ice.

Socioeconomic Impacts

Given that Alternative 1 is the status quo alternative and would not implement a rebuilding plan for EBS snow crab, the alternative is a violation of MSA 304(e)(4)(A) guidelines and should not be selected as the preferring alternative.

Relative to the Alternative 2/option 2 (the PPA), which would provide the possibility of a directed EBS snow crab fishery, the guaranteed loss of the directed EBS snow crab fishery during the six-year rebuilding period under Alternative 2/option 1 could result in severe impacts for those associated with the fishery. The assured loss of directed EBS snow crab for six years would represent substantial losses for vessel owners, crew, harvesting and processing quota shareholders, processors, CDQ groups, as well as the associated communities in addition to the communities where the shorebased processors are located. When combined with the potential of annual directed fishery closures of the BBRKC fishery, the socioeconomic impacts to the participants of the EBS snow crab fishery could be accentuated even further under Alternative 2/option 1. Under Alternative 2/option 2 (the PPA), which allows for a directed fishing during the six-year rebuilding period, the socioeconomic impacts of the rebuilding plan could be somewhat less but still substantial. The addition of a directed fishery under Alternative 2/option 2 (the PPA) for vessel owners, crew, and quota shareholders would provide critical ex-vessel payments from the harvesting of EBS snow crab that would keep the vessel and its crew active in the EBS snow crab fishery which would also provide benefits downstream for vessel owners, crew, quota shareholders, and the communities these participants reside. In addition, a directed fishery would provide valuable operating revenue to allow for paying processor wages, processor expenditures of goods and services in the local community, expenditures by processor workers in the communities, as well as remittances of wages to primary residences supported by processing workers who are working away from home. A directed fishery would also allow for CDQ earnings to continue to fund CDQ programs which in turn would continue the socioeconomic benefits to the CDQ communities. If, however, despite the projected removals noted in Table 3-6 under Alternative 2/option 2 (the PPA), the stock cannot support a directed fishery based on the State of Alaska harvest strategy, then the socioeconomic impacts under Alternative 2/option 2 (the PPA) could be similar or identical to the impacts under Alternative 2/option 1.

Major changes from December 2022

Provided below is a summary of changes to the document following the December 2022 Council review:

- Addition of projected removals based off the model projections (time frame 1982-2017). (SSC request)
 - Projected removals were used to inform the status of EBS snow crab throughout rebuilding and to evaluate potential economic effects throughout the rebuilding time frame as seen in section 3.2 and 3.6.
- Preliminary estimated gross exvessel and first wholesale revenue and price data were provided for the 2021/2022 EBS snow crab fishery as seen in Table 3-13. However, since the 2022 revenue and price data are estimates and not the official ADF&G COAR price data and CFEC Fish Ticket data for the 2022 year are not yet available, information on fishing dependent communities through the 2022 fishing year is not provided in the SIA tables. The existing SIA tables do show information on fishing dependent communities through the 2021 fishing year. (SSC request)
- Available information on CDQ ownership and participation was provided in Table 3-35 and Table 3-36. (SSC request)
- Alternative 1 was addressed and analyzed for all resource components in the section on expected effects of the alternatives. (Council request)

1 Introduction

This draft EA analyzes an amendment to the Bering Sea and Aleutian Island (BSAI) Crab FMP to rebuild the Eastern Bering Sea snow crab (*Chionoecetes opilio*) stock in compliance with section 304(e)(3) of the MSA. A range of alternative rebuilding time frames is considered based on whether or not directed fishing is permitted during stock recovery. The impacts of the alternatives considered on crab resources, fishery participants, habitat, and other components of the human environment are discussed in the analysis.

1.1 Purpose and Need

Pursuant to the MSA section 304(e)(4)(A) and the National Standard Guidelines, the purpose of this proposed action is to develop a rebuilding plan to prevent overfishing and to rebuild the EBS snow crab stock. Rebuilding time should be as short a time as possible, taking into account the status and biology of any overfished stocks of fish, the needs of fishing communities, and the interaction of the overfished stock of fish within the marine ecosystem. This action is necessary to facilitate compliance with the requirements of the MSA to end and prevent overfishing, rebuild overfished stocks, and achieve optimum yield. The Council approved the following purpose and need statement in June 2022.

The Eastern Bering Sea snow crab stock was determined to be overfished on October 19, 2021, because the estimated mature male biomass is below the minimum stock size threshold specified in the crab FMP. To comply with provisions of the Magnuson-Stevens Fishery Conservation and Management Act (MSA), a rebuilding plan must be implemented prior to the start of the 2023/2024 fishing season. This action is necessary to facilitate compliance with MSA requirements to rebuild overfished stocks and achieve optimum yield.

1.2 Requirements of the Magnuson-Stevens Act and National Standard Guidelines

This EA addresses alternatives for rebuilding the EBS snow crab stock as required under the MSA. Additionally, this action must be consistent with the ten National Standards of the MSA section 301(a)(1); fishery management plan provisions 303(a)(10) and 303(a)(14); rebuilding overfished fisheries 304(e); and national standard guidelines 50 CFR 600.310.

The EBS snow crab stock was declared overfished on October 19, 2021, because estimated mature male biomass was below the minimum stock size threshold specified in the crab FMP. Section 304(e)(3) of the MSA requires the Council and Secretary of Commerce (Secretary) to develop and implement a rebuilding plan within two years of receiving notification from the Secretary that the stock is overfished, approaching an overfished condition, or has not made adequate progress towards rebuilding.

In order to comply with provisions of the MSA, the EBS snow crab rebuilding plan must be implemented prior to the start of the 2023/2024 fishing season.

1.2.1 Magnuson-Stevens Act Language on Rebuilding Overfished Stocks

Rebuilding of overfished stocks is required by the MSA section 304. The applicable section of the Act is provided below.

(e) REBUILDING OVERFISHED FISHERIES--

- (1) The Secretary shall report annually to the Congress and the Councils on the status of fisheries within each Council's geographical area of authority and identify those fisheries that are overfished or are approaching a condition of being overfished. For those fisheries

managed under a fishery management plan or international agreement, the status shall be determined using the criteria for overfishing specified in such plan or agreement. A fishery shall be classified as approaching a condition of being overfished if, based on trends in fishing effort, fishery resource size, and other appropriate factors, the Secretary estimates that the fishery will become overfished within two years.

(2) If the Secretary determines at any time that a fishery is overfished, the Secretary shall immediately notify the appropriate Council and request that action be taken to end overfishing in the fishery and to implement conservation and management measures to rebuild affected stocks of fish. The Secretary shall publish each notice under this paragraph in the Federal Register.

(3) Within two years of an identification under paragraph (1) or notification under paragraphs (2) or (7), the appropriate Council (or the Secretary, for fisheries under section 302(a)(3)) shall prepare a fishery management plan, plan amendment, or proposed regulations for the fishery to which the identification or notice applies--

(A) to end overfishing in the fishery and to rebuild affected stocks of fish; or

(B) to prevent overfishing from occurring in the fishery whenever such fishery is identified as approaching an overfished condition.

(4) For a fishery that is overfished, any fishery management plan, amendment, or proposed regulations prepared pursuant to paragraph (3) or paragraph (5) for such fishery shall--

(A) specify a time period for ending overfishing and rebuilding the fishery that shall--

(i) be as short as possible, taking into account the status and biology of any overfished stocks of fish, the needs of fishing communities, recommendations by international organizations in which the United States participates, and the interaction of the overfished stock of fish within the marine ecosystem; and

(ii) not exceed 10 years, except in cases where the biology of the stock of fish, other environmental conditions, or management measures under an international agreement in which the United States participates dictate otherwise;

(B) allocate both overfishing restrictions and recovery benefits fairly and equitably among sectors of the fishery; and

(C) for fisheries managed under an international agreement, reflect traditional participation in the fishery, relative to other nations, by fishermen of the United States.

(5) If, within the 2-year period beginning on the date of identification or notification that a fishery is overfished, the Council does not submit to the Secretary a fishery management plan, plan amendment, or proposed regulations required by paragraph (3)(A), the Secretary shall prepare a fishery management plan or plan amendment and any accompanying regulations to stop overfishing and rebuild affected stocks of fish within 9 months under subsection (c).

(6) During the development of a fishery management plan, a plan amendment, or proposed regulations required by this subsection, the Council may request the Secretary to implement interim measures to reduce overfishing under section 305(c)) until such measures can be replaced by such plan, amendment, or regulations. Such measures, if otherwise in compliance with the provisions of this Act, may be implemented even though they are not sufficient by themselves to stop overfishing of a fishery.

(7) The Secretary shall review any fishery management plan, plan amendment, or regulations required by this subsection at routine intervals that may not exceed two years. If the Secretary finds as a result of the review that such plan, amendment, or regulations have not resulted in adequate progress toward ending overfishing and rebuilding affected fish stocks, the Secretary shall--

(A) in the case of a fishery to which section 302(a)(3) applies, immediately make revisions necessary to achieve adequate progress; or

(B) for all other fisheries, immediately notify the appropriate Council. Such notification shall recommend further conservation and management measures which the Council should consider under paragraph (3) to achieve adequate progress.

1.2.2 National Standard 1 Guidelines

Further clarification on stock rebuilding under the MSA for National Standard 1 (NS1) is provided in the excerpt below from the Final Rule on National Standard Guidelines published in the Federal Register on October 18, 2016 (81 FR 71858) and available on the NOAA Fisheries website:

<https://www.fisheries.noaa.gov/national/laws-and-policies/national-standard-guidelines>

Sec. 600.310 National Standard 1— Optimum Yield.

(j) *Council actions to address overfishing and rebuilding for stocks and stock complexes—*

(1) *Notification.* The Secretary will immediately notify in writing a Regional Fishery Management Council whenever the Secretary determines that:

(i) Overfishing is occurring;

(ii) A stock or stock complex is overfished;

(iii) A stock or stock complex is approaching an overfished condition; or

(iv) Existing remedial action taken for the purpose of ending previously identified overfishing or rebuilding a previously identified overfished stock or stock complex has not resulted in adequate progress (see MSA section 304(e)).

(2) *Timing of actions—*

(i) *If a stock or stock complex is undergoing overfishing.* Upon notification that a stock or stock complex is undergoing overfishing, a Council should immediately begin working with its science and statistical committee (SSC) (or agency scientists or peer review processes in the case of Secretariially-managed fisheries) to ensure that the ABC is set appropriately to end overfishing. Councils should evaluate the cause of overfishing, address the issue that caused overfishing, and reevaluate their ACLs and AMs to make sure they are adequate.

(ii) *If a stock or stock complex is overfished or approaching an overfished condition.* Upon notification that a stock or stock complex is overfished or approaching an

overfished condition, a Council must prepare and implement an FMP, FMP amendment, or proposed regulations within two years of notification, consistent with the requirements of section 304(e)(3) of the Magnuson-Stevens Act. Council actions should be submitted to NMFS within 15 months of notification to ensure sufficient time for the Secretary to implement the measures, if approved.

(3) *Overfished fishery.*—

(i) Where a stock or stock complex is overfished, a Council must specify a time period for rebuilding the stock or stock complex based on factors specified in Magnuson-Stevens Act section 304(e)(4). This target time for rebuilding (T_{target}) shall be as short as possible, taking into account: The status and biology of any overfished stock, the needs of fishing communities, recommendations by international organizations in which the U.S. participates, and interaction of the stock within the marine ecosystem. In addition, the time period shall not exceed 10 years, except where biology of the stock, other environmental conditions, or management measures under an international agreement to which the U.S. participates, dictate otherwise. SSCs (or agency scientists or peer review processes in the case of Secretarial actions) shall provide recommendations for achieving rebuilding targets (*see* Magnuson-Stevens Act section 302(g)(1)(B)). The above factors enter into the specification of T_{target} as follows:

(A) *The minimum time for rebuilding a stock (T_{min}).* T_{min} means the amount of time the stock or stock complex is expected to take to rebuild to its MSY biomass level in the absence of any fishing mortality. In this context, the term “expected” means to have at least a 50 percent probability of attaining the B_{MSY} where such probabilities can be calculated. The starting year for the T_{min} calculation should be the first year that the rebuilding plan is expected to be implemented.

(B) *The maximum time for rebuilding a stock or stock complex to its B_{MSY} (T_{max}).*

(1) If T_{min} for the stock or stock complex is 10 years or less, then T_{max} is 10 years.

(2) If T_{min} for the stock or stock complex exceeds 10 years, then one of the following methods can be used to determine T_{max} :

(i) T_{min} plus the length of time associated with one generation time for that stock or stock complex. “Generation time” is the average length of time between when an individual is born and the birth of its offspring,

(ii) The amount of time the stock or stock complex is expected to take to rebuild to B_{MSY} if fished at 75 percent of MFMT, or

(iii) T_{min} multiplied by two.

(3) In situations where T_{min} exceeds 10 years, T_{max} establishes a maximum time for rebuilding that is linked to the biology of the stock. When selecting a method for determining T_{max} , a Council, in consultation with its SSC, should consider the relevant biological data and scientific uncertainty of that data, and must provide a rationale for its decision based on the best scientific information available. One of the methods listed in subparagraphs (j)(3)(i)(B)(2)(ii) and (iii) may be appropriate, for example, if given data availability and the life history

characteristics of the stock, there is high uncertainty in the estimate of generation time, or if generation time does not accurately reflect the productivity of the stock.

(C) *Target time to rebuilding a stock or stock complex (T_{target}).* T_{target} is the specified time period for rebuilding a stock that is considered to be as short a time as possible, taking into account the factors described in paragraph (j)(3)(i) of this section. T_{target} shall not exceed T_{max} , and the fishing mortality associated with achieving T_{target} is referred to as F_{rebuild} .

(ii) Council action addressing an overfished fishery must allocate both overfishing restrictions and recovery benefits fairly and equitably among sectors of the fishery.

(iii) For fisheries managed under an international agreement, Council action addressing an overfished fishery must reflect traditional participation in the fishery, relative to other nations, by fishermen of the United States.

(iv) *Adequate Progress.* The Secretary shall review rebuilding plans at routine intervals that may not exceed two years to determine whether the plans have resulted in adequate progress toward ending overfishing and rebuilding affected fish stocks (MSA section 304(e)(7)). Such reviews could include the review of recent stock assessments, comparisons of catches to the ACL, or other appropriate performance measures. The Secretary may find that adequate progress is not being made if F_{rebuild} or the ACL associated with F_{rebuild} is exceeded, and AMs are not correcting the operational issue that caused the overage, nor addressing any biological consequences to the stock or stock complex resulting from the overage when it is known (see paragraph (g)(3) of this section). A lack of adequate progress may also be found when the rebuilding expectations of a stock or stock complex are significantly changed due to new and unexpected information about the status of the stock. If a determination is made under this provision, the Secretary will notify the appropriate Council and recommend further conservation and management measures, and the Council must develop and implement a new or revised rebuilding plan within two years (see MSA sections 304(e)(3) and (e)(7)(B)). For Secretariats-managed fisheries, the Secretary would take immediate action necessary to achieve adequate progress toward rebuilding and ending overfishing.

(v) While a stock or stock complex is rebuilding, revising rebuilding timeframes (i.e., T_{target} and T_{max}) or F_{rebuild} is not necessary, unless the Secretary finds that adequate progress is not being made.

(vi) If a stock or stock complex has not rebuilt by T_{max} , then the fishing mortality rate should be maintained at its current F_{rebuild} or 75 percent of the MFMT, whichever is less, until the stock or stock complex is rebuilt or the fishing mortality rate is changed as a result of the Secretary finding that adequate progress is not being made.

(4) *Emergency actions and interim measures.* If a Council is developing a rebuilding plan or revising an existing rebuilding plan due to a lack of adequate

progress (*see* MSA section 304(e)(7)), the Secretary may, in response to a Council request, implement interim measures that reduce, but do not necessarily end, overfishing (*see* MSA section 304(e)(6)) if all of the following criteria are met:

- (i) The interim measures are needed to address an unanticipated and significantly changed understanding of the status of the stock or stock complex;
- (ii) Ending overfishing immediately is expected to result in severe social and/or economic impacts to a fishery; and
- (iii) The interim measures will ensure that the stock or stock complex will increase its current biomass through the duration of the interim measures.

(5) *Discontinuing a rebuilding plan based on new scientific information.* A Council may discontinue a rebuilding plan for a stock or stock complex before it reaches B_{MSY} if the Secretary determines that the stock was not overfished in the year that the overfished determination (*see* MSA section 304(e)(3)) was based on and has never been overfished in any subsequent year including the current year.

(k) *International overfishing.* If the Secretary determines that a fishery is overfished or approaching a condition of being overfished due to excessive international fishing pressure, and for which there are no management measures (or no effective measures) to end overfishing under an international agreement to which the United States is a party, then the Secretary and/or the appropriate Council shall take certain actions as provided under Magnuson-Stevens Act section 304(i). The Secretary, in cooperation with the Secretary of State, must immediately take appropriate action at the international level to end the overfishing. In addition, within one year after the determination, the Secretary and/or appropriate Council shall:

(1) Develop recommendations for domestic regulations to address the relative impact of the U.S. fishing vessels on the stock. Council recommendations should be submitted to the Secretary.

(2) Develop and submit recommendations to the Secretary of State, and to the Congress, for international actions that will end overfishing in the fishery and rebuild the affected stocks, taking into account the relative impact of vessels of other nations and vessels of the United States on the relevant stock. Councils should, in consultation with the Secretary, develop recommendations that take into consideration relevant provisions of the Magnuson-Stevens Act and NS1 guidelines, including section 304(e) of the Magnuson-Stevens Act and paragraph (j)(3)(iii) of this section, and other applicable laws. For highly migratory species in the Pacific, recommendations from the Western Pacific, North Pacific, or Pacific Councils must be developed and submitted consistent with Magnuson-Stevens Reauthorization Act section 503(f), as appropriate.

(3) *Considerations for assessing “relative impact.”* “Relative impact” under paragraphs (k)(1) and (2) of this section may include consideration of factors that include, but are not limited to: Domestic and international management measures already in place, management history of a given nation, estimates of a nation's landings or catch (including bycatch) in a given fishery, and estimates of a nation's mortality contributions in a given fishery. Information used to determine relative impact must be based upon the best available scientific information.

(l) *Exceptions to requirements to prevent overfishing.* Exceptions to the requirement to prevent overfishing could apply under certain limited circumstances. Harvesting one stock at its optimum level may result in overfishing of another stock when the two stocks tend to be caught together (This can occur when the two stocks are part of the same fishery or if one is bycatch in the other's fishery). Before a

Council may decide to allow this type of overfishing, an analysis must be performed and the analysis must contain a justification in terms of overall benefits, including a comparison of benefits under alternative management measures, and an analysis of the risk of any stock or stock complex falling below its minimum stock size threshold (MSST). The Council may decide to allow this type of overfishing if the fishery is not overfished and the analysis demonstrates that all of the following conditions are satisfied:

- (1) Such action will result in long-term net benefits to the Nation;
- (2) Mitigating measures have been considered and it has been demonstrated that a similar level of long-term net benefits cannot be achieved by modifying fleet behavior, gear selection/configuration, or other technical characteristics in a manner such that no overfishing would occur; and
- (3) The resulting rate of fishing mortality will not cause any stock or stock complex to fall below its MSST more than 50 percent of the time in the long term, although it is recognized that persistent overfishing is expected to cause the affected stock to fall below its B_{MSY} more than 50 percent of the time in the long term.

1.2.3 Process for Determining Status of BSAI Crab Stocks

Under the crab FMP, each crab stock is annually assessed by the CPT and SSC to determine its status regarding whether (1) overfishing is occurring or the rate or level of fishing mortality for the stock is approaching overfishing, (2) the stock is overfished, or the stock is approaching an overfished condition, and (3) the catch has exceeded the ACL. If overfishing occurred or the stock is overfished, the MSA requires the NPFMC to immediately end overfishing and/or develop a plan to rebuild affected stocks. Status determination criteria for crab stocks are calculated using a five-tier system described below (Table 1-1 and Table 1-2) that accommodates varying levels of uncertainty of information. The five-tier system incorporates new scientific information and provides a mechanism to continually improve the status determination criteria as new information becomes available. Under the five-tier system, overfishing and overfished criteria and ABC levels for most stocks are annually formulated. The ACL for each stock equals the ABC for that stock. For crab stocks, the Overfishing Level (OFL) equals the maximum sustainable yield (MSY) and overfishing is determined by comparing the OFL with the catch estimates for that crab fishing year. Catch includes all fishery removals, including retained catch and discard mortality in the directed and non-directed fisheries. Discard mortality is determined by multiplying the appropriate handling mortality rate by observer-based estimates of discards.

To determine stock status, a stock is first assigned to one of the five tiers based on the availability of information for that stock and model parameter choices are made. Tier assignments and model parameter choices are recommended through the CPT process to the SSC. The SSC recommends tier assignments, stock assessment and model structure, and parameter choices, including whether the information is "reliable," for the assessment authors to use for calculating OFL and ABC.

For Tiers 1 through 4, once a stock is assigned to a tier (EBS snow crab is in Tier 3), the determination of stock status level is based on recent survey data and assessment models, as available. The stock status level determines the equation (Table 1-1) used in calculating the F_{OFL} . Three levels of stock status are specified and denoted by "a," "b," and "c" and the F_{MSY} control rule assigns F_{OFL} according to stock status level (Table 1-1). At stock status level "a," current stock biomass exceeds the B_{MSY} . For stocks in status level "b," current biomass is less than B_{MSY} but greater than a level specified as the "critical biomass threshold" (β). In stock status level "c," the ratio of current biomass to B_{MSY} (or a proxy for B_{MSY}) is below β . At stock status level "c," directed fishing is prohibited and an F_{OFL} at or below F_{MSY} would be determined for all other sources of fishing mortality in the development of the rebuilding plan.

1.2.4 Status Determination for Tier 3 Crab Stocks

As noted in Table 1-1 and Table 1-2, Tier 3 is for stocks like EBS snow crab for which reliable estimates of the spawner/recruit relationship are not available, but proxies for F_{MSY} and B_{MSY} can be estimated. For Tier 3, a designation of the form " F_X " refers to the fishing mortality rate associated with an equilibrium level of fertilized egg production (or its proxy such as mature male biomass at mating) per recruit equal to $X\%$ of the equilibrium level in the absence of any fishing. There exists sufficient information for modeling that captures the essential population dynamics of the stock as well as the performance of the fisheries.

The OFL and ABC calculation accounts for all losses to the stock not attributable to natural mortality. The OFL and ABC are total catch limits comprised of three catch components: (1) non-directed fishery discard losses; (2) directed fishery discard losses; and (3) directed fishery retained catch. To determine the discard losses, the handling mortality rate is multiplied by bycatch discards in each fishery. Calculations for OFLs and ABCs are done by the stock assessment author. OFLs are calculated by applying the F_{OFL} and using the most recent abundance estimates. The assessment authors calculate the proposed ABCs by applying the ABC control rule to the proposed OFL. Overfishing would occur if, in any year, the sum of all three catch components exceeds the OFL.

Table 1-1 Five-Tier System for setting overfishing limits (OFLs) and Acceptable Biological Catches (ABCs) for crab stocks. The tiers are listed in descending order of information availability.

Information available	Tier	Stock status level	F_{OFL}	ABC control rule
B, B_{MSY}, F_{MSY} , and pdf of F_{MSY}	1	a. $\frac{B}{B_{msy}} > 1$	$F_{OFL} = \mu_A$ = arithmetic mean of the pdf	$ABC \leq (1-b_y) * OFL$
		b. $\beta < \frac{B}{B_{msy}} \leq 1$	$F_{OFL} = \mu_A \frac{\frac{B}{B_{msy}} - \alpha}{1 - \alpha}$	
		c. $\frac{B}{B_{msy}} \leq \beta$	Directed fishery $F = 0$ $F_{OFL} \leq F_{MSY}^\dagger$	
B, B_{MSY}, F_{MSY}	2	a. $\frac{B}{B_{msy}} > 1$	$F_{OFL} = F_{msy}$	$ABC \leq (1-b_y) * OFL$
		b. $\beta < \frac{B}{B_{msy}} \leq 1$	$F_{OFL} = F_{msy} \frac{\frac{B}{B_{msy}} - \alpha}{1 - \alpha}$	
		c. $\frac{B}{B_{msy}} \leq \beta$	Directed fishery $F = 0$ $F_{OFL} \leq F_{MSY}^\dagger$	
$B, F_{35\%}^*, B_{35\%}^*$	3	a. $\frac{B}{B_{35\%}^*} > 1$	$F_{OFL} = F_{35\%}^*$	$ABC \leq (1-b_y) * OFL$
		b. $\beta < \frac{B}{B_{35\%}^*} \leq 1$	$F_{OFL} = F_{35\%}^* \frac{\frac{B}{B_{35\%}^*} - \alpha}{1 - \alpha}$	
		c. $\frac{B}{B_{35\%}^*} \leq \beta$	Directed fishery $F = 0$ $F_{OFL} \leq F_{MSY}^\dagger$	
B, M, B_{msy}^{prox}	4	a. $\frac{B}{B_{msy}^{prox}} > 1$	$F_{OFL} = \gamma M$	$ABC \leq (1-b_y) * OFL$
		b. $\beta < \frac{B}{B_{msy}^{prox}} \leq 1$	$F_{OFL} = \gamma M \frac{\frac{B}{B_{msy}^{prox}} - \alpha}{1 - \alpha}$	
		c. $\frac{B}{B_{msy}^{prox}} \leq \beta$	Directed fishery $F = 0$ $F_{OFL} \leq F_{MSY}^\dagger$	
Stocks with no reliable estimates of biomass or M.	5		OFL = average catch from a time period to be determined, unless the SSC recommends an alternative value based on the best available scientific information.	$ABC \leq 0.90 * OFL$

*35% is the default value unless the SSC recommends a different value based on the best available scientific information.

† An $F_{OFL} \leq F_{MSY}$ will be determined in the development of the rebuilding plan for an overfished stock.

Table 1-2. A guide for understanding the five-tier system.

• F_{OFL} — the instantaneous fishing mortality (F) from the directed fishery that is used in the calculation of the overfishing limit (OFL). F_{OFL} is determined as a function of: ○ F_{MSY} — the instantaneous F that will produce MSY at the MSY-producing biomass ▪ A proxy of F_{MSY} may be used; e.g., $F_{x\%}$, the instantaneous F that results in x% of the equilibrium spawning per recruit relative to the unfished value ○ B — a measure of the productive capacity of the stock, such as spawning biomass or fertilized egg production. ▪ A proxy of B may be used; e.g., mature male biomass ○ B_{MSY} — the value of B at the MSY-producing level ▪ A proxy of B_{MSY} may be used; e.g., mature male biomass at the MSY-producing level ○ β — a parameter with restriction that $0 \leq \beta < 1$. ○ α — a parameter with restriction that $0 \leq \alpha \leq \beta$. • The maximum value of F_{OFL} is F_{MSY}. $F_{OFL} = F_{MSY}$ when $B > B_{MSY}$. • F_{OFL} decreases linearly from F_{MSY} to $F_{MSY} \cdot (\beta - \alpha) / (1 - \alpha)$ as B decreases from B_{MSY} to $\beta \cdot B_{MSY}$ • When $B \leq \beta \cdot B_{MSY}$, $F = 0$ for the directed fishery and $F_{OFL} \leq F_{MSY}$ for the non-directed fisheries, which will be determined in the development of the rebuilding plan. • The parameter, β, determines the threshold level of B at or below which directed fishing is prohibited. • The parameter, α, determines the value of F_{OFL} when B decreases to $\beta \cdot B_{MSY}$ and the rate at which F_{OFL} decreases with decreasing values of B when $\beta \cdot B_{MSY} < B \leq B_{MSY}$. ○ Larger values of α result in a smaller value of F_{OFL} when B decreases to $\beta \cdot B_{MSY}$. ○ Larger values of α result in F_{OFL} decreasing at a higher rate with decreasing values of B when $\beta \cdot B_{MSY} < B \leq B_{MSY}$. • The parameter, b_y, is the value for the annual buffer calculated from a P^* of 0.49 and a probability distribution for the OFL that accounts for scientific uncertainty in the estimate of OFL. • P^* is the probability that the estimate of ABC, which is calculated from the estimate of OFL, exceeds the “true” OFL (noted as OFL') ($P(ABC > OFL')$).
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1.3 Scope of the Analysis

This EA relies heavily on the information and analysis contained in previous NEPA documents and Stock Assessment and Fishery Evaluation (SAFE) Reports. The annual SAFE Reports contain the status of the EBS snow crab stock as well as annual stock assessments for all ten BSAI crab stocks. This EA also utilizes specific information and analysis contained in the Council’s the [2022 EBS snow crab SAFE Report](#) (NPFMC 2022),

Relevant information from these documents are summarized in the appropriate chapters.

This EA also incorporates information from Amendment 40 to the BSAI groundfish FMP (NPFMC 2009), which established the current *C. opilio* Bycatch Limitation Zone (COBLZ) and set the formula for establishing the annual prohibited species catch (PSC) limit.

This analysis further incorporates information contained in the Bering Sea Aleutian Islands Crab Fisheries Final Environmental Impact Statement/Regulatory Impact Review/Initial Regulatory Flexibility Analysis/Social Impact Assessment (Crab EIS) (NMFS 2004) by reference and Ten-Year Program

Review for the Crab Rationalization Management Program in the Bering Sea/Aleutian Islands (NPFMC 2017).

The Council on Environmental Quality regulations encourage agencies preparing NEPA documents to, “tier their environmental impact statements to eliminate repetitive discussions of the same issues and to focus on the actual issues ripe for decision at each level of environmental review.” Specifically, 40 CFR 1502.20 states the following:

Whenever a broad environmental impact statement has been prepared (such as a program or policy statement) and a subsequent statement or environmental assessment is then prepared on an action included within the entire program or policy (such as a site specific action) the subsequent statement or environmental assessment need only summarize the issues discussed in the broader statement and incorporate discussions from the broader statement by reference and shall concentrate on the issues specific to the subsequent action. (40 CFR 1502.20)

The preparation of a Regulatory Impact Review (RIR) is required under Presidential Executive Order (E.O.) 12866 (58 FR 51735, September 30, 1993). The requirements for all regulatory actions specified in E.O. 12866 are summarized in the following Statement from the E.O.:

In deciding whether and how to regulate, agencies should assess all costs and benefits of available regulatory alternatives, including the alternative of not regulating. Costs and Benefits shall be understood to include both quantifiable measures (to the fullest extent that these can be usefully estimated) and qualitative measures of costs and benefits that are difficult to quantify, but nevertheless essential to consider. Further, in choosing among alternative regulatory approaches, agencies should select those approaches that maximize net benefits (including potential economic, environmental, public health and safety, and other advantages; distributive impacts; and equity), unless a statute requires another regulatory approach.

Because neither alternative would involve any changes to federal regulations, the preparation of an RIR is not necessary and is not a part of this analysis.

1.4 History of the Action

EBS snow crab biomass was below the threshold in 2021 and was declared overfished by NMFS on October 19, 2021. At the June 2022 meeting, the Council put forth a [motion](#) that identified draft alternatives for analysis. Subsequently, at the October 2022 meeting, the SSC reviewed several recommended model scenarios to aid in establishing model parameters and the selection of T_{min} and T_{max} , which establish the minimum and maximum time limits for rebuilding the snow crab stock under the rebuilding plan. The SSC recommended the rebuilding parameters of $T_{min}=6$ years, $T_{max}=10$, using the recruitment and mortality time frame scenarios from 1982-2017.

In addition, the Council requested, in their June 2022 motion, that additional information to help determine whether the following bycatch management measures would affect the rebuilding timeline:

- Remove the snow crab PSC floor
- Count all trawl PSC throughout the full range of the stock toward the PSC limit
- Limit on fixed gear PSC

At the December 2022 meeting, the Council reviewed the initial review analysis to rebuild the EBS snow crab stock in compliance with section 304(e)(3) of the Magnuson-Stevens Fishery Conservation and Management Act. The impacts of the alternatives considered on EBS snow crab stock, fishery participants, habitat, and other components of the human environment are described in the analysis.

After reviewing the initial review draft, the Council recommended scheduling the analysis for final action and selected Alternative 2, option 2 as their preliminary preferred alternative, which would allow for a directed fishery to open based on the state harvest strategy while the stock is rebuilding. In speaking to the motion, the Council noted that the preliminary preferred alternative is consistent with the mandate under the Magnuson Stevens Act to rebuild stocks and is also consistent with National Standard 1 in that it would establish a plan estimated to rebuild the stock in less than 10 years but allows directed fishing to occur under the state harvest strategy while the stock rebuilds to a level that achieves optimum yield. Maintaining the economic opportunity for a directed commercial fishery under the state harvest strategy is essential for harvesters, processors, and communities. Economic opportunities prove to be of grave importance during a time when a majority of commercial crab stocks are in a state of decline, and future openings are likely to be limited and/or closed. Selecting this option as the preliminary preferred alternative does not extend the proposed rebuilding timeline.

1.5 EBS Snow Crab Management

The analyses presented here consider the activities of directed EBS snow crab fisheries as well as other fisheries and non-fishing activities with the potential to impact the EBS snow crab stock. None of the actions to rebuild the EBS snow crab stock would redefine the fishery management unit in the FMP or introduce new or modified spatial management measures affecting EBS snow crab.

The Crab FMP in the BSAI was approved by the Secretary of Commerce on June 2, 1989. The FMP establishes a state/federal cooperative management regime that defers crab management to the State of Alaska with federal oversight. The Crab FMP divides management measures into three categories: (1) fixed in the Crab FMP and require an amendment to change, (2) frameworked in the Crab FMP which the state can change as outlined in the FMP, and (3) discretion of the State of Alaska (Table 1-3). State regulations are subject to the provisions of the FMP, including its goals and objectives, the Magnuson-Stevens Act National Standards, and other applicable federal laws.

Table 1-3. Crab FMP management measures by category.

Category 1 (Fixed in FMP)	Category 2 (Frameworked in FMP)	Category 3 (Discretion of State)
Legal Gear	Minimum Size Limits	Reporting Requirements
Permit Requirements	Guideline Harvest Levels	Gear Placement and Removal
Federal Observer Requirements	In-season Adjustments	Gear Storage
Limited Access	Districts, Subdistricts and Sections	Vessel Tank Inspections
Norton Sound Superexclusive Registration	Fishing Seasons	Gear Modifications
Essential Fish Habitat	Sex Restrictions	Bycatch Limits (in crab fisheries)
Habitat Areas of Particular Concern	Pot Limits	State Observer Requirements
	Registration Areas	Other
	Closed Waters	

The FMP authorizes the State of Alaska to define spatial management boundaries for crab stocks within the management unit. Under the State of Alaska's spatial management scheme, the EBS snow crab management area (Figure 1-1) is the Eastern Subdistrict (waters east of 173° W. long.) and the Western Subdistrict (waters east of 173° W. long. but west of 165° W. long.). As for season dates for the directed fisheries, Alaska department of fish and game (ADF&G) typically establishes the snow crab season start date for October 15 and, in general, the snow crab fishery ends on May 15 for the Eastern subdistrict and May 31 for the Western subdistrict. In contrast, Federal regulations specify the groundfish seasons to begin January 1 and end December 31, and the total allowable catch (TAC)-setting and specifications process are designed around this schedule.

In 2005, the BSAI Crab Rationalization (CR) Program was implemented to ensure allocation of BSAI crab resources among harvesters, processors, and coastal communities. For further information detailing the CR program, please refer to section 3.6.1.

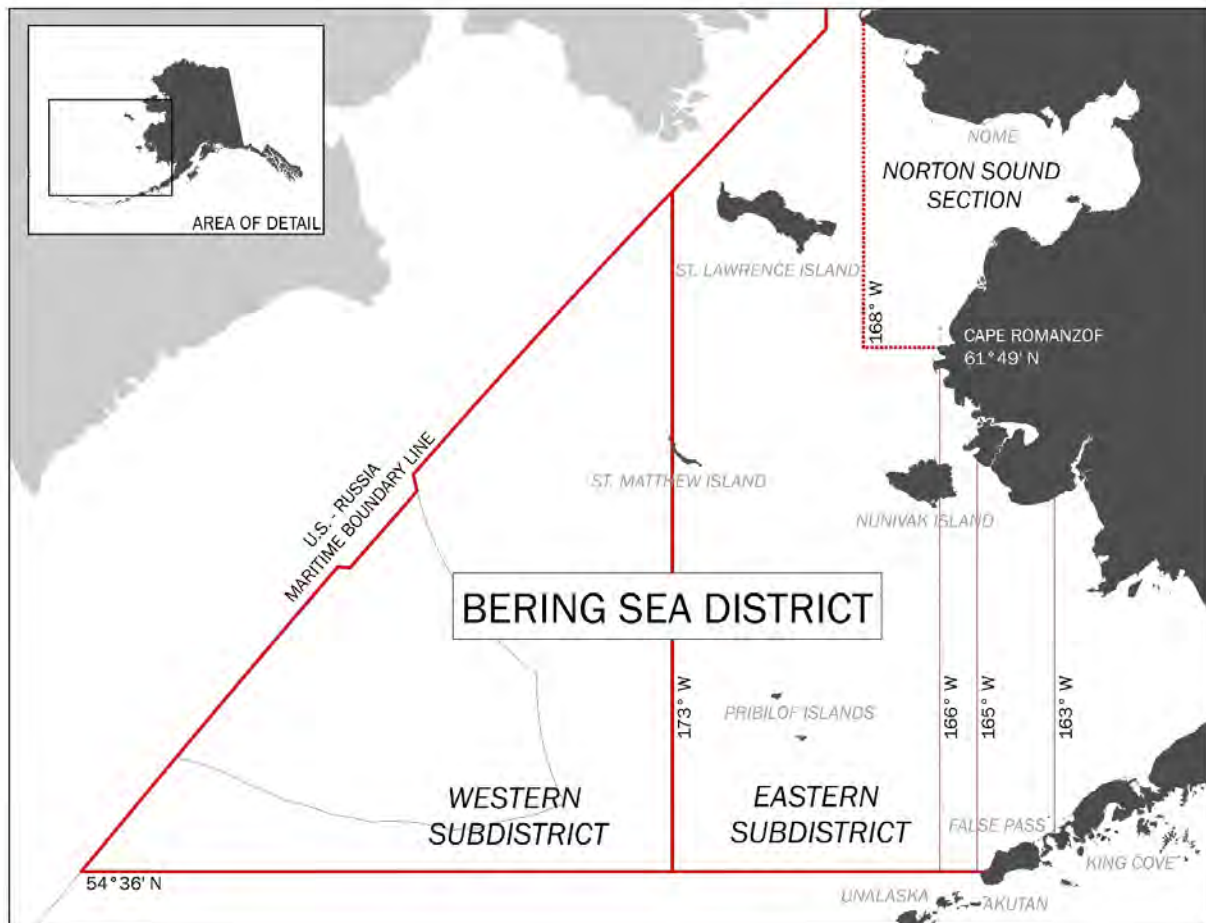


Figure 1-1 State of Alaska Bering Sea snow crab area which includes the Western Subdistrict and Eastern Subdistrict

2 Description of Alternatives

For all alternatives, the existing measures that minimize fishery impacts on EBS snow crab would be maintained throughout rebuilding. Additionally, since the alternatives propose to achieve rebuilding only by limiting direct harvest of EBS snow crab, no alternatives propose to create new management measures within the BSAI Crab FMP or any other FMP.

2.1 Alternative 1: No Action

Under Alternative 1, the Council would not develop a rebuilding plan, and no Federal management response to address an overfished stock would be undertaken. Importantly, taking no action to establish a rebuilding plan for an overfished stock is a violation of the MSA. Harvest under this alternative would continue to be defined by the State harvest strategy as in Alternative 2 / option 2 below. However, without a rebuilding plan, there would not be a mechanism to necessarily address operational issues that may constrain rebuilding, if they are present. Importantly, fishery management operations are not expected to constrain rebuilding under the existing ecosystem regime, which is the dominant limiting factor on stock productivity under any alternative. Alternative 1 is not feasible in order for the Council follow MSA 304(e)(4)(A) guidelines in establishing a rebuilding plan for an overfished stock. Therefore, in accordance with MSA guidelines, the Council has the option of selecting either proposed Alternative 2/option 1 or Alternative 2/option 2 (the PPA).

2.2 Alternative 2 (PPA): Establish a Rebuilding Plan

Under Alternative 2, a rebuilding plan would be established that would be consistent with the MSA and NS1 Guidelines on time for rebuilding, specifically rebuilding within a time (T_{target}) that is as short as possible, taking into account the status and biology of any overfished stocks of fish, the needs of fishing communities, recommendations by international organizations in which the United States participates, and the interaction of the overfished stock of fish within the marine ecosystems. The fastest rebuilding time (T_{min}), is calculated based on no fishing mortality ($F=0$). Based on projections described in section 2.2.1, the time with a greater than 50% probability of rebuilding to B_{MSY} at $F=0$ (T_{min}) is **6 years**. Because $T_{\text{min}} < 10$ years, $T_{\text{max}} = \mathbf{10 \text{ years}}$, based off of the guidelines outlined in NS1.

Under Alternative 2, a federal rebuilding plan for EBS snow crab will be incorporated into the BSAI King and Tanner Crab FMP by amending the FMP language to reflect the approved rebuilding alternative. The FMP authorizes the State to set preseason TACs and guideline harvest levels (GHLs) under State regulations, and seasons or areas are closed when the TAC is reached. The State must take into account the following factors, to the extent information is available, in developing harvest strategies or setting TACs (1) whether the ABC for that stock was exceeded in the previous year; (2) stock status relative to the OFL and ACL; (3) estimates of exploitable biomass; (4) estimates of recruitment; (5) estimates of thresholds; (6) market and other economic considerations; (7) additional uncertainty; and (8) any additional factors pertaining to the health and status of the stock or the marine ecosystem.

There are two options under Alternative 2 that consider alternative harvest scenarios during rebuilding:

Option 1: No directed fishing until the stock is rebuilt, allow bycatch removals only

Option 2 (PPA): Allow bycatch removals and a directed snow crab fishery under the current State of Alaska harvest strategy.

Under approval of Alternative 2/option 1, the FMP language would be amended to reflect a prohibition on directed harvest of EBS snow crab until the stock is declared rebuilt. This would prohibit the State of Alaska from setting a TAC under the State harvest strategy. Under Alternative 2/option 2 (the PPA), the amended FMP language could also: 1) include the specific formulae for opening the EBS snow crab

fishery provided in the current State harvest strategy or 2) state that no change in the harvest strategy should occur that would contribute to departure from the Council's intended rebuilding timeline.

For option 2, the State of Alaska's EBS snow crab harvest strategy (referred to as the "State harvest strategy" or SHS) is provided in the Alaska Administrative Code at 5 AAC 34.917 (below). The minimum legal size limit for snow crab is 78 mm, however, the snow crab market generally only accepts crab greater than 101 mm

The reader should note that B_{MSY} is defined differently for state and federal thresholds. As defined in the SAFE introduction, the federal definition for snow crab B_{MSY} is $B_{35\%}$, 35% of the long-term equilibrium biomass of morphometrically mature male crab in the absence of fishing, while the State defines B_{MSY} as the long-term average of total mature male and female biomass based on morphometric maturity (i.e., chela height in males and the abdominal flap in females). For the purposes of rebuilding, the threshold in determining rebuilt status, is defined in the federal assessment of B_{MSY} . Unless otherwise noted, the B_{MSY} defined in the analyses is referencing the federal definition.

Since 2000, the state harvest strategy sets harvest rate based on estimated mature biomass. The harvest rate scales with the status of the population relative to B_{MSY} , which is calculated as the average total mature biomass at the time of the survey from 1983 to 1997 and MSST is one half B_{MSY} . As noted in Figure 2-1, the harvest rate begins at 0.10 when total mature biomass exceeds 50% MSST (230 million lbs) and increases linearly to 0.225 when biomass is equal to or greater than B_{MSY} (Zheng et al. 2002). Where total mature biomass (TMB) and $TMB_{B_{MSY}}$ is the TMB associated with maximum sustainable yield. The maximum retained catch is set as the product of the exploitation rate, u , calculated from the above control rule and survey mature male biomass. If the retained catch in numbers is greater than 58% of the estimated number of new shell crabs greater than 101 mm plus 25% of the old shell crab greater than 101 mm, the catch is capped at 58%. Currently, the biological reference point for biomass is calculated using a spawning biomass per recruit proxy, $B_{35\%}$ (Clark, 1993). $B_{35\%}$ is the biomass at which spawning biomass per recruit is 35% of unfished levels and has been shown to provide close to maximum sustainable yield for a range of stock productivities (Clark, 1993). Consequently, it is an often used target when a stock recruit relationship is unknown or unreliable. The range of years of recruitment used to calculate biomass reference points is from 1982 to the present assessment year, minus 1.

As laid out under the BSAI Crab FMPs State/Federal cooperative management regime, the OFL and ABC for the Federal crab stocks are recommended to the Council by the Scientific and Statistical Committee (SSC).

The annual harvest levels and other management actions for the FMP crab stocks are determined by ADF&G according to State commercial fishery regulations. These regulations are established by the Alaska Board of Fisheries (BOF) and subject to the constraint that such harvest levels and management actions are consistent with provisions of the FMP, the National Standards of the MSA, and other applicable federal laws.

The FMP list out eight categories of factors the State of Alaska should take into account, to the extent information is available, in developing harvest strategies or setting TACs. This includes:

- (1) whether the ABC for that stock was exceeded in the previous year;
- (2) stock status relative to the OFL and ABC;
- (3) estimates of exploitable biomass;
- (4) estimates of recruitment;

- (5) estimates of thresholds;
- (6) market and other economic considerations;
- (7) additional uncertainty; and
- (8) any additional factors pertaining to the health and status of the stock or the marine ecosystem.

Additional uncertainty includes:

- (1) management uncertainty (i.e., uncertainty in the ability of managers to constrain catch so the ABC is not exceeded, and uncertainty in quantifying the true catch amount) and
- (2) scientific uncertainty identified and not already accounted for in the ABC (i.e., uncertainty in bycatch mortality, estimates of trends and absolute estimates of size composition, shell-condition, molt status, reproductive condition, spatial distribution, bycatch of non-target crab stocks, environmental conditions, fishery performance, fleet behavior, and the quality and amount of data available for these variables).

The FMP directs the State to establish an annual TAC for each crab stock at a level sufficiently below the ABC so that the sum of the total fishery removals and the State's assessment of additional uncertainty do not exceed the ABC. The State may establish the annual TACs below such a level to account for the other factors identified above. If an ABC is exceeded, the State will implement accountability measures in the fishing season following the overage to account for the overage through a downward adjustment to the TAC for that species by an amount sufficient to remedy the biological consequences of the overage.

Within these parameters laid out in the FMP, the State has further identified a process to establish annual harvest levels for each crab fishery. The process employed by the State begins with a review of stock status indicators derived from the recent assessments, including estimates of B_{MSY} (or its proxy), MSST, critical biomass threshold, and OFL (including a breakdown of the total OFL into subcomponents – estimates of future retained catch, discard mortality in directed fisheries, and non-target fishery bycatch). The State also relies on guidance provided in the annual NMFS stock status notification letter that is prepared for the Secretary of Commerce by the NMFS Alaska Region summarizing stock status relative to overfishing, OFLs for the 10 FMP crab stocks, and special concerns for stocks under rebuilding plans.

Annual biomass estimates of mature male biomass (MMB) provide a projection of stock status at the time of mating while the OFL estimate is a total catch level that may not be exceeded by the sum of all sources of fishing mortality. The OFL subcomponents provide additional information on the total catch OFL calculation for information relative to the directed fishing mortality estimate. The State has adopted harvest strategies for the crab fisheries which consist of rules in state regulation for computing TAC from survey and stock assessment data and identifying conditions under which the fishery would not open. Harvest strategy elements may include:

- a stock threshold for opening the fishery,
- rules for setting exploitation rate on abundance/biomass of mature-sized males,
- an exploitation rate dependent on stock index estimated from survey data,
- a cap on legal male exploitation rate, and
- a minimum TAC for fishery opening.

Both State harvest strategy thresholds and stock abundance or biomass estimates for computation of TACs reference stock biomass or abundance at the time of the NMFS trawl survey. State staff prepare annual assessments describing the requirements, process, and data needed to set TAC in manner that prevents overfishing. These assessments summarize stock status relative to OFL and document how the State sets TAC to account for uncertainty in stock biomass estimates and to ensure total removals remain below the ABC and OFL. The assessments are internal documents discussed with State, Federal, and Council staff during a series of teleconferences leading up to the announcement of TAC in early October. Details of the State TAC-setting process are publicly reviewed during an annual meeting with the BSAI crab industry after TACs are announced.

For EBS snow crab fishery to open, the preseason survey data must indicate that EBS snow crab is at least 25% of the B_{MSY} . The harvest strategy also includes thresholds for levels of exploitation based on different levels of estimated spawning biomass (ESB) relative to the B_{MSY} . The EBS snow crab harvest strategy was developed in 2002 (J. Zheng et al. 2002) and ADF&G uses the various abundance estimates available when implementing the harvest strategy. The State of Alaska's Bering Sea snow crab harvest strategy is provided in the Alaska Administrative Code at 5 AAC 35.517 (below).

5 AAC 35.517. Bering Sea *C. opilio* Tanner crab harvest strategy.

(a) In the Bering Sea District, the commercial *C. opilio* Tanner crab fishery may open only if the department's analysis of preseason survey data indicates the population of *C. opilio* Tanner crab

(1) contains an estimated spawning biomass of at least 25 percent of B_{MSY} ;

(2) repealed 6/10/2010.

(b) If the estimated spawning biomass of *C. opilio* Tanner crab is

(1) at least 25 percent of B_{MSY} , but less than B_{MSY} , the total allowable catch will be $(F_{MSY} / 3 + (B_t - 0.25 \times B_{MSY}) \times 0.417 \times F_{MSY} / (0.75 \times B_{MSY})) \times 100$ percent of the estimated mature male biomass or 58 percent of exploited legal males, whichever is less;

(2) at or above B_{MSY} , the total allowable catch will be $(0.75 \times F_{MSY}) \times 100$ percent of the estimated mature male biomass or 58 percent of the exploited legal males, whichever is less.

(c) In implementing this harvest strategy, the board directs the department to use the best scientific information available and to consider the reliability of estimates of *C. opilio* Tanner crab, the manageability of the fishery, and any other factors the department determines necessary to be consistent with the sustained yield principles.

(d) For the purposes of this section,

(1) " B_{MSY} " means the population biomass of mature male and female *C. opilio* Tanner crab that could produce maximum sustained yield under prevailing environmental conditions;

(2) " B_t " means the biomass of mature male and female *C. opilio* Tanner crab in a given year;

(3) "estimated mature male biomass" means the estimated biomass of all morphometrically mature male *C. opilio* Tanner crab;

(4) "estimated spawning biomass" means the estimated biomass of all morphometrically mature male *C. opilio* Tanner crab and all morphometrically mature female *C. opilio* Tanner crab;

(5) "exploited legal males" means 100 percent of the new-shell male *C. opilio* Tanner crab that are at least 102 millimeters (four inches) in width of shell, plus a percentage of old-shell male *C. opilio* Tanner crab that are at least 102 millimeters in width of shell estimated at the time of the survey; the percentage of old-shell male *C. opilio* Tanner crab will be based on the expected fishery selectivity for old-shell verses new-shell male *C. opilio* Tanner crab;

(6) "Fmsy" means the fishing mortality of the mature male *C. opilio* Tanner crab stock that could produce maximum sustained yield under prevailing environmental conditions.

There are various abundance estimates available for TAC-setting including raw survey area-swept estimates, model-based survey estimates, and model-based population estimates that account for survey selectivity. Because these estimates can vary greatly, the resulting TAC can vary depending which estimates are used as harvest strategy inputs. In a given year, it may be difficult to know which estimate is closer to the true population size and factors, such as those outlined in the FMP, are used when setting the TAC.

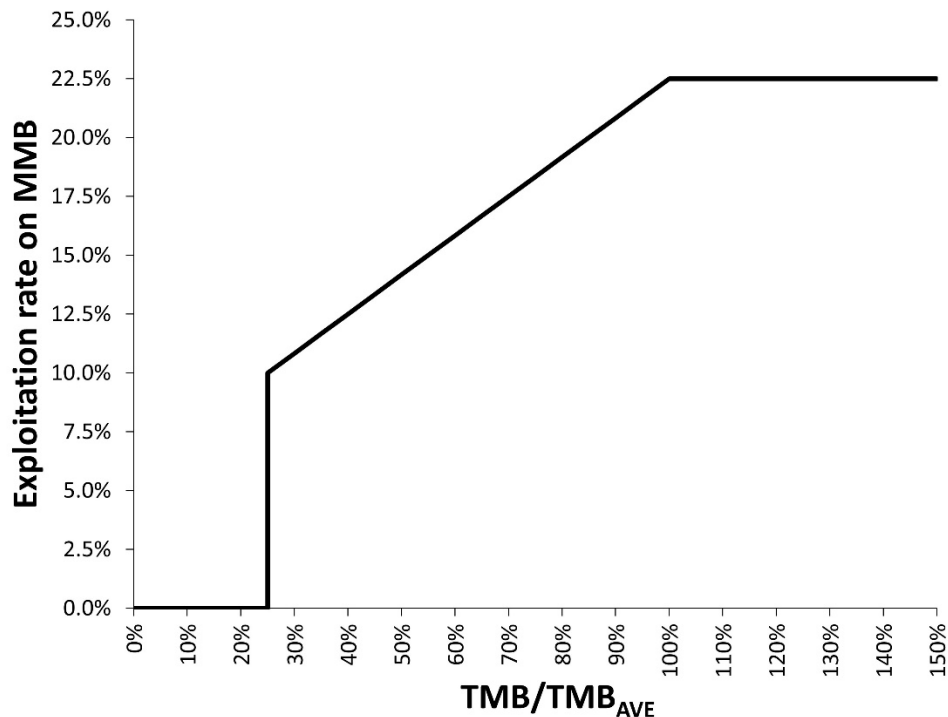


Figure 2-1. State harvest strategy for EBS snow crab.

Source: ADF&G

2.2.1 Rebuilding Projections

Quantitative estimation of rebuilding timelines under the alternatives was done through the SSC recommended modeled projections of population biomass. Each projection was identical in terms of starting population conditions, and multiple projection model parameterizations were explored to consider the range of stock productivity during rebuilding. The leading constraint on stock productivity is whether

future ecosystem conditions will be conducive to the successful recruitment of crab to the population. The protracted timeline for rebuilding under any alternative is a reflection of sub-optimal ecosystem conditions (summarized in section 3.4.1) both presently and throughout the rebuilding process.

2.2.2 Starting Population Conditions

All of the projections to estimate rebuilding times are consistent with regard to starting population conditions and the measure of biomass that defines a rebuilt stock, i.e., the end point for rebuilding. Population quantities and parameter values were based on the accepted 2022 stock assessment for EBS snow crab. The stock assessment uses the GMACS framework in order to facilitate an evaluation of model assumptions.

The projection model used for rebuilding projections was based on the SSC selected model 22.1a from the [2022 SAFE document](#). The population dynamics in this model track the number of crab of sex, s , maturity state, m , during year, y , at length, l , $N_{s,m,y,l}$. A terminal molt occurs in which crab move from an immature to a mature state, after which no further molting occurs. The mid-points of the size bins tracked in the model span from 27.5 to 132.5 mm carapace width, with 5 mm size classes. For the author-preferred model, 431 parameters were estimated. Parameters estimated within the assessment included those associated with the population processes recruitment, growth, natural mortality (subject to an informative prior and two years of additional 'mortality events' estimated in 2018 and 2019), fishing mortality, selectivity (fishery, survey, and BSFRF experiments), catchability, and maturity. Weight at length, discard mortality, bycatch mortality, and parameters associated with the proportion of recruitment allocated to size bin were estimated outside of the model or specified. See the assessment for a detailed description of the population dynamics model (Szuwalski, 2022).

Projections were performed by starting at the maximum likelihood estimates (MLE) of parameters in the population dynamics model. The MLE was identified by performing a jittering analysis in which the assessment was fit to the data starting from 100 parameter vectors randomly chosen given the specified prior distributions for all parameters. These are often uniform over the bounds for a given parameter, but some parameters have normal priors imposed (see assessment for description). Recruitment and natural mortalities were sampled from the estimated recruitments and natural mortalities based on user input range of years. Four future productivity scenarios were analyzed by crossing the periods 1982-2017 and 2005-2019 for sampling recruitment and natural mortality. The model was projected to 2040 in each of 2000 projections performed for each combination of recruitment and natural mortality scenarios.

Importantly, no stochasticity exists in the initial conditions of the projection—all projections start from the MLE. Furthermore, the projections models assume that the OFLs and ABCs are set with perfect estimates of the scale of the population, which have been historically uncertain.

Five fishing scenarios were performed within the productivity scenarios based on the SSC requests: zero fishing mortality, only bycatch mortality, an approximation of the State of Alaska's harvest with no bycatch, an approximation of the State of Alaska's harvest including bycatch, and the federally set acceptable biological catch (ABC).

Bycatch mortality was specified in the model as the average of the estimated bycatch fishing mortality over the last ten years. The State of Alaska's harvest control rule was approximated by averaging the ratio of the total allowable catch (TAC) set by the State and the ABC over the last 10 years (Daly, pers. comms.). The ratio (equal to 0.40) was used to scale the ABC calculated in the projections; the ABC was based on a 25% buffer of the OFL calculated using the current B_{MSY} proxy and F_{MSY} proxy ($B_{35\%}$ and $F_{35\%}$, respectively; see Szuwalski, 2022 for a detailed description of how these are calculated). Three proxies for B_{MSY} were calculated to evaluate rebuilding progress. These target biomasses correspond to the currently used $B_{35\%}$ (recruitment years 1982-2021), a target biomass calculated using expected

recruitment based on the years 1982-2017, and a target based on the recruitment estimates for the years 2005-2019. All biomass targets were calculated without incorporating the potential for mortality events to occur (i.e. the base estimate of natural mortality was used in projections).

2.2.2.1 Constraints on rebuilding from recruitment and natural mortality

Future recruitment and natural mortality are primary drivers of the rebuilding trajectory (in addition to fishing mortality) and two different productivity scenarios were explored for snow crab projections. Both productivity scenarios sampled the estimated recruitment and natural mortality from the stock assessment model from a pre-defined period of time to represent future conditions. The SSC selected the periods 1982-2017 (Figure 2-2) (to represent something similar to the current assumptions in the stock assessment, but without the possibility of mortality events) and 2005-2019 (Figure 2-3) (to represent recent environmental conditions and allow for mortality events to occur in the future).

Different assumptions about future recruitment and natural mortality have strong impacts on projected rebuilding trajectories and associated time to rebuilding. The average estimated recruitments of the period 1982-2017 and 2005-2019 are fairly similar, but when any mortality events are allowed in the projections, the ability of the stock to rebuild is hampered (compare Figure 2-2 to Figure 2-3; Table 2-1). Under no mortality events in the projection period (i.e. the 1982-2017 scenario) the median projected stock rebuilt by 2029 with no fishing mortality. The rebuilding time was extended to 2034 under no fishing if mortality events were imposed during 1 in 7 years (on average).

The CPT and SSC preferred the projection scenarios that draw recruitment and mortality from 1982-2017 (Figure 2-2), based on two factors. First, snow crab recruitment is projected to be lower in the future than it has been historically as a result of decreased sea ice extent in the Bering Sea (Szuwalski et al., 2021). The 1982-2017 scenario had the lowest average recruitment of the scenarios considered. Although this scenario was the lowest of the scenarios considered, the assessment author, CPT, and SSC allow for the possibility that even the lower recruitment scenario of the two reported could be too optimistic. Second, the mortality event that occurred in 2018 and 2019 appears to have been related to the unprecedented number of snow crab in the eastern Bering Sea and high bottom temperatures (section 3.2.3 and Appendix A to the [2022 snow crab SAFE](#)). Although bottom temperatures will likely be high at some point over the rebuilding period, crab densities will not be (and if they are high, the stock will have rebuilt).

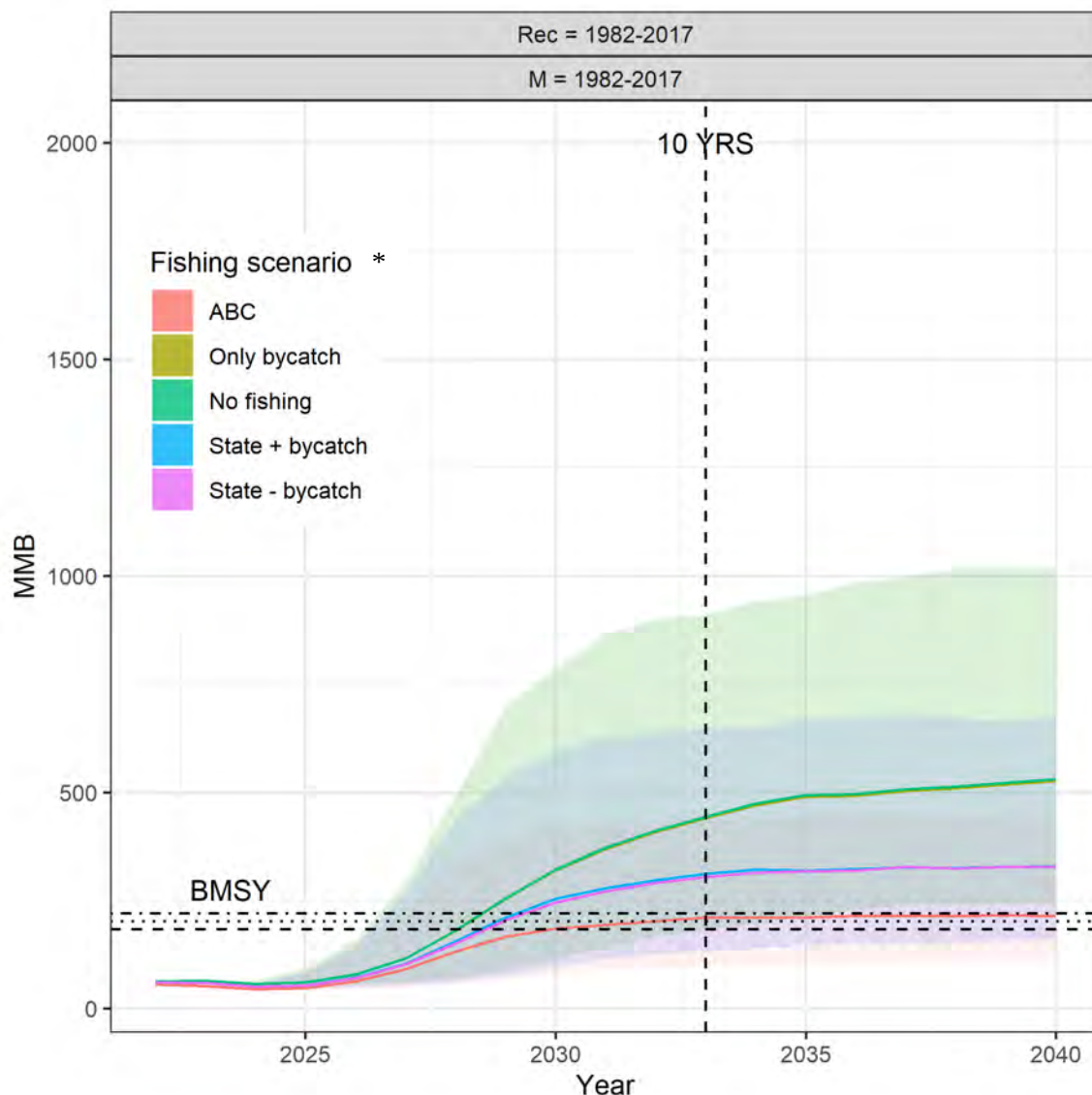
Consequently, the CPT and SSC recommended 1982-2017 as the preferred rebuilding scenario of the available options ([October 2022 SSC minutes](#)). Based on the SSC recommendations for the use of the recruitment and mortality scenarios from 1982-2017, T_{min} , under no fishing mortality=6, therefore T_{max} =10 (Table 2-1). Please note that under the 5 fishing mortality scenarios, the median projection times round to the same integer year (2029); however as shown in Table 2-1 there lies a difference between fishing mortality scenarios that is evidenced by the 5th and 95th inter-simulation quantiles for the uncertainty around biomass trajectories during the projection period. Thus, there is a slight difference in rebuilding projections influenced by different fishing mortality scenarios.

Table 2-1 Rebuilding time for scenarios described in the text. Median T_{min} is calculated relative to the status quo B_{MSY} proxy that uses average recruitment from 1982-2021. The '5%' and 95%' columns represent the 5th and 95th inter-simulation quantiles for the uncertainty around biomass trajectories during the projection period. "Inf" indicates that the stock never rebuilds to B_{MSY} .

<i>Projection specifications</i>					
Fishing Scenario*	Recruitment	Mortality	Median	5%	95%
<i>No fishing</i>	1982-2017	1982-2017	2029 (T_{min})	2027	2034
<i>Bycatch</i>	1982-2017	1982-2017	2029	2027	2034
<i>State + bycatch</i>	1982-2017	1982-2017	2029	2027	Inf
<i>State - bycatch</i>	1982-2017	1982-2017	2029	2027	Inf
<i>ABC</i>	1982-2017	1982-2017	2030	2027	Inf
<i>No fishing</i>	2005-2019	2005-2019	2034	2027	Inf
<i>bycatch</i>	2005-2019	2005-2019	2034	2027	Inf
<i>State + bycatch</i>	2005-2019	2005-2019	Inf	2027	Inf
<i>State - bycatch</i>	2005-2019	2005-2019	Inf	2027	Inf
<i>ABC</i>	2005-2019	2005-2019	Inf	2027	Inf

*Description of fishing scenarios:

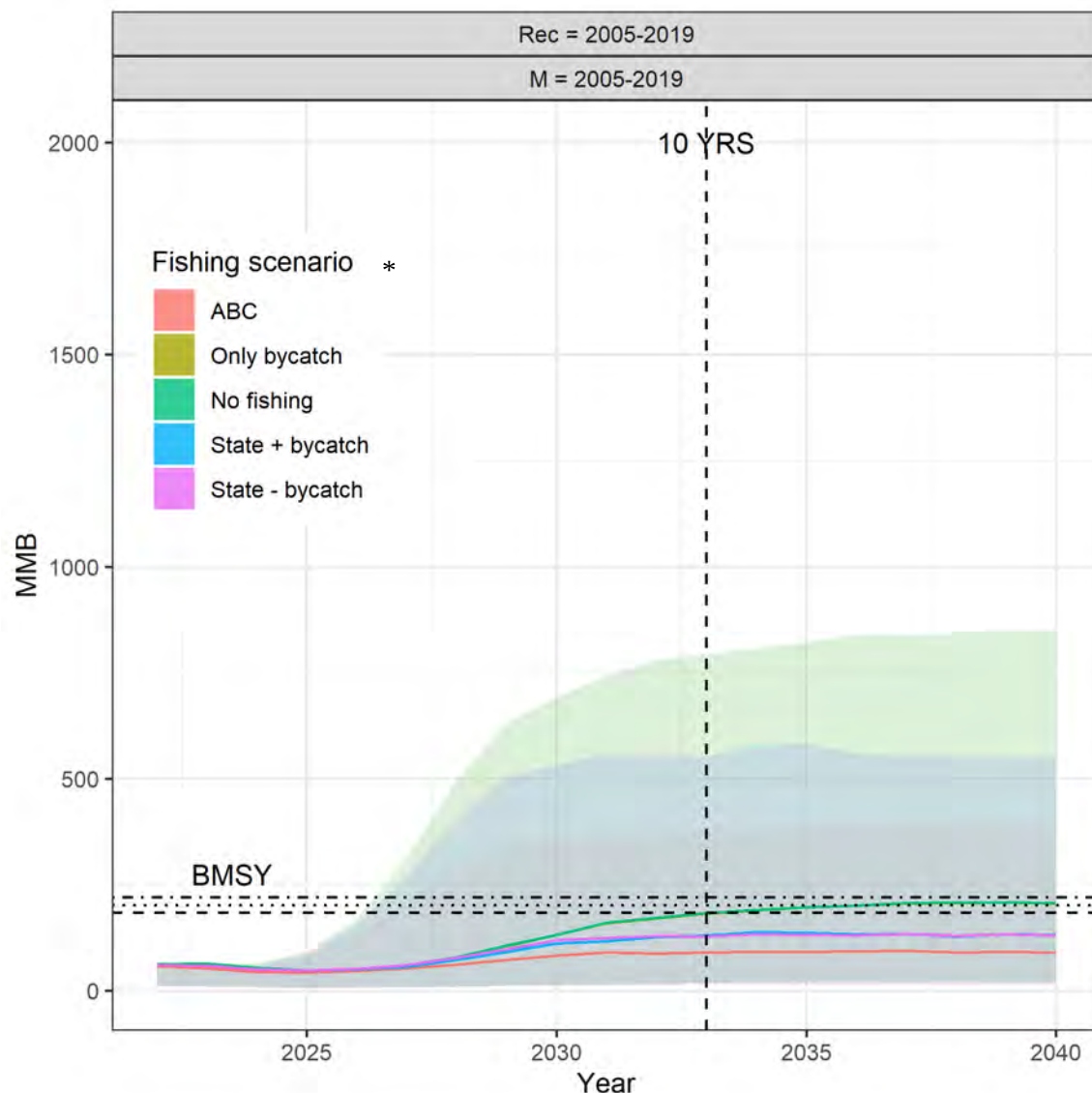
- No fishing = no EBS snow crab fishery and no BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- Bycatch scenario = Allow BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area. No EBS snow crab fishing.
- State + bycatch scenario = EBS snow crab fishing plus BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- State – bycatch scenario = EBS snow crab fishing. No BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- ABC = the ABC calculated in the 2022 Crab SAFE.



*Description of fishing scenarios:

- No fishing = no EBS snow crab fishery and no BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- Bycatch scenario = Allow BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area. No EBS snow crab fishing.
- State + bycatch scenario = EBS snow crab fishing plus BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- State – bycatch scenario = EBS snow crab fishing. No BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- ABC = the ABC calculated in the 2022 Crab SAFE.

Figure 2-2 Rebuilding trajectories under different productivity assumptions associated with the years 1982-2017 and various fishing strategies. Shaded areas represent the 5th and 95th inter-simulation interval and the timing with which the median (solid line), upper, and lower intervals cross the line representing B_{MSY} are reported in Table 2-1 as measures of uncertainty around the minimum time to rebuilding. B_{MSY} thresholds are represented by three horizontal lines corresponding to 1982-2021, 1982-2017, and 2005-2019, in ascending order. Median T_{min} is calculated relative to the status quo B_{MSY} proxy that uses average recruitment from 1982-2021.



*Description of fishing scenarios:

- No fishing = no EBS snow crab fishery and no BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- Bycatch scenario = Allow BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area. No EBS snow crab fishing.
- State + bycatch scenario = EBS snow crab fishing plus BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- State – bycatch scenario = EBS snow crab fishing. No BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- ABC = the ABC calculated in the 2022 Crab SAFE.

Figure 2-3 Rebuilding trajectories under productivity assumptions associated with the years 2005-2019 and various fishing strategies. Shaded areas represent the 5th and 95th inter-simulation interval and the timing with which the median (solid line), upper, and lower intervals cross the line representing B_{MSY} are reported in Table 2-1 as measures of uncertainty around the minimum time to rebuilding. B_{MSY} thresholds are represented by three horizontal lines corresponding to 1982-2021, 1982-2017, and 2005-2019, in ascending order. Median T_{min} is calculated relative to the status quo B_{MSY} proxy that uses average recruitment from 1982-2021.

3 Environmental Assessment

This chapter evaluates the potentially affected environment and the degree of the impacts of the alternatives and options on the various resource components, together with relevant past, present, and reasonable foreseeable future actions (RFFA).

Recent and relevant information, necessary to understand the affected environment for each resource component, is summarized in the relevant section below. For each resource component, the analysis identifies the potential impacts of each alternative, and evaluates these impacts. If significant impacts are likely to occur, preparation of an EIS is required. Although an EA should evaluate economic and socioeconomic impacts that are interrelated with natural and physical environmental effects, economic and social impacts by themselves are not sufficient to require the preparation of an EIS (see 40 CFR 1508.14).

3.1 Methods

3.1.1 Resource Components Addressed in the Analysis

Table 3-1 shows the components of the environment and whether the proposed action and its alternatives have the potential to impact that resource component and thus require further analysis. Extensive environmental analysis on all resource components is not needed in this document because the proposed action is not anticipated to have environmental impacts on all resource components.

The proposed action is to establish a rebuilding plan for EBS Snow Crab. Table 3-1 identifies the components of the human environment that would be affected by establishing a rebuilding plan.

Table 3-1. Resources potentially affected by the proposed action and alternatives.

Potentially affected resource component					
Eastern Bering Sea Snow Crab	Non-Target Species	Marine Mammals	Seabirds	Habitat	Social and Economic
Y	Y	Y	N	Y	Y

N = no impact anticipated by each alternative on the component.

Y = an impact is possible if each alternative is implemented.

3.1.2 Effects of Aggregate Past, Present, and Reasonably Foreseeable Future Actions

This EA analyzes the effects of each alternative and the effects of past, present, and RFFA. Based on Table 3-1, the resources with potentially meaningful effects are EBS snow crab, non-target species, marine mammals, habitat, and social and economic components. The aggregate effects and the impacts of this proposed action and alternatives on seabirds is thought to be minimal, therefore there is no need to conduct an additional aggregate impacts analysis.

Each section below provides a review of the relevant past, present, and RFFA that may result in aggregate effects on the resource components analyzed in this document. A complete review of the past, present, and RFFAs are described in the prior NEPA documents incorporated by reference (section 1.3).

Actions are understood to be human actions (e.g., a designation of northern right whale critical habitat in the Pacific Ocean), as distinguished from natural events (e.g., an ecological regime shift). CEQ regulations require consideration of actions, whether taken by a government or by private persons, which are reasonably foreseeable. This requirement is interpreted to indicate actions that are more than merely possible or speculative. In addition to these actions, this aggregate effects analysis includes the effects of climate change.

Actions are considered reasonably foreseeable if some concrete step has been taken toward implementation, such as a Council recommendation or NMFS's publication of a proposed rule. Actions only "under consideration" have not generally been included, because they may change substantially or may not be adopted, and so cannot be reasonably described, predicted, or foreseen. Identification of actions likely to impact a resource component within this action's area and time frame will allow the public and Council to make a reasoned choice among alternatives.

3.2 Eastern Bering Sea Snow Crab

Snow crab (*Chionoecetes opilio*) are a circumpolar species. While commercial catches in Alaska are concentrated in the Bering Sea (Figure 3-1), the species is also found on the Chukchi Sea and Beaufort Sea continental shelves. The eastern Bering Sea population within U.S. waters is managed as a single stock; however, the distribution of the population may extend into Russian waters to an unknown degree. In the Bering Sea, snow crab are distributed widely over the shelf and are common at depths less than ~200 meters. Primiparous female snow crab appear to track near-bottom temperature during a northeast to southwest ontogenetic migration to warmer waters near the shelf break (Ernst et al., 2005; Parada et al., 2010). Shifts in centers of distribution of mature female snow crab relative to prevailing currents may affect larval supply to nursery areas (Zheng and Kruse, 2006) and thermal occupancy patterns of snow crab depend on the availability of cold water habitat (Fedewa et al., 2020).

Spatial patterns in juvenile and adult snow crab distribution are determined largely by ontogenetic migrations linked to size- and thermal requirements. Immature snow crab concentrate in colder, shallow waters of the northern Bering Sea (NBS) and EBS middle shelves. (Kolts et al., 2015). 2°C may represent a critical temperature threshold for immature snow crab (Murphy, 2020), negative effects on metabolic processes are not apparent in mature snow crab until temperatures exceed 7°C (Foyle et al., 1989). Temperature also influences molt timing (Dutil et al., 2010), growth rates (Yamamoto et al., 2015), energy stores (Hardy et al., 2000), and body condition (Dutil et al., 2010) of snow crab in the laboratory.

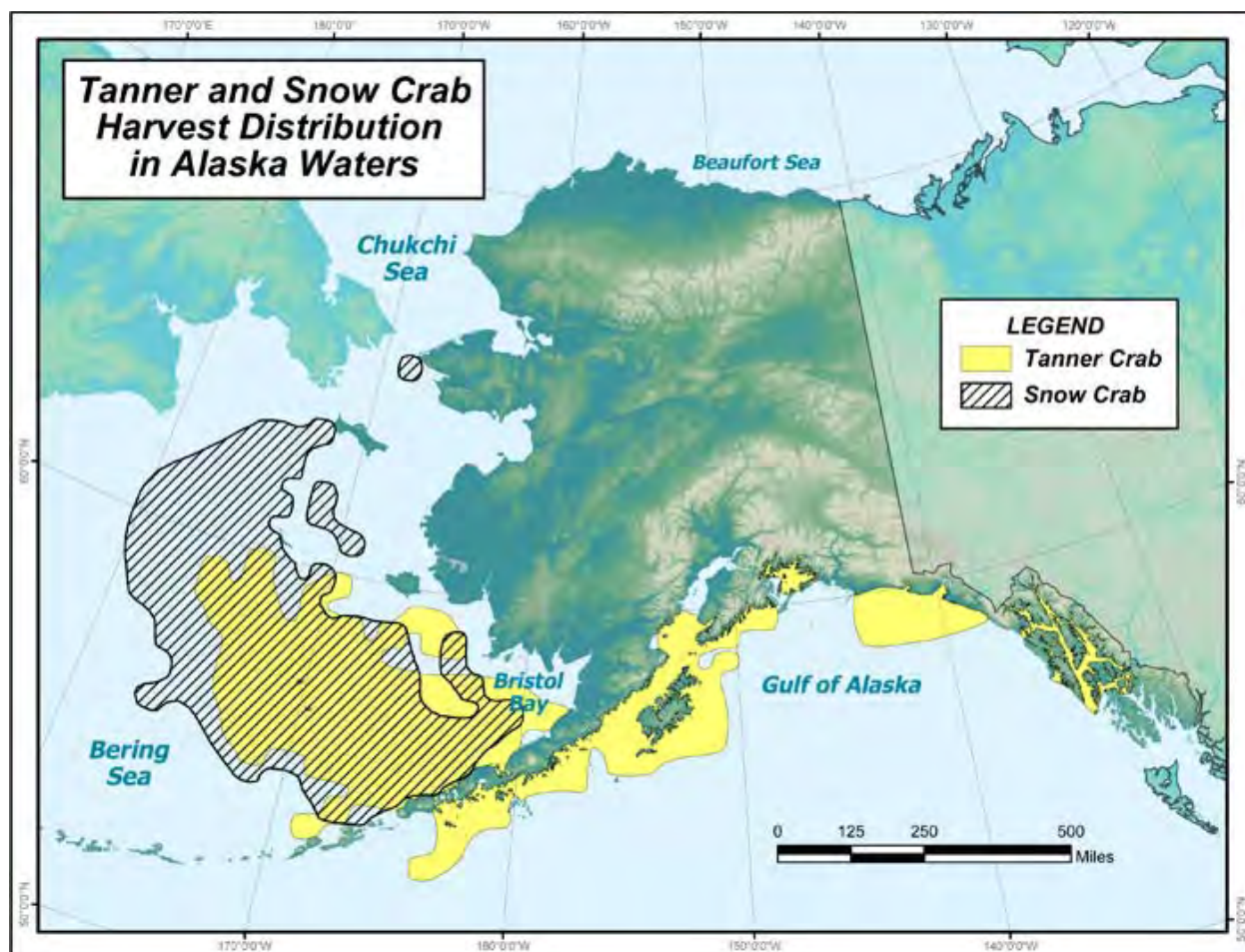


Figure 3-1 Snow crab and Tanner Crab distribution within the waters of Alaska.

3.2.1 Eastern Bering Sea Molting and Mating Cycle

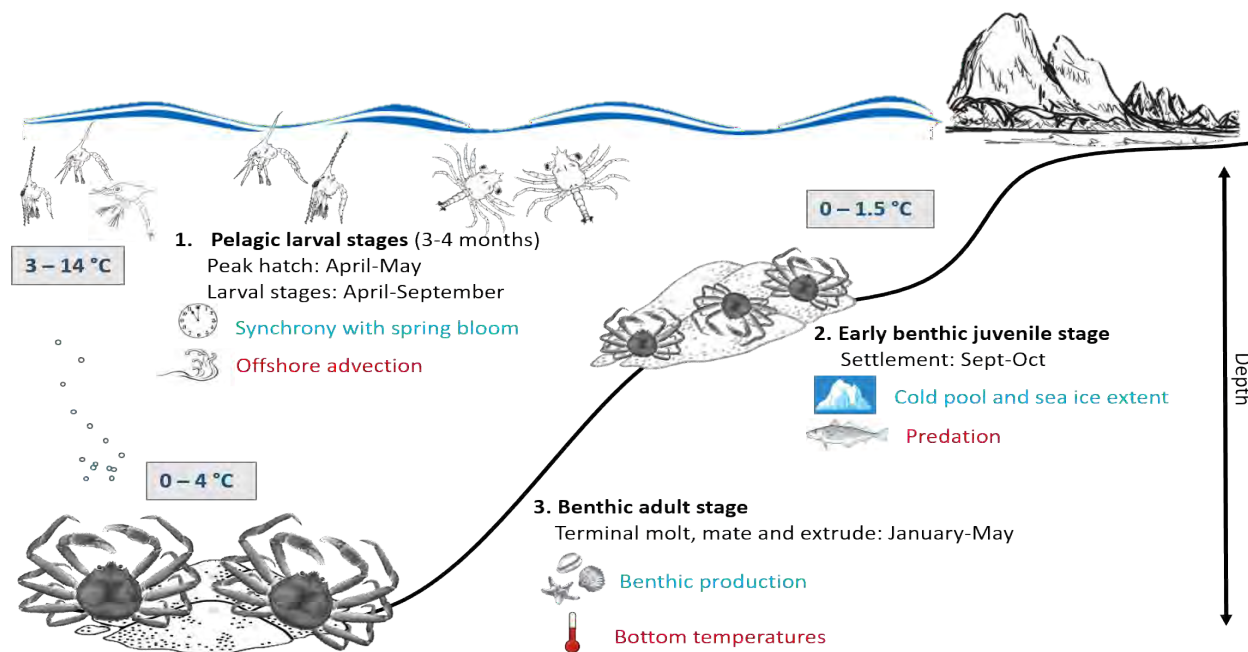
As detailed in the [2022 Snow crab Ecosystem and socioeconomic profile](#), EBS snow crab undergo distinct life cycle stages (Source: 2022 EBS snow crab ESP

Figure 3-2, Table 3-2) associated with molting and mating cycles that correlate with spatial, environmental and temporal preferences. As mature females molt to maturity, female snow crab mate and extrude new egg clutches each spring, which remain attached to pleopods on the female's abdomen for a full year prior to hatching. Peak hatching of snow crab larvae occurs in April (Armstrong et al., 1981) and phyto-detritus may act as a chemical cue for larval release (Starr et al., 1994). Larval duration for each of the two zoeal stages is approximately 30 days (Incze et al., 1982). A longer larval stage associated with cooler temperatures may leave larvae more vulnerable to pelagic predators for a prolonged period. Furthermore, historical larval year-class failures have coincided with low zooplankton abundance over the middle shelf and low water column stability, suggesting that increased larval mortality is related to less favorable feeding conditions (Incze et al., 1987) and mismatches between larval release and the spring bloom (Somerton 1982).

Snow crab larvae settle from late August to the end of October (Conan et al., 1992). Early benthic instars are cryptic and concentrate in shallow, cold-water habitats (Lovrich et al., 1995; Murphy et al., 2010). Previous laboratory studies have shown that adequate energetic stores are prerequisites for molting, growth, and survival in snow crab early life history stages (e.g. Lovrich and Ouellet, 1994), indicating that variability in energetic reserves could represent a potential recruitment bottleneck in snow crab. Both

settlement intensity and early benthic survival are likely critical determinants of year-class strength in snow crab (Sainte-Marie et al., 1996), and successful advection to areas of suitable temperature and muddy substrate are thought to be critical criteria for juvenile survival (Dionne et al., 2003). Density-dependence may also play a regulatory role due to high rates of cannibalism (Lovrich and Sainte-Marie 1997) and potential prey resource limitation in juvenile nurseries. Previous studies have shown that Pacific cod, sculpin, skates and halibut are major predators of juvenile snow crab (Livingston et al., 1993; Livingston and deReynier, 1996; Lang et al., 2003) and the cold pool may provide refuge from predators like Pacific cod that avoid waters less than 2°C (Ciannelli and Bailey, 2005). Juvenile snow crab are especially vulnerable to predation and cannibalism during and immediately following molting. Table 3-2 summarizes the ecological preferences and information utilized by EBS snow crab across ontogeny.

Generation time, or average age of spawning individuals within the EBS snow crab population, is estimated to be 7 years (E. Fedewa pers. Comm.). Minimum generation time is thought to be determined by primiparous females, rather than males as males undergo molt to maturity at a wider range of benthic instar stages. It is thought that females are 6-8 years post hatch when they terminally molt to maturity (Sainte-Marie et al., 1996), while quantitative assessment of instar composition and size frequency distribution similarly suggests approximately 7 years as the mean time lapsed between egg extrusion by females of a pseudocohort and terminal molt of primipara originating from those eggs (Ernst et.al., 2012).



Source: [2022 EBS snow crab ESP](#)

Figure 3-2 Life history conceptual model for EBS snow crab summarizing ecological information and key ecosystem processes affecting survival by life history stage. Red text means increases in process negatively affect survival, while blue text means increases in process positively affect survival.

Table 3-2 Ecological information by life history stage for EBS snow crab.

Stage	Habitat & Distribution	Phenology	Age, Length, Growth	Energetics	Diet	Predators/Competitors
Egg	Clutch of embryos brooded under the female's abdomen until hatching	240 days at 6°C to 353 days at -1°C; cold temperatures trigger a 2-year reproductive cycle(1)	Egg diameter: 644.4-772.1 µm(2)	Optimal: 0°C – 3°C(3)	Yolk	Nemertean worms and amphipods feed on egg clutches
Larvae	Pelagic; concentrated in the upper 20m over the middle shelf(4)	April-June hatch	Mean carapace length: 1.25mm	Optimal: 6.9°C – 9.1°C(5)	Diatoms, small copepods	Jellyfish, juvenile pollock and Pacific salmon
Juvenile	Benthic; found in mud and gravel habitat in 1°C bottom temperatures (50-100m depth)	Peak settlement in October, later benthic stages molt annually in the spring	10-12 benthic instar stages until final molt to maturity(6)	Growth indices highest at 5°C(7)	Crustaceans, bivalves, polychaetes(8)	Pacific cod, flatfish, sculpins, crab(9)
Adult	Benthic: sand and mud bottoms (70-200m depth)	6-7+ years, migration to shallow waters in spring to mate	Average size range at terminal molt: females 47-59 mm CW, males 73-101mm CW(10)	Growth is optimum at 4°C(11)	Polychaetes, crustaceans, echinoderms, mollusks(12)	Pacific cod, halibut, skates(13)

Source: [2022 EBS snow crab ESP](#)

Note: Subscripts in table correspond to the following citations in sequential order 1. Webb et al., 2006, 2. Moriyasu and Lanteigne, 1998, 3. Webb et al., 2007, 4. Armstrong et al., 1981, 5. Yamamoto et al., 2017, 6. Sainte-Marie et al., 1995, 7. Yamamoto et al., 2015, 8. Kolts et al., 2013, 9. Lang et al., 2003, 10. Murphy 2021, 11. Foyle et al., 1989, 12. Divine et al., 2017, 13. Livingston et al., 1993

3.2.2 EBS Snow Crab Fishery Stock Biomass and Catches

This section is intended to detail historical snow crab biomass and catches throughout the duration of fishery operations. Snow crab were harvested in the Bering Sea (BS) by the Japanese from the 1960s until 1980 when the MSA prohibited foreign fishing. After the closure to foreign fleets, retained catches increased from relatively low levels in the early 1980s (e.g. retained catch of 11.85 kt during 1982) to historical highs in the early and mid-1990s (retained catches during 1991, 1992, and 1998 were 143.02, 104.68, and 88.09 kt, respectively; Table 3-3).

In 1999 the stock was declared overfished, at which time retained catches dropped to levels similar to the early 1980s (e.g. retained catch during 2000 was 11.46 kt). In 2000, under amendment 14, the EBS snow crab rebuilding plan contained three components to improve the status of the stock: (1) a harvest strategy, (2) bycatch control measures, and (3) habitat protection measures. The rebuilding plan was estimated to allow the BS snow crab stock to rebuild to the B_{MSY} level, with a 50 percent probability, in seven to ten years. The snow crab stock remained low for some time after the rebuilding plan was implemented.

In 2009, the NMFS Alaska Region notified the Council that the EBS snow crab stock would not be rebuilt by the end of the rebuilding time period, 2009/10, and that a revised rebuilding plan must be developed for that stock and implemented within two years of that notification. Amendment 39 passed in 2010 modifying the snow crab rebuilding plan to define the stock as rebuilt the first year the stock biomass was above the necessary level to produce maximum sustainable yield (MSY). The stock was declared rebuilt in 2011 and had expanded substantially, supporting some of the largest annual BSAI crab fisheries until the most recent decline in 2019/2020 prompting a new snow crab rebuilding plan when the stock was declared overfished in 2021.

Current estimates of biomass and size frequency data are calculated based off the NMFS trawl survey, total catch data from the directed fishery, bycatch data from the trawl fishery, size frequency data for male-retained catch in the directed fishery, and male and female bycatch in the directed and non-target fisheries. Additional information of biomass estimates from the 2009/2010 BSFRF surveys are incorporated into the stock assessment model. It should be noted that the 2020 survey data is not included, as the survey season was cancelled due to the Covid-19 pandemic. The current stock assessment of EBS snow crab is based on a size- and sex-structured model in which crabs are categorized as immature or mature, and account is taken of a terminal molt. The model takes into account all aforementioned estimates of biomass and size frequency to best update the stock assessment on an annual basis. Updated data in the [2022 assessment](#) include retained catch, total catch and length frequencies from the 2021/22 directed fishery, discard catch and length frequencies from the 2021/22 groundfish fisheries, and biomass and length frequencies from the 2022 NMFS bottom trawl survey. Results from the 2022 NMFS bottom trawl survey indicated a decrease in biomass estimates for all size classes, with the exception of a slight increase in estimated industry preferred biomass (+9%) from the 2021 survey. 2022 observed EBS snow crab female mature and male mature biomass is the lowest value in history Table 3-4.

Based on the [2022 BSAI Crab SAFE](#), the model estimate of mature male biomass for the 2021/22 fishing season (41.2 kt) was below the MSST; 91.6 kt, and so the stock remains in an overfished status. The 2019/2020 season was the first time a mass mortality event appears to have occurred for snow crab since the survey began and the biomass of important size categories of crab are at historic lows. The observed biomass of males greater than 101mm carapace width was 13.36 kt in 2022, second lowest aside from 2021 (12.43 kt) (Table 3-4). The third lowest the biomass has ever been was 20,740 t in 2016. For context, when the stock was declared overfished in 1999, the observed biomass was 52,042 tons. Females are also currently at historic lows, observed at 23.89 kt in 2022. For the first time in history, the fishery will remain closed for the 2022/2023 fishing season.

Historical retained catch from the directed snow crab pot fishery from survey years 1982 to 2021 are presented in Table 3-3 and Table 3-5. Discards and bycatch mortality estimates are derived from at-sea observer data and assume a 30 percent handling mortality rate, 80 percent for trawl gear, and 50 percent in fixed gear groundfish fisheries. Figure 3-3 and Figure 3-4 illustrate the spatial distribution of the directed snow crab fisheries in the BSAI overlayed on the shaded COBLZ area. Figure 3-3 shows the statistical areas with retained catch from the 2021/22 season (with statistical areas that include at least three vessels) and Figure 3-4 demonstrates the weighted center of catch over time. The footprint of the directed snow crab fishery has remained fairly consistent over time. Snow crab fishing occurs over a wide distribution on and near the shelf edge and north toward Saint Matthew Island.

Table 3-3. EBS Snow crab directed pot fishery characteristics from 1990/91-2021/22 The Guideline Harvest Level and Total Allowable Catch (GHL/TAC) are equivalent and are in millions of pounds.

Year	GHL/TAC (lb)	Retained Catch (lb)	Retained Catch (crab)	Pot lifts	CPUE	avg_wt (lb)	avg CW (mm)
1990	315,000,000	328,648,169	265,124,637	1,382,908	191.7	1.24	110.1
1991	333,000,000	315,302,034	227,376,582	1,278,502	177.8	1.39	114.1
1992	207,200,000	230,754,145	169,531,168	969,209	174.9	1.36	113.4
1993	105,800,000	149,792,718	114,810,186	716,524	160.2	1.30	111.9
1994	55,700,000	75,309,187	60,591,399	507,603	119.4	1.24	110.2
1995	50,700,000	65,696,173	52,892,320	520,685	101.6	1.24	110.1
1996	117,000,000	119,589,339	100,013,816	754,140	132.6	1.20	108.8
1997	234,100,000	252,339,284	193,618,550	930,794	208.0	1.30	111.9
1998	196,000,000	194,363,869	151,183,798	945,533	159.9	1.29	111.4
1999	28,500,000	33,291,344	25,081,681	182,634	137.3	1.33	112.5
2000	27,300,000	25,256,384	18,612,605	191,200	97.3	1.36	113.3
2001	30,820,000	32,633,210	25,155,221	326,977	76.9	1.30	111.7
2002	25,610,000	28,316,923	23,252,904	153,862	151.1	1.22	109.4
2003	20,831,000	23,942,373	18,669,591	123,709	150.9	1.28	111.3
2004	20,932,000	24,892,128	17,985,745	75,095	239.5	1.38	114.0
2005	37,184,000	36,923,482	24,520,279	117,375	208.9	1.51	117.2
2006	36,566,000	36,243,989	29,536,398	86,328	342.1	1.23	109.7
2007	63,034,000	63,002,304	50,307,812	140,857	357.2	1.25	110.4
2008	58,550,000	58,547,849	45,945,092	163,537	280.9	1.27	111.0
2009	48,017,000	48,014,089	35,289,022	137,292	257.0	1.36	113.4
2010	54,281,000	54,263,200	37,758,496	147,478	256.0	1.44	115.4
2011	88,894,000	88,830,652	60,555,105	270,602	223.8	1.47	116.2
2012	66,350,000	66,254,528	47,455,883	225,627	210.3	1.40	114.4
2013	53,983,000	53,978,074	41,923,152	225,245	186.1	1.29	111.4
2014	67,950,000	67,939,253	55,027,927	279,183	197.1	1.23	109.9
2015	40,611,000	40,594,509	29,603,375	202,526	146.2	1.37	113.7
2016	21,570,000	21,570,915	16,412,386	118,548	138.4	1.31	112.2
2017	18,961,000	18,888,112	15,637,993	114,673	136.4	1.21	109.1
2018	27,581,000	27,501,780	22,408,836	119,484	187.5	1.23	109.7
2019	34,020,000	34,024,553	28,626,114	188,958	151.5	1.19	108.6
2020	45,000,000	45,001,190	37,492,237	171,678	218.4	1.20	108.9
2021	5,600,000	5,523,226	4,575,974	36,878	124.1	1.21	109.1

Table 3-4 Observed mature male and female biomass (1000 t) at the time of the survey and coefficients of variation.

Survey year	Female mature biomass	Female CV	Mature male biomass	Male CV	Males >101mm (kt)	Males >101mm (million)
1982	144.4	0.15	176.8	0.14	33.34	60.91
1983	90.13	0.2	161.6	0.13	38.09	70.09
1984	42.32	0.19	177.7	0.12	88.73	151.8
1985	6.12	0.2	71.84	0.11	43.39	72.84
1986	15.74	0.18	89.81	0.11	46.7	77.91
1987	122.6	0.16	194.6	0.11	74.44	128.6
1988	169.9	0.17	259.4	0.15	104.7	173.1
1989	264.2	0.25	299.2	0.11	92.31	158.9
1990	182.9	0.19	443.8	0.14	224.7	386.4
1991	214.9	0.19	466.6	0.15	292.2	452.9
1992	131.4	0.18	235.5	0.09	143.9	227.3
1993	132.1	0.16	183.9	0.1	78.11	126.7
1994	126.2	0.15	171.3	0.08	44.78	72.57
1995	168.7	0.14	220.5	0.13	37.75	65.18
1996	107.3	0.14	288.4	0.12	87.57	155.2
1997	103.8	0.2	326.8	0.1	168.7	280.6
1998	72.73	0.25	206.4	0.09	126.7	209.7
1999	30.89	0.21	95.85	0.09	52.53	85.2
2000	96.46	0.52	96.39	0.14	41.88	69.83
2001	77.24	0.28	136.5	0.12	41.51	70.69
2002	30.22	0.28	93.17	0.23	36.56	64.16
2003	41.71	0.31	79.07	0.12	32.57	55.61
2004	50.16	0.26	79.57	0.14	35.99	57.42
2005	64.85	0.17	123.5	0.11	40.67	63.26
2006	51.93	0.17	139.3	0.26	71.13	120.9
2007	55.89	0.22	153.1	0.15	73.62	127.5
2008	57.15	0.19	142	0.1	66.56	113.6
2009	52.16	0.21	148.2	0.13	78.92	129.9
2010	98.01	0.17	162.8	0.12	88.35	138.3
2011	175.8	0.18	167.1	0.11	94.67	147.6
2012	149.4	0.2	122.2	0.12	53.17	85.35
2013	131.4	0.17	97.46	0.12	42.93	71.79
2014	119.7	0.19	163.5	0.16	81.39	138.8
2015	85.13	0.17	80.04	0.12	35.77	56.11
2016	55.39	0.21	63.21	0.11	21.96	36.51
2017	106.8	0.21	83.96	0.13	20.52	35.02
2018	165.9	0.18	198.4	0.17	26.75	48.08
2019	110.4	0.2	169.1	0.17	28.12	51.27
2021	31.66	0.43	62.25	0.13	12.43	23.17
2022	22.44	0.31	37.5	0.15	13.36	23.89

Table 3-5 Observed retained catches, discarded catch, and bycatch. Discards and bycatch have assumed mortalities applied.

Survey year	Retained Catch (kt)	Discarded females (kt)	Discarded Males (kt)	Trawl bycatch (kt)
1982	11.85	0.02	1.47	0.37
1983	12.16	0.01	1.43	0.47
1984	29.94	0.01	3.2	0.5
1985	44.45	0.01	4.65	0.43
1986	46.22	0.02	4.92	0
1987	61.4	0.03	6.4	0
1988	67.79	0.04	6.74	0
1989	73.4	0.05	7.74	0.1
1990	149.1	0.05	17.62	0.71
1991	143	0.06	13.9	1.5
1992	104.7	0.12	17.06	2.28
1993	67.94	0.08	5.32	1.57
1994	34.13	0.06	4.03	2.67
1995	29.81	0.02	5.75	1.01
1996	54.22	0.07	7.44	0.66
1997	114.4	0.01	5.73	0.82
1998	88.09	0.01	4.67	0.54
1999	15.1	0	0.52	0.47
2000	11.46	0	0.62	0.41
2001	14.8	0	1.89	0.31
2002	12.84	0	1.47	0.17
2003	10.86	0	0.57	0.46
2004	11.29	0	0.51	0.63
2005	16.77	0	1.36	0.2
2006	16.49	0	1.78	0.42
2007	28.59	0.01	2.53	0.18
2008	26.56	0.01	2.06	0.18
2009	21.78	0.01	1.23	0.47
2010	24.61	0.01	0.62	0.14
2011	40.29	0.18	1.69	0.15
2012	30.05	0.03	2.32	0.22
2013	24.49	0.07	3.27	0.11
2014	30.82	0.17	3.52	0.13
2015	18.42	0.07	2.96	0.13
2016	9.67	0.02	1.31	0.06
2017	8.6	0.02	1.93	0.04
2018	12.51	0.02	2.86	0.23
2019	15.43	0.02	5.07	0.24
2020	20.41	0	5.8	0.07
2021	2.48	0	1.16	0.06

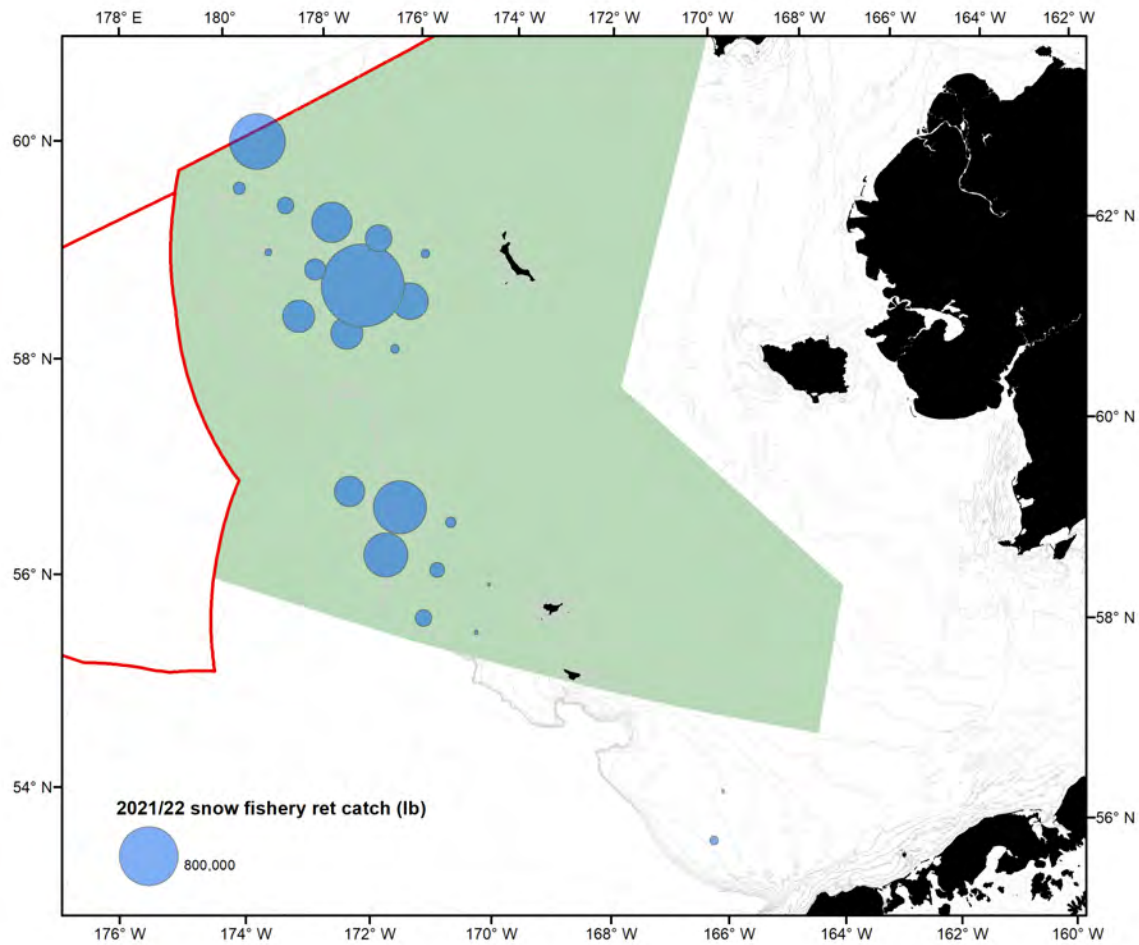


Figure 3-3 Retained catch of EBS snow crab in the directed fishery, 2021/22, where size of the blue dot corresponds to the magnitude of catch in each ADF&G statistical area (grid). Shaded area represents the COBLZ.

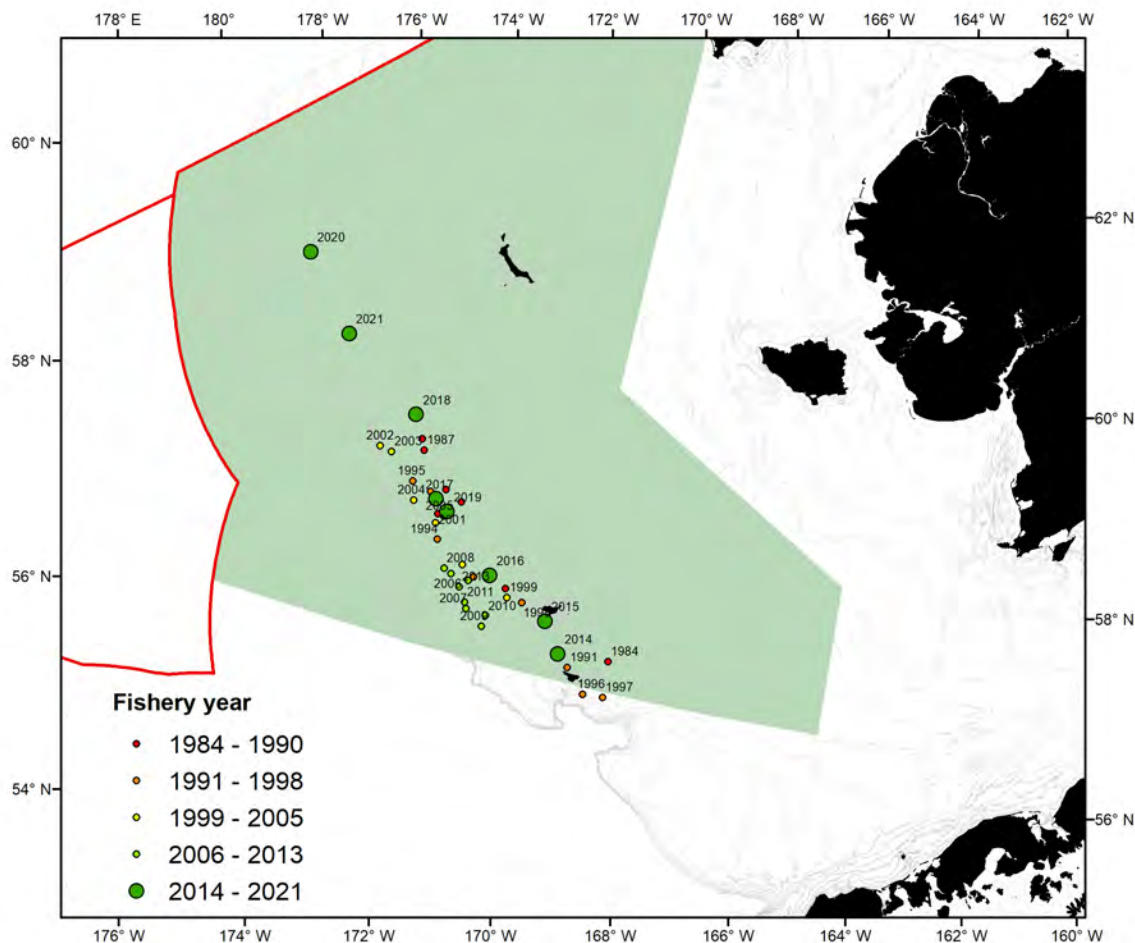
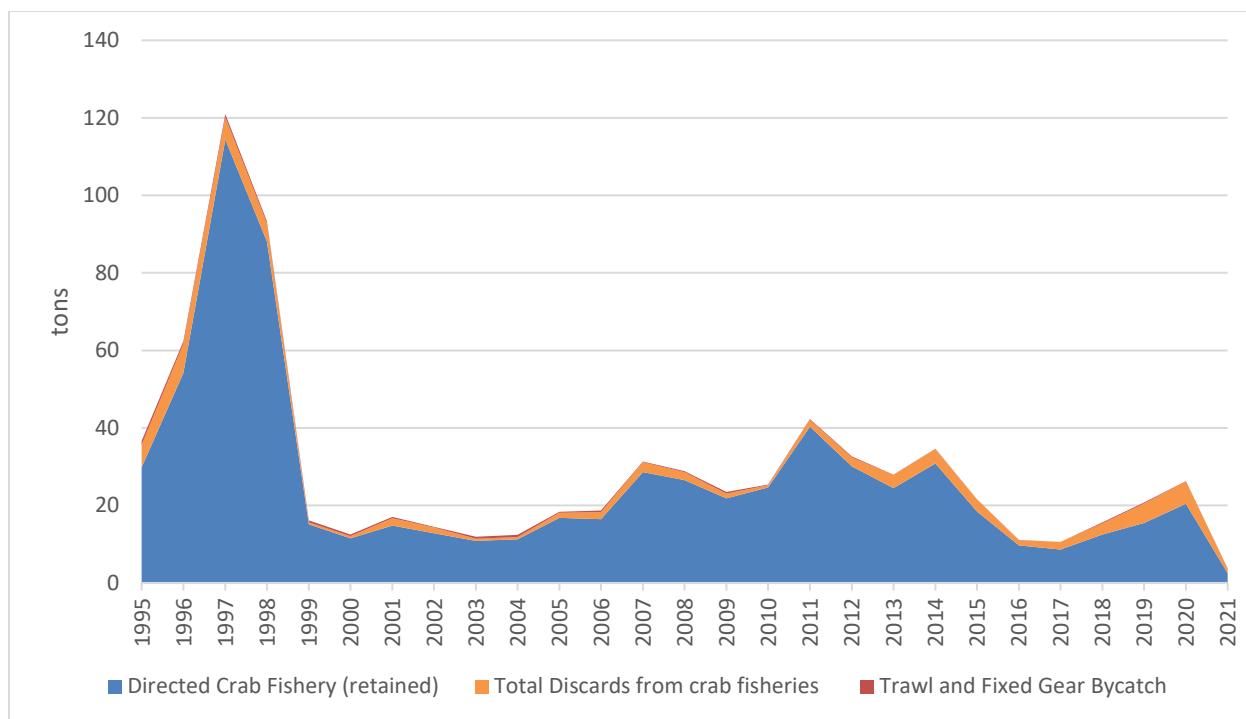


Figure 3-4 Weighted center of EBS snow crab catch in the directed fishery for 1984-2021. Shaded area represents the COBLZ.

Discard mortality in the directed fishery is the next largest source of mortality after retained catch. The highest estimated discard mortality occurred during 1992 at 17.06 kt, which was 16% of the retained catch. Discard from the directed pot fishery has been estimated from observer data since 1992 and has ranged from 11-100% of the magnitude of retained catch by numbers. Regulatory female discard catch has been very low compared to male discard catch and has not been a substantial source of mortality.



Source: EBS Snow Crab SAFE, September 13, 2022 (Table 3-5)

Figure 3-5 Directed snow crab catch and other sources of mortality including discards from directed crab fisheries and trawl and fixed gear bycatch in groundfish fisheries, 1995 through 2020

3.2.3 EBS Snow Crab and Climate Change

As shown in the [2022 EBS snow crab ESP](#), ecosystem indicators prove beneficial when assessing snow crab populations. In the 2022 ESP, authors highlighted the potential loss of cold-water habitat available to snow crab, evidenced by record-low cold pool extent coinciding with dramatic increases in temperature in areas occupied by immature snow crab in recent years. Temperature has shown to be a primary ecosystem indicator when assessing snow crab populations. Fluctuations in temperature influence molt timing (Dutil et al., 2010), growth rates (Yamamoto et al., 2015), energy stores (Hardy et al., 2000), metabolic rate (Foyle et al., 1998) and body condition (Dutil et al., 2010) of snow crab in the laboratory. Declines in sea ice extent also pose negative consequences for spring cold pool formation, carbon flux to the benthos and spatiotemporal mismatches between snow crab larvae and spring blooms.

The snow crab collapse coincided with rapid warming in the EBS during 2014-2020. The peak years of this warming event were 2016, 2018, and 2019, when annual mean sea surface temperature (SST) was well outside the range of previous observations and 1.9 - 2.3°C above the pre-1950 mean (Figure 3-6). The goals of this section are to: 1) review the evidence for warm temperatures as a cause of the snow crab collapse; 2) review the evidence for human-induced climate change as the mechanism leading to the warming event; and 3) evaluate the risk for similar warming events during snow crab rebuilding.

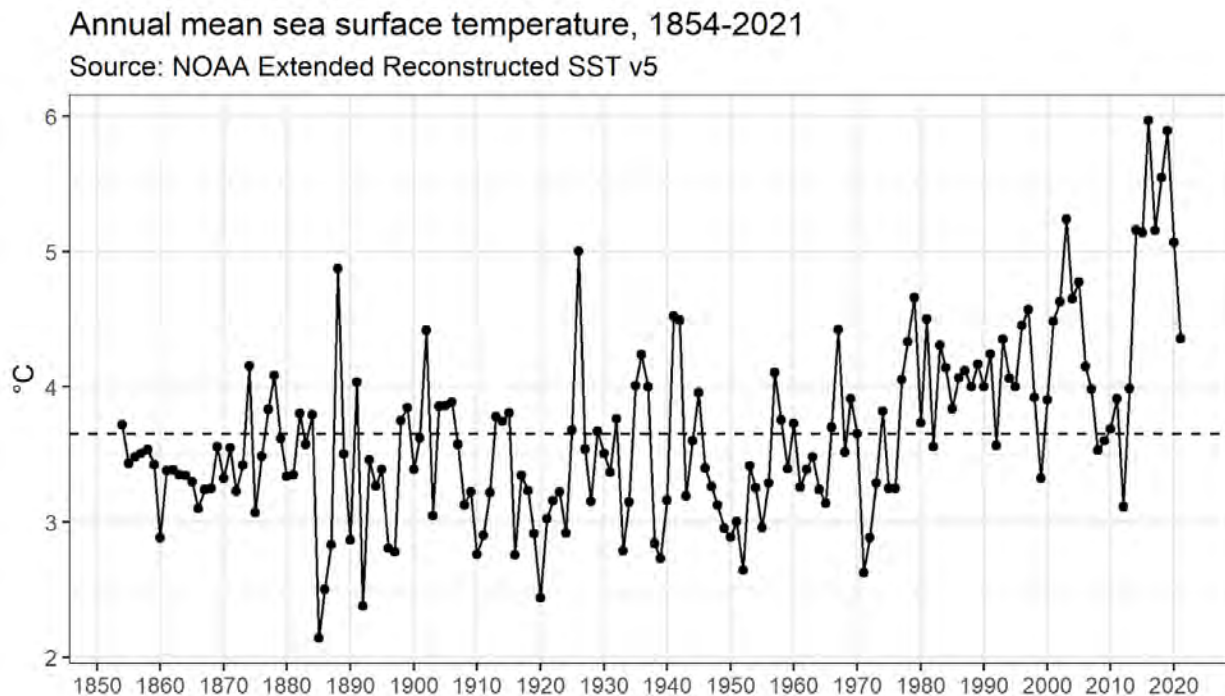


Figure 3-6 Annual mean sea surface temperature (SST) in the EBS. Dashed line indicates pre-1950 mean.

3.2.3.1 Warming as a cause of the snow crab collapse

Snow crab are a cold-water, Arctic species that is primarily found in seasonally ice-covered areas of the Bering sea (BS) with summer bottom water temperatures $< 2^{\circ}\text{C}$. Because of this cold-water, ice-associated climate envelope, snow crab have long been proposed as a species that is likely to be negatively impacted by climate warming and the loss of sea ice in the BS (Mueter and Litzow 2008).

Two ongoing studies, not yet published, have investigated the causes of the post-2018 collapse of snow crab, and both have independently identified warming as a major contributor. Szuwalski et al. (in review) used annual mortality estimates from a population dynamics model run on males between 30 and 95 mm as the response variable for evaluating the role of a wide range of possible causes of the collapse. Candidate covariates included bycatch, directed fishing discard mortality, cannibalism, Pacific cod predation, the incidence of bitter crab syndrome, temperature, and population density. This model showed robust evidence for temperature and density-dependence as the causes of the collapse, with no evidence for an effect of the other covariates.

Borealization - the switch from an Arctic ecosystem state to a subarctic state - has long been proposed as the most important consequence of climate change for Arctic marine communities and fisheries (Mueter and Litzow 2008, Fossheim et al. 2015, Polyakov et al. 2020, Mueter et al. 2021, Emblemstvag et al. 2022). Litzow et al. (in prep.) evaluated the role that borealization of the southeast BS played in the snow crab collapse. This analysis used time series for thirteen physical and biological variables to create a borealization index that tracks the transition of the region from an Arctic, seasonally ice-covered state to a subarctic, ice-free state over the years 1972-2022 (Figure 3-7). This borealization index identifies 2018 and 2019 as the years when subarctic conditions were strongest in the southeast BS, and the analysts found a robust statistical relationship between the borealization index and the snow crab collapse.

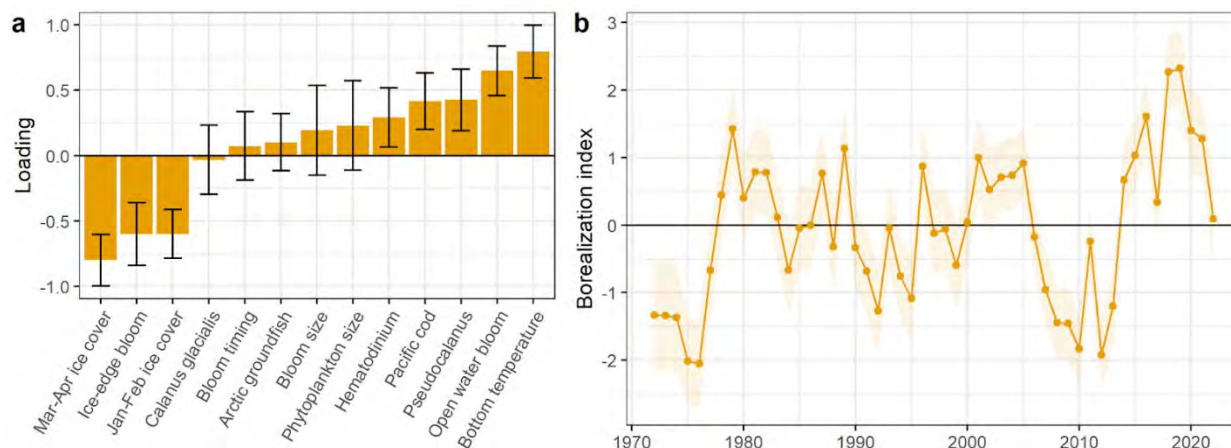


Figure 3-7 An index of borealization (the transition from Arctic to subarctic conditions) in the southeast BS, 1972-2022. a) Loadings for thirteen physical and biological time series onto a shared trend identified using Dynamic Factor Analysis (DFA). Time series with negative loadings are associated with more Arctic conditions, time series with positive loadings are associated with more subarctic conditions, and time series with 95% confidence intervals (error bars) that include 0 cannot be assigned to either state. b) Borealization index, defined as the shared trend from the DFA model (estimate with 95% confidence interval). Negative values indicate more Arctic conditions, positive values indicate more subarctic conditions.

3.2.3.2 The role of human-induced climate change

An important question regarding recent warming in the BS is the extent to which these events are the result of internal climate variability (e.g., Di Lorenzo and Mantua 2016), in which case the warming event could be understood as a one-off event, or the product of human-caused global warming (Broecker 1975), in which case the warming event could be understood as a part of a trend. Extreme event attribution is a branch of climate science that can address questions such as this, by quantifying the contribution of human-induced climate change to individual climate and weather events (Otto 2017).

Several attribution studies have identified a human role in recent North Pacific and BS climate extremes. Laufkötter et al. (2020) found that the 2013-2015 “Blob” marine heatwave that marked the onset of extreme BS temperatures could be attributed to human activities. Walsh et al. (2018) and Thoman et al. (2020) came to the same conclusion about 2016 BS sea surface temperature / ocean heat content and 2018 BS ice cover, respectively. In an unpublished study, Litzow et al. (in review) applied Extreme Event Attribution techniques to annual BS sea surface temperature observations for 1950-2021. Using outputs from 23 CMIP6 (Coupled Model Intercomparison Project, Phase 6) climate models, these authors calculate the Fraction of Attributable Risk (the proportion of risk for a given event that can be ascribed to human activities, FAR) and the Risk Ratio (how much more likely a given event is due to human activity). This analysis found FAR values ≥ 0.998 for 2018-2019 sea surface temperature values, and Risk Ratio values that indicated that 2018-2019 temperatures were ~500-1500 times more likely to occur due to human activities (Figure 3-8). These results provide compelling evidence that warming associated with the snow crab collapse was the result of human-caused climate change. This in turn indicates the likelihood that the Bering Sea will continue to warm, requiring the use of a forward-looking perspective for managing snow crab and other Bering Sea fisheries, based on the expectation that current trends will continue (Pershing et al. 2019).

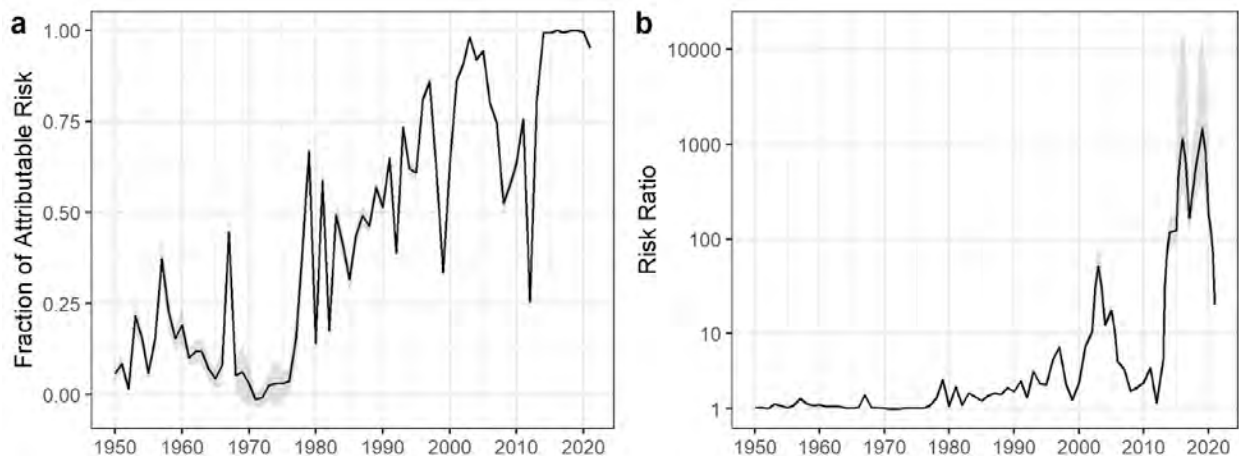


Figure 3-8 Attribution of change in BS sea surface temperature to human activities, 1950-2021. a) Fraction of Attributable Risk (proportion of risk for a given temperature that can be ascribed to human activities). b) Risk Ratio (how much more likely an observed temperature is due to human activities when compared with the preindustrial climate). Both time series are multi-model estimates from 23 climate models, plotted as posterior means and 95% credible intervals.

3.2.3.3 Risk for continued warming events during rebuilding

In order to make projections of the risk for extreme warming events during rebuilding, Litzow et al. (in prep.) identified the critical threshold in sea surface temperature that was associated with rapid borealization and the snow crab collapse, and then used outputs from the same 23 CMIP6 climate models to estimate the probability of temperatures as great as or greater than the critical threshold. This threshold was identified as the minimum SST value associated with the value of the borealization index observed to date (2016, 2018-2020; Figure 3-7b). Relative to preindustrial (pre-1950) temperatures, this critical threshold is a 3.85 standard deviation anomaly. (Anomalies are used rather than original units in order to ease comparison among observations and models; the 2019 anomaly was slightly higher at 4.9 standard deviations.)

The risk for temperatures exceeding this threshold was estimated for a range of different North Pacific warming levels (preindustrial to 0.5° warming, 0.5° - 1.0° warming, 1.0° - 1.5° warming [the current climate], and 1.5° - 2.0° warming). Multi-model estimates of the probability of temperatures as great as or greater than the critical threshold indicate that these temperatures were nearly unknown to rare historically (0% of years in the preindustrial climate, 0.1% from 1950 to 0.5° warming, 2.3% at 0.5° - 1.0° warming), can be expected as common events in the current climate (17% of years), and will become more common between 1.5° and 2.0° warming (32% of years; Figure 3-9a). Under the SSP2-4.5 emissions scenario (corresponding to ~3° global warming by 2100), the North Pacific is expected to exceed 1.5° warming some time in the 2040s (Figure 3-9b).

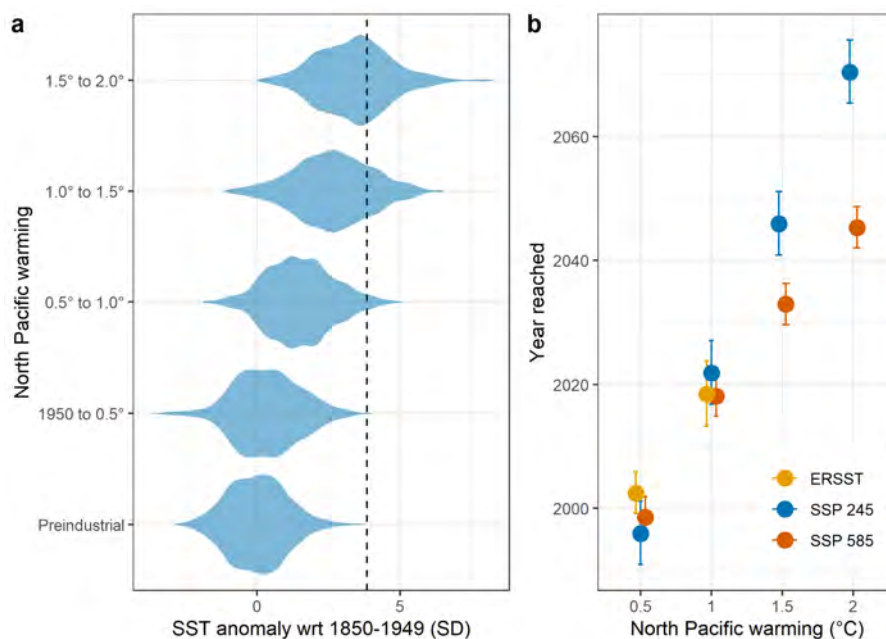


Figure 3-9 Global warming and climate risk for EBS snow crab. a) Bering Sea sea surface temperature probability densities projected for different levels of North Pacific warming. Dashed vertical line indicates critical threshold corresponding to observed borealization in 2020. **b)** Timing of observed and projected warming of the North Pacific warming from observations (ERSST) and two model scenarios (SSP5-8.5, corresponding to no mitigation, and SSP2-4.5, corresponding to ~3°C global warming by 2100). Plotted values are posterior means with 95% credible intervals.

3.2.4 Effects of the Alternatives on EBS snow crab

It is possible that rebuilding to B_{MSY} may not occur under any alternative, given that the predominant constraint on stock productivity is likely ecosystem conditions and recruitment. A complex suite of variables affects mortality of all life stages, but ecosystem conditions especially impact survival from age zero to approximately age 7, when male crab recruit to the mature male component of the population. As noted in section 3.2.3, the EBS snow crab mass mortality event is likely a result of the warming event associated with anthropogenic climate change. The warming event in the BS has been a contributor in the decline of snow crab stock.

The collapse of snow crab in the BS is thought to be one of the largest losses of marine macrofauna that can be attributed to the marine global heatwave ([2022 BSAI Crab SAFE](#)). Due to the uncertain nature of the snow crab stock in recent years; there are several possibilities that may influence the effectiveness of rebuilding the snow crab stock as detailed below:

- Highly specific thermal optimums and habitat requirements of EBS snow crab may alter physiological demands as a response to warmer than average bottom temperatures.
- Warmer temperatures may alter prey-predator relationships and predator distribution, resulting in a shift in predator-prey interactions, and food web dynamics.
- Constraints on recruitment will likely persist for an extended period of time despite the implementation of a rebuilding plan.

Alternative 1: No action, would not follow the federal mandates required per MSA, and would allow the fishery to operate at status quo. Whereby, an annual stock assessment is presented through the Council

process, the council sets the specifications and delegates to the state to set the TAC per the state harvest strategy (section 2.2).

Alternative 2: Establish a rebuilding plan would implement a rebuilding plan under the parameters $T_{\min}=6$ and $T_{\max}=10$. The Council has yet to select a target time for rebuilding (T_{target}), but it will likely fall between 6 and 10 years.

Alternative 2/option 1 would designate no directed EBS snow crab fishery, with bycatch removals only, and implications to the stock will be similar to those seen as a result of the 2022/2023 fishery closure. The 2022/23 fishing season closure is the first closure in the history of the fishery; therefore, it is hard to determine at this time what the effects of no fishery would be on the EBS snow crab stock. Given the current biomass and abundance estimates, it is likely that with no directed fishery and bycatch removals only there would be an increased opportunity for the stock to continue an upward tick in recruitment. However, there existed no difference in median rebuilding time under the bycatch-only fishing mortality scenario when compared to the state harvest strategy scenario.

Alternative 2/option 2 (the PPA) would allow for directed harvest if an opening is triggered by threshold survey catches under the State harvest strategy. The EBS snow crab stock is likely to have similar rebuilding trajectories under Alternative 1 and Alternative 2/option 2 (the PPA), due to the nature of the current FMP delegation of TAC setting to the state of Alaska. Because a rebuilding plan would be in place, constraints on fishing mortality could be made more conservative by further restricting fishery operations if necessary to ensure adequate progress. For the EBS snow crab fishery to open under option 2, the model estimate must indicate that EBS snow crab is at least 25% of the B_{MSY} .

Uncertainty in stock growth persists under all fishing mortality scenarios (Figure 2-2) and are likely related to delays in the onset of increases in recruitment. The mature male biomass (MMB) remains low in all fishing mortality projection scenarios for approximately the first 4 years, Figure 2-2 and Table 2-1 illustrate the average projected response of MMB under both Alt 2, options 1 and 2, with the associated variability for each estimated MMB trend.

Given the variability in biomass in the last few years, the estimates in any given projection year indicate that future stock productivity is highly uncertain. The area where the 95% credible intervals overlap for all fishing mortality options illustrates the high variability associated with the biomass projections. The effects of stock productivity, a function of future ecosystem conditions, may overwhelm the effect of harvest under the Alternative 2 options. Nevertheless, this only serves to illustrate the uncertainty in future conditions. Under any productivity scenario, the median values suggest that rebuilding would be the same under Alternative 2/ option 1 and Alternative 2/option 2 (the PPA) (6 years).

Under the proposed rebuilding parameters, the projected removals under both Alternative 2/option 1, bycatch removals only, and Alternative 2/option 2 (the PPA), a directed fishery and bycatch removals predict low removals for the first three to four years of the proposed rebuilding time (Figure 3-10 and Table 3-6). It is important for the reader to note that there is substantial uncertainty surrounding the projected removal values as the TAC setting process is determined by the state of Alaska and varies annually based on numerous factors, which will influence the number of removals annually. The values presented in Figure 3-10 and Table 3-6 are in no way a definitive prediction of removals throughout rebuilding. However, the values provide better evidence of potential fishing activity throughout rebuilding under Alternative 2/option 2 (the PPA).

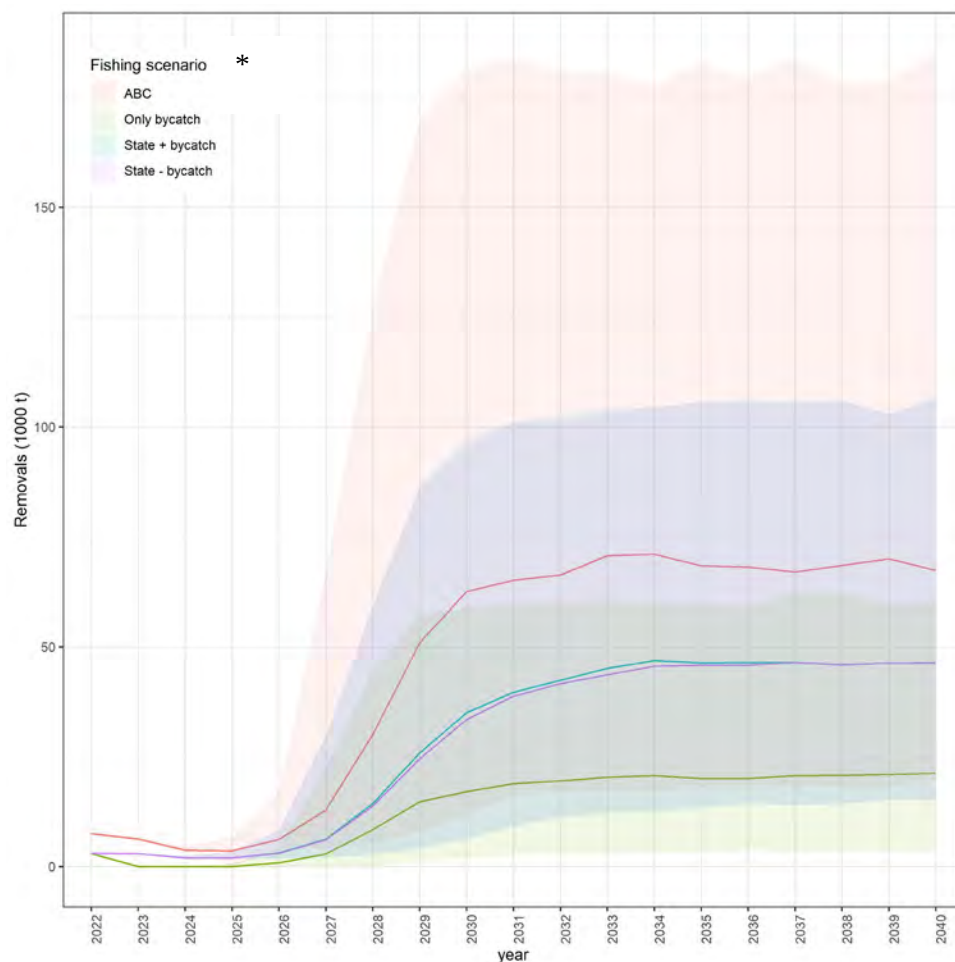
The predicted removals further exemplify the low recruitment into the population exhibited by predicted low removals. The status of the EBS snow crab directed fishery is likely to be uncertain for the first three to four years as the stock begins to recruit into the population and become mature.

Table 3-6 Projected removals using recruitment and mortality scenarios from 1982-2017 for bycatch only fishing removals and removals from a directed fishery and bycatch in million lbs of crab.

Sum of removals (mil lbs)		
Year	bycatch only*	state + bycatch*
2022	6.59	6.73
2023	0.05	6.53
2024	0.02	4.43
2025	0.03	4.33
2026	1.93	6.89
2027	6.37	13.83
2028	18.57	31.62
2029	32.56	57.01
2030	37.59	77.13
2031	41.62	87.56
2032	43.06	93.55
2033	44.93	99.51
2034	45.68	103.43
2035	44.30	102.24
2036	44.30	102.50
2037	45.65	102.41
2038	45.76	101.31
2039	46.27	102.22
2040	46.85	102.30

*Description of fishing scenarios:

- Bycatch only scenario = Allow BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area. No EBS snow crab fishing.
- State + bycatch scenario = EBS snow crab fishing plus BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.



*Description of fishing scenarios:

- Bycatch scenario = Allow BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area. No EBS snow crab fishing.
- State + bycatch scenario = EBS snow crab fishing plus BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- State – bycatch scenario = EBS snow crab fishing. No BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- ABC = the ABC calculated in the 2022 Crab SAFE.

Figure 3-10 Projected removals using recruitment and mortality scenarios from 1982-2017 for bycatch only fishing removals and removals from a directed fishery and bycatch in 1,000t of crab.

Conclusions

If the speed of rebuilding is the primary metric for benefits to the EBS snow crab stock, Alternative 2 / option 1 and option 2 provide no difference in rebuilding timeframe metrics. The main driver in speed of rebuilding is likely related to recruitment and the conditions that allow for increased recruitment into the population. Ecosystem conditions (section 3.2.3) may improve, and improvements would result in reduced natural mortality and increased production and are addressed during rebuilding through continued monitoring of ecosystem indicators. The allowance, in the projections, for recruitment to eventually increase and contribute to stock growth assumes that existing ecosystem conditions or other constraints on production will not continue indefinitely. Alternatively, if current ecosystem conditions prevail, and recruitment remains at low levels, the population may take substantially longer to show rebuilding progress and may never reach the designated B_{MSY} level.

3.3 Impacts of snow crab bycatch in other fisheries on rebuilding

Under the proposed alternatives, all of the existing measures that minimize fishery impacts on EBS snow crab would be expected to be maintained throughout the rebuilding period. The rebuilding projections presented in 2.2 provide evidence that median rebuilding time does not greatly differ under the various fishing mortality scenarios. Below the analysts detail non-target fishery interactions with EBS snow crab.

3.3.1 Bycatch of EBS Snow Crab in other fisheries

Bycatch of snow crab occurs in the BBRKC and EBS Tanner crab fishery, and in the groundfish fisheries. Bycatch in fisheries other than the groundfish has historically been relatively low (Table 3-7). Bycatch mortality estimates in Table 3-7 are derived from at-sea observer data and assume a 30% handling mortality rate.

Table 3-7 Bycatch of snow crab in the Tanner and BBRKC fisheries from 2005/06 season to the 2021/22 season

Year	Tanner fishery (mt)		BBRKC (mt)	
	Bycatch	Bycatch mortality	Bycatch	Bycatch mortality
2005/06	49	15	9	3
2006/07	15	5	6	2
2007/08	76	23	12	4
2008/09	10	3	7	2
2009/10	0	0	11	3
2010/11	0	0	5	1
2011/12	0	0	4	1
2012/13	0	0	8	2
2013/14	275	82	1	0
2014/15	1,785	535	1	0
2015/16	3,214	964	1	0
2016/17	0	0	3	1
2017/18	201	60	6	2
2018/19	697	209	2	1
2019/20	0	0	1	0
2020/21	484	145	3	1
2021/22	230	69	0	0

Source: Ben Daly, ADF&G, Oct 25, 2022

BBRKC = Bristol Bay Red King Crab

In the groundfish fisheries, crab bycatch management measures exist for the protection of BBRKC, EBS Tanner crab, and EBS snow crab stocks in the BSAI and include triggered area closures for BSAI groundfish trawl fisheries. Retention of crab bycatch is prohibited, so crab bycatch is also referred to as Prohibited Species Catch (PSC). For BBRKC, EBS snow crab, and EBS Tanner crab, triggered crab PSC limits exist for non-pelagic trawl fishing within specified areas. Non-pelagic trawl PSC accrues within these areas and these areas are closed to non-pelagic trawl directed fishing for groundfish in the fishery/sector that reaches its specified PSC limit. An area closure for EBS snow crab is triggered if the groundfish trawl fisheries by target/sector reach their allocated PSC limit for the COBLZ (Figure 3-11). PSC limits are based on a calendar year and not a crab year (July 1 – June 30).

The limit accrues only for EBS snow crab PSC taken within the COBLZ. No measures limiting PSC are currently in place for any pelagic trawl and fixed gear fisheries, nor are there overall limits placed on bycatch of snow crab species outside of COBLZ.

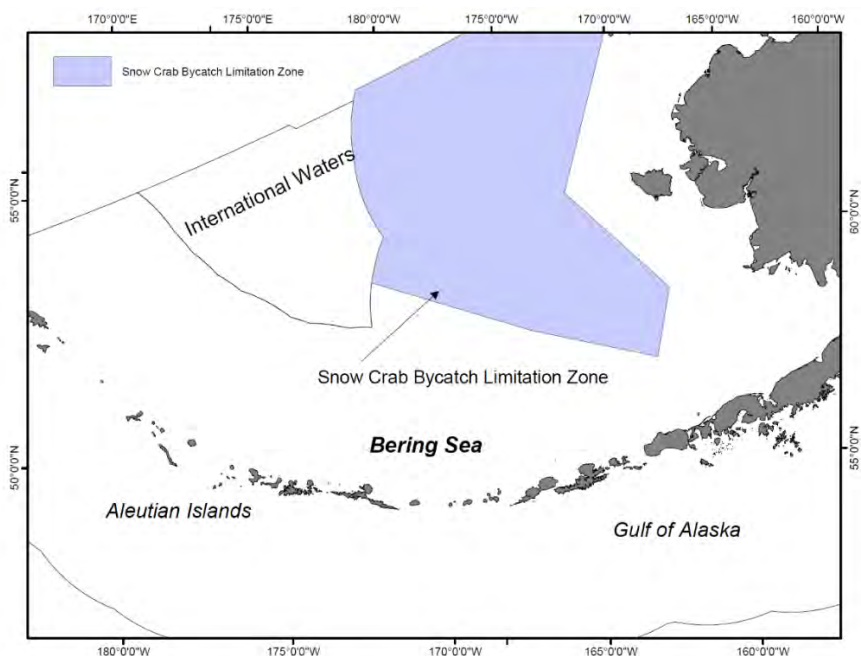


Figure 3-11 Map of COBLZ labeled as snow crab bycatch limitation zone in figure

The crab PSC limits are set each year in December during the groundfish harvest specifications process and apportioned across groundfish sectors. To determine PSC limits, crab stock assessment authors provide NMFS Inseason Management and/or Council staff with the abundance or biomass values necessary to compare to PSC thresholds established in Federal regulations.

EBS snow crab PSC limits are based on total abundance of snow crab as indicated by the model estimate of survey abundance. The limit in COBLZ is set annually at 0.1133% of the snow crab accepted assessment modeled abundance estimate from the NMFS standard summer trawl survey minus 150,000 crab, unless a minimum or maximum abundance threshold is reached.

- If 0.1133% multiplied by the total abundance is less than 4.5 million, then the minimum PSC limit will be 4.350 million animals.
- If 0.1133% multiplied by the total abundance is greater than 13 million, then the maximum PSC limit will be 12.850 million animals.¹

Snow crab bycatch that occurs outside COBLZ does not accrue towards the COBLZ limit.

The total abundance or biomass values are calculated differently for each stock. Table 3-8 provides the estimates of abundance for snow crab and the historical snow crab COBLZ PSC limits from 2006 through 2023. Historically, these values were derived from area-swept estimates of the NMFS bottom trawl survey. Presently, they are derived from model-based survey estimates.

Prior to implementation of Amendment 40, snow crab PSC limits did not exist for BSAI groundfish trawl fisheries. The Final Rule for Amendment 40 (62 FR 66829) explains that bottom trawl survey data from

¹ 50 CFR 679.21(e)(1)(iii)

1996 was indicating an increasing abundance of adult males, but females and pre-recruits (males that have not reached legal commercial size) were becoming less abundant. This trend was troubling because it could indicate declining abundance over a longer term. The Council relied on an industry work group to review proposed PSC limits for snow crab. The group met November 6–7, 1996, and came to a consensus on a PSC limit for snow crab. The group negotiated the PSC limit control rule based on the range included in Amendment 37 (0.005% to .25% of the total snow crab population) and past PSC use at different abundance levels.

Based on industry recommendations and Council and Secretary approval, Amendment 40 established a snow crab PSC limit as a rate that fluctuated with snow crab abundance and was applied within the newly defined area of the COBLZ. The PSC limit was established as 0.1133% of the total abundance under Amendment 40. However, the rule also included a lower bound (4.5 million animals) and an upper bound (13 million animals). Upon attainment of the snow crab bycatch limit as apportioned to a particular trawl fishery category, the COBLZ would be closed to directed fishing for species in that trawl fishing category, except for pollock with pelagic trawl gear. Snow crab PSC limits were later adjusted under Amendment 57 by reducing snow crab PSC limits in COBLZ by 150,000 animals.

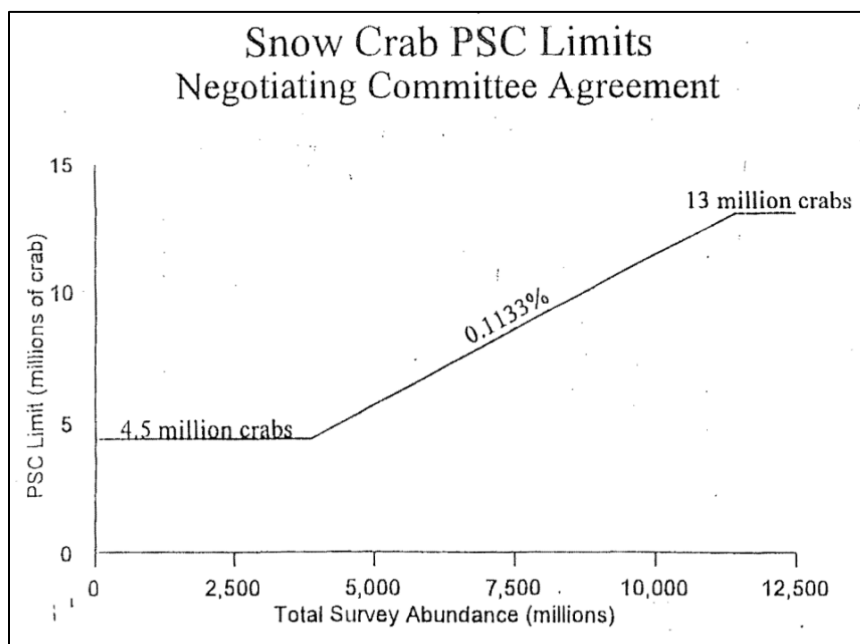


Figure 3-12 PSC limits for EBS snow crab within the COBLZ, as implemented under Amendment 40 (later amended to the current limits as described in this section)

The Council requested at their June 2022 meeting that the analysis include additional information surrounding the removal of the PSC floor, count all trawl PSC throughout the full range of the stock toward the PSC limit, and limit on fixed gear PSC. To address this request, analysts provided information in Table 3-8 detailing PSC limit calculations without the PSC limit floor, PSC (number of snow crab), and percent of PSC limit inside and outside the COBLZ in the BSAI non-pelagic trawl fisheries and pelagic trawl fisheries² from 2006 through 2023. The table also provides PSC for pot and hook-and-line gear inside and outside the COBLZ. PSC for pot and hook-and-line inside the COBLZ is not available prior to 2010. Table 3-9 provides PSC by gear in metric tons.

² There is one PSC limit exception for trawl vessels using pelagic gear. When a bycatch allowance, or seasonal apportionment thereof, specified for the pollock/Atka mackerel/"other species" fishery category is reached, only directed fishing for pollock is closed to trawl vessels using non-pelagic trawl gear (50 CFR 679.21(e)(7)(i)&(iv)).

The EBS snow crab abundance estimate has been historically low enough to result in a PSC limit floor (4,350,000 crab) six times since 2006 for non-pelagic trawl gear (Table 3-8). The current regulations surrounding PSC limits have been traditionally non-constricting for non-pelagic and pelagic trawl gear. Concerning PSC calculations without the PSC limit floor, for the years 2007-2010, 2022, and 2023, the abundance estimates would be low enough that the PSC limit would be lower than 4,350,000 million animals. Nevertheless, as noted in Table 3-8 for those years, the PSC limit without a floor and factoring in PSC inside and outside the COBLZ would have been non-constraining for non-pelagic and pelagic trawl gear.

For snow crab, non-pelagic trawl PSC occurs primarily in the southeast portion of the COBLZ and extends northwest throughout the zone and to the north, east, and south of the zone (Figure 3-13). For pelagic trawl gear, snow crab PSC follows a similar spatial pattern but to a much smaller magnitude and does not spread as far around the southeast border of the COBLZ. PSC in pot gear is distributed throughout the southern two-thirds of the COBLZ and beyond the southeast border of the COBLZ along the Aleutian peninsula. Snow crab PSC in hook and line (HAL) gear seems to have the largest spatial distribution, which is likely due to the spatial distribution of effort in the HAL fisheries in these areas.

Table 3-8 Snow crab PSC limit with the PSC floor, a hypothetical scenario detailing PSC limits without the PSC floor, and PSC for BSAI non-pelagic and pelagic trawl, and PSC for pot and hook and line fisheries from 2006 through 2022 (# of crabs)

Year	Snow crab abundance estimate ¹ (billions of animals)	Snow crab PSC limit	Snow crab PSC limit without floor	Nonpelagic trawl PSC (# of crabs)			Pelagic trawl PSC (# of crabs)			Total trawl PSC as a % of PSC limit	Total trawl PSC as a % of PSC limit without a floor	Pot PSC (# of crabs)*			H&L PSC (# of crabs)*			Groundfish total (# of crabs)	Total groundfish PSC for all gears relative to
				PSC inside COBLZ	PSC outside COBLZ	Total PSC (from both COBLZ and outside COBLZ)	PSC inside COBLZ	PSC outside COBLZ	Total PSC (from both COBLZ and outside COBLZ)			PSC inside COBLZ	PSC outside COBLZ	Total PSC	PSC inside COBLZ	PSC outside COBLZ	Total PSC		
2006	5.22	5,761,674	5,761,674	947,380	63,343	1,010,723	480	2,423	2,903	18%	18%	N/A	N/A	333,050	N/A	N/A	49,597	1,396,273	0.4188%
2007	3.25	4,350,000	3,532,250	1,821,672	78,201	1,899,874	357	2,580	2,936	44%	54%	N/A	N/A	1,536,818	N/A	N/A	48,545	3,488,173	0.3920%
2008	3.33	4,350,000	3,622,890	677,361	112,455	789,816	1,482	3,483	4,965	18%	22%	N/A	N/A	695,159	N/A	N/A	83,162	1,573,101	0.2847%
2009	2.60	4,350,000	2,795,800	436,051	87,256	523,307	162	3,048	3,209	12%	19%	N/A	N/A	605,631	N/A	N/A	54,638	1,186,785	0.2375%
2010	3.06	4,350,000	3,316,980	1,677,389	25,164	1,702,552	5,227	383	5,610	39%	51%	62,247	460,357	522,605	22,432	13,765	36,197	2,266,964	0.1918%
2011	7.47	8,310,480	8,310,480	741,568	20,670	762,238	4,444	670	5,113	9%	9%	16,647	85,029	101,676	27,602	20,764	48,366	917,394	0.1177%
2012	6.34	7,029,520	7,029,520	600,223	22,767	622,990	2,721	501	3,222	9%	9%	1	16,536	16,536	21,163	25,951	47,114	689,862	0.1054%
2013	9.40	10,501,333	10,501,333	673,966	15,285	689,251	3,670	395	4,065	7%	7%	0	14,784	14,784	17,660	11,552	29,212	737,312	0.0946%
2014	10.01	11,185,892	11,185,892	466,885	14,280	481,165	2,823	508	3,331	4%	4%	2,159	82,808	84,967	24,536	14,591	39,127	608,590	0.0867%
2015	9.85	11,011,976	11,011,976	484,297	4,367	488,664	2,906	55	2,961	4%	4%	20,390	101,171	121,561	19,956	7,347	27,303	640,489	0.0806%
2016	4.29	4,708,314	4,708,314	163,878	2,211	166,090	765	119	884	4%	4%	6,039	13,998	20,037	27,356	14,215	41,570	228,581	0.0741%
2017	8.17	9,105,477	9,105,477	153,101	6,243	159,343	253	81	334	2%	2%	2,567	142,891	145,457	18,658	18,501	37,158	342,293	0.0688%
2018	8.18	9,120,539	9,120,539	1,577,907	4,242	1,582,149	247	30	277	17%	17%	399	52,136	52,535	9,090	15,886	24,976	1,659,938	0.0646%
2019	10.65	11,916,450	11,916,450	936,578	4,650	941,228	48	21	69	8%	8%	19,726	52,447	72,174	11,231	14,699	25,930	1,039,401	0.0443%
2020	7.71	8,580,898	8,580,898	756,559	22,296	778,855	1,672	42	1,714	9%	9%	24,037	118,021	142,059	9,490	12,181	21,671	944,298	0.0346%
2021	6.48	7,191,840	7,191,840	234,590	12,105	246,695	451	71	522	3%	3%	15,903	51,549	67,452	10,948	10,238	21,186	335,855	0.0223%
2022	1.42	4,350,000	1,458,860	172,299	3,036	175,336	29	13	42	4%	12%	23,484	21,820	45,304	12,285	10,152	22,437	243,118	0.0171%
2023**	2.58	4,350,000	2,777,672																

Source: NMFS Alaska Region Catch Accounting System, data compiled by AKFIN file name Snow Crab PSC Tables

Bold text indicates the PSC limit was set to its lowest limit

¹Abundance estimate is based survey results from the previous year.

* Denotes sectors that do not have a PSC limit

**Denotes estimated PSC limit and PSC is not yet available for the 2023 groundfish fisheries.

NA - PSC for pot and hook-and-line gears prior to 2010 was not broken out by inside/outside COBLZ.

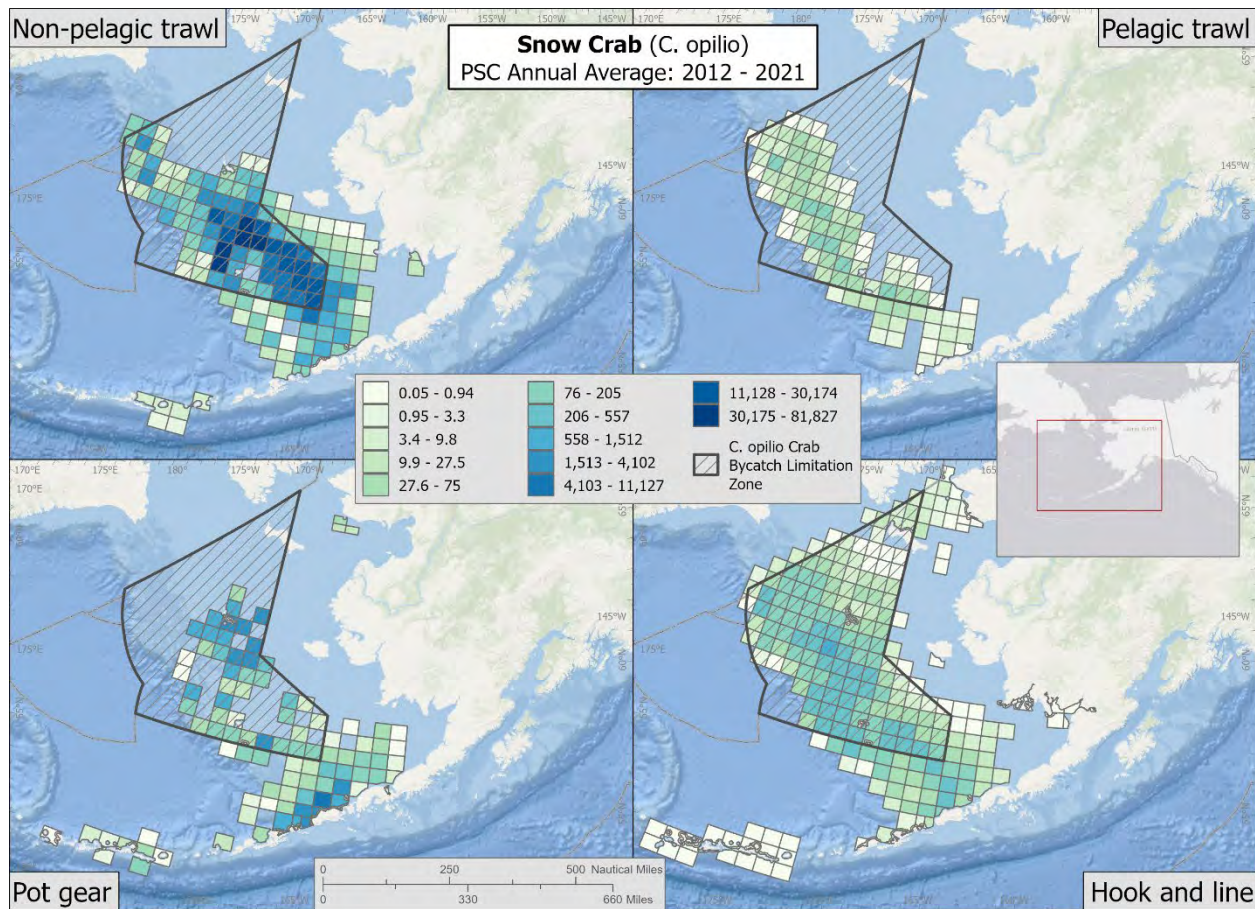
Table 3-9 Estimated PSC for non-pelagic and pelagic trawl, pot, and hook and line fisheries from 2006 through 2022 (mt). The PSC limit only applies to non-pelagic and pelagic trawl gear. All other gear types do not have a PSC limit in place.

Year	Nonpelagic trawl PSC (mt)			Pelagic trawl PSC (mt)			Pot PSC (mt)*			H&L PSC (mt)*			Groundfish total (mt)
	PSC		Total PSC	PSC inside COBLZ	PSC outside COBLZ	Total PSC	PSC inside COBLZ	PSC outside COBLZ	Total PSC	PSC inside COBLZ	PSC outside COBLZ	Total PSC	
	PSC inside COBLZ	outside COBLZ											
2006	351.33	23.52	374.85	0.19	0.95	1.13	N/A	N/A	127.38	N/A	N/A	19.00	522.36
2007	639.78	27.10	666.87	0.10	0.73	0.83	N/A	N/A	545.83	N/A	N/A	15.81	
2008	187.37	30.48	217.85	0.43	0.99	1.42	N/A	N/A	190.75	N/A	N/A	23.30	433.31
2009	119.23	23.88	143.11	0.04	0.78	0.82	N/A	N/A	152.69	N/A	N/A	14.18	310.80
2010	310.38	4.66	315.03	1.38	0.10	1.49	26.58	196.58	223.16	12.28	7.53	19.81	559.49
2011	161.03	4.49	165.51	1.55	0.23	1.79	10.22	52.22	62.45	16.22	12.20	28.42	258.16
2012	133.56	5.07	138.62	0.84	0.15	0.99	0.00	10.59	10.59	13.08	16.04	29.11	179.32
2013	171.95	3.90	175.85	0.47	0.05	0.52	0.00	8.60	8.60	9.67	6.33	16.00	200.96
2014	102.00	3.12	105.12	0.60	0.11	0.71	1.23	47.04	48.26	11.20	6.66	17.86	171.94
2015	77.10	0.70	77.79	0.42	0.01	0.43	10.78	53.49	64.27	8.82	3.25	12.07	154.57
2016	28.58	0.39	28.96	0.15	0.02	0.17	2.56	5.94	8.50	11.91	6.19	18.09	55.73
2017	27.45	1.12	28.57	0.05	0.02	0.06	0.76	42.16	42.92	7.76	7.69	15.45	87.00
2018	343.48	0.92	344.40	0.05	0.01	0.05	0.16	20.67	20.83	3.47	6.07	9.54	374.82
2019	173.34	0.86	174.20	0.00	0.00	0.01	3.04	8.09	11.13	4.07	5.33	9.40	194.75
2020	135.19	3.98	139.18	0.24	0.01	0.25	6.12	30.04	36.16	3.37	4.33	7.70	183.29
2021	49.92	2.58	52.50	0.07	0.01	0.08	5.56	18.02	23.58	4.23	3.96	8.19	84.34
2022	43.88	0.77	44.65	0.01	0.00	0.01	9.48	8.81	18.29	5.12	4.23	9.36	72.30

Source: NMFS Alaska Region Catch Accounting System, data compiled by AKFIN file name Snow Crab PSC Tables

* Denotes sectors that do not have a PSC limit

N/A - PSC for pot and hook-and-line gears prior to 2010 was broken out by inside/outside COBLZ.



Source: NMFS Alaska Region Catch Accounting System, data compiled by AKFIN in Comprehensive_PSC Maps produced by PSFMC.

Figure 3-13 EBS snow crab PSC (average annual #crab) by gear type, 2012-2021

Table 3-10 demonstrate that non-pelagic trawl fisheries accounted for the greatest levels of snow crab in groundfish fisheries from 2012 through 2022. Bycatch of EBS snow crab in the groundfish fisheries has been highest in the yellowfin sole fishery, followed by the flathead sole fishery.

Table 3-10 Snow crab PSC in COBLZ by gear type and target species from 2012 through 2022 (# of crab)

Gear type	Target Species	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
HAL	Halibut	-	-	-	-	10	19	92	24	62	22	0
NPT	Alaska Plaice - BSAI	2,098	7,258	-	21,117	2,519	140	5,279	1,925	1,964	13,266	2
	Arrowtooth Flounder	2,518	8,892	6,440	3,786	2,761	33,442	465	6,205	30,646	1,557	27
	Flathead Sole	17,166	67,239	79,887	20,802	10,537	30,510	279,286	217,865	197,199	53,071	10,858
	Greenland Turbot - BSAI	-	-	-	-	117	2,002	78	38	3,008	162	0
	Kamchatka Flounder - BSAI	-	-	-	-		0	457	1,188	190	0	0
	Pacific Cod	415	6,170	6,657	4,464	1,869	900	6	45,175	1,567	115	82
	Pollock - bottom	C	1,888	15,301	5,296	190	3,058	4,866	6,006	38,288	3,944	1,539
	Rock Sole - BSAI	-	1,807	8,024	6,058	27,468	19,118	2,454	10,421	18,191	7,830	6,410
	Rockfish	-	C	-	-	17	0	14,408	652	92	487	0
	Yellowfin Sole - BSAI	559,559	550,261	329,488	420,528	115,127	61,049	1,268,997	644,006	460,446	151,531	101,047
Pot	Pacific Cod	1	-	-	-	-	1,396	25	-	47	1	-
	Sable	-	-	-	-	-	-	-	-	28	0	-
PTR	Pollock - bottom	67	135	-	C	51	C	-	-	-	-	-
	Pollock - midwater	2,453	3,380	2,811	2,887	682	202	247	48	1,647	449	0

Source: NMFS Alaska Region Catch Accounting System, data compiled by AKFIN in Comprehensive_PSC [Crab_PSC_AREA(11-13-20)]

C indicates confidential data, HAL = hook and line, NPT = non-pelegic trawl, pot = pot gear, PTR = pelagic trawl

3.3.1.1 Unobserved Mortality

Fishing activities lead to crab mortality in ways that are not directly observed. This includes both post-release mortality of discarded crab (which is estimated through a discard mortality rate) and crab that are never captured by fishing gear but die due to gear interactions or sustained damages that cause mortality. The potential for unobserved mortality of crabs that interact with gear but are not captured has long been a concern for the management of groundfish fisheries in the BS (Witherell and Pautzke, 1997; Witherell and Woodby, 2005). However, unobserved mortality is not accounted for in crab stock assessments and is not accrued towards trawl PSC limits.

Published studies on the impacts of trawl gear on crab have generally focused on non-pelagic gear, including studies in the BS and in the shrimp fishery off the east coast of Canada. Studies have utilized bottom and wing recapture nets to collect impacted crab that would not have ended up in the trawl net, cameras to visualize crab that were avoiding the trawl net, and even submersible camera-equipped vehicle dives to compare damage to crabs before and after trawling in an area. Rose (1999) cites an earlier study (Donaldson 1990) as a “preliminary estimate” of the rate of unobserved crab injuries, wherein RKC were tethered to the seafloor, a trawl net was towed over the area, and divers attempted to recover the crab. Of 169 crab, 21 percent were captured in the net, 46 percent were recovered by the divers, and 33 percent could not be located. While only two of the 78 recovered crabs were injured, Rose noted the ambiguity posed by the fate of the unrecovered crabs relative to the sample size. An unpublished video study (Rose 1995) found that sweep diameter was the main factor in whether crab could escape over the sweep (note that sweeps were not elevated during this period). The study was not able to determine the frequency, nature, or severity of injuries to crabs that went under the sweep. The Rose (1999) study in Bristol Bay used a recapture net to assess injury rates to crab that pass under different types of footrope. Eight experimental tows yielded injury rates of between 5 percent and 10 percent of the recaptured crab.

Subsequent work by Rose et al. (2013) provided estimates of the unobserved mortality rates of crabs swept over by trawl gear common to bottom trawl fisheries in the BS. This study again recaptured crab after encountering trawl sweeps and footropes, but also used a reflex-assessment method³, calibrated on mortalities of crabs held onboard the vessel, to predict the delayed mortality of recaptured crab impacted by, but not captured in, the trawl. This study also evaluated crab caught in a control net where they did not encounter the trawl gear to adjust observed mortality rates for the effects of capture and handling. This research demonstrated that mortality rates were higher for RKC than either snow or tanner crab and depended substantially on which part of the trawl system crabs encountered. Additionally, reduction of crab mortality rates by altering specific gear designs showed that gear modifications can mitigate unobserved mortality (Hammond et al. 2013). Raised sweeps essentially eliminated the 5% snow crab mortality rates measured using on-bottom sweeps, while not reducing flatfish herding (Rose et al 2010). This partly justified regulations requiring raised sweeps for flatfish fisheries (Amendment 94). Footrope modifications also reduced mortality rates for crabs encountering footropes. Relative to an older-style footrope, one with less weight and more clearance underneath reduced snow crab mortality rate from 10 to 5%. One supposition was that effective herding by sweeps can reduce overall crab mortality because it reduces the amount of footrope-swept area needed to catch the same number of flatfish.

Further follow-up research (Rose et al. 2014) used the same methods to estimate unobserved mortality rates, but also used recapture nets covering the full area behind the footrope, allowing assessment of the proportion of crabs captured to those passing under the net. It also included footrope designs better matching current fishery practice. While the final report graphically provided mortality rates and proportion-captured for snow and Tanner crabs, it did not combine these values to estimate a multiplier to estimate unobserved crab mortalities from observed crab bycatch. The primary author calculated such multipliers for snow crab from the original data and has provided them and a description of the

³ Reflex Action Mortality Predictor (RAMP); see Davis and Ottmar (2006) and Stoner et al. (2008).

calculations as a comment to the October 2022 SSC meeting (Rose 2022). Multipliers for different footropes ranged from 0.5 to 3.9 unobserved snow crab mortalities per observed crab in the catch and multipliers were higher for males than for females.

The remote-video study of shrimp trawl interactions with snow crab off Saint Mary's Bay in southeastern Canada only assessed areas swept by the trawl footrope (Dawe et al. 2007). The study did not collect a large sample of direct post-trawl observations but ultimately did not report any dead crab in the trawl corridor or crab with carapace damage. The study did not find a reduced density of snow crab in the trawled bays after trawling occurred. However, the study concluded that intensive trawling could increase crab leg-loss by about 10 percent.

A trawl-mounted video study in the same part of Canada looked at how snow crab physically reacted to shrimp bottom trawls (Nguyen et al. 2014). This study was also limited to the footrope portion of the trawl and concluded that about 54 percent of observable crab interacted with the footgear (e.g., elevating discs, spacers, or chains). The majority of video-observed crabs actively responded to the approaching trawl and tried to escape. The study was unable to estimate the severity or likelihood of mortality after passing under footgear. This study, and references to herding in Rose et al. (2013), highlights the relevance of crab shell condition to susceptibility to unobserved trawl mortality. In a time/area where crab are likely to be in a soft-shell condition and less mobile, unobserved mortality rates could be higher than the ranges estimated in the studies available.

The topic of unobserved mortality was addressed in a Council analysis when crab PSC limits for trawl fisheries were reviewed in February 2021 prior to taking no action (see section 3.4.6 and Appendix 4 in NPFMC 2021a). The SSC's February 2021 report noted that including any future estimation of unobserved crab mortality (from both groundfish and directed crab fishing) in a stock assessment would require extensive evaluation to understand how the assessment's parameters for factors like catchability, natural mortality and reference points would be affected. The SSC noted that "unobserved mortality is a source of both assessed and unassessed uncertainty throughout the history of the assessments (e.g., currently attributed to natural mortality), and that the ABC/TAC buffers in place are an appropriate process to account for sources of uncertainty that cannot be explicitly described in the assessment." Finally, at the October 2022 meeting, the SSC recommended a working group to develop a framework for how to estimate the magnitude of unobserved mortality for crab stocks and how these estimations may be utilized in BSAI crab stock assessments.

3.3.2 Effects of the alternatives on Snow Crab Bycatch in Other Fisheries

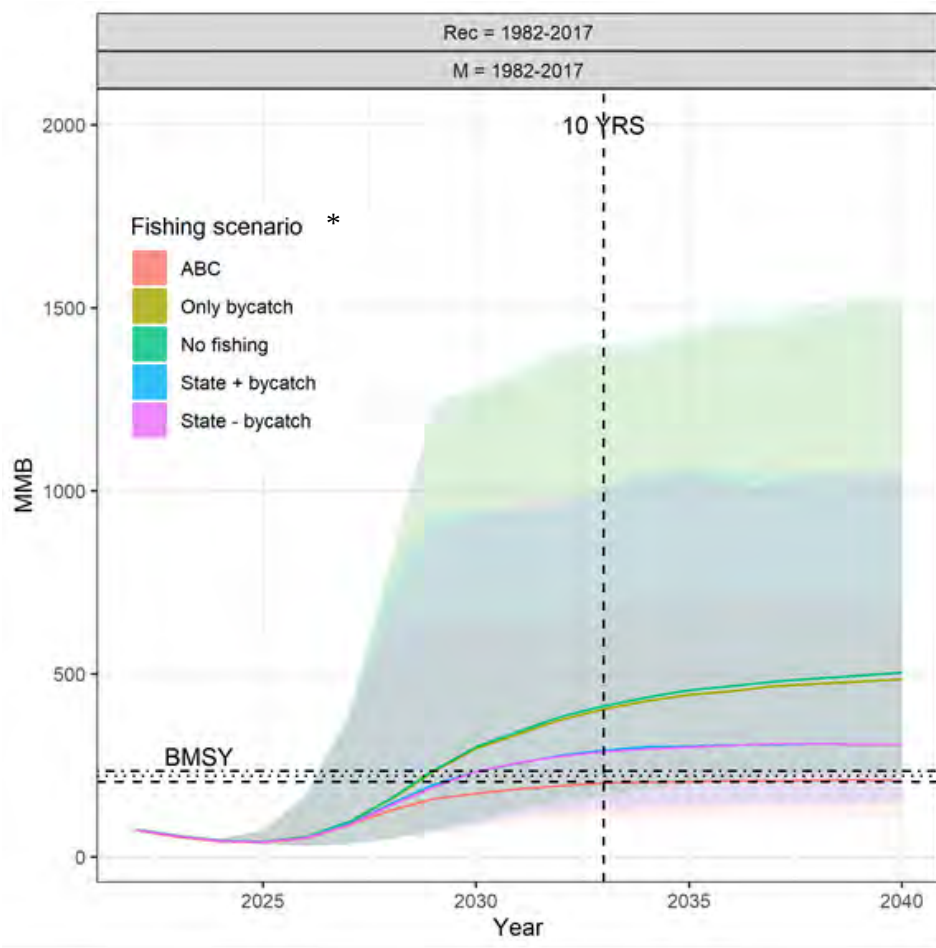
Under the proposed alternatives, all the existing measures that minimize fishery impacts on EBS snow crab would be expected to be maintained throughout rebuilding period. These measures include an established EBS snow crab PSC limit for non-pelagic trawl gear within COBLZ. No alternative proposes new management measures to reduce PSC of EBS snow crab in the groundfish fisheries.

The impacts of EBS snow crab bycatch were thoroughly evaluated as a potential factor in stock recovery and analyses of rebuilding times factoring in bycatch by groundfish fisheries at both average and 5x and 100x observed bycatch levels. Bycatch mortality was specified in the projection model as the average of the estimated bycatch fishing mortality over the last ten years. The State of Alaska's harvest control rule was approximated by averaging the ratio of the total allowable catch (TAC) set by the State and the ABC over the last 10 years (Daly, personal communication). The ratio (equal to 0.40) was used to scale the ABC calculated in the projections. In these projections, the time for stock recovery with average bycatch and with no bycatch are not differentiable (no effect). The reason for no discernible difference in the rebuilding time with or without bycatch is that EBS snow crab mortality due to bycatch is very small relative to total abundance of EBS snow crab and the effects of stock productivity, a function of future ecosystem conditions, overwhelms the effect of bycatch under the Alternative 2 options.

To better assess the impacts to non-target species, the Council requested, in their [June 2022 motion](#), that analysts include additional information surrounding the removal of the PSC floor. To address this request, analysts detailed historical PSC limits with and without the PSC floor, PSC (number of snow crab), and percent of PSC limit inside the COBLZ and percent of PSC limit inside and outside the COBLZ in the BSAI non-pelagic and pelagic trawl fisheries in Table 3-8. The table also provides the total PSC for pot and hook-and-line fisheries inside and outside the COBLZ from 2006 through 2021 since these gear/fisheries are not limited by PSC. Overall, changing PSC management to remove the PSC limit floor and include PSC outside the COBLZ for the non-pelagic and pelagic trawl gear, which has been non-constraining for the gear since 2006, would not have any measurable effect on the rebuilding time for EBS snow crab since the rebuilding projections with and without bycatch were indiscernible. This assessment also applies to changes in PSC management to include PSC limits for pot and hook-in-line gears. The addition of PSC limits for pot and hook-and-line gears would not change the rebuilding time for EBS snow crab since model projections indicate no discernible difference in the median rebuilding time with or without bycatch.

Additionally, sensitivities about the assumptions of unobserved mortality were explored in which the observed time series of bycatch was multiplied by 5 and 100 before the model was fit to the data. Model results of these assumptions using the selected recruitment and natural mortality parameters are provided in Figure 3-12 and Figure 3-13 and their associated projection specifications in Table 3-11 and Table 3-12. The general trends of the timing of rebuilding under a given productivity scenario and zero fishing mortality were similar to the results from modeling when no additional unobserved mortality was modeled. The lack of differences in the rebuilding time with and without unobserved mortality is again likely due to the effects of stock productivity overwhelming the effect of unobserved mortality.

As noted in the September 13, 2022, EBS snow crab SAFE, Appendix D by the stock author, there must be some unobserved mortality on snow crab in the BS by the groundfish fleets (section 3.3.2). However, it is difficult to make a case for large impacts of non-directed fisheries on the recent population dynamics of snow crab. If the non-directed fisheries were a larger driver of population dynamics, it is hard to explain how the largest cohort ever observed would have occurred recently and developed through the size ranges that are impacted by the non-directed fleets. Still, managers only have two levers for impacting the population dynamics of snow crab in the BS: adjusting fishing mortality in the directed fishery or adjusting fishing mortality in the non-directed fleets. The other apparent drivers of snow crab dynamics (e.g., sea ice) are outside of the control of managing bodies.



*Description of fishing scenarios:

- No fishing = no EBS snow crab fishery and no BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- Only bycatch scenario = Allow BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area. No EBS snow crab fishing.
- State + bycatch scenario = EBS snow crab fishing plus BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- State – bycatch scenario = EBS snow crab fishing. No BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- ABC = the ABC calculated in the 2022 Crab SAFE.

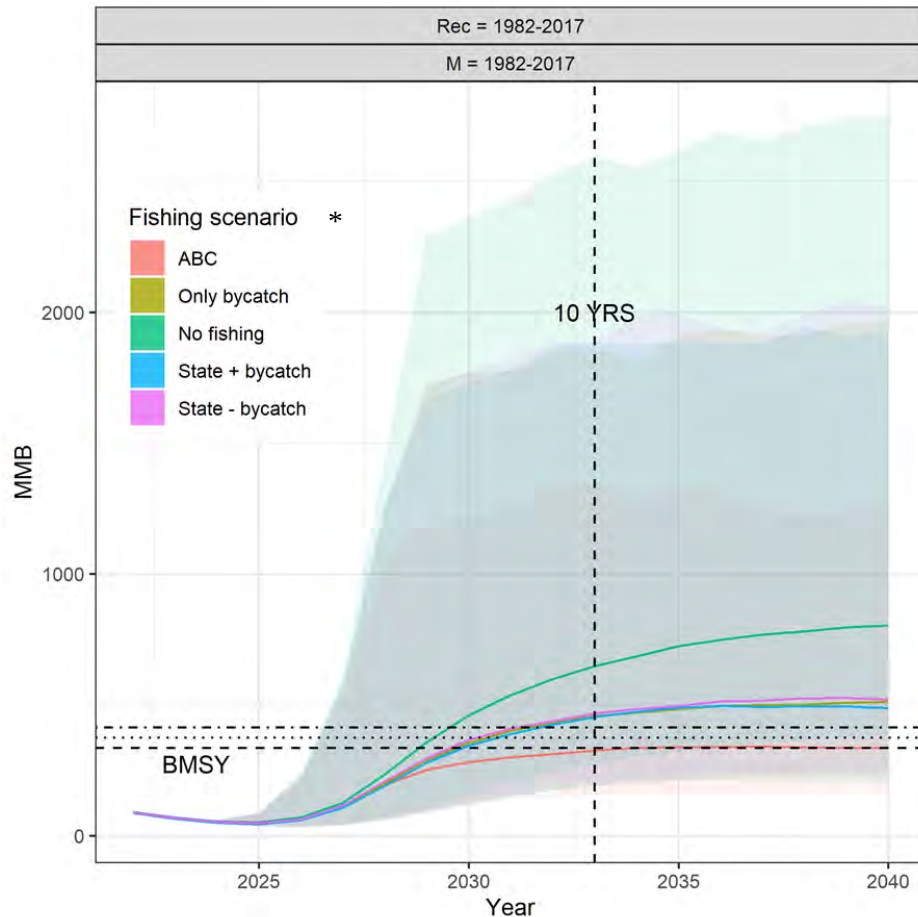
Figure 3-14 Projections of rebuilding trajectories under 1982-2017 productivity scenario (recruitment and mortality), fishing strategies, and target biomasses (three horizontal lines corresponding to 1982-2021, 1982-2017, and 2005-2019, in ascending order).

Table 3-11 Rebuilding time for the 1982-2017 productivity scenario. Median T_{min} is calculated relative to the status quo B_{MSY} proxy that uses average recruitment from 1982-2021. The '5%' and 95%' columns represent the 5th and 95th inter-simulation quantiles for the uncertainty around biomass trajectories during the projection period. Bycatch time series are 5x those used in the status quo assessment.

Projection specifications – 5x			T_{min}		
Fishing Scenario*	Recruitment	Mortality	Median	5%	95%
<i>No fishing</i>	1982-2017	1982-2017	2029	2027	2035
<i>bycatch</i>	1982-2017	1982-2017	2029	2027	2036
<i>State + bycatch</i>	1982-2017	1982-2017	2030	2027	Inf
<i>State - bycatch</i>	1982-2017	1982-2017	2030	2027	Inf
<i>ABC</i>	1982-2017	1982-2017	2035	2027	Inf

*Description of fishing scenarios:

- No fishing = no EBS snow crab fishery and no BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- Bycatch scenario = Allow BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area. No EBS snow crab fishing.
- State + bycatch scenario = EBS snow crab fishing plus BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- State – bycatch scenario = EBS snow crab fishing. No BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- ABC = the ABC calculated in the 2022 Crab SAFE.



*Description of fishing scenarios:

- No fishing = no EBS snow crab fishery and no BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- Only bycatch scenario = Allow BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area. No EBS snow crab fishing.
- State + bycatch scenario = EBS snow crab fishing plus BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- State – bycatch scenario = EBS snow crab fishing. No BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- ABC = the ABC calculated in the 2022 Crab SAFE.

Figure 3-15 Projections of rebuilding trajectories under 1982-2017 productivity scenario (recruitment and mortality), fishing strategies, and target biomasses (three horizontal lines corresponding to 1982-2021, 1982-2017, and 2005-2019, in ascending order). Bycatch time series are 100x those used in the status quo assessment.

Table 3-12 Rebuilding time for the 1982-2017 productivity scenario. Median T_{min} is calculated relative to the status quo B_{MSY} proxy that uses average recruitment from 1982-2021. The '5%' and '95%' columns represent the 5th and 95th inter-simulation quantiles for the uncertainty around biomass trajectories during the projection period. Bycatch time series are 100x those used in the status quo assessment.

Projection specifications – 100x			Tmin		
Fishing Scenario*	Recruitment	Mortality	Median	5%	95%
<i>No fishing</i>	1982-2017	1982-2017	2029	2027	2036
<i>bycatch</i>	1982-2017	1982-2017	2030	2027	Inf
<i>State + bycatch</i>	1982-2017	1982-2017	2030	2027	Inf
<i>State - bycatch</i>	1982-2017	1982-2017	2030	2027	Inf
<i>ABC</i>	1982-2017	1982-2017	2035	2027	Inf

*Description of fishing scenarios:

- No fishing = no EBS snow crab fishery and no BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- Bycatch scenario = Allow BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area. No EBS snow crab fishing.
- State + bycatch scenario = EBS snow crab fishing plus BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- State – bycatch scenario = EBS snow crab fishing. No BBRKC, Tanner crab, and groundfish fishing in the EBS snow crab stock area.
- ABC = the ABC calculated in the 2022 Crab SAFE.

The NMFS trawl survey would continue throughout rebuilding, and the BSAI Crab Plan Team would continue to report stock status and progress towards the rebuilt level in the annual SAFE Report. Additionally, ADF&G and NMFS monitor catches and bycatch of EBS snow crab in other fisheries. ADF&G also requires observer coverage on catcher vessels and catcher processing vessels to monitor catch and landings. ADF&G reports harvest from the commercial fishery and those data are included annually in the SAFE.

State and federal observer programs monitor bycatch with State coverage of the crab fisheries and federal monitoring of the groundfish trawl, pot, and longline fisheries. Estimates of crab bycatch from all fisheries will be reported annually in the SAFE. The BSAI Crab Plan Team will assess bycatch relative to the expectations and assumptions of the rebuilding plan. Additionally, if bycatch were to increase substantially, inseason actions could be taken to restrict harvest or area in the groundfish fisheries, if necessary, to reduce bycatch.

In conclusion, because fishing mortality is not the primary driver of the current snow crab population status, either option does not substantively change the projection of T_{MIN} , and under both options it is assumed that the stock would rebuild within ten years. Bycatch rates in non-target fisheries showed no substantial effect on rebuilding time. In projections that apply average bycatch levels during rebuilding, the median time for stock recovery was not differentiable from the no fishing scenario. Additionally, the time for stock recovery was minimally affected in projections that used 5x and 100x level of observed bycatch as seen in the rebuilding projections. Therefore, analysts concluded that recovery of the EBS snow crab stock is likely to not be affected by current or predicted bycatch levels, based on average historical bycatch.

Additionally, Alternative2/option 1 or Alternative/option 2, would maintain all of the existing measures that minimize fishery impacts on EBS snow crab throughout the rebuilding period. Alternative 1 would likely produce similar results as Alternative 2/option 1; however, selection of this alternative is not feasible as it is not in compliance with MSA guidelines. No alternative proposes new management measures to reduce PSC of EBS snow crab in the groundfish fisheries. Therefore, bycatch in groundfish fisheries or other-directed crab fisheries are not expected to directly impact the success in rebuilding under any of the proposed alternatives.

3.4 Habitat

3.4.1 Prevailing Ecosystem Conditions

EBS Snow Crab rebuilding will occur within the context of prevailing ecosystem conditions, which are most recently characterized in the “Ecosystem and Socioeconomic Profile” (ESP) included in the [2022 BSAI Crab SAFE](#). The ESP uses data collected from a variety of sources to generate ecosystem and socioeconomic indicators that may help explain trends for a given stock.

Effects of climate change on EBS snow crab (section 3.2.3) was evident and likely contributed to the mass mortality event in snow crab. Increased warming and declines in sea ice are expected to decrease benthic juvenile snow crab prey resources supplied to the benthos through decreased benthic-pelagic flux (Copeman et al., 2021). Both settlement intensity and early benthic survival are likely critical determinants of year-class strength in snow crab (Sainte-Marie et al., 1996), and successful advection to areas of suitable temperature and muddy substrate are thought to be critical criteria for juvenile survival (Dionne et al., 2003)

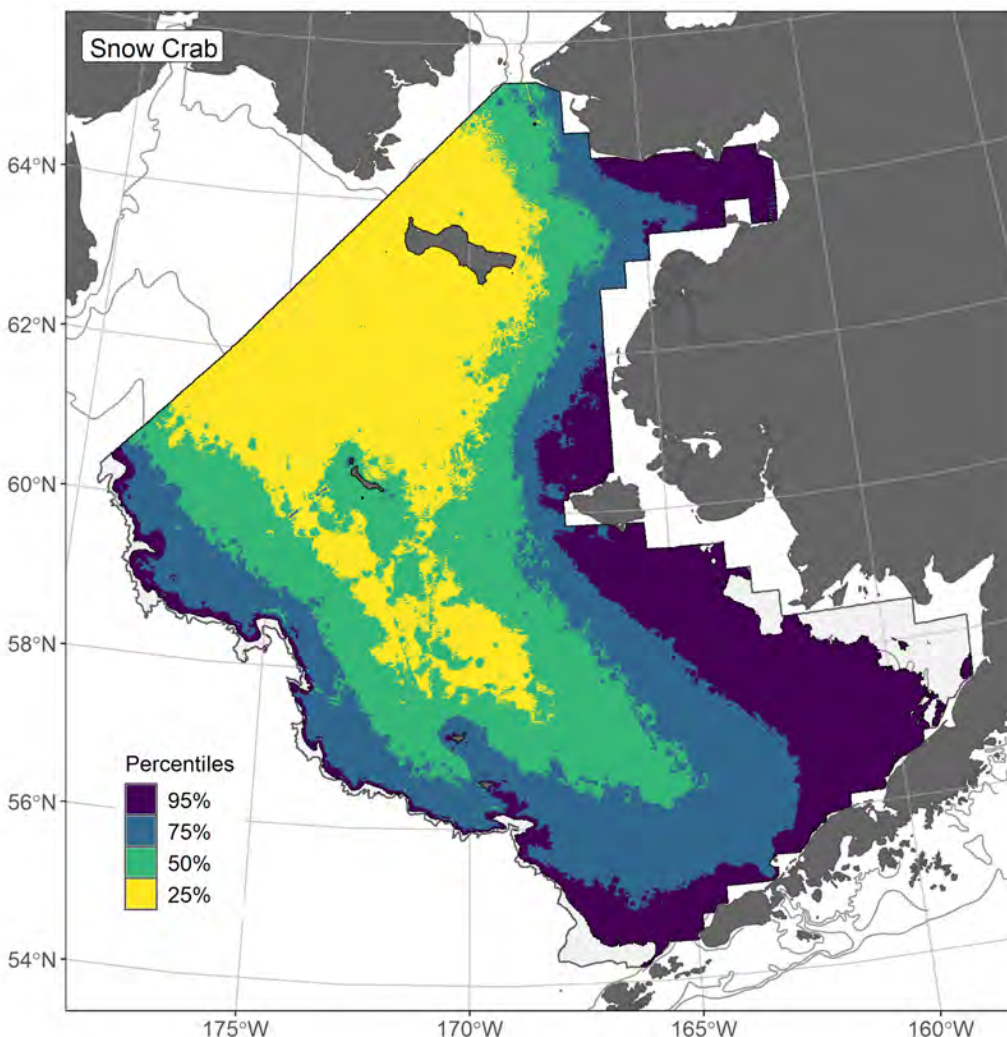
Physical indicators reveal much more favorable conditions for snow crab in 2022 following the 2018-2019 heat wave. Sub-0°C temperatures occupied by immature snow crab suggest that survival may be optimal for a new cohort of juveniles evident in the 2022 NOAA bottom trawl survey (Zacher et al., in review). Likewise, above-average chlorophyll-a biomass and benthic invertebrate density may be indicative of increased prey resources for larval and benthic stages of snow crab. Pacific cod consumption and bitter crab syndrome prevalence reached all-time highs in 2016 and may have been contributed to 2018-2019 mortality events, although both indices have returned to near-average in recent years. Northerly shifts in male snow crab centers of abundance in 2021-2022 have coincided with continued declines in mature male biomass, and may be a distributional response to recent warming in the BS.

Spatial patterns in juvenile and adult snow crab distribution are determined largely by ontogenetic migrations linked to size- and thermal requirements. Therefore, a better understanding of EBS snow crab physiological and biological response to the rapidly changing ecosystem conditions in the BS is necessary. Recent, dramatic population declines emphasize the importance of understanding proximate causes and mechanisms for mortality including predator-prey interactions, disease dynamics, shifts in benthic production, and responses to thermal stress.

3.4.2 Essential Fish Habitat

The Council and NMFS are currently evaluating updates to essential fish habitat (EFH) in the FMPs, including revisions to the model-based maps of EFH for BS, Aleutian Islands (AI), and Gulf of Alaska (GOA) groundfish and BSAI crab species, and updated output from the Fishing Effects (FE) model developed to assess the effects of fishing activities on EFH. The 2022 EFH preliminary results of the habitat mapping and fishing effects (FE) were presented to the Council in February and October 2022.

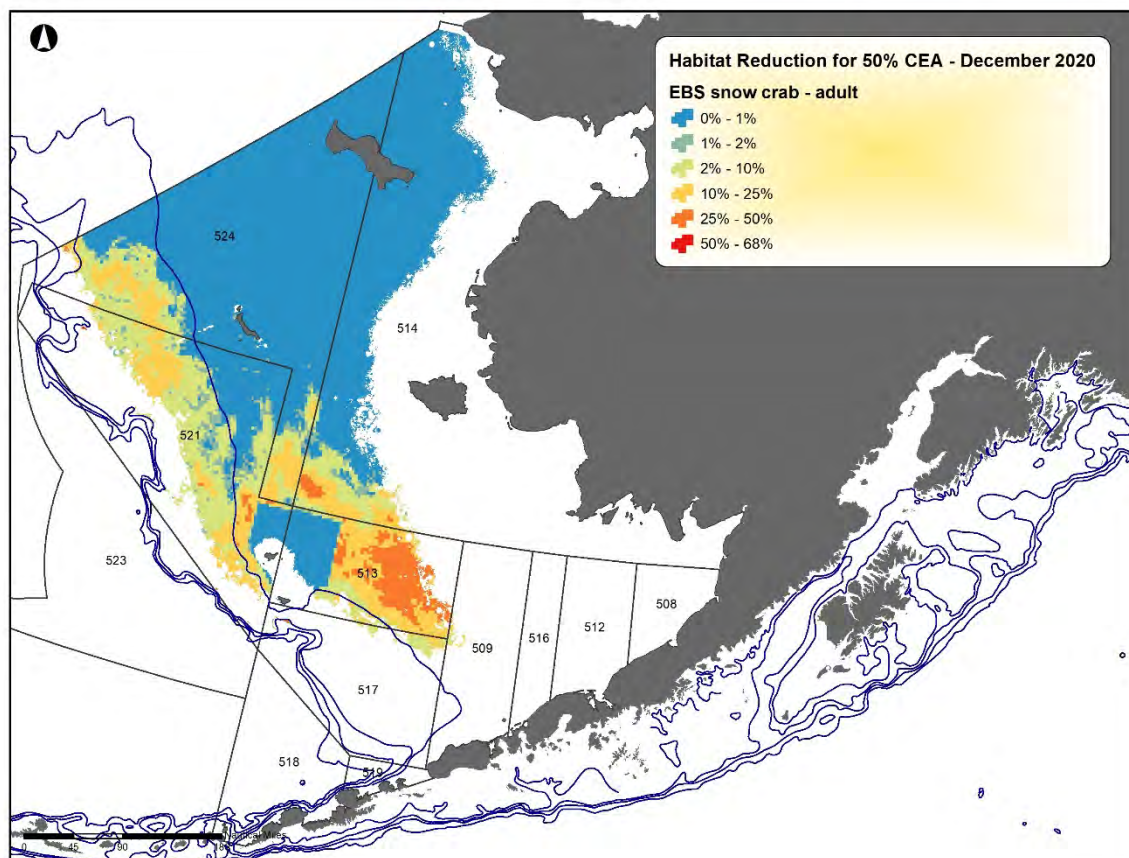
A level 2 habitat map for BS snow crab (Figure 3-16) whereby distribution data is available for the species as well as habitat-related densities or relative abundance of the species are available. The information provided is for all BS snow crab. Figure 3-16 presented below exhibited the most up to date habitat map for snow crab.



Source: [EFH Eastern Bering Sea FE species \(NPFMC October 2022\)](#)

Figure 3-16 Habitat map for EBS snow crab. EFH is the area containing the top 95% of occupied habitat (defined as model estimated encounter probabilities greater than 5%) from a habitat-based ensemble fitted to EBS snow crab distribution and abundance in AFSC RACE-GAP summer bottom trawl surveys; within the EFH area map are the subareas of the top 25% (EFH hot spots), top 50% (core EFH area (CEA)), and top 75% (principal EFH area).

An additional component of EFH is assessing the FE on species. Assessing FE on habitat through the EFH process uses the FE model, developed by the Fisheries, Aquatic Sciences, & Technology (FAST) Lab at Alaska Pacific University (APU), and was recently published (Smeltz et. al, 2019). The FE model is a cumulative representation of the impact of all gears on benthic habitat, accounting for not only bottom contact but also the susceptibility of biological and geological habitats and recovery from fishery disturbance. The 2022 EFH 5 year review process is still underway; however, preliminary presentation of the FE results at the October 2022 SSC meeting indicated that snow crab in the EBS experienced 3.8% habitat disturbance ([EFH FE Evaluation Discussion Paper](#)) in 2020 (Figure 3-17).



Source: [EFH FE Evaluation Discussion Paper](#)

Figure 3-17 Proportion of habitat disturbance, December 2020 - EBS Snow crab.

3.4.3 Fishing impacts to Snow crab habitat

The FE model acts as a unique tool outside of the EFH process. The primary output of the FE model is an estimate of cumulative habitat impacts. While these results are useful for assessing habitat impacts, they do not directly address the year-to-year pressure on seafloor habitat. However, an intermediate data product of the FE model workflow is the estimate of bottom contact explicit over time and space. The analysts collaborated with the APU FAST lab to detail the estimates of bottom contact areas by gear type in areas with high EBS snow crab distribution in the Bering sea (Figure 3-1), and historic areas that contain a high abundance of snow crab ([2022 BSAI Crab SAFE](#)). It is important to note that the estimated bottom contact area is not directly equivalent to EBS snow crab bycatch, mortality or impacts on the ability of EBS snow crab to reproduce and recruit into the fishery. It is the analysts' goal to present bottom contact area estimates to provide the historical fishing footprint and estimated bottom contact areas in areas with historically high snow crab abundance to estimate how fishing activity has historically influenced snow crab habitat.

The full results of the analysis are found in Appendix 1 and highlight bottom contact area by gear type. Analysts concluded that estimated bottom contact areas, and the historical fishing footprint has not drastically altered snow crab habitat in recent years. There are no proposed regulatory management measures under either of the alternatives; therefore, it is unlikely that fishing activity will vary from the

historical average. In addition, the presence of fishing, and the current estimates of bottom contact are an unlikely cause of the EBS snow crab decline, which further emphasizes the lack of impact to snow crab habitat by fishing activity.

3.4.4 Effects of the alternatives on snow crab habitat

Given the current variability in climate, it is likely that snow crab habitat will undergo changes in the next decade (section 3.2.3). However, it is unlikely that under any of the proposed alternatives, there will be a negative effect on habitat as a direct result of implementing a rebuilding plan. Continuous monitoring of the environmental conditions detailed in the snow crab ESP, and summarized in section 3.4.1, should occur throughout the rebuilding plan. The snow crab ESP is brought forth on an annual basis and reviewed by the CPT. Monitoring ecosystem predictors for the duration of rebuilding may prove beneficial in monitoring snow crab habitat and using environmental indicators to aid in predicting stock status. Additionally, the EFH 5-year review process will act as a tool to monitor long-term effects on snow crab habitat in the BS. The current EFH 5-year review cycle will likely be completed in 2023, thus the next 5-year review cycle would be up for review in 2028. The timing of the next 5-year review cycle will occur just before $T_{\min}$ (6 years) is predicted to be reached based off the rebuilding projections. The timing in the next iteration of the EFH 5-year review will provide a retrospective look at snow crab habitat and how the BS habitat fluctuated from 2023 to 2028.

In summary, there are likely no negative effects on habitat as a direct result of implementing a rebuilding plan under proposed Alternative 2, and continued monitoring of habitat should occur throughout the duration of the rebuilding plan. If no rebuilding plan was in place, as suggested under Alternative 1, there would likely be no negative effects on habitat as the outcome of Alternative 1 is strikingly similar to Alternative 2/option 1.

3.5 Marine Mammals

The bearded seal, *Erignathus barbatus*, is distributed among much of the BS, and is known to forage on invertebrates, specifically snow crab. Snow crab has been present in 54%-91% frequency of occurrence in biosampled stomachs (Lowry et.al, 1980, Antonelis et. al., 1994, Crawford et.al., 2015, Oxtoby et.al., 2017). Early stomach content analysis of bearded seal diet, indicated that crab consumed were smaller (mean carapace width of 57 mm) crab (Lowry et. al. 1980. The minimum legal size limit for snow crab is 78 mm, however, the snow crab market generally only accepts crab greater than 101 mm. It is likely that bearded seals are consuming snow crab prior to them reaching a maturity size class that is acceptable for commercial fishing. Given bearded seals' reliance on snow crab, it is possible that the food web dynamics between these species have shifted given the recent decline in snow crab abundance. With the decreased abundance, bearded seals are likely to have to switch food sources which may result in additional interspecific competition that did not exist prior to the collapse of snow crab.

3.5.1 Effects of the alternatives on Marine Mammals

Under the proposed alternatives, there are likely no negative effects on marine mammals. The foreseen impacts are dependent on the state of snow crab stock. Alternative 1, status quo, would not make an effort to rebuild the EBS snow crab stock, and may result in further declines in prey availability for marine mammals. However, under any alternative 2, implementing a rebuilding plan with intentions to rebuild the EBS snow crab stock will likely have positive implications for marine mammals, as some marine mammals are dependent on snow crab as a food source. Implications for bearded seals as a result of the snow crab stock decline may include: varying food web interactions and potential increased resource partitioning interspecifically. Interspecific resource competition may add additional strain to many species that are mitigating the effects of loss of sea ice habitat (Oxtoby et. al., 2017). The bearded seal has a wide variety of prey items, so it is likely that the effects of a decreased in snow crab availability resulted in the bearded seal switching prey consumption to accommodate. It is unlikely that the decrease in snow crab

abundance had substantial effects on bearded seal fitness, as they are reliant on additional benthic prey to subsidize their diet. The larger trophic level effects for bearded seals and snow crab likely require future work especially given the unprecedented state of the EBS snow crab population, and the current climate mediated changes in the phenology of arctic sea ice.

3.6 Economic and Social Effects

This section provides background information on the fishery as well as the economic and social impacts of the alternatives including identification of the individuals or groups that may be affected by the action, and the nature, direction and magnitude of impacts if possible. The objective of this amendment is to rebuild the EBS snow crab stock to B_{MSY} . The action alternative (Alternative 2) would either not allow a directed fishery during the rebuilding period (option 1) or provide for the possibility of a directed fishery under the current State of Alaska harvest strategy (option 2). The alternatives were discussed in section 2.

3.6.1 Description of the BSAI Crab Rationalization

The BSAI Crab Rationalization (CR) Program was implemented in 2005. The CR Program is a “voluntary three pie cooperative” program which allocates BSAI crab resources among harvesters, processors, and coastal communities. The CR Program was designed to address conservation, social, economic, and management issues associated with the previous over-capitalized derby fishery, as well as increased the safety of crab fishermen by ending the race for fish. The program issued harvest quota shares to vessel owners (License Limitation Program license holders) and captains and crew, as well as processor quota shares to processors based on historic participation to protect investment in and reliance on the program fisheries.

There are nine large crab fisheries in the BSAI rationalization under the program⁴, specifically:

- BRR Bristol Bay red king crab
- BBS Bering Sea snow crab (*C. opilio*)
- EBT Eastern Bering Sea Tanner crab (*C. bairdi*) – East of 166° W
- WBT Western Bering Sea Tanner crab (*C. bairdi*) – West of 166° W
- PIK Pribilof Islands blue and red king crab
- SMB Saint Matthew Island blue king crab
- WAG Western Aleutian Islands (Adak) golden king crab – West of 174° W
- EAG Eastern Aleutian Islands (Dutch Harbor) golden king crab – East of 174° W
- WAI Western Aleutian Islands (Petrol Bank District) red king crab – West of 179° W

Each program fishery is managed with a TAC, which sets a specific catch limit. Once the TAC is set for the fishery, 10 percent of this amount is available for the Community Development Quota (CDQ) Program and 90 percent of the TAC is converted into individual fishing quota (IFQ) for harvest under the CR Program.

Below is a brief description of the CR Program including harvester shares, processor shares, regional share designation, catcher processor shares, and crew shares that was included in the January 2017 Ten-Year Program Review for the Crab Rationalization Management Program in the Bering Sea/Aleutian Islands (NPFMC 2017). For further information about these elements and other elements of the CR

⁴ Some crab fisheries are considered one unit stock for assessment purposes but are managed as more than one fishery. For example, Eastern and Western Aleutian Islands golden king crab are assessed as one stock but are managed as distinct fisheries with separate TACs.

Program, refer to the January 2017 Ten-Year CR Program Review (NPFMC 2017) for the Crab Rationalization Management Program in the Bering Sea/Aleutian Islands.

Harvesting shares

Harvesting quota shares (QS) were created in each crab fishery of the program (Figure 3-18). QS are a revocable privilege that allow the holder to harvest a specific percentage of the annual TAC in a program fishery. The corresponding annual allocations, which are expressed in pounds, are referred to as IFQ. The size of each annual IFQ allocation is based on the amount of QS held in relation to the QS pool in a program fishery – a person holding one percent of the QS pool receives IFQ to harvest one percent of the annual TAC in the fishery. IFQ TACs do not include pounds that have been set aside for the CDQ Program. All crab that is sold or kept for personal use, and all deadloss is debited against the IFQ amount of the allocation holder. Legal discards, however, are not counted against an IFQ holder's account, but are accounted for in total fishery removals.

QS is designated as either catch vessel QS or catcher processor QS, depending on whether the vessel that created the privilege processed the qualifying landings on board. Approximately 97 percent of the QS (referred to as "owner QS") in each program fishery was initially allocated to LLP license holders based on their catch histories in the fishery. The remaining 3 percent of the QS (referred to as "C shares" or "crew QS") were initially allocated to captains based on their catch histories in the fishery.

Catcher vessel owner IFQ are issued in two classes, Class A IFQ and Class B IFQ. Crab harvested using Class A IFQ must be delivered to a processor holding unused individual processing quota (IPQ). In addition, Class A IFQ are subject to regional share designations, whereby harvests are required to be delivered within an identified region. The delivery restrictions of Class A IFQ are intended to add stability to the processing sector by protecting processor investment in program fisheries and to preserve the historic distribution of landings and processing between regions.

Crab harvested using Class B IFQ can be delivered to any processor that is a registered crab receiver (except a catcher processor) regardless of whether the processor holds unused IPQ. In addition, Class B IFQ are not regionally designated. The absence of delivery restrictions on a portion of the catch is intended to provide harvesters with additional market leverage for negotiating prices for landings of crab.

Crab Fishery TAC - 10% for CDQ & Adak = IFQ allocation

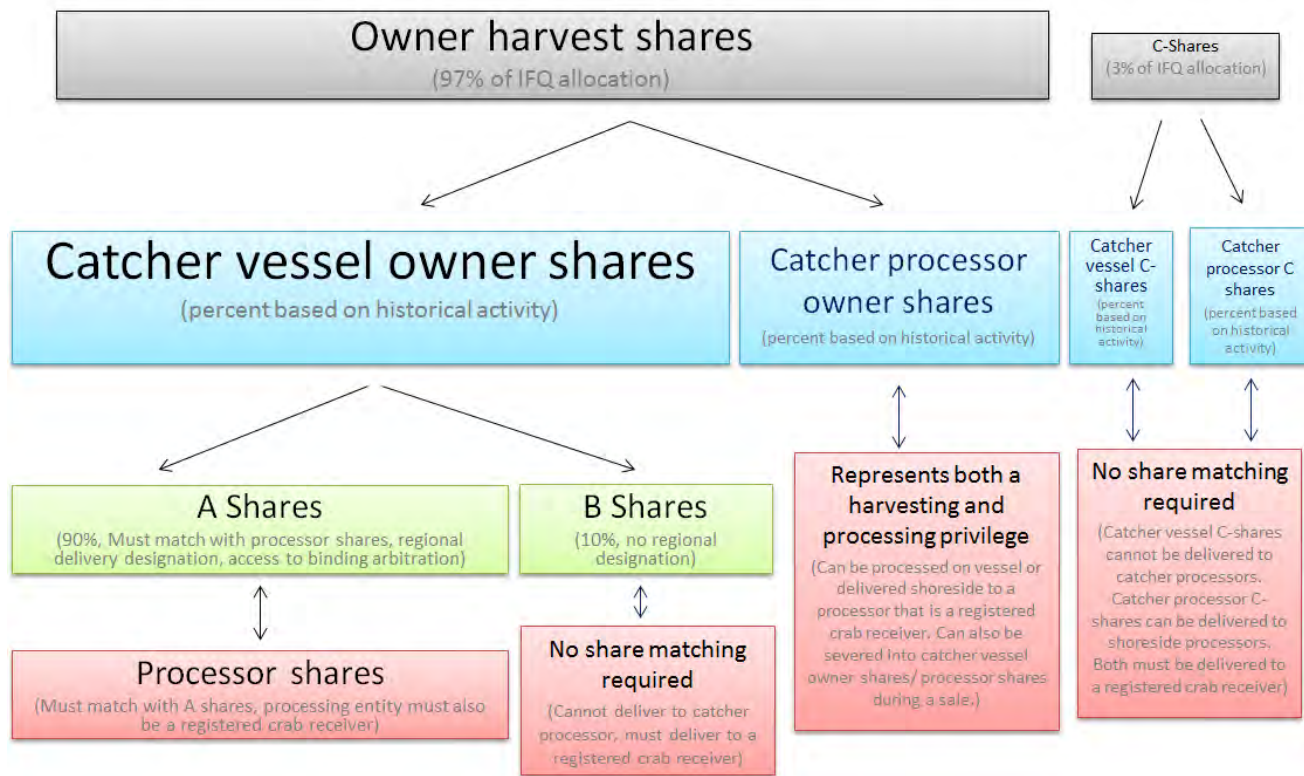


Figure 3-18 Diagram of quota shares in the CR Program

QS and IFQ are transferrable under the CR Program, subject to limits on the amounts of shares a person may own or use. Transferability of shares among eligible purchasers of QS and IFQ may promote production efficiency in the harvest sector and provides a means for compensated removal of excess harvesting capacity in the program fisheries. In addition, transferability may be used to avoid overages, in the event a harvester exceeds its available IFQ, and IFQ transfers post-delivery are also authorized to remedy a harvest overage. For further details on the limits of QS and IFQ transferability, see section 2.3.2 of the CR Program Review (NPFMC 2017).

Processing Shares

The CR Program also created processing quota shares (PQS), which are allocated to processors and are analogous to the QS allocated to harvesters. PQS are a revocable privilege to receive deliveries of a fixed percentage of the annual TAC from a program fishery. These annual allocations are referred to as individual processing quota (IPQ). IPQ is issued for 90 percent of the catcher vessel owner IFQ pool, corresponding to the 90 percent allocation for catcher vessel IFQ issued as Class A IFQ. As with Class A IFQ, PQS and IPQ are designated for processing by region.

IPQ landing requirements do not apply to the remaining 10 percent of the owner IFQ, corresponding to the 10 percent of the owner IFQ allocated as Class B IFQ, as these Class B IFQ are intended to provide harvesters with additional bargaining power. In addition, Class B IFQ may provide an opportunity for the entry of new processors in the program fisheries. Alternatively, new processors can enter a fishery by purchasing PQS or IPQ or by purchasing landings of CDQ crab. To ensure harvesters the latitude to use

their Class B IFQ to pursue the best markets, processors are not permitted to leverage their IPQ to acquire crab harvested using Class B IFQ; the penalty is forfeiture of all of the processor's IPQ.

As in the harvester sector, processors received initial allocations of PQS based on processing history during a specified qualifying period for each fishery. A processor's PQS allocation, as a percentage of the pool, in a program fishery was equal to its share of qualified processing in the qualifying period (i.e., pounds processed by the processor divided by pounds processed by all qualified processors).

Processing shares are transferable, including leasing of PQS (or equivalently, the sale of IPQ) subject to use caps. As with harvesting shares, transferability of processing shares is intended to promote efficiency and facilitate compensated reduction of excess capacity. In addition, IPQ transfers may aid in the coordination of deliveries from the fisheries.

A right of first refusal was granted to community groups and CDQ groups from communities with substantial crab processing history on the sale of any processing share for use outside of the community of origin. The intention of the right of first refusal is to allow the community of origin the opportunity to keep PQS in a community under the same terms and condition the seller of PQS would have offered another buyer. For further details on the limits of PQS and IPQ transferability, see section 2.3.3 of the January 2017 Ten-Year Program Review for the Crab Rationalization Management Program in the Bering Sea/Aleutian Islands.

Regional Share Designation

The allocation to regions is accomplished by regionally designating all Class A (delivery restricted) harvest shares and all corresponding processing shares. In most CR Program fisheries including BS snow crab, regionalized shares are either North or South, with North shares designated for delivery in areas on the BS north of 56°20' north latitude and South shares designated for any other areas, including Kodiak and other areas on the Gulf of Alaska. In the Western Aleutian Islands (Adak) golden king crab fishery, the designation is based on an east/west line to accommodate a different distribution of activity in that fishery. Share designations are based on the history location of the landings and processing that gave rise to the shares.

Effective June 14, 2013, the Council approved an amendment that established a process whereby holders of regionally designated IFQ and IPQ in six CR Program fisheries, including BS snow crab, may receive an exemption from regional delivery requirements in the North or South Region. This regulatory action establishes a process that can mitigate disruptions in a CR Program fishery that prevent participants from complying with regional delivery requirements. For example, in the event of a strong ice pack around Saint Paul Island, North-designated harvested crab might be stranded if there is not flexibility to allow processing to occur elsewhere. A privately signed framework agreement stipulates the circumstance under which relief is granted from regional delivery requirements.

Catcher Process Shares

Catcher processors participate in both the harvest and processing sectors and therefore have a unique position in the program. Catcher processors are allocated catcher processor QS and issued corresponding catcher processor IFQ. These shares carry both a harvest privilege and an accompanying onboard processing privilege. To be eligible for the initial allocation of catcher processor QS, a person must have been eligible for a harvest allocation by holding a permanent, fully transferable catcher processor License Limitation Program (LLP) license. In addition, the catcher processor must have processed crab in either 1998 or 1999. These requirements parallel the harvester QS and processor PQS eligibility requirements, respectively. Persons meeting these eligibility requirements were allocated catcher processor QS in accordance with the allocation rules for harvest shares for all qualified catch that was processed onboard.

Since catcher processor IFQ provide both harvesting and on-board processing privileges, a person holding those shares may harvest and process crab onboard under the allocation. In addition, holders of catcher processor IFQ may choose not to process harvested crab, instead delivering their catch to any other processor. Use of catcher processor IFQ in this manner is akin to the use of Class B IFQ, which do not require the receiving processor to hold unused IPQ. Catcher/processor shares do not have regional designations.

Holders of catcher processor QS may also sever the harvesting and processing privileges, thereby creating separate QS and PQS. These newly severed interests create a privilege to annual IFQ allocations and IPQ allocations, which can be held by different persons. When severed, the resulting QS and PQS must be designated for a region with both shares taking the same regional designation. Allowing the conversion of shares permits a catcher processor shareholder to realize the maximum value of shares and provides greater flexibility in using the privileges.

Some catcher processors historically accepted delivery of crab from catcher vessels for processing. PQS are allocated based on this activity to the extent that processing vessels met processor eligibility requirements and had qualifying processing history. In addition, catcher processors are permitted to purchase and use additional IPQ. All processing of deliveries by catcher processors is required to take place within three miles of shore in the applicable region. The requirement of processing within three miles of shore is intended to ensure that the regional benefits of processing activity occur. Catcher processors may not purchase for processing crab harvested with Class B shares.

Crew Shares

To protect captains' historical interests in the program fisheries, 3 percent of the initial allocation of QS were issued to eligible captains. These "C shares" are to be held only by active captains and crew and are intended to provide additional leverage to those captains and crew when negotiating contracts with vessel owners. The Council chose to exempt C shares from all IPQ and regional landing requirements, as it recognized the logistical complications that would likely arise under the program as a result of the interaction of active participation requirements, fleet contraction, and the IPQ and regional landing requirements.

To be eligible for the initial allocation of C share QS, a captain was required to demonstrate both historical dependence on a program fishery and recent participation. Allocations to captains were based on participation in landings during the same qualifying years applicable to owner QS allocations. To ensure C shareholders are an integral part of the program, C shareholders are permitted to join cooperatives. IFQ attributable to C share QS of cooperative members are allocated directly to the cooperative and are harvested in accordance with the applicable cooperative agreement.

To ensure that C shares benefit active participants in the program fisheries, C share QS and IFQ may be acquired by transfer only by persons who are active in one of the program fisheries in the 365 days prior to the application for transfer. Under current rules, individuals who hold C share IFQ are required to be on board the vessel harvesting those IFQ. However, C shareholders who choose to join a cooperative are effectively exempted from the 'owner on board' rule, since the IFQ are held by the cooperative.

Under Amendment 31, annual C share IFQ are issued only to C share QS holders who meet an active participation requirement of being on board a vessel for one landing in the three years preceding the IFQ allocation. In addition, C share QS is revoked from persons who are not active in at least one of the crab fisheries for 4 consecutive years. The Council also included a transition period for persons who would be deprived of IFQ or QS by these active participation requirements. Under this transition period, no IFQ would be withheld until 3 years after implementation of the amendment and no QS would be revoked until 5 years after the implementation of the amendment. This amendment became effective May 1, 2015.

In 2022, an emergency rule to temporarily suspend the active participation requirement for captains and crew holding crew quota or C shares under the Bering Sea and Aleutian Islands Crab Rationalization Program (CR Program) was implemented. The emergency rule is in effect from July 15, 2022 through January 11, 2023 (87 FR 42390).

Individual C share holdings and use are capped at the same level as the vessel use caps applicable to owner IFQ (i.e., twice the owner QS cap level). A “grandfather” provision exempted initial allocations of Class C shares in excess of the cap. C share IFQ are not considered in determining a vessel’s compliance with the vessel use caps applicable to owner IFQ.

Catcher processor captains are allocated catcher processor C share QS that include both a harvesting and onboard processing privilege. Harvests with catcher processor C share IFQ may also be delivered to shoreside or stationary floating processors. Harvests with catcher vessel C share IFQ must be delivered to shoreside or stationary floating processors (i.e., they cannot be delivered to catcher processors)

3.6.2 Economic Status and Trends in the EBS Snow Crab Fishery

Table 3-13 provides general statistics on the harvest and processing of EBS snow crab fishery with corresponding Figure 3-19 through Figure 3-21 to visualize these trends. As seen in Table 3-13 and Figure 3-19, the number of vessels participating in the EBS snow crab fishery has generally trended downward between the 2005/2006 season and the 2020/2021 season. However, during the 2021/2022 EBS snow crab fishery, only 42 vessels were active in the fishery which was a substantial decline in the number of active vessels. As for retained catch of EBS snow crab, during the 2005/2006 through 2011/2012 period it generally trended upward from a low of 37 million pounds in 2005/2006 to high of 88 million pounds in 2011/2012 followed by an overall decline with temporary peaks during the 2014/2015 season at 68 million pounds and 2020/2021 season at 45 million pounds. Following the 2020/2021 retained catch of 45 million pounds, retained catch declined substantially the following year to record low of 5.5 million pounds due to the collapse of the population in 2021 (see Figure 3-20). Gross ex-vessel revenue also slowly increased from a low of \$56 million in 2005/2006 season to a high of \$224 million in the 2011/2012 EBS snow crab season, followed by a downward trend for most years to a low of \$76 million in 2016/2017 and 2017/2018 seasons. Following the decline in gross ex-vessel revenue in 2016/2017 and 2017/2018 seasons, gross ex-vessel revenue for the fishery trended upward to a high of \$219 million during the 2020/2021 season followed by dramatic decline in 2021/2022 season with a preliminary estimate of \$34 million (see Table 3-13 and Figure 3-21).

From the production perspective, the number of active shore plants that have received EBS snow crab deliveries has, in general, slowly trended downward from a high of 18 during the 2006/2007 season to a low of nine starting in the 2015/2016 season and continuing every season except during the 2016/2017 season when 11 shore processors received EBS snow crab deliveries (see Table 3-13). Gross first wholesale revenue shows a similar trend as the revenue gradually increased from a low \$96 million during the 2005/2006 season to a high of \$324 million during the 2011-2012 season followed by a gradual decline to \$86 million during the 2017/2018 season (see Figure 3-21). Following the 2017/2018 season, gross first whole revenue for the EBS snow crab fishery recovered during the subsequent three seasons to a high of \$286 million during the 2020/2021 season. For 2021/2022 season, preliminary estimates show a dramatic decline in the gross first wholesale revenue of \$40 million.

Table 3-13 EBS snow crab retained catch, gross ex-vessel value and price, and gross first wholesale revenue and price, 2005-2008 through 2021-2022

Snow crab year	TAC/GHL 1000t	TAC/GHL million lbs	Vessels	Retained catch 1000t	Retained catch million lbs	Gross ex-vessel revenue \$ million	Ev-vessel price \$/lb	Plants	Gross first wholesale revenue \$ million	Gross first wholesale price \$/lb
2005-2006	16.86	37.18	78	16.77	36.97	\$55.79	\$1.51	13	\$96.27	\$2.60
2006-2007	16.59	36.57	69	16.47	36.31	\$72.67	\$2.00	18	\$118.89	\$3.27
2007-2008	28.59	63.03	78	28.59	63.02	\$133.69	\$2.12	17	\$209.31	\$3.32
2008-2009	26.56	58.55	77	26.55	58.54	\$101.24	\$1.73	16	\$163.64	\$2.80
2009-2010	21.78	48.02	69	21.69	47.82	\$76.48	\$1.60	11	\$125.92	\$2.63
2010-2011	24.62	54.28	68	24.61	54.26	\$164.80	\$3.04	14	\$231.51	\$4.27
2011-2012	40.32	88.89	72	39.99	88.16	\$224.00	\$2.54	13	\$323.57	\$3.67
2012-2013	30.1	66.35	70	29.71	65.49	\$173.96	\$2.66	12	\$259.17	\$3.96
2013-2014	24.48	53.98	70	24.49	53.98	\$144.25	\$2.67	10	\$222.86	\$4.13
2014-2015	30.82	67.95	70	30.79	67.88	\$157.49	\$2.32	11	\$227.73	\$3.36
2015-2016	18.42	40.61	69	18.41	40.60	\$119.89	\$2.95	9	\$174.12	\$4.29
2016-2017	9.78	21.57	63	9.76	21.53	\$75.82	\$3.52	10	\$127.00	\$5.90
2017-2018	8.6	18.96	63	8.60	18.95	\$76.25	\$4.02	9	\$86.39	\$4.56
2018-2019	12.51	27.58	61	12.47	27.50	\$109.49	\$3.98	9	\$154.02	\$5.60
2019-2020	15.4	33.95	59	15.43	34.02	\$133.53	\$3.92	9	\$177.81	\$5.23
2020-2021	20.4	44.97	62	20.41	45.00	\$219.16	\$4.87	9	\$286.37	\$6.36
2021-2022*	2.5	5.51	42	2.5	5.51	\$33.56	\$6.09	Not available	\$40.61	\$7.37

Source: AKFIN (ADF&G fish ticket data and ADF&G COAR data). 2021/2022 retained catch from 2021/2022 Snow Crab SAFE. Source for 2021/2022 vessel count from September 2022 Ecosystem & Socioeconomic Profile for Eastern Bering Sea Snow Crab.

Data includes CDQ harvest

All price data is in real 2021 dollars

*Note that 2022 official ex-vessel prices are not yet available so preliminary prices were estimated using in-season ex-vessel prices inflated for post-season adjustment and first wholesale prices were estimated using the 2021/2022 preliminary ex-vessel price plus the average difference between ex-vessel price and first wholesale price 2005/2006 - 2020/2021 seasons.

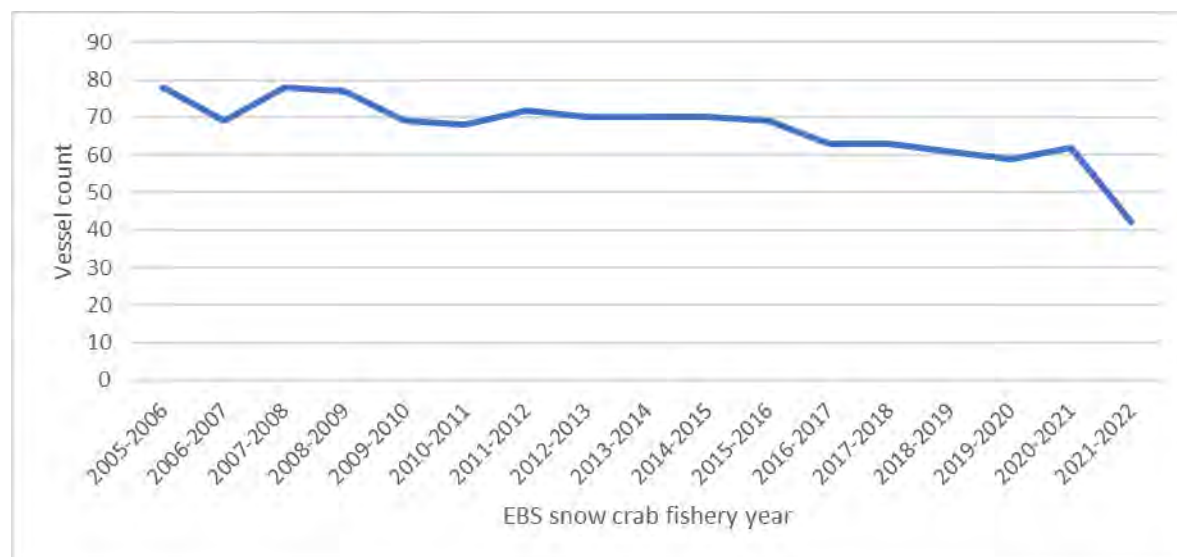


Figure 3-19 Annual number of vessels active in the EBS snow crab fishery, 2005/2006 through 2021/2022

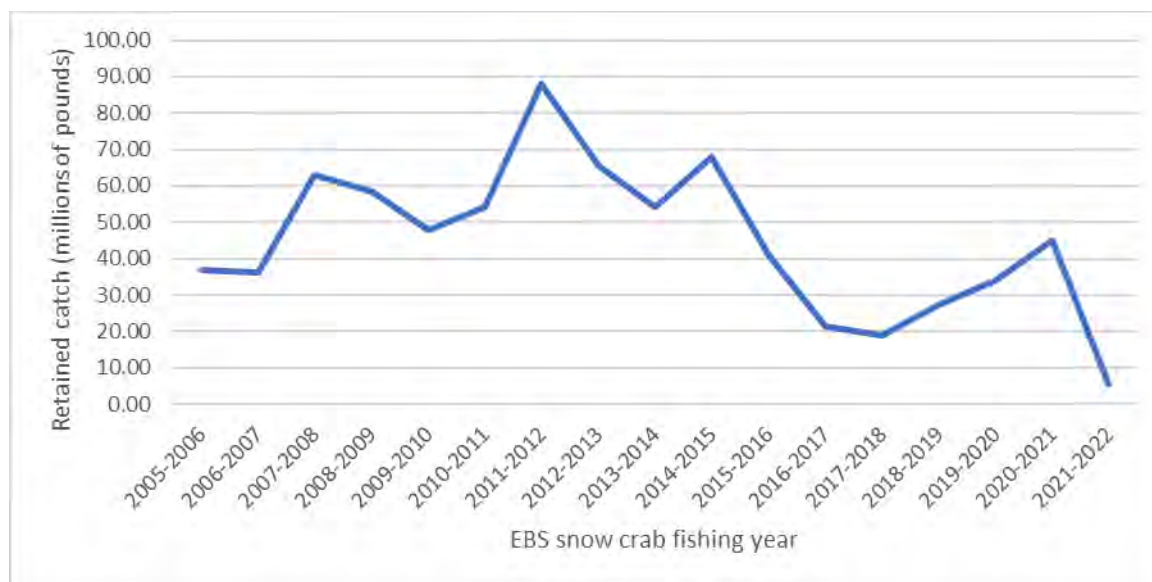


Figure 3-20 Annual retained catch (millions of lbs.) in the EBS snow crab fishery, 2005/2006 through 2021/2022

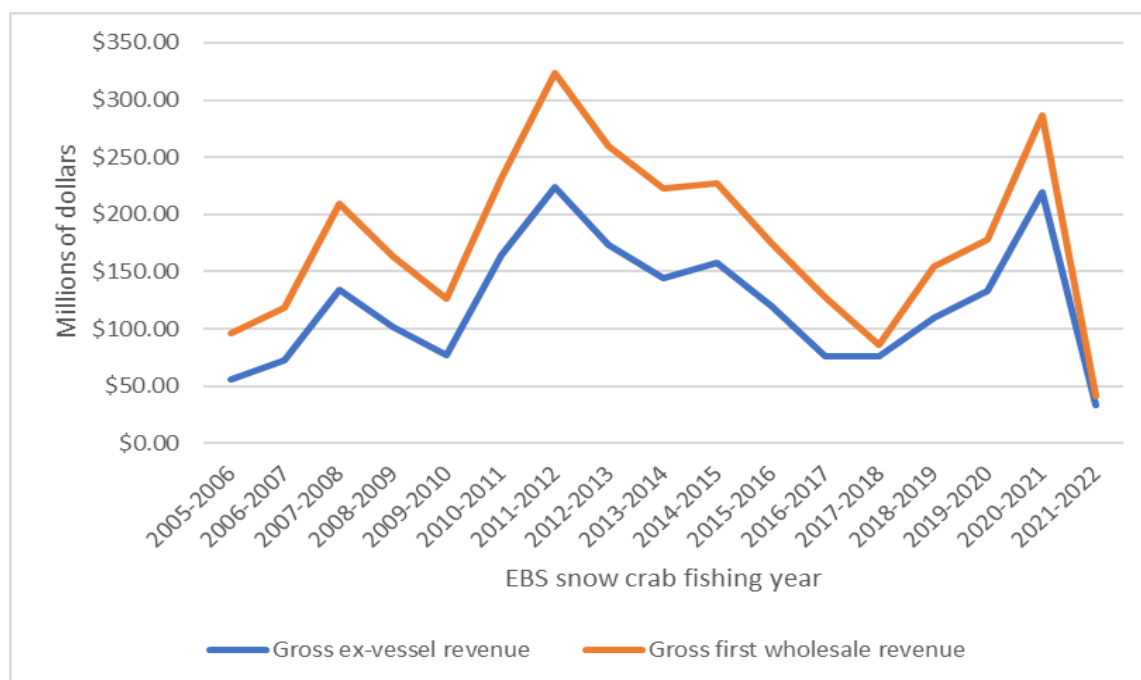


Figure 3-21 Annual gross ex-vessel and first whole revenue (millions of \$) for the EBS snow crab fishery, 2005/2006 through 2021/2022

A summary of selected indicators from the most recent employment data available for the EBS snow crab fishery (up to 2019) is provided in Table 3-14. Data reported in Table 3-14 are by calendar year which is different than the data reported in Table 3-13 which is reported by EBS snow crab fishery season. As result, vessel and processor counts may differ between the tables. There were an estimated 428 total crew positions on 61 vessels in the EBS snow crab fishery for 2019 which is the third lowest number of crew positions since 2006 and 2007. Crew earnings for 2019 was \$14.7 million up the previous year of \$10.3 million, while captain earnings were \$6.4 million up from \$4.4 million the previous year.

Crab processing employment in 2019, as measured by hours of processing labor input at plants that received EBS snow crab IFQ and CDQ landings, is estimated at 333 thousand labor hours, an increase of 44 percent from 2018. Aggregate wages paid to crab processing line employees in the EBS snow crab fishery was \$4.6 million an increase of 58 percent than the previous year.

Table 3-14 EBS snow crab fisheries crew and processing sector employment and earnings

Year	Crew positions ^a			Crew share ^b		Captain share	Processor labor hours ^c			Processing labor payment ^d			
	Vessels	Total crew positions	Mean per vessel	Total \$ million	Per vessel, median \$1,000	Total \$ million	Vessel median \$1,000	Plants	Total 1,000 hrs	Plant median 1,000 hrs	Median \$/hour	Total \$ million	Plant median \$1,000
2006	74	418	5.6	\$6.58	\$76.82	3.29	\$40.00	10	445	49.45	\$11.80	\$5.14	\$582
2007	65	377	5.8	\$9.71	\$128.01	4.62	\$66.65	10	442	41.29	\$12.22	\$5.57	\$512
2008	74	447	6.0	\$17.86	\$221.82	8.5	\$113.19	12	712	30.52	\$12.19	\$9.94	\$570
2009	77	536	7.0	\$14.27	\$162.90	6.32	\$80.58	14	600	58.41	\$11.69	\$7.61	\$349
2010	68	444	6.5	\$10.29	\$136.20	4.62	\$65.16	11	534	50.9	\$11.18	\$6.22	\$411
2011	68	453	6.7	\$21.99	\$310.21	9.84	\$143.95	14	555	45.69	\$11.64	\$6.79	\$393
2012	72	502	7.0	\$29.73	\$409.14	13.38	\$192.10	13	1087	77.94	\$11.41	\$13.15	\$671
2013	71	481	6.8	\$24.10	\$310.04	10.97	\$154.69	12	774	63.55	\$10.99	\$8.74	\$527
2014	70	480	6.9	\$19.13	\$255.64	8.58	\$118.48	10	590	76.01	\$11.47	\$6.85	\$495
2015	70	491	7.0	\$18.12	\$253.75	\$8.20	\$120.25	10	747	95.42	\$11.56	\$9.21	\$857
2016	68	463	6.8	\$15.46	\$200.05	\$6.91	\$99.06	8	447	69.4	\$12.62	\$5.98	\$567
2017	63	441	7.0	\$12.84	\$171.06	\$5.45	\$79.56	8	266	34.61	\$12.34	\$3.35	\$218
2018	63	436	6.9	\$10.25	\$140.53	\$4.39	\$65.89	8	232	30.48	\$12.23	\$2.89	\$166
2019	61	428	7.0	\$14.67	\$199.90	\$6.35	\$97.38	8	333	45.7	\$12.92	\$4.58	\$306

Source: 2021 Crab Econ SAFE (ADF&G fish ticket data, eLandings, ADF&G COAR data, and EDR data)

Notes: Data show for snow crab fishery by calendar year. All dollar values are adjusted for inflation to 2019-equivalent value.

^a Crew positions total and mean summary statistics are calculated from vessel-level observations derived from eLandings crew size reporting, average over all landings in the respective fishery reported by each active vessel. Calculations do not include CP numbers from 2006-2008 due to confidentiality.

^b Crew and captain payments reflect amounts paid for labor during the crab fishery and include all post-season adjustments, bonuses, and deductions for shared expenses such as fuel, bait, and food and provisions; payments for IFQ royalties, labor outside of crab fishery, health/retirement or other benefits are excluded.

^c Processing labor hours reflect hours worked by processing-line employees working at shoreside and floating processor sectors only, excluding processing employees on catcher/processors and salaried workers employed in the processing sectors.

^d Pay per hour statistics reflect only the shoreside and floating processing sectors; all other processing labor pay statistics are reported inclusive of catcher/processor.

3.6.2.1 Socioeconomic Fishery Performance and Economic Indicators for EBS Snow Crab

This section provides the latest fishery performance and economic indicators from the [2022 Snow crab Ecosystem and socioeconomic profile \(ESP\)](#).

Fishery performance and economic indicators (Figure 3-22):

- Annual number of active vessels in the snow crab fishery, representing the level of fishing effort assigned to the fishery
- Annual CPUE, expressed as mean number of crabs per potlift, in the snow crab fishery, representing relative efficiency of fishing effort
- Center of gravity, expressed in latitude, as an index of spatial distribution for the snow crab fishery to monitor spatial shifts in fishery behavior
- Annual incidental catch of snow crab in EBS groundfish fisheries
- Annual total potlifts in the snow crab fishery, representing the level of fishing effort expended by the active fleet
- Percentage of the annual EBS snow crab TAC (GHL prior to 2005) that was harvested by active vessels, including deadloss discarded at landing

- g.) Annual snow crab ex-vessel price per pound, representing per-unit gross economic returns to the harvest sector, as a principal driver of fishery behavior
- h.) Annual snow crab ex-vessel revenue share, expressed as vessel-average proportion of annual gross landings revenue earned from the EBS snow crab fishery
- i.) Annual snow crab ex-vessel value of the snow crab fishery landings, representing gross economic returns to the harvest sector, as a principal driver of fishery behavior
- j.) Annual snow crab TAC in pounds.

Fishery performance indicators are reported through calendar year 2022 (corresponding to the 2021-2022 crab season), with the exception of incidental catch in the (currently ongoing) EBS groundfish fisheries, reported through 2021. The active snow crab fleet during 2022 declined to 42 vessels, the lowest level since 1977 at the beginning of the time series, and approximately 68% of the average number of vessels participating during the previous five years. Relative to the substantially reduced TAC (less than 13% of the previous year and less than 20% of the previous five-year average), less consolidation of fishing activity occurred than would be expected based on economic efficiency, and it is unclear if other factors driving this level of vessel participation will persist if TAC levels remain comparably low. CPUE in the fishery declined from 218 the previous year to a post-rationalization low of 124 legal crab per potlift, and total potlifts declined from 172 thousand in 2021 to 37 thousand, with both indicators approaching the lower bound of one standard deviation below the long term (1991-current) average, respectively. The latitude of the center of gravity of fishing activity during 2022 shifted somewhat south compared to the previous year but remained approximately two standard deviations greater than the long-term average. Incidental catch in EBS groundfish fisheries during 2021 declined for a fourth consecutive year to 77 thousand kg, approaching the lower bound of the long-term range of variation. TAC utilization reached 99% for the 2021-2022 snow crab fishery, however, fishing extended later than usual, with four vessels making landings later than May 15.

Economic performance indicators included in this ESP are reported through calendar year 2021, the most recent year for which data are available. With a TAC of 18.37 thousand metric tons, the highest since the 2014-2015 crab season, combined with historically high market values for snow crab driven by high consumer demand during the first two years of the covid-19 pandemic, estimated ex-vessel revenue in the snow crab fishery during 2021 exceeded \$219 million, approaching the upper bound of one standard deviation above the long-term (1991-2021) average. Average ex-vessel price per pound reached a historical high in 2021, increasing by 25% from 2020, to \$4.97 per pound, greater than two standard deviations higher than the historical average since 1991 (adjusted for inflation). As a result of the historically high ex-vessel value of the snow crab fishery during 2021, combined with the closure or reduced TAC levels in most crab and other fisheries targeted by the snow crab fleet, ex-vessel revenue share increased to an unprecedented 85% of total annual ex-vessel landings revenue, summed across all fisheries in which snow crab vessel landed catch during the 2021 calendar year. Although 2022 data is not yet available for economic performance indicators, news reports and other information indicate that market demand for crab and other premium seafood products contracted sharply in 2022, suggesting that economic returns for most or all of the fleet active during the 2021-2022 snow crab season were poor and many vessels likely operated at a loss.

All of the socioeconomic indicators associated with the snow crab target fishery included in the ESP exhibited substantial deviation from historical patterns during the most recent period for which data are available. During 2022, the number of active vessels in the fishery fell to 42, the lowest level since 1977, but, with the historically low TAC set for the 2021-2022 season and early evidence of sharply reduced market value, likely exceeding the number of vessels that could be financially sustained at similarly reduced production levels. Results from an industry skipper survey highlight concerns with perceived low abundance on the fishing grounds and changes in fishing behavior attributed to reductions in the 2021/2022 snow crab TAC. Historically low CPUE in 2022, and the continued spatial shift of fishing

activity far to the north of historical fishing grounds, reflected adverse fishing conditions. A historically high ex-vessel price during 2021, combined with a relatively high TAC level, contributed to strong economic performance in the snow crab fishery during 2021. However, the ex-vessel revenue share indicator increased for 2021 to an unprecedented 85% share of the fleet's total gross landings revenue for the year, reflecting increased dependence on the snow crab fishery. The continued limited availability of alternative fishing targets for the fleet, combined with high operating costs associated with adverse fishery performance indicators noted above, without the mitigating (though limited) effect of high ex-vessel price observed for the previous year, is evidence of severe economic stress on the snow crab fleet and dependent stakeholders and communities during the 2021/2022 season.

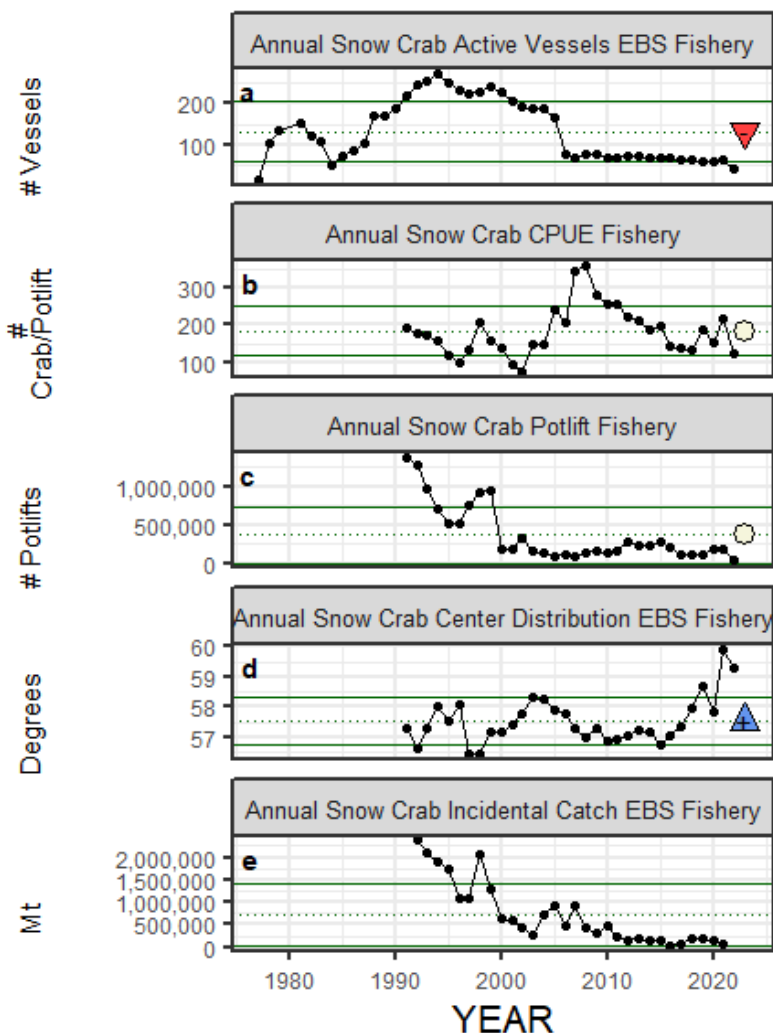


Figure 3-22 Selected socioeconomic indicators for EBS snow crab with time series ranging from 1977 – present. Upper and lower solid green horizontal lines represent 1 standard deviation of the time series mean. Dotted green horizontal line is the mean of the time series. A symbol appears when current year data are available and follows the traffic light status table designations (triangle direction represents if above or below 1 standard deviation from the time series mean, color represents proposed relationship for stock, white circle for neutral).

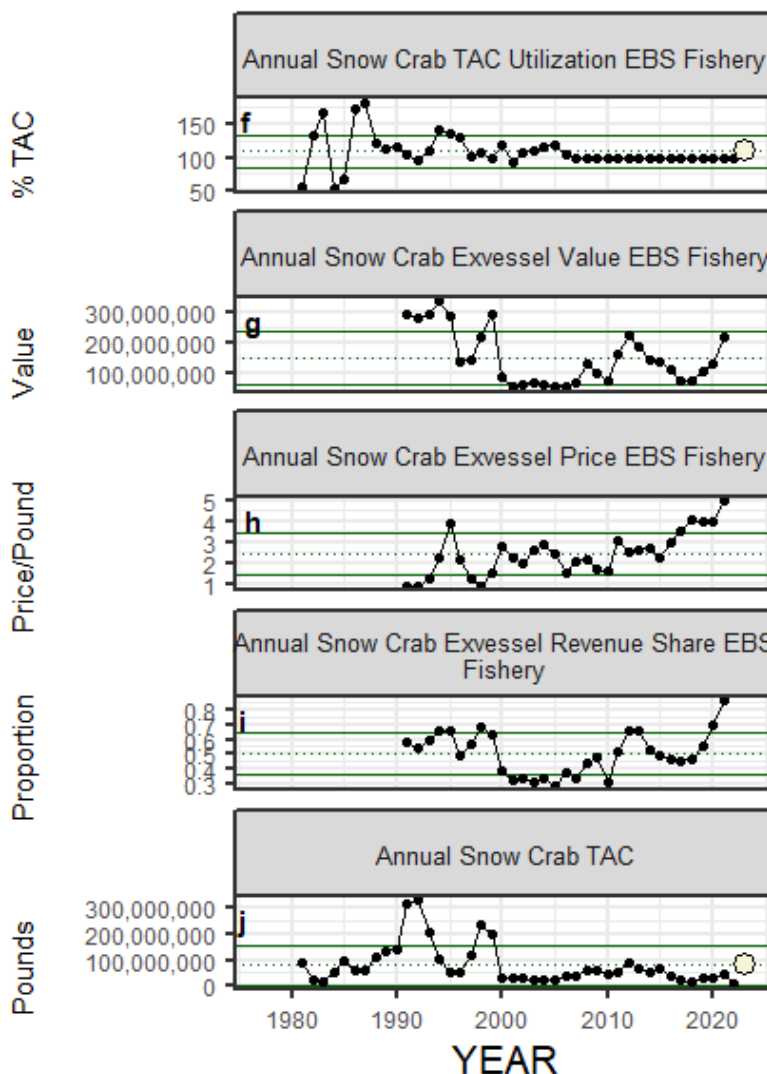


Figure 3-22 (cont.). Selected socioeconomic indicators for EBS snow crab with time series ranging from 1977 – present. Upper and lower solid green horizontal lines represent 1 standard deviation of the time series mean. Dotted green horizontal line is the mean of the time series. A symbol appears when current year data are available and follows the traffic light status table designations (triangle direction represents if above or below 1 standard deviation from the time series mean, color represents proposed relationship for stock, white circle for neutral).

3.6.2.2 Local and Traditional Knowledge and Subsistence Specific to BS Snow Crab

When preparing this analysis, staff used the Local Knowledge (LK), Traditional Knowledge (TK), and Subsistence search engine to look for sources of information containing LK and TK specific to Bering Sea snow crab. The search engine contains scientific articles in peer-reviewed journals, white papers, archival references, and other sources of information related to LK, TK, the social science of LK and TK, and subsistence information. No results based on LK from the Bering Sea snow crab fleet or communities substantially engaged in, or dependent on, snow crab were returned. Likewise, no results were based on TK were returned.

LK is based on the observations and experience of local people in a region with significant in-situ expertise related to particular species, environments, and practices (Martin et al., 2007). LK holders, such

as long-term crab skippers or crew members, may be some of the earliest observers of environmental and/or fishery changes because of their long-term experience working and harvesting specific areas (Johannes & Nies 2007). For these reasons, the authors reached out to the Alaska Bering Sea Crabber's Association (ABSC) regarding their new skipper survey.

ABSC's survey was administered online to all Bering Sea snow crab skippers that participated in the 2020/2021 and 2021/2022 seasons. The results reported in the analysis draw from the 2022 skipper survey administered at the close of the 2021/2022 season. Forty-two vessels participated in 2021, of which 13 skippers fully completed the survey in 2022, yielding a 31% response rate. Mailed or online surveys can collect data from larger numbers of people that can be extrapolated to a sample population (i.e., the entire fleet of snow crab skippers during the 2020/2021 and 2021/2022 fishing seasons) and minimize response effects based on direct interviewer/surveyor interactions (Bernhard 2006; Salant & Dillman 1994). However, it is important to apply some level of caution when extrapolating the results yield from the surveyed population to the sample population. The skipper survey results do not reflect the full observations of the Bering Sea snow crab fleet as only 31% of the active 2021/2022 skippers completed the most recent survey. Fishermen make different operational choices, work under different business plans, and therefore are likely to have diverse observations about the fishery.

A subset of survey questions compared observations between the 2021/2022 and 2020/2021 snow crab fishing seasons. This approach provides ABSC an opportunity to establish a time-series of observations that can be compared to the prior year, each year the survey is administered. Specifically, the survey collects skipper's observations of changes in the amount of industry preferred males, sub-legal males, immature males and females, changes in fishing behavior (e.g., fishing deeper waters, longer pot soak times), factors that motivated changes in fishing behavior (e.g., weather, empty pots, etc.), and more.

The 2022 skipper survey results show participating skippers reported a decrease in their encounters with commercial sized male crab, as did their encounters with sub-commercial sized males. Participating skippers also reported they fished deeper compared to the prior fishing season while others reported no significant changes in fishing behavior. Those skippers that did change their fishing behavior attributed it to smaller TACs in the 2021/2022 snow crab season.

Results from the skipper survey also highlight skipper's concerns about low abundance on the snow crab fishing grounds, and the subsequent changes in fishing behavior were attributed to reductions in the 2021/2022 snow crab TAC and that there was greater sea ice extent southward in the Bering Sea. Historically low CPUE in 2022, and the continued spatial shift of fishing activity far to the north of historical fishing grounds, reflected adverse fishing conditions. Despite a relatively low Bering Sea snow crab TAC in the 2021/2022 fishing season, the fishery had a strong economic performance because of the higher ex-vessel price for the fishery (see Table 3-13). However, as noted in section 3.6.2.1, the continued limited availability of alternative fishing targets for the fleet, combined with higher operating costs (e.g., increased fuel costs, fishing further north, etc.), suggest the fleet, and communities that are substantially engaged in or dependent on snow crab for shoreside processing, are experiencing overall negative economic impacts, despite a higher ex-vessel price in 2021 (2021 Bering Sea Snow Crab ESP; C. Lescher).

It is the author's understanding that snow crab is not a historically important species for subsistence uses among Bering Sea communities because the fishing grounds are far offshore as compared to other subsistence species, such as Red King Crab. However, it is possible that Alternative 2 could negatively impact indirect personal use harvest of snow crab by reducing the opportunities for home pack.

3.6.3 Social Impact Assessment of the EBS Snow Crab Fishery

3.6.3.1 Social Impact Assessment Context

This community-level social impact assessment (SIA) of the proposed action is guided largely by National Standard 8 – Communities under the provisions of the Magnuson-Stevens Act; the National Environmental Policy Act (NEPA); and Executive Order (EO) 12898, Federal Action to Address Environmental Justice in Minority Population and Low-Income Populations, EO 13175, a recent Presidential Memorandum on consultation and coordination with Indian Tribal Governments were available to the NPFMC for their consideration. Additionally, potentially relevant sections of new (2021) Executive Orders on (1) advancing racial equity and support for underserved communities, (2) tackling the climate crisis, and (3) advancing equity, justice, and opportunity for Asian Americans, Native Hawaiians, and Pacific Islanders contain embedded direction on economic and environmental justice and serving disadvantaged and underserved communities are noted below.

3.6.3.1.1 Magnuson-Stevens Act National Standard 8

National Standard 8 (50 CFR [Code of Federal Regulations] 600.345) specifies that conservation and management measures shall, consistent with the conservation requirements of the Magnuson-Stevens Act, take into account the importance of fishery resources to fishing communities by utilizing economic and social data that are based on the best scientific information available in order to (1) provide for the sustained participation of such communities, and (2) to the extent practicable, minimize adverse economic impacts to such communities.

Per National Standard 8, the term “fishing community” means a community that is substantially dependent on or substantially engaged in the harvest or processing of fishery resources to meet social and economic needs, and includes fishing vessel owners, operators, and crew, and fish processors that are based in such communities. A fishing community is a social or economic group whose members reside in a specific location and share a common dependency on commercial, recreational, or subsistence fishing or directly related fisheries-dependent services and industries (for example, boatyards, ice suppliers, tackle shops).

Also, per National Standard 8, the term “sustained participation” means continued access to the fishery within the constraints of the condition of the resource. Per the guidelines for National Standard 8:

FMPs [Fishery Management Plans] must examine the social and economic importance of fisheries to communities potentially affected by management measures. For example, severe reductions of harvests for conservation purposes may decrease employment opportunities for fishermen and processing plant workers, thereby adversely affecting their families and communities. Similarly, a management measure that results in the allocation of fishery resources among competing sectors of a fishery may benefit some communities at the expense of others (50 CFR 600.345⁵).

3.6.3.1.2 Magnuson-Stevens Act National Standard 4

Under National Standard 4 (50 CFR 600.325), conservation and management measures shall not discriminate between residents of different states. If it becomes necessary to allocate or assign fishing privileges among various U.S. fishermen, such an allocation shall be: (1) fair and equitable to all such

⁵ The National Standard 8 guidelines referenced in this SIA, current as of February 12, 2021, are from the Electronic Code of Federal Regulations (CFR) Title 50, Chapter VI, Part 600, Subpart D, Section 600.345 (cited as 50 CFR 600.345) are available at https://www.ecfr.gov/cgi-bin/retrieveECFR?gp=&SID=6b0acea089174af8594db02314f26914&mc=true&r=SECTION&n=se50.12.600_1345 accessed 2/17/2021.

fishermen; (2) reasonably calculated to promote conservation; and (3) carried out in such a manner that no particular individual, corporation, or other entity acquires an excessive share of such privileges. Among other National Standard 4 guidelines:

Definition. An “allocation” or “assignment” of fishing privileges is a direct and deliberate distribution of the opportunity to participate in a fishery among identifiable, discrete user groups or individuals. Any management measure (or lack of management) has incidental allocative effects, but only those measures that result in direct distributions of fishing privileges will be judged against the allocation requirements of Standard 4.

An allocation of fishing privileges may impose a hardship on one group if it is outweighed by the total benefits received by another group or groups. An allocation need not preserve the status quo in the fishery to qualify as “fair and equitable,” if a restructuring of fishing privileges would maximize overall benefits. The Council should make an initial estimate of the relative benefits and hardships imposed by the allocation, and compare its consequences with those of alternative allocation schemes, including the status quo. Where relevant, judicial guidance and government policy concerning the rights of treaty Indians and aboriginal Americans must be considered in determining whether an allocation is fair and equitable (50 CFR 600.325⁶).

3.6.3.1.3 Social and Economic Analysis Under NEPA

Under NEPA, “economic” and “social” effects are specific environmental consequences to be examined (40 CFR 1502.16 and 1508.8). Economic effects are examined primarily in the Environmental Assessment (EA), while social effects (and community-level economic effects) are examined primarily in this SIA.

3.6.3.1.4 EO 12898 Environmental Justice

EO 12898 of February 11, 1994, Federal Actions to Address Environmental Justice in Minority Populations and Low-Income Populations (59 Federal Register [FR] 7629; February 16, 1994), directs Federal agencies “to make achieving environmental justice part of its mission by identifying and addressing, as appropriate, disproportionately high and adverse human health or environmental effects of its programs, policies, and activities on minority populations and low-income populations.”

The EO directs the development of agency strategies to include identification of differential patterns of consumption of natural resources among minority populations and low-income populations; Council on Environmental Quality (CEQ) environmental justice guidance under NEPA also specifically calls for consideration of potential disproportionately high and adverse impacts to Indian tribes⁷ beyond a more general consideration of potential disproportionately high and adverse impacts to minority populations (Council on Environmental Quality 1997).

3.6.3.1.5 Tribal Consultation and Collaboration

EO 13175 of November 6, 2000, Consultation and Coordination with Indian Tribal Governments (65 FR 67249; November 9, 2000), was promulgated:

⁶ The National Standard 4 guidelines referenced in this SIA, current as of February 12, 2021, are from the Electronic Code of Federal Regulations (CFR) Title 50, Chapter VI, Part 600, Subpart D, Section 600.325 (cited as 50 CFR 600.325) are available at https://www.ecfr.gov/cgi-bin/retrieveECFR?gp=&SID=6b0acea089174af8594db02314f26914&mc=true&r=SECTION&n=se50.12.600_1325 accessed 2/17/2021.

⁷ The term “Indian tribe” is retained due to its use in both the EO and CEQ guidance; the provisions of the EO and CEQ guidance are understood to apply to federally recognized Alaska Native tribes in the region potentially affected by the proposed action alternatives. The ANCSA status, ANCSA regional corporation, ANCSA village corporation, federal tribal recognition status, and CDQ membership status of each potentially substantially engaged and/or substantially dependent EBS snow crab fishing community is noted in the Table 3-31.

“...in order to establish regular and meaningful consultation and collaboration with tribal officials in the development of Federal policies that have tribal implications, to strengthen the United States government-to-government relationships with Indian tribes, and to reduce the imposition of unfunded mandates upon Indian tribes.”

The Presidential Memorandum of January 26, 2021, Tribal Consultation and Strengthening Nation-to-Nation Relationships (86 FR 7491, January 29, 2021) affirms that the current Administration:

“...is committed to honoring Tribal sovereignty and including Tribal voices in policy deliberation that affects Tribal communities. The Federal Government has much to learn from Tribal Nations and strong communication is fundamental to a constructive relationship.”

The Presidential Memorandum on Tribal Consultation and Strengthening Nation-to-Nation Relationships does not change the definition of a Federal agency as specified under EO 13175, and as such, the National Marine Fisheries Service (NMFS) is the agency responsible for carrying out Tribal Consultations.

Additionally, on February 8, 2021, the NPFMC unanimously adopted a motion relative to the Community Engagement Committee that recommended, among other actions, that the Council work “with NMFS to receive and understand results of Tribal Consultation meetings as early in the process as possible, preferably prior to Council final action.” The Council’s February 8, 2021 motion also recommended that the Executive Director assign responsibilities of a Rural Fisheries Community/Tribal Liaison position to staff. It is possible that the addition of these responsibilities will help facilitate more information related to Tribal Consultation and engagement at the regional level in a timely manner into the Council’s process.

3.6.3.1.6 EO 13985 Advancing Racial Equity and Support for Underserved Communities Through the Federal Government

EO 13985 of January 20, 2021, Advancing Racial Equity and Support for Underserved Communities Through the Federal Government (86 FR 7009; January 25, 2021), addresses issues of equity for Indigenous and Native American persons, persons who live in rural areas, and persons otherwise adversely affected by persistent poverty or inequality, among other groups, as well as underserved communities in general. Specifically, under Section 2, Definitions:

For purposes of this order: (a) The term “equity” means the consistent and systematic fair, just, and impartial treatment of all individuals, including individuals who belong to underserved communities that have been denied such treatment, such as Black, Latino, and Indigenous and Native American persons, Asian Americans and Pacific Islanders and other persons of color; members of religious minorities; lesbian, gay, bisexual, transgender, and queer (LGBTQ+) persons; persons with disabilities; persons who live in rural areas; and persons otherwise adversely affected by persistent poverty or inequality.

(b) The term “underserved communities” refers to populations sharing a particular characteristic, as well as geographic communities, that have been systematically denied a full opportunity to participate in aspects of economic, social, and civic life, as exemplified by the list in the preceding definition of “equity.”

Section 8, Engagement with Members of Underserved Communities, specifies that:

In carrying out this order, agencies shall consult with members of communities that have been historically underrepresented in the Federal Government and underserved by, or subject to discrimination in, Federal policies and programs.

3.6.3.1.7 EO 14008 Tackling the Climate Crisis at Home and Abroad

EO 14008 of January 27, 2021, Tackling the Climate Crisis at Home and Abroad (86 FR 7619; February 1, 2021), under Part II, Taking a Government-Wide Approach to the Climate Crisis, includes language on securing environmental justice and spurring economic opportunity. Specifically, Section 219 states:

To secure an equitable economic future, the United States must ensure that environmental and economic justice are key considerations in how we govern. That means investing and building a clean energy economy that creates well-paying union jobs, turning disadvantaged communities—historically marginalized and overburdened—into healthy, thriving communities, and undertaking robust actions to mitigate climate change while preparing for the impacts of climate change across rural, urban, and Tribal areas.

Agencies shall make achieving environmental justice part of their missions by developing programs, policies, and activities to address the disproportionately high and adverse human health, environmental, climate-related and other cumulative impacts on disadvantaged communities, as well as the accompanying economic challenges of such impacts.^{8,9}

As noted in Section 220, EO 14008 also amends Section 1-102 of EO 12898 Environmental Justice (Creation of an Interagency Working Group on Environmental Justice), replacing it with the creation, within the Executive Office of the President, a White House Environmental Justice Interagency Council.

3.6.3.1.8 EO 14031 Advancing Equity, Justice, and Opportunity for Asian-Americans, Native Hawaiians, and Pacific Islanders

EO 14031 of May 28, 2021, Advancing Equity, Justice, and Opportunity for Asian-Americans, Native Hawaiians, and Pacific Islanders (86 FR 29675; June 3, 2021), builds upon EO 13985 to advance equity and racial justice for underserved communities, which include Asian American, Native Hawaiian, and Pacific Islander communities, and the Presidential Memorandum of January 26, 2021 (Condemning and Combating Racism, Xenophobia, and Intolerance Against Asian Americans and Pacific Islanders in the

⁸ In the July 20, 2021 *Interim Implementation Guidance for the Justice40 Initiative*, Memorandum for the Heads of Departments and Agencies (M-21-28, Executive Office of the President, Office of Management and Budget, <https://www.whitehouse.gov/wp-content/uploads/2021/07/M-21-28.pdf>), an “Interim Definition of Disadvantaged Communities” is provided that includes several variables that may apply singly or in varying combinations to some of the fishing communities that may be directly or indirectly impacted by one or more of the proposed action alternatives or the no action alternative. These include low income, high and/or persistent poverty; high unemployment and underemployment; linguistic isolation; high housing cost burden and substandard housing; high transportation cost burden and/or low transportation access; disproportionate environmental stressor burden and high cumulative impacts; limited water and sanitation access and affordability; disproportionate impacts from climate change; high energy cost burden and low energy access; and access to health care, among others. This same interim implementation guidance defines communities as “either a group of individuals living in geographic proximity to one another, or a geographically dispersed set of individuals (such as migrant workers or Native Americans), where either type of group experiences common conditions.”

⁹ In September 2021, the United States Environmental Protection Agency (EPA) published *Climate Change and Social Vulnerability in the United States: A Focus on Six Impacts* (EPA 430-R-21-003, www.epa.gov/cira/social-vulnerability-report). As noted on page 4 of that document, however, “due to data limitations, this report does not analyze the impacts of climate change on socially vulnerable populations living in Hawai’i or Alaska.” Primary climate change impacts that were analyzed in the document are: air quality and health; extreme temperature and health; extreme temperature and labor; coastal flooding and traffic; coastal flooding and property; inland flooding and property.

United States [86 FR 7485; January 29, 2021]), which articulates the policy of the current administration to address and confront racism, xenophobia, and intolerance. Specifically, under Section 1, Policy:

The purpose of this order is to build upon those policies by establishing the President's Advisory Commission on Asian Americans, Native Hawaiians, and Pacific Islanders and the White House Initiative on Asian Americans, Native Hawaiians, and Pacific Islanders. Both will work to advance equity, justice, and opportunity for AA and NHPI communities in the United States.

Under Section 2, the EO establishes in the Department of Health and Human Services the President's Advisory Commission on Asian Americans, Native Hawaiians, and Pacific Islanders. Under Section 3, the EO establishes a White House Initiative on Asian Americans, Native Hawaiians, and Pacific Islanders, a federal Interagency working group.

3.6.3.2 Methodology

A two-part approach was used in characterizing the communities engaged in or dependent on the EBS snow crab fishery in ways that may be affected by the proposed action. First, tables based on existing quantitative fishery information were developed and are presented in section 3.6.3.3 to identify patterns of engagement in and dependency on the EBS snow crab fishery based on the distribution across communities of the sector most likely to be directly affected by one or more of the proposed alternatives. This is consistent with the National Standard 8 guidelines.

To address the sustained participation of fishing communities that will be affected by management measures, the analysis first identifies affected fishing communities and then assesses their differing levels of dependence on and engagement in the fishery being regulated (50 CFR 600.345).

The second approach involved selecting a subset of communities that, based on the results of the first approach, appeared to be substantially engaged in or substantially dependent on the EBS snow crab fishery and provides a more qualitative description of the community and its relationship with the EBS snow crab fishery for characterization of the specific community context of the fishery. This is consistent with the portion of the National Standard 8 guidelines that states:

The best available data on the history, extent, and type of participation in these fishing communities in the fishery should be incorporated into the social and economic information presented in the FMP. The analysis does not have to contain an exhaustive listing of all communities that might fit the definition; a judgment can be made as to which are primarily affected (50 CFR 600.345).

This two-part approach provides a more wholistic context for the subsequent analysis of potential community impacts that may occur due to the rebuilding alternative selected by the Council (section 3.6.4). The characterization of the relevant communities, appearing in section 3.6.3 incorporates existing and easily accessible community descriptive information by reference to the extent feasible, which has been supplemented with limited phone and email contacts with individuals and entities to update existing information where needed.

3.6.3.3 Quantitative Indicators of Community Fishery Engagement and Dependency

The sections below provide quantitative participation information, within the bounds of confidentiality restrictions, for the communities most directly engaged in and dependent on the relevant sectors of the EBS snow crab fishery. Specifically, the individual sections include a series of tables containing a range of quantitative information describing the distribution of sector-specific community engagement (or participation) in and dependency (or reliance) on the EBS snow crab fishery for the following sectors:

- EBS snow crab CVs and CPs owners

- EBS snow crab vessel crew
- CV and CP EBS snow crab quota share owners
- Shore-based processors operating in Alaska accepting EBS snow crab deliveries

Within this quantitative characterization of fishery participation, several simplifying assumptions were made. First, assignment of vessels to a region or community has been made based upon ownership address information as listed in the CFEC vessel registration files. Thus, some caution in the interpretation of this information is warranted. It is not unusual for vessels to have complex ownership structures involving more than one entity in more than one region. Further, the community of ownership address does not directly indicate where a vessel spends most of its time, purchases services, or hires its crew as, for example, some of the vessels with ownership addresses in the Pacific Northwest spend a great deal of time in Alaska ports and hire at least some crew members from these ports. The region or community of ownership address does, however, provide a rough indicator of the direction or nature of ownership ties and is a proxy for associated economic activity, as no existing datasets provide consistently collected time-series information on where CV expenditures on support services are made, especially when patterns are viewed at the sector or vessel class level. The vessel discussion includes the crew data that is available for this sector, which is useful in understanding the geographic footprint of sector employment and earnings and potentially where earnings are at least in part spent.

Vessel ownership address that is reported in CFEC data has been chosen for this analysis as the link of vessels to communities rather than other indicators, such as vessel homeport information. Previous Council FMP SIA experience (e.g., AECOM 2010) has highlighted the problematic nature of existing homeport data, i.e., that it does not have a consistent meaning across vessels.

Community designation for EBS snow crab quota shareholders has been made based on ownership address from NMFS, Restricted Access Management (RAM) data and for crew of EBS snow crab vessels ownership addresses is based on ADF&G crew licenses linked with crab Economic Data Report (EDR) data.

For shore-based processors, community designation was based on the operating location of the plant (rather than ownership address) to provide a relative indicator of the local volume of processing-related economic activity, which can also serve as a rough proxy for the relative level of associated employment, income, and local government revenues. There are, however, considerable limitations with the data that can be utilized for these purposes, because of confidentiality restrictions. A prime example of this is where a community is the site of one or two shore-based processors active in a community in a given year. No information can be disclosed about the volume and/or value of landings in those communities.

3.6.3.3.1 EBS Snow Crab Vessels

The following tables provide a series of quantitative indicators of sector engagement in and dependency on the EBS snow crab fishery, by community and/or regional geography depending on data confidentiality restrictions, for EBS snow crab CVs and one CP with local ownership addresses, as noted in the following paragraphs. Overall community vessel fleet dependency is also shown to the extent possible within data confidentiality restrictions. Where appropriate, ex-vessel gross revenue and first wholesale gross revenue from harvesting and processing of CDQ EBS snow crab allocations are included in the vessel harvesting tables and shore-based processor tables.

Table 3-15 provides a count, by community of historical ownership address and year (2006-2021), of EBS snow crab for all Alaska communities with any vessels active in the fishery in any given year during this time, as well as for the Seattle metropolitan area, as defined by the Seattle Metropolitan Statistical

Area (Seattle MSA¹⁰); Washington communities outside of the Seattle MSA combined; and Oregon/other states combined. For each geography, annual average counts and percentages of the grand total are also provided, along with a count of unique vessels, which may be indicative of continuity of participation (or lack thereof) at the vessel level. As shown, vessel ownership among states is concentrated in Washington, specifically within the Seattle MSA, Alaska, concentrated within Kodiak, Anchorage and Homer/Seldovia, and Oregon/other states.

Table 3-15 Vessels harvesting EBS snow crab by community of vessel historic ownership address, 2006-2021 (number of vessels)

Community	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Annual Average 2006-2021 (number)	Annual Average 2006-2021 (percent)	Unique Vessels 2006-2021 (number)
Homer	3	2	3	4	5	4	5	5	4	4	4	3	3	3	3	3	3.6	5.48%	7
Seldovia	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	0.9	1.42%	1
Homer/Seldovia	4	3	4	4	6	5	6	6	5	5	5	4	4	4	4	4	4.6	6.90%	8
Kodiak	10	8	10	11	9	8	8	8	8	8	8	8	7	7	7	7	8.3	12.48%	18
Anchorage	5	7	9	8	8	9	9	8	8	9	7	7	6	6	6	10	7.6	11.53%	17
Unalaska	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.1	0.09%	1
Ketchikan	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0.1	0.19%	1
Wasilla	0	0	0	1	0	0	0	0	0	1	1	1	1	1	1	1	0.5	0.76%	2
Other AK	6	8	10	9	8	9	9	8	8	10	8	8	7	7	7	11	8.3	12.57%	21
Alaska	20	19	24	24	23	22	23	22	21	23	21	20	18	18	18	22	21.1	31.95%	44
Seattle MSA**	39	34	37	38	34	35	36	34	33	33	35	29	31	30	31	27	33.5	50.66%	64
Other WA	4	2	4	3	2	3	3	4	4	4	4	4	4	4	2	3	3.4	5.10%	11
Washington	43	36	41	41	36	38	39	38	37	37	39	33	35	34	33	30	36.9	55.77%	72
Oregon	9	8	8	8	6	6	6	7	8	7	6	7	7	7	6	6	7.0	10.59%	13
Other States	2	1	1	1	1	1	1	1	2	1	1	1	1	0	1	2	1.1	1.70%	4
Oregon/Other States	11	9	9	9	7	7	7	8	10	8	7	8	8	7	7	8	8.1	12.29%	17
Grand Total	74	64	74	74	66	67	69	68	68	68	67	61	61	59	58	60	66.1	100.00%	111

Source: ADFG/CFEC Fish Tickets, data compiled by AKFIN in Comprehensive_FT

**Seattle MSA includes all communities in King, Pierce, and Snohomish counties.

Table 3-16 and Figure 3-23 provides EBS snow crab ex-vessel gross revenue by ownership address at the community and year (2006-2021) level to the extent possible within data confidentiality restrictions, along with annual averages in terms of inflation-adjusted dollars and percentages of the grand total for all geographies combined. Given the few vessels with historical ownership addresses outside of Kodiak, Homer/Seldovia, and Seattle MSA, little can be shown at the community level. The overall pattern of distribution of revenue is clear with Alaska ownership address vessels accounting for about 32 percent of the total of annual average ex-vessel gross revenue, Washington at 57 percent, and Oregon/other states at 11 percent, respectively.

¹⁰ The Seattle MSA encompasses all communities in King, Pierce, and Snohomish counties, Washington.

Table 3-16 Ex-vessel gross revenue for vessels harvesting EBS snow crab by community of vessel historic ownership address, 2006-2021 (thousands of real 2021 dollars)

Community	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Annual Average 2006-2021 (thousands)	Annual Average 2006-2021 (percent)	Unique Vessels 2006-2021 (number)
Homer/Seldovia	\$1,149	\$1,206	\$2,923	\$3,027	\$3,902	\$7,727	\$9,155	\$7,449	\$6,058	\$6,010	\$4,374	\$1,667	\$1,514	\$2,377	\$2,841	\$4,630	\$4,126	3.39%	8
Kodiak	\$3,657	\$6,934	\$18,607	\$13,042	\$6,997	\$18,357	\$25,922	\$19,047	\$15,203	\$16,134	\$11,701	\$6,041	\$5,510	\$7,943	\$11,535	\$16,003	\$12,665	10.42%	18
Other AK	\$5,171	\$15,886	\$24,623	\$16,669	\$13,063	\$27,823	\$36,638	\$26,489	\$23,250	\$30,687	\$18,757	\$13,466	\$12,927	\$18,660	\$25,042	\$44,097	\$22,078	18.16%	21
Alaska	\$9,977	\$24,026	\$46,152	\$32,738	\$23,963	\$53,907	\$71,714	\$52,985	\$44,511	\$52,832	\$34,832	\$21,175	\$19,951	\$28,980	\$39,418	\$64,730	\$38,868	31.97%	44
Seattle MSA**	\$28,010	\$30,356	\$56,529	\$48,335	\$40,932	\$81,716	\$111,563	\$85,458	\$66,789	\$72,710	\$61,878	\$38,670	\$40,867	\$58,014	\$73,531	\$105,750	\$62,569	51.47%	64
Other WA	\$3,610	\$2,382	\$5,832	\$3,427	\$2,959	\$7,419	\$9,567	\$10,760	\$9,822	\$9,739	\$7,090	\$6,654	\$5,130	\$7,296	\$5,142	\$12,764	\$6,850	5.63%	11
Washington	\$31,620	\$32,738	\$62,361	\$51,762	\$43,891	\$89,135	\$121,130	\$96,219	\$76,611	\$82,448	\$68,968	\$45,324	\$45,997	\$65,310	\$78,673	\$118,514	\$69,419	57.10%	72
Oregon/Other States	\$8,947	\$9,485	\$12,863	\$8,046	\$6,294	\$14,740	\$23,333	\$18,107	\$16,910	\$17,027	\$10,043	\$7,112	\$7,576	\$11,222	\$14,303	\$26,479	\$13,280	10.92%	17
Grand Total	\$50,544	\$66,249	\$121,376	\$92,546	\$74,149	\$157,782	\$216,178	\$167,310	\$138,032	\$152,307	\$113,843	\$73,611	\$73,523	\$105,512	\$132,393	\$209,723	\$121,567	100.00%	111

Source: ADFG/CFEC Fish Tickets, data compiled by AKFIN in Comprehensive_FT (BSS_SIA(10-12-22_1)).

**Seattle MSA includes all communities in King, Pierce, and Snohomish counties.

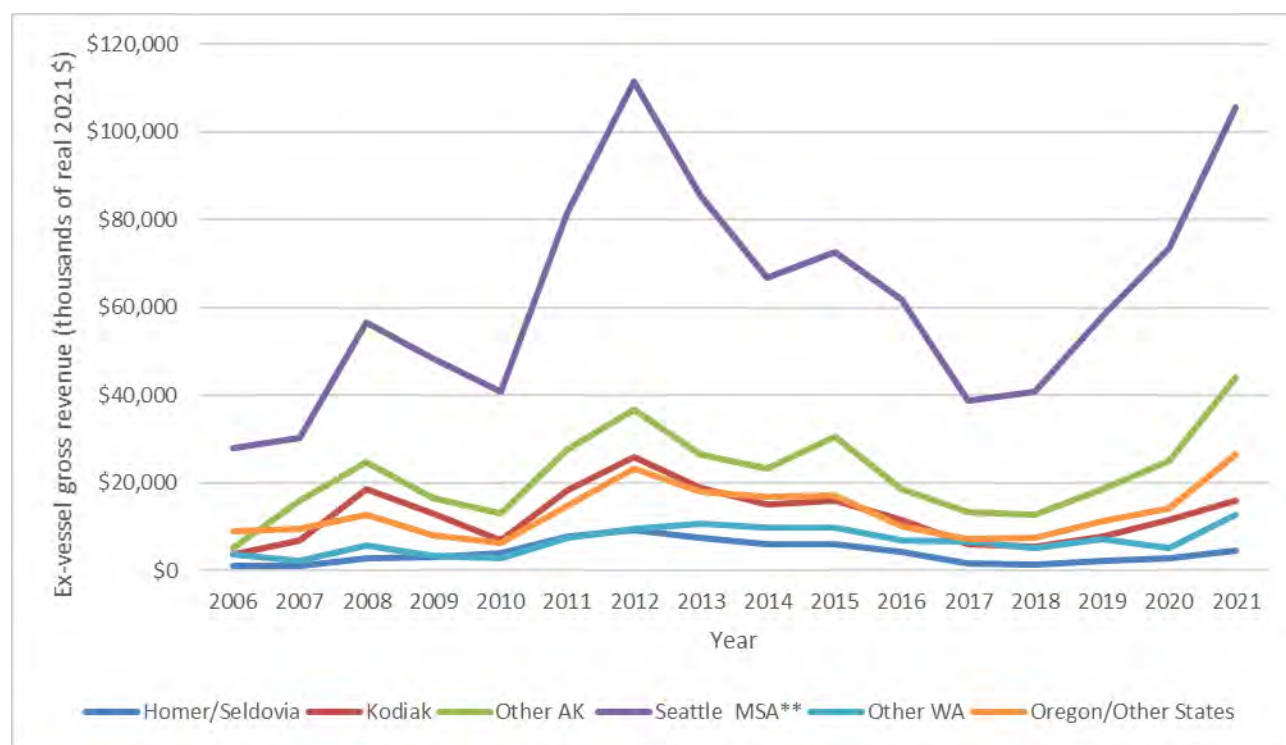


Figure 3-23 Ex-vessel gross revenue for vessels harvesting EBS snow crab by community of vessel ownership address, 2006-2021 (thousands of real 2021 dollars)

Table 3-17 provides information on EBS snow crab vessel dependency on EBS snow crab compared to all other areas, gear types, and species fished by those same vessels, as measured by percentage contribution to annual average ex-vessel gross revenue. As shown, dependency on EBS snow crab is relatively important for Alaska address vessels as well as vessels with addresses in Washington and Oregon/other states with EBS snow crab contributing between 53 percent to 50 percent of the annual average total ex-vessel gross revenue from all areas, gear types, and species fished.

Table 3-17 Revenue diversification for vessels harvesting EBS snow crab by ex-vessel gross revenue diversification, 2006-2021

Geography	Annual Average Number of Vessels	Annual Average Ex-Vessel Gross Revenues from Bering Sea Snow Crab Only (millions 2021 real \$)	Annual Average Total Ex-Vessel Gross Revenues from All Area, Gear, and Species Fisheries	EBS Snow Crab Ex-Vessel Revenue as a Percentage of Total Ex-Vessel Gross Revenue Annual Average
Homer/Seldovia	4.6	\$4.1	\$8.2	50.4%
Kodiak	8.3	\$12.7	\$25.0	50.8%
Other AK	8.3	\$22.1	\$40.4	54.6%
Alaska	21.1	\$38.9	\$73.5	52.9%
Seattle MSA**	33.5	\$62.6	\$127.3	49.2%
Other WA	3.4	\$6.8	\$11.7	58.3%
Washington	36.9	\$69.4	\$139.0	49.9%
Oregon/Other States	8.1	\$13.3	\$29.2	45.4%
Grand Total	66.1	\$121.6	\$241.8	50.3%

Source: ADFG/CFEC Fish Tickets, data compiled by AKFIN in Comprehensive_FT (BSS_SIA(10-12-22_1)).

**Seattle MSA includes all communities in King, Pierce, and Snohomish counties.

Table 3-18 provides information on overall community vessel dependency on EBS snow crab. This table includes all commercial fishing vessels, not just vessels that participate in the EBS snow crab fishery for those communities that had at least local ownership address EBS snow crab vessel participating in the fishery in any year 2006-2021. It compares the ex-vessel revenue from vessel caught EBS snow crab to ex-vessel revenue from all other areas, gear types, and species fished by all commercial fishing vessels with ownership addresses in that same community. As shown, Seattle MSA ownership address community fleet is relatively more dependent on the EBS snow crab fishery, as measured by contribution to total ex-vessel gross revenues, than the other communities or aggregation of communities other than Oregon/other states. In addition, Kodiak, Alaska ownership address community fleet is also more dependent on the EBS snow crab fishery than other Alaska communities.

Table 3-18 Revenue diversification for communities with vessels harvesting EBS snow crab by ex-vessel gross revenue diversification, 2006-2021

Geography	Annual Average Number of Vessels	Annual Average Number of All Commercial Fishing Vessels in those Same Communities	Annual Average Ex-Vessel Gross Revenues from Bering Sea Snow Crab Only (millions 2021 real \$)	Annual Average Total Ex-Vessel Gross Revenues from All Areas, Gears, and Species Fisheries for the Community Fleet (millions 2021 real \$)	EBS Snow Crab Community Ex-Vessel Gross Revenue as a Percentage of Total Community Ex-Vessel Gross Revenue Annual Average
Homer/Seldovia	4.6	377.4	\$4.1	\$96.9	4.3%
Kodiak	8.3	247.8	\$12.7	\$128.3	9.9%
Other AK	8.3	1149.9	\$22.1	\$216.6	10.2%
Alaska	21.1	1775.2	\$38.9	\$441.8	8.8%
Seattle MSA**	33.5	332.3	\$62.6	\$443.5	14.1%
Other WA	3.4	250.9	\$6.8	\$76.1	9.0%
Washington	36.9	583.2	\$69.4	\$519.6	13.4%
Oregon/Other States	8.1	63.1	\$13.3	\$67.5	19.7%
Grand Total	66.1	2421.4	\$121.6	\$1,029.0	11.8%

Source: ADFG/CFEC Fish Tickets, data compiled by AKFIN in Comprehensive_FT (BSS_SIA(10-12-22_1)).

**Seattle MSA includes all communities in King, Pierce, and Snohomish counties.

Table 3-19 provides information on EBS snow crab vessel dependency on EBS snow crab fishery, BBRKC fishery, and aggregated groundfish fisheries by comparing annual average ex-vessel gross revenue from these fisheries to annual average total ex-vessel gross revenue from all other areas, gear types, and species from 2006-2021. The largest share of annual average gross ex-vessel revenue for EBS snow crab vessels is from the EBS snow crab fishery followed by the BBRKC fishery, while annual average gross ex-vessel revenue from aggregate groundfish fisheries contributed a relatively small portion of the annual average total ex-vessel revenue from all areas, gears, and species fished.

Table 3-19 Revenue sources for vessels harvesting EBS snow crab by ex-vessel revenue, 2006-2021

Geography	Annual Average Ex-Vessel Gross Revenues from Bering Sea Snow Crab (millions 2021 real \$)	Annual Average Ex-Vessel Revenues from Bristol Bay Red King Crab (millions 2021 real \$)	Annual Average Ex-Vessel Revenues from All Crab (millions 2021 real \$)	Annual Average Ex-Vessel Revenues from Groundfish (millions 2021 real \$)	Annual Average Total Ex-Vessel Revenues from All Areas, Gears, and Species Fisheries (millions 2021 real \$)
Homer/Seldovia	\$4.1	\$2.9	\$7.4	\$0.6	\$8.2
Kodiak	\$12.7	\$8.5	\$22.0	\$1.3	\$25.0
Other AK	\$22.1	\$12.4	\$39.0	\$1.3	\$40.4
Alaska	\$38.9	\$23.7	\$68.4	\$3.1	\$73.5
Seattle MSA**	\$62.6	\$38.4	\$99.0	\$9.1	\$127.3
Other WA	\$6.8	\$3.4	\$10.8	\$0.3	\$11.7
Washington	\$69.4	\$41.9	\$109.7	\$9.3	\$139.0
Oregon/Other States	\$13.3	\$1.0	\$23.3	\$2.3	\$29.2
Grand Total	\$121.6	\$66.5	\$201.4	\$14.8	\$241.8

Source: ADFG/CFEC Fish Tickets, data compiled by AKFIN in Comprehensive_FT (BSS_SIA(10-12-22_1)).

**Seattle MSA includes all communities in King, Pierce, and Snohomish counties.

Table 3-20 provides information on annual EBS snow crab vessel dependency on the EBS snow crab fishery relative to the total gross ex-vessel revenue from all other areas, gear types, and species fished by these vessels. In general, during 2006 to 2010, largest number of vessels received between 10 percent to 50 percent of their total revenue from the EBS snow crab fishery but starting in 2011 there was a slight shift in EBS snow crab dependency where the largest group of EBS snow crab vessels received between 30 percent to 70 percent of their total revenue from this fishery. In 2021, the majority of the EBS snow crab vessels' total gross ex-vessel revenue was from the EBS snow crab fishery, which is likely due primarily to the closure of the directed BBRKC fishery.

Table 3-20 EBS snow crab revenue as a percent of total revenue for vessels harvesting EBS snow crab, 2006-2021 (number of vessels)

BSS Crab Rev as a % of Total	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Annual Average 2006-2021
0-10%	1	1	0	1	1	0	0	0	1	0	0	0	0	0	0	0	0.31
10-20%	11	8	2	2	7	2	1	0	0	0	1	1	1	1	0	0	2.31
20-30%	19	17	6	3	24	0	0	1	2	3	2	3	3	0	0	0	5.19
30-40%	16	14	19	16	18	5	1	1	7	11	16	13	13	6	2	1	9.94
40-50%	14	15	23	25	11	16	4	2	16	12	25	19	18	9	0	0	13.06
50-60%	2	4	16	17	1	25	11	16	24	25	14	14	14	13	6	2	12.75
60-70%	2	2	3	1	1	14	22	24	14	9	6	7	6	18	15	0	9.00
70-80%	2	0	1	1	0	2	19	15	2	3	0	0	2	6	20	7	5.00
80-90%	0	2	0	1	0	0	8	2	0	1	1	0	0	2	7	10	2.13
90-100%	7	1	4	7	3	3	3	7	2	4	2	4	4	4	8	40	6.44
Grand Total	74	64	74	74	66	67	69	68	68	68	67	61	61	59	58	60	66.13

Source: ADFG/CFEC Fish Tickets, data compiled by AKFIN in Comprehensive_FT (BSS_SIA(12-5-22))

3.6.3.3.2 Crew Licenses Harvesting EBS Snow Crab

Table 3-21 provides information on the number of crew licenses harvesting EBS snow crab by community for the years 2012 through 2020¹¹. As shown, the largest number of crew licenses harvesting EBS snow crab were from Washington state with an annual average of 242 licenses, of which 91 on average originate from Seattle and the other 151 remaining licenses originate from other Washington communities. Of the Alaska communities, Kodiak had the largest number of annual average crew licenses harvesting EBS snow crab at 61 licenses. Crab CVs harvesting EBS snow crab typically employ six to seven crew members including the captain (Garber-Yonts & Lee 2021). On average in the given time period, this represented 668 people employed annually harvesting EBS snow crab.

¹¹ EDR crew license data was redesigned in 2012 that included improvements to determine crew license location, so data was limited to 2012-2020.

Table 3-21 Crew licenses harvesting EBS snow crab by community, 2012-2020

Community	2012	2013	2014	2015	2016	2017	2018	2019	2020	Annual Average 2012-2020 (number)	Annual Average 2012-2021 (percent)
Akutan	1	2	2	2	3	0	1	0	0	1.2	0.18%
Anchorage/Palmer/Wasilla	45	49	46	55	41	35	37	44	33	42.8	6.41%
Dutch Harbor/Unalaska	23	22	22	28	20	12	18	19	3	18.6	2.78%
Homer/Seldovia	37	29	31	39	27	22	24	26	18	28.1	4.21%
King Cove	4	2	4	8	9	6	9	6	3	5.7	0.85%
Kodiak	70	70	76	83	60	62	54	50	24	61.0	9.14%
Other Ak	52	42	45	50	39	40	32	35	151	54.0	8.09%
Saint Paul	0	0	3	2	1	1	2	1	0	1.1	0.17%
Alaska	232	216	229	267	200	178	177	181	232	211.2	31.64%
Seattle MSA*	57	49	47	185	43	145	121	107	48	89.1	13.35%
Other Washington	224	214	208	120	219	84	92	88	96	149.4	22.39%
Washington	281	263	255	305	262	229	213	195	144	238.6	35.74%
Oregon	63	61	65	80	71	52	53	55	21	57.9	8.67%
Other States	143	136	134	196	201	148	139	167	175	159.9	23.95%
Grand Total	719	676	683	848	734	607	582	598	572	667.6	100.00%

Source: Economic Data Reports, data compiled by AKFIN source file BSS_SIA_Crew(9-15-22)

*Seattle MSA includes all communities in King, Pierce, and Snohomish counties.

3.6.3.3 EBS Snow Crab Harvest Shareholders

Table 3-22 provides information on EBS snow crab harvester quota share by community from 2012-2022. The state with largest concentration of EBS snow crab quota shares is Washington at an annual average of 569 million quota shares followed by Alaska at 321 million quota shares. Of the quota shares with a Washington address, Seattle MSA accounts for the largest portion with an annual average of 427 million quota shares. For Alaska, Anchorage/Wasilla/Palmer has the largest concentration of quota shares at annual average of 145 million shares followed by Kodiak at 81 million shares. Shares by community have remained relatively stable over the 2012-2022 period.

The majority of EBS snow crab IFQ is leased. Between 2012 and 2021, 81-89 percent of the total EBS snow crab IFQ was leased, with a slightly increasing trend (Garber-Yonts & Lee 2021). Table 3-23 provides information on EBS snow crab IFQ lease prices and total value of those leases. During the 2012 through 2021 period, lease prices were highest in 2021 at \$2.28 per pound of quota share for a total value of \$90 million for that year. Quota lease rates (i.e., the per-pound lease cost as a percentage of ex-vessel value) have remained quite stable overtime. The median lease rate for arms-length transactions of EBS snow crab have been about 46 percent, which tends to be lower than IFQ lease rates for BBRKC (about 63 percent) but higher than Tanner crab (about 29 percent). Further information about EBS snow crab IFQ leasing by quota share type is available from Garber-Yonts & Lee (2021).

Table 3-22 EBS snow crab quota shareholders by community, 2012-2022 excluding 2020 (millions of shares)

Community	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	Annual Average 2012-2022
Anchorage/Palmer/Wasilla	116.47	140.75	138.92	168.88	139.88	138.80	138.69	141.02	169.71	162.92	163.36	147.22
Dillingham	*	*	*	*	*	*	*	*	*	47.92	54.26	51.09
Homer/Seldovia	30.52	29.91	24.87	24.17	24.17	18.22	18.22	18.22	18.24	18.24	18.00	22.07
Kodiak	89.61	86.59	80.31	76.35	81.62	87.90	82.78	75.99	76.74	75.25	76.63	80.89
Petersburg	12.10	12.07	*	12.24	12.24	*	12.32	12.32	6.16	12.32	12.32	11.57
Other AK*	41.76	34.28	75.39	33.65	62.76	74.60	62.89	62.89	0.00	41.03	36.58	52.58
Alaska	290.45	303.60	319.48	315.29	320.68	319.52	314.90	310.45	355.34	357.68	361.16	324.41
Seattle MSA**	489.86	480.09	455.68	442.45	441.10	444.84	419.40	408.97	320.24	340.48	347.05	417.29
Other Washington	118.55	112.58	120.57	137.13	125.60	131.36	152.97	167.90	204.96	184.72	168.24	147.69
Washington	608.41	592.67	576.25	579.58	566.70	576.20	572.37	576.87	525.20	525.20	515.28	564.98
Oregon	77.76	81.15	77.08	78.00	78.24	77.91	78.62	79.00	76.40	76.40	76.40	77.91
Other States	29.33	29.50	33.40	34.13	39.90	31.10	36.99	40.73	43.94	43.94	54.20	37.92
Grand Total	1005.95	1006.92	1006.22	1007.00	1005.53	1004.73	1002.88	1007.04	1000.88	1003.22	1007.05	1005.22

Source: NMFS Restricted Access Management (RAM) division sourced through AKFIN source file BSS_SIA_QS(11-10-22)

*Includes listed communities when data is confidential

**Seattle MSA includes all communities in King, Pierce, and Snohomish counties.

Table 3-23 EBS snow crab IFQ leasing, 2012-2021 (2021 real dollars)

Crab Fishery	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Annual Average 2012-2021
Price Per Pound	\$1.05	\$1.15	\$1.13	\$0.96	\$1.30	\$1.90	\$1.80	\$2.09	\$1.80	\$2.28	\$1.55
Total Value	\$62,062,982	\$58,600,842	\$49,165,576	\$43,836,861	\$38,370,276	\$33,569,980	\$27,948,171	\$48,525,660	\$50,724,828	\$89,947,222	\$50,275,240
Total QS Leased	70,074,652	59,237,504	49,679,334	51,695,480	33,011,671	19,426,419	16,666,954	24,465,746	29,435,361	39,366,169	39,305,929

Source: Economic Data Reports, data compiled by AKFIN source file BSS_SIA_QS_Leasing(10-7-22)

3.6.3.3.4 EBS Snow Crab Processor Shareholders

Table 3-24 provides information on EBS snow crab processor quota share by community from 2012-2022. The state with largest concentration of EBS snow crab processor quota shares is Washington at an annual average of 703 million quota shares followed by Alaska at 300 million quota shares. Of the quota shares with a Washington address, Seattle MSA accounts for the largest portion with an annual average of 464 million quota shares. For Alaska, Anchorage/Wasilla/Palmer has the largest concentration of quota shares at annual average of 112 million shares followed. Other Alaska communities with concentrations of processor quota shareholders include Saint Paul, Ninilchik, and Juneau.

Table 3-24 EBS snow crab processing quota shareholders by community, 2012-2022 (millions of shares)

Community	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	Annual Average 2012-2022
Anchorage/Palmer/Wasilla	91.28	145.64	91.28	263.57	91.28	91.28	91.28	91.28	91.28	91.36	91.36	111.90
Other AK*	115.17	58.25	229.27	56.99	229.27	229.27	229.27	229.27	229.27	229.27	229.27	187.69
Alaska	206.46	203.89	320.55	320.55	320.55	320.55	320.55	320.55	320.55	320.63	320.63	299.59
Seattle MSA**	690.81	689.55	571.62	571.62	571.62	571.70	286.81	286.81	286.81	286.73	286.73	463.71
Other Washington	104.82	108.65	109.92	109.92	109.92	109.92	394.80	394.80	394.80	394.80	394.80	238.83
Washington	795.64	798.20	681.54	681.54	681.54	681.62	681.62	681.62	681.62	681.54	681.54	702.55
Other States	0.08	0.08	0.08	0.08	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.04
Grand Total	1002.17	1002.17	1002.17	1002.17	1002.17	1002.17	1002.17	1002.17	1002.17	1002.17	1002.17	1002.17

Source: NMFS Restricted Access Management (RAM) division sourced through AKFIN; source file is BSS_SIA_QS_PRO(12-20-22)

*Other AK communities include St. Paul, Ninilchik, Juneau

**Seattle MSA includes all communities in King, Pierce, and Snohomish counties.

3.6.3.3.5 Shore-based Processors Accepting EBS Snow Crab

The following tables provide a series of quantitative indicators of sector engagement in and dependency on the EBS snow crab fishery. Engagement is shown through participation by community and/or regional geography depending on data confidentiality constraints, for shore-based processors operating in Alaska and Washington, as noted in the following paragraphs. Overall community shore-based processor dependency (as measured in percentage of total ex-vessel value paid for all deliveries from all fisheries made to the relevant processors) is also shown to the extent possible within confidentiality constraints.

Table 3-25 provides information on the distribution of relevant shore-based processors in Alaska and Washington communities active in the period 2006-2021. For the purpose of this portion of the analysis, relevant shore-based processors are defined as those shore-based entities accepting EBS snow crab deliveries. As shown, five Alaska communities and Seattle MSA were the locations of relevant shore-based processing over this period accepting EBS snow crab deliveries in each year included in the data (Alaska communities were Unalaska/Dutch Harbor, Akutan, King Cove, Kodiak, Saint Paul), with one community (Unalaska/Dutch Harbor) having multiple processors accepting EBS snow crab deliveries in each year.

Table 3-25 also includes some custom processing activity of EBS snow crab at shoreside processors in St. Paul and Dutch Harbor. This EBS snow crab custom processing is on the behalf of companies with addresses in Anchorage, Juneau, and Savoonga. As a result, the first wholesale revenue from custom processing is appropriately accrued to the company requesting the custom processing that were based in Anchorage, Juneau, and Savoonga rather than the shoreside processors as indicated in the table. Table 3-25 also includes processing of EBS snow crab on floating processors with addresses in Edmonds and Seattle.

Table 3-25 Processors accepting EBS snow crab by community of operation, 2006-2021 (number of processors)

Community	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Annual Average 2006-2021 (number)	Annual Average number of processors 2006-2021 (percent)	Unique Processors 2006-2021 (number)
Akutan	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1.0	8.29%	1
King Cove	1	2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1.1	8.81%	3
Kodiak	2	2	3	2	1	3	2	2	1	2	1	1	1	1	1	1	1.6	13.47%	4
St Paul	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1.0	8.29%	1
Akutan/King Cove/Kodiak/St Paul	5	6	6	5	4	6	5	5	4	5	4	4	4	4	4	4	4.7	38.86%	9
Dutch Harbor	3	4	4	5	3	3	3	3	3	3	3	4	3	3	3	3	3.3	27.46%	6
Anchorage	0	0	0	1	1	2	3	3	2	1	1	1	1	2	2	2	1.4	11.40%	4
Edmonds	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0.1	0.52%	1
Juneau	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	0	0.8	6.22%	1
King Salmon	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0.1	1.04%	1
Savoonga	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0.1	0.52%	1
Seattle MSA*	2	4	3	2	2	2	2	1	1	2	1	1	1	1	1	1	1.7	13.99%	6
Other	2	5	4	4	4	5	6	5	4	4	3	4	3	4	4	4	4.1	33.68%	14
Grand Total	10	15	14	14	11	14	14	13	11	12	10	12	10	11	11	11	12.1	100.00%	29

Source: ADFG/CFEC Fish Tickets, data compiled by AKFIN in Comprehensive_FT source file BSS_SIA_Proc(10-13-22).

*Seattle MSA includes all communities in King, Pierce, and Snohomish counties.

Table 3-26 and Figure 3-24 provides information on the first wholesale gross revenue for EBS snow crab by shore-based processors associated with EBS snow crab deliveries by community and year (2006-2021) to the extent possible within data confidentiality constraints. The decision to group Akutan with King Cove, Kodiak, and Saint Paul was due to confidentiality constraints. As shown, Akutan, King Cove, Kodiak, and Saint Paul combined account for approximately 46 percent of the average annual first wholesale gross revenue of EBS snow crab delivered to shore-based processors from 2006 through 2021, while Unalaska/Dutch Harbor accounts for 30 percent of the average annual first wholesale gross revenue for EBS snow crab. The remaining 26 percent of the average annual first wholesale gross revenue from EBS snow is included in other communities.

Table 3-26 EBS snow crab first wholesale gross revenue for shored-based processors accepting EBS snow crab by community of operation, 2006-2021 (thousands of 2021 real dollars)

Community	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Annual Average 2006-2021 (thousands)	Annual Average 2006-2021 (percent)	Unique Processors 2006-2021 (number)
Akutan/King Cove/Kodiak/St Paul	\$35,584	\$35,188	\$73,080	\$65,074	\$45,211	\$93,798	\$139,523	\$109,355	\$103,892	\$105,757	\$78,598	\$60,019	\$41,010	\$71,294	\$79,118	\$134,199	\$79,419	45.56%	9
Dutch Harbor	*	*	\$58,268	\$55,082	\$41,156	\$64,178	\$83,849	\$71,699	\$61,369	\$58,497	\$47,487	\$31,675	\$21,672	\$32,773	\$37,999	\$66,203	\$50,439	28.94%	6
Other	*	*	\$34,726	\$29,816	\$28,981	\$63,589	\$89,752	\$68,518	\$48,164	\$53,833	\$39,364	\$30,938	\$20,740	\$35,863	\$43,761	\$73,911	\$44,449	25.50%	14
Grand Total	\$88,682	\$106,442	\$166,073	\$149,972	\$115,348	\$221,564	\$313,124	\$249,573	\$213,425	\$218,088	\$165,449	\$122,631	\$83,422	\$139,930	\$160,879	\$274,314	\$174,307	100.00%	25

Source: ADFG/CFEC Fish Tickets, data compiled by AKFIN in Comprehensive_FT source file BSS_SIA_Proc(10-13-22).

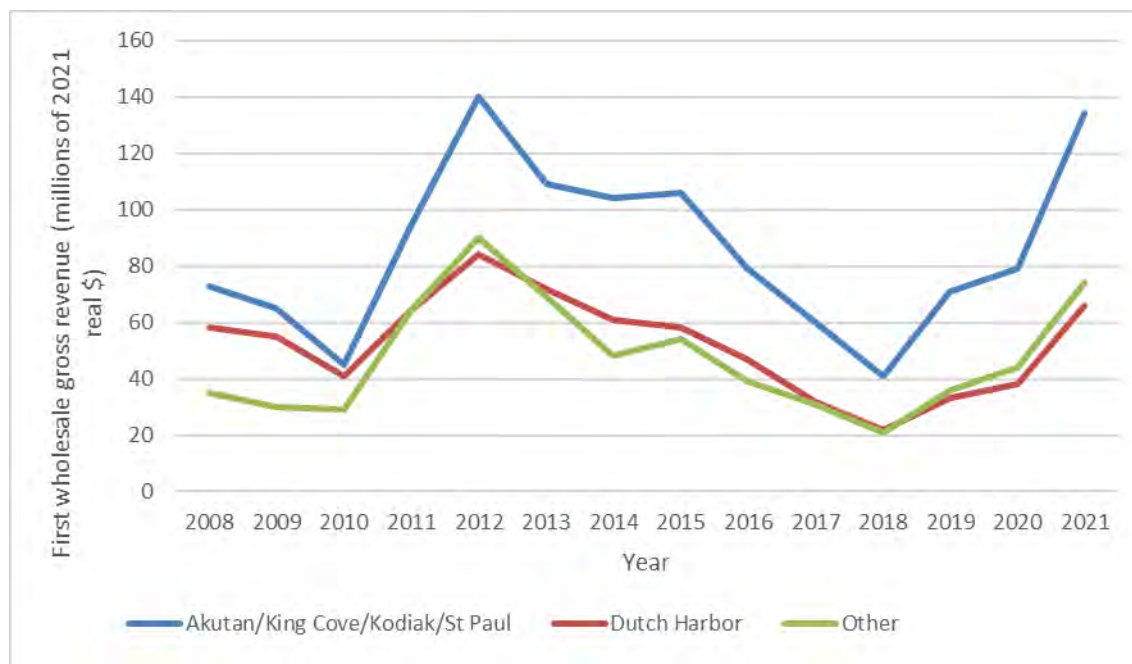


Figure 3-24 EBS snow crab first wholesale gross revenue for shore-based processors accepting EBS snow crab by community of operation, 2008-2021 (millions of 2021 real dollars)

Table 3-27 provides information on average annual shore-based processor dependency on deliveries of EBS snow crab compared to all area and species fisheries landings processed by those same processors for the years 2006-2021, as measured in percentage of first wholesale gross revenue associated with deliveries made to these processors. As shown, of the deliveries made to the combined Akutan, King Cove, Kodiak, and Saint Paul shore-based processors, approximately 13 percent of total first wholesale gross revenue of landings of all species were associated with EBS snow crab deliveries over the period. For the processors in Unalaska/Dutch Harbor, that figure was also approximately 13 percent. Due to confidentiality restrictions, aggregating communities could hide the level of dependency on the EBS snow crab fishery for some communities like Saint Paul. However, this community-level context is qualitatively described more in section 3.6.4.

Table 3-27 Revenue diversification for shore-based processors accepting EBS snow crab by first wholesale gross revenue, 2006-2021

Geography	Annual Average Number of Processors	Annual Average First Wholesale Revenues from Bering Sea Snow Crab Only (millions 2021 real \$)	Annual Average Total First Wholesale Revenues from All Area, Gear, and Species Fisheries	Bering Sea Snow Crab First Wholesale as a Percentage of Total First Wholesale Gross Revenue Annual Average
Akutan/King Cove/Kodiak/St Paul	9	\$79.4	\$630.4	12.60%
Dutch Harbor	6	\$50.4	\$400.1	12.61%
Other	14	\$44.4	\$97.7	45.51%
Grand Total	25	\$174.3	\$1,128.1	15.45%

Source: ADFG/CFEC Fish Tickets, data compiled by AKFIN in Comprehensive_FT source file BSS_SIA_Proc(10-13-22).

Table 3-28 provides information on average annual total shore-based processor dependency on EBS snow crab (all shore-based processors in the communities that had at least one shore-based processor that accepted EBS snow crab deliveries, not just the shore-based processors that participate in that fishery) compared to all area and species fishery landings processed by all processors in the community(ies) for the years 2006-2021, within the constraints of confidentiality restrictions, as measured by first wholesale gross revenue associated with those landings. As shown, for the span of years provided, EBS snow crab first wholesale gross revenue of landings accounted for about 7 percent of all shore-based processor first wholesale gross revenue of landings for Akutan, King Cove, Kodiak, and Saint Paul combined, while for Unalaska/Dutch Harbor that figure was 8 percent for all shore-based processor first wholesale gross revenue.

Table 3-28 Revenue diversification for communities with shore-based processors that processed EBS snow crab by first wholesale gross revenue by these shore-based processors, 2006-2021

Geography	Annual Average Number of Vessels	Annual Average Number of All Commercial Fishing Processors in those Same Communities	Annual Average First Wholesale Revenues from Bering Sea Snow Crab Only (millions 2021 real \$)	Annual Average Total First Wholesale Revenues from All Areas, Gears, and Species Fisheries for the Community Fleet (millions 2021 real \$)	EBS Snow Crab First Wholesale Revenue as a Percentage of Total Community Ex-Vessel Gross Revenue Annual Average
Akutan/King Cove/Kodiak/St Paul	9	23.8	\$79.4	\$947.2	8.38%
Dutch Harbor	6	7.4	\$50.4	\$582.4	8.66%
Other	14	310.6	\$44.4	\$585.2	7.60%
Grand Total	25	341.8	\$174.3	\$2,114.7	8.24%

Source: ADFG/CFEC Fish Tickets, data compiled by AKFIN in Comprehensive_FT source file BSS_SIA_Proc(10-13-22).

Table 3-29 provides information on EBS snow crab processor dependency on the EBS snow crab fishery, BBRKC fishery, and aggregated groundfish fisheries by comparing annual average first wholesale gross revenue from these fisheries to annual average total first wholesale gross revenue from all other areas, gear types, and species from 2006-2021. As shown, the largest share of annual average first wholesale gross revenue for EBS snow crab processors is from the aggregated groundfish fisheries followed by the EBS snow crab fishery.

Table 3-29 Revenue sources for processors accepting EBS snow crab for shore-based processors accepting EBS snow crab by gross first wholesale revenue, 2006-2021

Geography	Annual Average First Wholesale Revenues from Bering Sea Snow Crab (millions 2021 real \$)	Annual Average First Wholesale Revenues from Bristol Bay Red King Crab (millions 2021 real \$)	Annual Average First Wholesale Revenues from All Crab (millions 2021 real \$)	Annual Average First Wholesale Revenues from Groundfish (millions 2021 real \$)	Annual Average Total First Wholesale Revenues from All Areas, Gears, and Species Fisheries (millions 2021 real \$)
Akutan/King Cove/Kodiak/St Paul	\$79.4	\$34.2	\$115.9	\$332.0	\$630.4
Dutch Harbor	\$50.4	\$37.5	\$116.8	\$257.5	\$400.1
Other	\$44.4	\$13.3	\$62.7	\$4.6	\$97.7
Grand Total	\$174.3	\$85.0	\$295.4	\$594.1	\$1,128.1

Source: ADFG/CFEC Fish Tickets, data compiled by AKFIN in Comprehensive_FT source file BSS_SIA_Proc(10-13-22).

Table 3-30 provides information on annual EBS snow crab processor dependency on the EBS snow crab fishery relative to the total first wholesale gross revenue from all other areas, gear types, and species processed by those same processors. In general, during 2006 to 2021, largest number of processors received up to 20 percent of their total first wholesale gross revenue from EBS snow crab. The table also indicates that a few EBS snow crab processors routinely received between 70 percent to 100 percent of their annual total first wholesale gross revenue from EBS crab.

Table 3-30 Annual average EBS snow crab first wholesale gross revenue as a percentage of annual average total first wholesale gross revenue, 2006-2021 (number of processors)

BSS Crab Revenue as a % of Total	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Annual Average 2006-2021
0-10%	6	8	6	6	4	6	2	3	3	4	4	6	5	4	4	3	4.63
10-20%	0	0	2	1	1	1	3	4	2	2	1	0	1	1	0	2	1.31
20-30%	0	2	1	0	0	0	2	1	0	0	0	0	0	1	2	1	0.63
30-40%	1	3	1	0	3	0	0	0	1	2	1	1	2	1	0	0	1.00
40-50%	0	0	0	2	1	2	1	0	2	0	1	1	1	0	1	0	0.75
50-60%	1	1	1	2	1	2	0	2	1	2	2	1	1	1	0	2	1.25
60-70%	1	0	1	0	0	2	2	2	1	0	0	2	0	1	0	0	0.75
70-80%	0	0	1	1	1	0	2	0	0	0	1	1	1	1	2	0	0.69
80-90%	1	0	0	1	0	1	1	1	0	1	0	0	0	1	0	1	0.50
90-100%	0	1	1	1	0	0	1	0	1	1	0	0	0	0	2	2	0.63
Grand Total	10	15	14	14	11	14	14	13	11	12	10	12	10	11	11	11	12.06

Source: ADFG/CFEC Fish Tickets, data compiled by AKFIN in Comprehensive_FT source file BSS_SIA_Proc(10-13-22).

3.6.3.4 Overview of EBS Snow Crab Communities

The information included in this section originates from four sources: 1) [Community Profiles for North Pacific Fisheries](#)¹²; 2) Appendix A BSAI Crab Rationalization Ten-Year Program Review, Social Impact Assessment; 3) Annual Community Engagement and Participation Overview (ACEPO), March 19, 2021, and 4) Regulatory Impact Review/Environmental Assessment for Proposed Amendment 122 to the Fishery Management Plan for Groundfish of the BS and Aleutian Islands Management Area, BSAI Pacific Cod Trawl Cooperative Program.

A summary of the institutional structure for the Alaska communities relevant to this SIA analysis is shown in Table 3-31. Narrative summaries of the historic context of each Alaska community listed are presented in the following sections along with Seattle MSA.

¹² <https://www.fisheries.noaa.gov/resource/document/profiles-fishing-communities-alaska>

Table 3-31 Institutional indicators for selected Alaska communities engaged in the EBS snow crab rebuilding

Community	Traditional Community Name and Translation	Borough	Municipal Government (Incorporation Status, Date)	ANCSA Regional Corporation*	ANCSA Village Corporation	Federally Recognized Tribe and Tribal Government
Akutan	Achan-ingiiga (Unangan Aleut)	Aleutians East Borough	City of Akutan (2 nd Class City, 1979)	Aleut Corporation	Akutan Corporation	Native Village of Akutan
King Cove	Agdaagux*	Aleutians East Borough	City of King Cove (1 st Class City, 1947)	Aleut Corporation	The King Cove Corporation	Agdaagux Tribe of King Cove, Native Village of Belkofski
Kodiak	Sun'aq	Kodiak Island Borough	Home Rule City (2 nd Class City, 1940)	Koniag, Incorporated	Natives of Kodiak	Sun'aq Tribe of Kodiak
Saint Paul	Tanax* Amix*	Unorganized Borough	City of Saint Paul (2 nd Class City, 1971)	Aleut Corporation	Tanadgusix Corporation (TDX)	Saint Paul Island (Pribilof Islands Aleut Communities of St Paul and St. George Islands)
Unalaska	Lluulux (Unangan Aleut)	Unorganized Borough	City of Unalaska (1 st Class City, 1942)	Aleut Corporation	Ounalashka Corporation	Qawalangin Tribe of Unalaska

Notes:

* Regional ANCSA corporations are listed only for those communities where they are affiliated with an ANCSA village corporation, but they also serve shareholders in other communities. All of the KPB communities listed as "not an ANCSA community" are within the regional boundaries of Cook Inlet Region, Inc., except Seward, which is within the regional boundaries of the Chugach Alaska Corporation.

** Seldovia Village, an unincorporated CDP first appearing in the U.S. Census in 2000, is adjacent to, but outside of, the city limits of the City of Seldovia.

*** Eklutna is a small ANCSA village located within the much larger boundaries of the Unified Home Rule Municipality of Anchorage and is one of the villages within the Cook Inlet Region, Inc. family of villages; Anchorage itself is not an ANCSA village.

Source: Alaska Department of Commerce, Community, and Economic Development (2020a).

3.6.3.4.1 Unalaska/Dutch Harbor

Unalaska is located within the Aleutians West Census Area and is not under the jurisdiction of a borough. In 2010, there were 4,376 residents in Unalaska, making it the 26th largest of 352 total Alaskan communities with recorded populations that year. In a survey conducted by NOAA's Alaska Fisheries Science Center in 2011, community leaders estimated that approximately 2,500 seasonal or transient workers come to Unalaska each year. Community leaders also noted that the population of Unalaska reaches its annual peak between January 15th and April 1st each year (during Pollock "A" Season), and that the annual peak in population is "entirely" driven by employment in the fishing sectors.

Unalaska, traditionally an Aleut community, has become a relatively large, plural community with population growth that has accompanied port and fisheries-related development. Despite being an ANCSA village and having a federally recognized tribe, Unalaska did not qualify for CDQ membership based in part on having previously developed harvesting or processing capability sufficient to support substantial groundfish participation in the BSAI. It is, however, an *ex-officio* member of the Aleutian Pribilof Island Community Development Association (APICDA) CDQ group, a status that facilitates the participation of Unalaska residents in a range of APICDA programs. While the Unalaska/Dutch Harbor local commercial fishing fleet is typically represented in the Council and other regulatory processes by the Unalaska Native Fishermen's Association which, according to tribal leadership has a close working relationship with the Qawalangin Tribe of Unalaska, membership is not limited to those residents of Alaska Native descent.

Unalaska's economy is based on commercial fishing, fish processing, and fleet services, such as fuel, repairs, maintenance, trade, and transportation. The community enjoys a strategic position as the center of a rich fishing area and is used for transferring cargo between Pacific Rim trading partners. The Great Circle shipping route from major U.S. west coast ports to the Pacific Rim passes within 50 mi of Unalaska, and Dutch Harbor provides natural protection for fishing vessels. Onshore and offshore processors provide some local employment. However, non-resident workers are usually brought in during the peak season. In 2010, 31 residents held commercial fishing permits.

With respect to local economies, the importance of commercial fishing for Unalaska/Dutch Harbor cannot be overstated, as Unalaska/Dutch Harbor has ranked as the number one U.S. port in volume of landings since 1992 and has ranked second in value of landings (behind New Bedford, Massachusetts) since 2000. In recent years, employment statistics for Unalaska/Dutch Harbor have shown that the top three employers in the community were seafood processing companies, and that their employees accounted for over half of all employment in the city. The support service sector for the commercial fishing fleet is by far the most developed in the BSAI region, and Unalaska and firms dependent on the fisheries, such as stevedoring and shipping, regularly rank as some of the largest employers. There is no other community in the region with the level of development or the range of support services provided to the various fishery sectors operating in the BSAI region, which include accounting and bookkeeping, banking, construction and engineering, diesel sales and service, electrical and electronics services, freight forwarding, hydraulic services, logistical support, marine pilots/tugs, maritime agencies, gear replacement and repair, vessel repair, stevedoring, vehicle rentals, warehousing, and welding, among others (AECOM 2010; NOAA 2014).

As noted in the ACEPO document, Unalaska has a total of 12 seafood processing plants, five of which process groundfish. The vast majority of landings in Unalaska is pollock at 90 percent, Pacific cod at 7 percent, and crab at 2 percent. Although the percent of crab processing in Unalaska is only 2 percent, the community nevertheless is highly engaged in the crab processing sector, as noted in the ACEPO document. During 2019, Unalaska processed 15.7 million pounds of crab with an associated value of \$66.9 million. This marks a 15 percent increase (2 million pounds) in volume landed since 2018, and a 10 percent decreased in landed value (down \$7.8 million). Compared to the previous five-year average both volume processed and landed value in 2019 has decreased. The amount of BSAI crab processed in the region reached a peak of 35.4 million pounds (with a value of \$112 million) in 2015, then began a steep decline.

The first wholesale gross revenue for processors accepting EBS snow crab has ranged from \$22 million in 2018 to \$84 million in 2012, with an average annual first wholesale gross revenue of \$50 million from 2006 through 2021 (see Table 3-26). Relatively to the annual average gross revenues of only EBS snow crab deliveries to shore-based processors, Unalaska accounted for 29 percent of this value. Relatively to the average annual total first wholesale gross revenue of all fisheries processed by these Unalaska EBS snow crab processors from 2006 through 2021, EBS snow crab accounted for 13 percent of that total (see Table 3-27). Expanding to include all shore-based processors in Unalaska, the average annual EBS snow crab first wholesale gross revenue from 2006 through 2021 accounted for 8 percent (see Table 3-28).

Unalaska/Dutch Harbor EBS snow crab crew employment has been variable. As demonstrated in Table 3-21, between 2012 through 2020, EDR data indicate that between 3 (2020) and 28 (2015) crew from EBS snow crab vessels provided Unalaska/Dutch Harbor addresses on their ADFG commercial fishing crew licenses.

Table 3-32 provides information on City of Unalaska tax revenue deriving from sources directly related to fishing activities (the city raw seafood tax, the state shared fisheries business tax, and the state shared fisheries resource landing tax) compared to all general fund revenues received by the city for fiscal years 2010-2019. As shown, for the City of Unalaska, between roughly 37 percent and 50 percent of all general fund revenues in any given year derive from direct fishery revenue sources. These figures do not include revenue from other taxes and fees from activities in the community that are fishing related (e.g., property taxes paid by fisheries businesses, fuel transfer tax revenue, and harbor fee revenue, among others).

Table 3-32 City of Unalaska selected fisheries-related general fund revenues, fiscal years 2010-2019

Fiscal Year	Revenue (dollars) by Direct Fishery Revenue Source				All General Fund Revenue	Direct Fishery Revenue Source Total as a Percent of All General Fund Revenue
	Direct Fishery Revenue Source			Direct Fishery Revenue Source Total		
	City Raw Seafood Tax	Shared State Fisheries Business Tax	Shared State Fisheries Resource Landing Tax			
FY 2010	\$3,594,173	\$4,547,084	\$4,676,603	\$12,817,860	\$29,604,371	43.3%
FY 2011	\$5,371,768	\$3,199,290	\$3,531,739	\$12,102,797	\$29,152,912	41.5%
FY 2012	\$5,260,999	\$4,143,777	\$3,469,263	\$12,874,039	\$31,634,417	40.7%
FY 2013	\$4,784,198	\$4,398,441	\$4,898,543	\$14,081,182	\$32,609,892	43.2%
FY 2014	\$4,449,921	\$4,377,934	\$6,974,887	\$15,802,742	\$34,376,971	46.0%
FY 2015	\$4,981,770	\$3,639,448	\$5,014,309	\$13,635,527	\$34,525,170	39.5%
FY 2016	\$5,123,372	\$4,099,315	\$3,034,438	\$12,257,125	\$30,723,626	39.9%
FY 2017	\$4,657,385	\$4,276,287	\$8,272,661	\$17,206,333	\$34,371,441	50.1%
FY 2018	\$4,475,150	\$4,014,323	\$4,532,106	\$13,021,579	\$30,300,957	43.0%
FY 2019	\$4,761,506	\$3,528,499	\$5,220,958	\$13,510,963	\$36,419,248	37.1%

Source: City of Unalaska, Alaska. Comprehensive Annual Financial Reports, Fiscal Years 2010-2019.
<https://www.commerce.alaska.gov/dca/dcrarepoext/Pages/FinancialDocumentsLibrary.aspx>. Accessed 4/25/2020.

3.6.3.4.2 Akutan

Akutan is located on Akutan Island in the eastern Aleutians, one of the Krenitzin Islands of the Fox Island group. It is 35 mi east of Unalaska and 766 mi southwest of Anchorage. The area occupies 14.0 sq mi of land and 4.9 sq mi of water. The community was incorporated as a Second-class city in 1979 and is under the jurisdiction of the Aleutians East Borough.

Akutan, a community within the Aleutians East Borough, is somewhat unique demographically since it is the home of a large shore-based processor and the population size of the processing workforce residing in company housing at the plant site¹³ is larger than the predominately Alaska Native population residing within the traditional community footprint. The dual nature of the community demographics and socioeconomic attributes is reflected in the history of the community involvement in the CDQ program. Initially (in 1992), Akutan was deemed not eligible for participation in the CDQ program as the community was home to “previously developed harvesting or processing capability sufficient to support substantial groundfish participation in the BSAI...” though the community met other qualifying criteria. The Akutan Traditional Council subsequently initiated action to show that large industrial enclave-style development of the locally operating shore-based processor was essentially socially and economically separate and distinct from the traditional community of Akutan. With the support of APICDA and others, Akutan was successful in obtaining CDQ community status in 1996 and became a member community of APICDA.

As of 2010, there were 1,027 residents, ranking Akutan 64th of 352 Alaskan communities in terms of population size (this population count captured a large number of non-permanent residents). In a survey conducted by NOAA’s Alaska Fisheries Science Center (AFSC) in 2011, community leaders reported that there were an estimated 85 permanent and 900 seasonal residents living in Akutan in 2010. Seasonal workers typically live in Akutan from January through April with the population peaking in May. Peaks in Akutan’s population are mostly driven by seasonal employment in fisheries sectors.

Akutan has seen little in the way of fishery support service development, but the local processing operation accounts for a large percentage of local private sector employment and income opportunities.

¹³ According to the owner/operator’s website, the Akutan shoreplant has become the largest seafood production facility in North America with more than 1,400 company-housed employees present during peak seasons (accessed 6/30/2021).

Akutan is home to a large processing operation (Trident Seafoods) that was a major crab processing plant prior to rationalization and has remained so post-rationalization. Confidentiality restrictions do not allow disclosure of processing volumes or values.¹⁴ Given the lack of processor quota movement from the community, however, it is assumed that net processing volumes as a percentage of total fishery processing volumes have not decreased substantially, and it can be assumed with an increase in custom processing since the time of the 5-year program review that it is likely that locally processed percentage of overall fishery volume has increased. According to 2008 interviews with Akutan community leaders, no long-term residents of the community work at the plant other than a few individuals who came to the community for employment at the plant, a situation that existed prior to rationalization. According to local leadership, at present (2016) there is one local individual who has worked at the plant “for the last couple of years.”

As noted in the ACEPO document, Akutan, King Cove, and Sand Point crab processing activity which were aggregated due to confidentiality concerns, are highly engaged in the crab processing sector with six processing facilities in the region. In 2019, these communities processed 8.1 million net pounds of crab with an associated value of \$35.8 million. Compared to the previous five-year average, the volume decreased by 5 million pounds (down 39 percent) and the value decreased by \$17.6 million (down 33 percent). The amount of BSAI crab processed in the region reached a peak of 24.5 million pounds in 2015, quickly dropping to 16.3 million pounds the following year (down 33 percent). Comparatively, the associated value dropped by \$5.4 million or 7 percent during the same year. Both volume and landed value continued a steady decline since. In addition to crab processing, these communities are also highly engaged in the groundfish processing sector. Walleye pollock accounts for 76 percent of the landed value within the processing sector in Akutan, King Cove, and Sand Point. Other processing activity include Pacific cod at 13 percent and salmon at 8 percent.

As demonstrated in Table 3-21, EDR data indicated that one to three EBS snow crab crew members who provided Akutan addresses on their ADFG commercial fishing crew licenses between the period of 2012 through 2016, in addition to one crew member in 2018.

Table 3-33 provides information on City of Akutan tax revenues deriving from direct fishery revenue sources (the city raw seafood tax, the state shared fisheries business tax, and the state shared fisheries resource landing tax) compared to all general fund revenues received by the city for fiscal years 2010-2019. As shown, for the City of Akutan, between roughly 75 percent and 99 percent of all general fund revenues in any given year derive from direct fishery revenue sources.

¹⁴ In terms of processing quota shares held, at the time of initial allocation, Trident held processor quota shares with Akutan as the designated boundary for right of first refusal and the Aleutians East Borough designated as the cooling off boundary under the community protection measures built into the program for the Bristol Bay red king crab, Bering Sea snow crab, EAI golden king crab, Saint Matthew blue king crab, and Pribilof Islands blue and red king crab fisheries. At the time of the 5-year program review, these share holdings were unchanged from initial allocations and right of first refusal provisions are unchanged as well; the same holds true at present (2016).

Table 3-33 City of Akutan selected fisheries-related general fund revenues, fiscal years 2010-2019

Fiscal Year	Revenue (dollars) by Direct Fishery Revenue Source				All General Fund Revenue	Direct Fishery Revenue Source Total as a Percent of All General Fund Revenue
	Direct Fishery Revenue Source			Direct Fishery Revenue Source Total		
	City Raw Seafood Tax	Shared State Fisheries Business Tax	Shared State Fisheries Resource Landing Tax			
FY2010	\$753,127	\$1,088,369	\$307,561	\$2,149,057	\$2,588,527	83.0%
FY 2011	\$1,222,653	\$827,408	\$154,758	\$2,204,819	\$2,926,637	75.3%
FY 2012	\$1,385,057	\$853,570	\$244,134	\$2,482,761	\$3,077,710	80.7%
FY 2013	\$1,663,209	\$1,186,396	\$178,611	\$3,028,216	\$3,831,293	79.0%
FY 2014	\$1,715,128	\$1,217,118	\$157,540	\$3,089,786	\$3,602,184	85.8%
FY 2015	\$1,774,963	\$1,029,663	\$69,412	\$2,874,038	\$3,418,630	84.1%
FY 2016	\$2,098,763	\$943,814	\$173,049	\$3,215,626	\$3,253,634	98.8%
FY 2017	\$2,044,698	\$1,082,206	\$210,114	\$3,337,018	\$3,784,609	88.2%
FY 2018	\$1,985,328	\$1,358,949	\$4,916	\$3,349,193	\$3,796,184	88.2%
FY 2019	\$2,101,784	\$1,097,955	\$163,372	\$3,363,111	\$3,887,032	86.5%

Note: in 2013, the City of Akutan raised its local fish tax from 1.0 to 1.5 percent.

Source: City of Akutan, Alaska Basic Financial Statements, Required Supplementary Information, Additional Supplementary Information, and Compliance Reports, fiscal years 2010-2013 and 2015-2019; Certified Financial Statement, fiscal year 2014. <https://www.commerce.alaska.gov/dcral/dcrarepoext/Pages/FinancialDocumentsLibrary.aspx>. Accessed 9/15/2020.

3.6.3.4.3 King Cove

King Cove is located on the south side of the Alaska Peninsula, on a sand spit fronting Deer Passage and Deer Island. It is 18 miles southeast of Cold Bay and 625 miles southwest of Anchorage. King Cove is in the AEB and the Aleutian Islands Recording District. King Cove's area encompasses 25.3 square miles of land and 4.5 square miles of water.

In 2010, there were 938 residents in King Cove, making it the 70th largest of 352 total Alaskan communities with recorded populations that year. Overall, between 1990 and 2010, the population increased by 108 percent. Between 2000 and 2010, there was an average annual growth rate of 18.4 percent, which was substantially above the statewide average of 0.75 percent. It is important to note that, in a survey conducted by AFSC in 2011, community leaders reported that the 2010 Decennial Census population count may have been inflated, including approximately 400 individuals that do not reside in the community permanently. Community leaders estimated 450 permanent residents in 2010.

In the 2011 AFSC survey, community leaders indicated that 75 percent of economic activity in King Cove is based on direct and indirect fishing activity. Between 2000 and 2010, the number of commercial fishery permit holders, crew license holders, and vessel owners residing in King Cove all declined, but all of these numbers represented a large percentage of the population. On average over the 2000-2010 period, the number of state permit holders was equivalent to 7.8 percent of the local population, the number of vessel owners was equivalent to 10.2 percent, and the number of crew license holders was equivalent to 19.2 percent.

King Cove is one of the leading processing communities in Alaska, ranking 7th in landings and 11th in ex-vessel revenue out of 67 Alaskan ports that received landings in 2010. As noted in the Akutan community profile of the ACEPO document, when combined with King Cove and Sand Point, these communities are highly engaged in the crab processing sector with six processing facilities in the region.¹⁵

¹⁵ Akutan, King Cove, and Sand Point processing activity was aggregated due to confidentiality concerns.

In the 2011 AFSC survey, community leaders indicated that vessels homeporting in King Cove range in size from 35 feet to over 125 feet in length, and utilize gear types including trawl, pots, longline, gillnet, purse seine, and troll gear. They also indicated that most of the decline in fishing vessels in recent years has been a loss of the larger fleet – those vessels ranging from 60 to more than 125 feet in length. Specifically, they reported that one source of this decline is fewer BS crab boats coming to King Cove as a result of crab rationalization.

King Cove is home to a large processing operation (Peter Pan Seafoods) that was a major crab processing plant prior to rationalization and has remained so post-rationalization. Confidentiality restrictions do not allow disclosure of processing volumes or values, but city officials on multiple occasions, have noted that local fish taxes, while varying from year-to-year are often a rough balance between crab, salmon, and groundfish.¹⁶ Given the lack of processor quota movement from the community, however, it is assumed that net processing volumes as a percentage of total fishery quota processed have not changed substantially. Additionally, as reported in the 5-year program review (and confirmed with subsequent correspondence with Peter Pan Seafoods management), the plant has benefited from a consolidation of processor history within the AEB (and within the same firm) that was originally associated with processing activity during the qualification period that took place in both False Pass and Port Moller.¹⁷ This consolidated processing has continued to take place in King Cove through the present (2016). Further, according to interviews with plant management conducted for the 5-year program review, employment levels and the annual activity fluctuations at the plant have remained consistent with the patterns seen before rationalization was implemented.

Also, as reported in the 5-year program review, according to interviews, no long-term residents of the community work at the plant other than a few individuals who originally came to the community for employment at the plant, a situation that existed prior to rationalization; according to more recent (2016) interview information, this situation remains unchanged, although it is reported also that there have always been a few local teenagers who take the opportunity provided by the summer break to work at the plant.

As previously reported in the 5-year program review, changing processor ownership patterns have required divestiture, and resulted in the transfer (through sale) of some King Cove-based processor quota from Peter Pan Seafoods to Aleutia, a regional-based (Aleutian East Borough-based) entity. These shares have continued to be processed in King Cove under a series of annual custom processing agreements.¹⁸

¹⁶ Percentage dependency for major species groups ranged widely on an annual basis between FY 2000 and FY 2015, based on relative fishing success and variable market (price) conditions. During this time span, crab ranged between roughly 30 and 50 percent, salmon accounted for between roughly 15 and 40 percent, and groundfish between roughly 25 and 50 percent of total local landing taxes in any given year.

¹⁷ Qualifying crab processing history associated both False Pass and Port Moller resulted exclusively from floating processors operating within those communities. All False Pass associated processing history was derived from Peter Pan Seafoods operations. In the case of Port Moller, Peter Pan Seafoods was one of three firms with qualifying history associated with that community (with the other two being Snopac and Icicle Seafoods). According to Peter Pan Seafoods management, the Peter Pan Seafoods processor quota associated with False Pass and Port Moller have been processed annually in the Peter Pan Seafoods plant in King Cove; the Port Moller associated processing quota shares owned by the other two firms (or the successor owners of the processor quota originally owned by those other firms, APICDA and CBSFA/57 Degrees North, respectively), has not been custom processed at the Peter Pan Seafoods King Cove plant.

¹⁸ At the time of initial allocation, Peter Pan Seafoods held processor quota shares with King Cove designated as the right of first refusal boundary and the Aleutians East Borough designated as the cooling off boundary under the community protection measures built into the program for the Bristol Bay red king crab, Bering Sea snow crab, Pribilof blue and red king crab, and Saint Matthew blue king crab fisheries. At the time of the 5-year program review (2010), Peter Pan Seafoods retained ownership of all of these shares, except for a portion of the Bristol Bay red king crab shares that had been acquired by Aleutia. For the shares owned by Aleutia (including EBS Tanner and WBS Tanner crab, in addition to the Bristol Bay red king crab quota obtained from Peter Pan Seafoods), no right of first refusal provisions existed; for all other shares, right of first refusal provisions were unchanged. At the time of 2015/2016 IPQ allocation, Aleutia's holdings were unchanged from the time of the 5-year program review. For Peter Pan Seafoods, all remained the same as at the time of the 5-year program review except for their Saint Matthew blue king crab processor quota shares which, according to the dataset, were listed under a different holder (B&N Fisheries Company) and had no right of first refusal provisions attached.

According to city staff, this situation has remained unchanged in more recent years, with the city and the borough continuing to support Aleutia's efforts through a favored tax status.

As demonstrated in Table 3-21, EDR data indicated that two to nine Alaska residents who provided King Cove addresses on their ADFG commercial fishing crew licenses have crewed on EBS snow crab vessels each year from 2012 through 2020, with an average during this period of 6 crew licenses.

In 2020, King Cove total tax revenue was \$1,319,017 of which \$695,368 was from the 2 percent raw fish tax and \$623,649 was from a 6 percent sales tax. King Cove also received \$12,844.87 in shared fisheries business tax for that year. In 2021, King Cove received \$2,152,507 in total tax revenue of which \$743,413 originated from a 2 percent raw fish tax and \$1,409,094 was from a 6 percent sales tax. The community also received \$7,237 in shared fisheries business tax for that year. Community leaders reported that a variety of public services are at least partially funded by fisheries-related taxes and fees, including harbor maintenance, the health clinic, roads, the police force and fire protection, the recreation center, social services such as libraries, and general city administration. King Cove has local fishing-related fee programs that charge the fishing industry specifically to support public services and infrastructure.

3.6.3.4.4 Saint Paul Island

The community of Saint Paul Island is located on a narrow peninsula on the southern tip of Saint Paul Island, the largest of the five Pribilof Islands. It lies 47 miles north of Saint George Island, 240 miles north of the Aleutian Islands, 300 miles west of the Alaska mainland, and 750 air miles west of Anchorage. Saint Paul Island is located in the Aleutian Islands Recording District. The community encompasses 40.3 square miles of land and 255.2 square miles of water.

In 2010, the U.S. Census determined that there were 479 residents in Saint Paul Island, making it the 122nd largest of 352 total Alaskan communities with recorded populations that year. However, the 2006-2010 American Community Survey (ACS) estimated that Saint Paul Island had 1,065 residents in 2010. Another census by the city reports there were 344 full time residents as of October 2022. In a survey conducted by AFSC in 2011, community leaders reported that an estimated 300 seasonal workers or transients live in Saint Paul Island for part of the year, mostly working in the construction, tour guide, and fishing industries (e.g., snow crab, halibut, and king crab). They also indicated that the population of Saint Paul Island reaches an annual peak in January when transient processors are present, and that the population peak is entirely driven by fisheries-related employment.

Saint Paul Island is home to one large onshore processing operation (Trident Seafoods), which was a major crab processing plant prior to rationalization and has remained so post-rationalization. As noted in the ACEPO document, most processing activity in Saint Paul Island is for crab (94 percent of landed revenue). Halibut accounts for 6 percent of landed revenue. As noted in Table 3-25 in section 3.6.3.3.4, only one facility in Saint Paul has received EBS snow crab deliveries since 2006. In addition, as noted in Table 9 of the ACEPO document, Saint Paul Island has the second highest commercial crab processing engagement score since 2008, second only to Unalaska/Dutch Harbor. This index is an indicator of the degree of crab processing participation in a community relative to the crab processing participation of other communities.

Saint Paul Island had also historically been the site of a number of mobile processing operations over the years either inside the harbor (with larger operations including UniSea and Icicle) or in the area but outside the harbor (including Norquest and a number of others) as the nature of the fishery and its economic incentives dictated, and by limitations imposed by weather. While the floating processors did not typically employ any Saint Paul Island residents, a handful of long-term residents were employed at the Trident Seafoods shore plant, mostly as dock workers or crane operators. These employees typically worked the entire year, which includes the BSAI crab season in the fall and winter months, and the halibut season in the spring and summer months.

According to CBSFA staff, in each year since the 5-year program review, the Icicle Seafoods floating processor R.M. Thorstenson processed BSAI crab within the municipal boundaries but outside of the harbor up through the 2014/2015 season. It did not do so in the 2015/2016 season and, with the sale of Icicle Seafoods crab assets consisting of both IFQ and IPQ to 57 Degrees North, it is not expected to return to the community as a crab processor in the future. The Trident Seafoods shoreplant has continued to provide employment to a number of residents, mostly in dock crew and crane operator positions, but CBSFA staff report that a few local residents have worked line and other inside jobs at the plant as well in recent years. With the local presence of a major processing operation, CBSFA staff also point out that indirect employment is generated across a range of businesses, including at the fuel dock, the grocery store, and the air carriers, among others, not to mention administrative positions at the CBSFA as well as 57 Degrees North, Saint Paul Fish Company, and other CBSFA subsidiaries.

In terms of direct participation, local fishermen are almost exclusively engaged in the halibut fishery. With CBSFA investments in multiple crab vessels, Saint Paul residents interested in obtaining a crew position on a crab vessel have ready access through the CBSFA. At the time of the 5-year program review, officials from the CBSFA report, however, that this is not common because of (1) the relative ability of halibut fishermen to receive income throughout the year due to a phased payment for the halibut harvest that continues through the fall and winter, (2) relatively ample alternate employment opportunities on-island during typical crabbing months, and (3) the less attractive nature of the BSAI crab fishery when compared to the halibut fishery.

More recently, however, CBSFA leadership reported that this situation has changed somewhat. While there is still a strong local focus on the halibut fishery, the vitality of which is largely attributable to the CDQ program, a number of residents have reportedly crewed on CBSFA's 58-foot cod vessels, and at least a few individuals in the most recent seasons have worked on crab vessels in which CBSFA or its subsidiaries have an ownership interest. Local officials are optimistic that continuing opportunities for experience in the other two fisheries will, in turn, provide others the opportunity to take similar routes into the crab fishery, should they choose to do so.

As noted in Table 3-21, EDR data indicated that between one to three Alaska residents with Saint Paul Island addresses on their ADFG commercial fishing crew license crewed on EBS snow crab vessels from 2013 to 2019; none did so in 2012 and 2020.

In 2021, Saint Paul Island total tax revenue was \$4,188,257 of which \$3,347,430 was from the local fish tax on the sale of seafood which is composed of 3.5 percent on rationalized shares and species of crab and 2 percent on non-rationalized shares and species of crab. Saint Paul Island also received \$840,827 in shared state fish tax for that year. In 2022, Saint Paul Island received \$2,049,748 in total tax revenue of which \$503,856 originated from fish tax on the sale of seafood and \$1,545,892 from shared state fish tax. The city is now facing a decline in fisheries related revenue, due to a sharp decline in EBS snow crab TAC in 2021/2022 season and no directed fishing for EBS snow crab in the 2022/2023 season, by far the most economically important fishery overall to the community, combined with the closure of the BBRKC fishery, and an ongoing decline in halibut exploitable biomass available to the directed halibut fishery, the most substantial fishery pursued by the local fleet. Slowdowns in these fisheries are also felt throughout the range of support services provided in the community, such as fuel sales and marine usage, which also underpin the local economy.

3.6.3.4.5 Kodiak

Kodiak is located near the northwestern tip of Kodiak Island in the Gulf of Alaska. Kodiak Island (aka: "the emerald isle") is the largest island in Alaska and is the second largest island in the United States. Kodiak National Wildlife Refuge (KNWR) encompasses nearly 1.9 million acres on Kodiak and Afognak

Islands. It is 252 mi south of Anchorage (a 45-minute flight) and is a 4-hour flight from Seattle. Kodiak was first incorporated in 1940 and is now a Home Rule City and the seat of the Kodiak Island Borough.

In 2010, there were 6,130 residents in Kodiak, ranking it the 16th largest of 352 total Alaskan communities with recorded populations that year. Between 1990 and 2010, the population declined by 3.7%. Between 2000 and 2009, the population increased by 4.6% with an average annual growth rate of 0.51%, which was similar to statewide average of 0.75% and indicative of modest growth. In a survey conducted by the AFSC in 2011, community leaders reported that there were an estimated 6,000 permanent residents, and 600 transient residents living in Kodiak in 2010. According to community leaders, seasonal workers live in Kodiak from July through September, with annual population peaks typically occurring in July and August. Peaks in population are mostly driven by employment in fisheries sectors.

Commercial fishing, seafood processing, and commercial fishing support services are the major industries contributing to the local economy in Kodiak. The U.S. Coast Guard station is also a substantial employer. Other industries include retail services and government. Tourism is growing, and recreational fishing, hiking, and kayaking are increasing in popularity. In a survey conducted by the AFSC in 2011, community leaders reported that Kodiak's economy is reliant on logging, fishing, ecotourism, and sport hunting and fishing.

As noted in the ACEPO document, Kodiak has a diversified fisheries profile, with groundfish making up about 40 percent of total fisheries harvest by weight. Pacific cod has shown a consistent decline in recent years, as has halibut, crab, and to some degree pollock. In 2019, the volume of groundfish harvest was 241.6 million pounds with an associated value of \$44.1 million. Compared to the previous five-year average, both the harvest volume and associated value showed declines of 18 million pounds (down 7 percent) and \$3.8 million (down 8 percent) respectively.

For EBS snow crab fishery, eight Kodiak-owned vessels participated in the fishery from 2011 through 2017 while the remaining years through 2021 had seven vessels participating in the fishery (see Table 3-15). The annual average ex-vessel revenue from EBS snow crab fishery by the Kodiak-owned EBS snow crab vessels was \$12.7 million, which accounts for 10.4 percent of the total of annual average EBS snow crab ex-vessel gross revenue for the fishery (see Table 3-16). Relatively to annual average total ex-vessel gross revenues from all areas, gear, and species for these Kodiak-owned EBS snow crab vessels, the annual average EBS snow crab ex-vessel gross revenue accounted for 50.8 percent (see Table 3-17). From the perspective of the annual average total EBS snow crab ex-vessel gross revenue from 2006 through 2021 for these Kodiak-owned vessels as a percentage of total community ex-vessel gross revenue from all areas, gears, and species from the Kodiak-owned fleet during the same years, Kodiak-owned EBS snow crab vessels accounted for 9.9 percent of that community total (see Table 3-18).

Kodiak is also home to several crab processing plants. As noted in the BSAI Crab Rationalization Ten-Year Program Review, during the second five years of the rationalization program (2010/2011 through 2014/2015) there were between three and five Kodiak plants that processed BBRKC and between one and three Kodiak plants that processed EBS snow crab in any given year. This trend has continued through 2021 for EBS snow crab with only one shorebased processor accepting deliveries of EBS snow crab since 2015 (Table 3-25). The annual average number of plants processing BBRKC was the same during the first five and the second five years of the rationalization program (3.6 plants), while the annual average number of Kodiak plants processing EBS snow crab declined slightly (from 2.0 to 1.8 plants).

As noted in Table 3-21, EDR data, for years 2012 through 2020, indicate that between 24 in 2020 and 83 in 2015 crew provided Kodiak addresses on their ADFG commercial fishing crew licenses crewed on EBS snow crab vessels each year.

3.6.3.4.6 Seattle MSA

As described in the Seattle community profile in the *BSAI Crab Fisheries Final Environmental Impact Statement Social Impact Assessment* (NOAA 2004, Appendix 3), Seattle is the community most engaged in the BSAI crab fisheries, if gauged by the sheer number of locally

owned vessels participating in the fisheries as a whole. Seattle is also the location of regional if not company headquarters for a number of the processing firms engaged in the BSAI crab fisheries. It is also a major support service center for the fleet, both in terms of providing services directly and as the headquarters for a number of firms that provide support services out of Alaskan ports. As described in the in section 3.6.3.3, crab fishery support activity takes a variety of forms and does not appear to be heavily concentrated in any one area of Seattle.

For EBS snow crab fishery, the number of vessels harvesting EBS snow crab from 2006 to 2021 with ownership address of Seattle MSA ranged from a low of 27 in 2021 to a high of 39 in 2006 (see Table 3-15). The annual average ex-vessel revenue from EBS snow crab fishery by the Seattle MSA-owned EBS snow crab vessels was \$63 million, which accounts for 51 percent of the total of annual average EBS snow crab ex-vessel gross revenue for the fishery (see Table 3-16). Relatively to annual average total ex-vessel gross revenues from all areas, gear, and species for these Seattle MSA-owned EBS snow crab vessels, the annual average EBS snow crab ex-vessel gross revenue accounted for 49 percent (see Table 3-17). From the perspective of the annual average total EBS snow crab ex-vessel gross revenue from 2006 through 2021 for these Seattle MSA-owned vessels as a percentage of total community ex-vessel gross revenue from all areas, gears, and species from the Seattle MSA-owned fleet during the same years, Seattle MSA-owned EBS snow crab vessels accounted for 14.1 percent of that community total (see Table 3-18).

EDR data for years 2012 through 2020 indicate that between 43 (2016) and 185 (2015) EBS snow crab crew members provided Seattle MSA addresses on their ADFG commercial fishing crew licenses (Table 3-21). This represents on average 13.35 percent of the EBS snow crab crew and the community with the largest share of crew employment.

3.6.3.5 Community Development Quota

Regulations establishing the CDQ Program were first implemented in 1992. The CDQ Program was incorporated into the MSA in 1996, through the Sustainable Fisheries Act (Pub. L. 104-297). Since the inception of the program, CDQ fisheries management regulations have continued to be developed and amended.

Particularly in fitting with National Standard 8,¹⁹ MSA §305(i)(1) describes the intent of the CDQ Program:

- (i) to provide eligible western Alaska villages with the opportunity to participate and invest in fisheries in the Bering Sea and Aleutian Islands Management Area;
- (ii) to support economic development in western Alaska;
- (iii) to alleviate poverty and provide economic and social benefits for residents of western Alaska;
- (iv) to achieve sustainable and diversified local economies in western Alaska.

The CDQ is an economic development program associated with federally managed fisheries in the BSAI. The purpose of the program is to provide these 65 western Alaska communities the opportunity to

¹⁹ Conservation and management measures shall, consistent with the conservation requirements of this Act (including the prevention of overfishing and rebuilding of overfished stocks), take into account the importance of fishery resources to fishing communities by utilizing economic and social data that meet the requirements of paragraph (2), in order to (A) provide for the sustained participation of such communities, and (B) to the extent practicable, minimize adverse economic impacts on such communities. (MSA §301(a)(8)).

participate and invest in BSAI fisheries, to support economic development in western Alaska, to alleviate poverty, and provide economic and social benefits for residents of western Alaska, and to achieve sustainable and diversified local economies in western Alaska.

The MSA allocates a portion of the annual catch limit for each directed fishery of the BS and Aleutian Islands management area among six entities representing 65 western Alaska villages. The six entities (CDQ groups) and the villages associated with each of those entities are specifically named in the MSA. The CDQ groups include the Aleutian Pribilof Island Community Development Association (APICDA), the Bristol Bay Economic Development Corporation (BBEDC), the Central BS Fishermen's Association (CBSFA), the Coastal Villages Region Fund (CVRF), the Norton Sound Economic Development Corporation (NSEDC), and the Yukon Delta Fisheries Development Association (YDFDA). The CDQ groups are nonprofit corporations whose board of directors and staff manage and administer CDQ allocations, investments, and economic development projects. CDQ groups use the revenue derived from the harvest of their fisheries allocations to fund economic development activities and provide employment opportunities.

Prior to the implementation of the CR Program, the CDQ groups received an allocation of 7.5 percent of the GHL in the BPRKC, Pribilof red and blue king crab, Norton Sound red king crab, EBS snow crab, and BS Tanner crab fisheries. The CR Program increased the program allocation up to 10 percent and was expanded to all crab fisheries included under the CR Program. In a similar design to the CDQ Program, all allocation of 10 percent of the Aleutian Islands golden king crab TAC was granted to the community of Adak at the onset of the CR Program. The CDQ and Adak community allocation, are exempt from the CR Program management, but are subject to separate CDQ/Adak community allocation regulations.

Table 3-34 illustrates the breakout of the 10 percent CR Program allocation by CDQ group as a percentage of program allocation. The CDQ group makes internal management decisions about how to harvest their program allocations. Some of the allocations under the CDQ groups are focused towards providing directed harvest opportunities for members of the CDQ communities. For example, some fisheries are relatively easier and safer to access with smaller vessels and have represented historical sources of employment and income for residents. Other allocations under the CDQ program, particularly in those fisheries that are more industrialized like EBS snow crab require greater access to capital and specialized gear, and/or prosecuted in remote areas of the BSAI, are often harvested on larger, more efficient vessels. The revenues from these types of operations can aid in funding other types of economic development opportunities.

Table 3-34 CDQ and Adak community allocations by fishery and group

Fishery	Group Allocation (as a % of program allocation)						Adak allocation	Program allocation (% of TAC)
	APICDA	BBEDC	CBSFA	CVRF	NSEDC	YDFA		
BBR	17%	19%	10%	18%	18%	18%	-	10%
BSS	8%	20%	20%	17%	18%	17%	-	10%
EBT	10%	19%	19%	17%	18%	17%	-	10%
WBT	10%	19%	19%	17%	18%	17%	-	10%
WAG	-	-	-	-	-	-	100%	10%
EAG	8%	18%	21%	18%	21%	14%	-	10%
WAI	8%	18%	21%	18%	21%	14%	-	10%
STB	50%	12%		12%	14%	12%	-	10%
PIK	-	-	100%	-	-	-	-	10%

Both before and after implementation of the CR Program, CDQ groups made substantial investments in the BSAI crab fisheries. While these entities do not meet the requirements to hold C shares, community

groups may, and have, invested in both catcher vessel owner (CVO) and catcher processor owner (CPO) QS. Some CDQ groups and wholly owned subsidiaries were granted CR Program QS at initial allocation; however, much of holdings have been acquired since program implementation. In some cases, QS is purchased outright, and in some cases groups may acquire a company or equity in a company that might include QS, vessels, or other assets. Allowing for QS and PQS acquisition by CDQ groups was in line with the Program's intention to:

(4) *[Promote] economic stability for harvesters, processors, and coastal communities;*

(6) *Address the social and economic concerns of communities;*

Table 3-35 provides the annual percent of EBS snow crab catcher processor owner shares, catcher vessel owner shares, and processors shares owned by all CDQ groups combined. In 2021, the CDQ groups owned 54.1 percent of the catcher processor owner shares, 20 percent of the catcher vessel owner shares, and 23 percent of the processor shares. Table 3-36 provides ownership percentage for catch processor owner shares, catcher vessel owner shares, and processor shares by CDQ group for 2021. CDQ group name has been masked in Table 3-36 due to the potential of confidential information associated with subsidiary ownership percentages. As noted in the table, four CDQ groups own EBS snow crab catcher processor owner shares with one CDQ group owning 26.2 percent in 2021. All six CDQ groups own a share of the EBS snow crab catcher vessel owner shares, while only two CDQ groups own EBS snow crab processor shares with one CDQ group owning 17.3 percent of the processor shares in 2021.

Table 3-35 Annual percent of EBS snow crab catch processor owner shares, catcher vessel owner shares, and processor shares owned by all CDQ groups in aggregate

Year	Percent of EBS snow catcher processor owner shares owned by all CDQ groups combined	Percent of EBS snow crab catcher vessel owner shares owned by all CDQ groups combined	Percent of EBS snow crab processor shares owned by all CDQ groups combined
2005	19.8%	6.9%	0.1%
2006	19.8%	7.4%	0.1%
2007	19.8%	10.7%	0.0%
2008	23.9%	12.9%	0.0%
2009	43.8%	14.1%	11.5%
2010	43.8%	14.5%	11.5%
2011	43.8%	16.6%	11.5%
2012	43.8%	16.6%	11.5%
2013	43.8%	16.6%	11.1%
2014	54.1%	18.4%	11.1%
2015	54.1%	20.0%	22.9%
2016	54.1%	20.0%	22.9%
2017	54.1%	20.0%	23.0%
2018	54.1%	20.0%	22.9%
2019	54.1%	20.0%	22.9%
2020	54.1%	20.0%	22.9%
2021	54.1%	20.0%	23.0%

Source: AKFIN; source file Crab_QS_by_CDQ_BSS(12-29--22)

Table 3-36 Percent of EBS snow crab catch processor catch processor owner shares, catcher vessel owner shares, and processor shares owned by all CDQ groups for 2021

Groupings	Percent of EBS snow crab catcher processor owner shares owned by individual CDQ groups	Percent of EBS snow crab catcher vessel owner shares owned by individual CDQ groups	Percent of EBS snow crab processor shares owned by individual CDQ groups
CDQ group 1	0.0%	0.8%	5.7%
CDQ group 2	0.0%	6.2%	0.0%
CDQ group 3	9.7%	2.9%	17.3%
CDQ group 4	10.6%	4.4%	0.0%
CDQ group 5	7.7%	2.4%	0.0%
CDQ group 6	26.2%	3.2%	0.0%
Non-CDQ Group	45.9%	80.0%	77.0%

Source: AKFIN; source file Crab_QS_by_CDQ_BSS(12-29--22)(1)

Provided below is a summary of each CDQ group. The information provided in this section originates from the most current annual report available for each group and where available additional information from the CDQ group.

APICDA (2020 Annual Report)

APICDA represents 6 communities: Akutan, Atka, False Pass, Nelson Lagoon, Nikolski, Saint George, and Unalaska. The region APICDA encompasses is along the remote Aleutian Chain and Saint George Island of the Pribilof Islands. The total population represented by APICDA is 1,257, which represents 15% of the region's population. APICDA's for-profit subsidiary is APICDA Joint Ventures, which manages the majority of their partnerships and assets. APICDA owns three processing companies, Atka Pride Seafoods, Bering Pacific Seafoods, and Cannon Fish Company. The pollock trawl catcher/processor Starbound was lengthened in 2016 to install a fish meal plant in its factory. The Starbound harvests and processes 80% of APICDA's CDQ quota. As noted in their 2021 Annual report, in late 2021, APICDA sold its interest in Starbound, which at the time was a 20 percent interest. APICDA still has a strong and successful relationship with Aleutian Spray Fisheries. The Aleutian Longling Company, a fleet of cod freezer longliners that includes the F/V Arctic Prowler, F/V Ocean Prowler, F/V Siberian Sea, and the F/V U.S. Liberator, is jointly owned by Prowler Fisheries (a subsidiary of APICDA Joint Ventures) and Aleutian Spray Fisheries.

BBEDC (2020 Annual Report)

BBEDC represents 17 communities in the Bristol Bay watershed: Aleknagik, Clark's Point, Dillingham, Egegik, Ekuk, Ekwok, King Salmon, Levelock, Manokotak, Naknek, Pilot Point, Port Heiden, Portage Creek, South Naknek, Togiak, Twin Hills, and Ugashik. The major hub is Dillingham. The total population represented by BBEDC is 5,380, which represents 72% of the region's population.

The region is known for its sockeye salmon runs in its expansive watershed and BBEDC uses its royalties to provide several finance programs to their fishing fleet, including a vessel upgrade grant and refrigerated sea water support. BBEDC had 20 fishermen participate in its small boat halibut fishery in 2016, harvesting 77,306 lbs. Bristol Bay Science and Research Institute is a non-profit research entity and subsidiary of BBEDC that conducts fisheries research and monitoring for the region. BBEDC engages in several partnerships to harvest and process its CDQ quota. The Defender, a 195' American Fisheries Act (AFA) vessel, was converted and upgraded in 2016 and began fishing in 2016. BBEDC owns 50% of Ocean Beauty Seafoods, one of the largest seafood buyers in Alaska. In 2016, BBEDC-contracted vessels harvested 87 million lbs. of CDQ allocations. They received \$15.0 million in CDQ allocation royalties in 2016 and an additional \$3.1 million in IFQ program royalties.

As noted in the BBEDC 2020 Annual Report, in 2020, 15 communities from BBEDC and 15 communities from CVRF, along with their CDQ groups acquired all the sellers' partnership interests in the Mariner Companies. The buy-out provides participating communities with snow and red king crab quota, equaling 3 percent of the total crab fishery. The BBEDC acquired full ownership of four crabbing vessels which it will operate under a new company called Bristol Group. The boats acquired are the Aleutian Mariner, Bristol Mariner, Nordic Mariner, and Pacific Mariner.

CBSFA (2021 Annual Report combined with updated information from the group)

CBSFA's fleet of BS catcher vessels includes the 165' F/V Fierce Allegiance (crab/pollock), 124' F/V Starward and F/V Starlite (pollock/cod), 108' F/V Early Dawn (crab), 100' F/V Adventure (crab/tendering, through 2021), 58' F/V Saint Paul and F/V Saint Peter (halibut, cod, sablefish, tendering). Saint Paul Fishing Company (SPFC), a wholly owned subsidiary of CBSFA, owns 9.9% of Bering Sea Partners, LLC, in partnership with Unisea Seafoods. SPFC also holds crab, cod, and pollock permits, as well as sablefish and crab Individual Fishing Quota for CBSFA. CBSFA has two longtime business partners, Unisea and Rick and Mary Mezich.

CBSFA is heavily invested in all BS crab fisheries and has used its revenue from crab investments as well as CDQ crab to help benefit the community of Saint Paul. In 2021, two of CBSFA's crab vessels, the Fierce Allegiance and Adventure landed 2.8 million pounds and received \$5.00 per pound on average. Unexpectedly, in late 2021, the annual crab stock assessment indicated a near-total collapse of snow crab stocks. As a result, the F/V Fierce Allegiance and F/V Early Dawn had only 492,000 pounds of EBS snow crab to harvest in 2022 – an 82% reduction. Closure of the BBRKC fishery and severely reduced snow crab quotas in 2021 necessitated consolidation within CBSFA's crab fleet. CBSFA sold the F/V Adventure in early 2022. Only CBSFA's smallest crab vessel, the F/V Early Dawn, will remain in operation, harvesting Eastern Aleutian Islands Golden king crab and a small amount of Tanner crab.

CVRF (2021 Annual Report)

CVRF represents 20 communities: Chefornak, Chevak, Eek, Goodnews Bay, Hooper Bay, Kipnuk, Kongiganak, Kwigillingok, Mekoryuk, Napakiak, Napaskiak, Newtok, Nightmute, Oscarville, Platinum, Quinhagak, Scammon Bay, Toksook Bay, Tuntutuliak, and Tununak. CVRF communities are in the Kuskokwim River Delta and on Nelson Island. The total population represented by CVRF is 9,429 in 2016, which represents 51% of the region's population. The for-profit subsidiary, Coastal Alaska Premier Seafoods, owns 6 vessels and harvested and processed 126 million pounds of seafood in 2016. In a combined single platform since 2014, the fleet harvests pollock, crab, and cod with vessels that are wholly-owned by CVRF. CVRF was the first CDQ group to catch its pollock, crab, and cod CDQ allocation aboard vessels that they own and operate. In 2001, CVRF entered into a partnership with NSEDC and Maruha Nichiro Corporation to form BSAI Partners to manage five in-shore pollock catcher vessels and their quota. As noted in the Coastal Villages Region Fund 2021 Annual Report, in 2021, the CVRF Bering Sea fleet caught more than 126 million pounds of pollock, Pacific cod, and crab and generated more than \$100 million in revenue from sales of these fish products of which 60.09 percent was from pollock, 14.41 percent was from Pacific cod, and 20.50 percent was from crab.

NSEDC (2021 Annual Report)

The most northern CDQ group, NSEDC, represents 15 communities: Brevig Mission, Diomede, Elim, Gambell, Golovin, Koyuk, Nome, Savoonga, Shaktoolik, Saint Michael, Stebbins, Teller, Unalakleet, Wales, and White Mountain. Communities in NSEDC stretch as far north as Diomede Island and as far south as Stebbins in the Norton Sound region. The major hub is Nome. The 2016 total population represented by NSEDC is 9,310, which represents 98% of the region's population. NSEDC receives 50% of the Norton Sound Red King Crab CDQ allocation and opens the fishery to its residents. NSEDC provides a loan program to residents of Norton Sound. Norton Sound Seafood Products, a processing

subsidiary with plants in Nome, Unalakleet, and Savoonga, purchases salmon, crab, halibut, and herring from the resident fleet. Crab and salmon were worth \$5.7 million in 2016 for local residents. NSEDC created Siu Alaska Corporation in 2009 to manage its fishing-related ventures and fishing partnerships. NSEDC owns a 37.5% stake in BSAI Partners, a fishing company venture jointly owned with CVRF and Maruha. NSEDC owns outright or has a stake in 14 vessels. With the revenues received from their CDQ and IFQ royalties, NSEDC received \$17.7 million in CDQ revenues in 2016 and an additional \$6.9 million in IFQ fishing revenues.

YDFDA (2020 Annual Report)

YDFDA represents six communities located along the Yukon River delta: Alakanuk, Emmonak, Grayling, Kotlik, Mountain Village, and Nunam Iqua. The total population represented by YDFDA is 3,434, which represents 26% of the region's population. As noted in their 2020 Annual Report, YDFDA's crab portfolio consists of all the major species, including BBRKC, snow and Tanner crab, and Golden King crab. YDFDA's crab investments are substantial as it generates nearly 1/3 of the YDFDA's total royalty revenue. YDFDA receives annual allocation of crab CDQ and has purchases substantial amounts of crab IFQs that have brought YDFDA closer to meeting the maximum allocation allocations cap under the Crab Rationalization Program. To date, YDFDA owns up to the cap in both the eastern and western Tanner crab fisheries. YDFDA's other crab caps are BBRKC at 80.40 percent, snow crab at 97.45 percent, and Western golden crab at 59.94 percent. To harvest its crab quota, YDFDA engages with vessels, the C/P Baranof, and the F/V Kiska Sea, as well as 3rd party harvesters. Yukon Delta Fisheries, Inc. (YDFI), YDFDA's for-profit subsidiary, owns 45% of the F/V Kiska Sea, a crab catcher vessel. YDFDA, in coordination with the harvesting vessels, enters into profit/margin sharing agreements with various shoreside processors to boost royalties off of the unrestricted crab shares to maximize crab revenues

3.6.4 Expected Effects of the Alternatives

This section provides an overview of the potential socioeconomic effects of EBS snow crab rebuilding alternatives. This section will provide qualitative discussion of the likely impacts of the two rebuilding alternatives on vessel owners, crew, quota shareholders and the communities associated with these EBS snow crab participants separate from the impacts of the alternatives on processors and the communities where the processing activity is located. Additionally, since both EBS snow crab rebuilding Alternative 2/option 1 and Alternative 2/option 2 (the PPA) allow for bycatch of EBS snow crab while targeting other crab fisheries or groundfish fisheries and does not adjust PSC limits or change the management of the COLBZ, this socioeconomic effect section focuses solely on impacts to the directed EBS snow crab fishery participants and communities from a directed fishery closure perspective. For impacts on bycatch of EBS snow crab, see section 3.3.1.

The rebuilding plan alternatives affect the TAC to be set and thereby the opportunity for any directed catch of EBS snow crab. Since allocation of directed catch is determined by quota share holdings, the action alternatives would directly affect holders of all classes of quota share (A, B, and C), processor shares, as well as CDQ organizations and those whose fish the CDQ allocations. There are also identifiable indirect effects. For example, changes in payments to labor or diminished revenue for vessel owners or quota shareholders would then affect expenditures by labor, crew, vessel owners, or quota shareholders in landing ports and in their place of residence. Finally, there are also induced impacts from direct and indirect expenditures in a community. Induced impacts are money recirculated through the household spending patterns causing further local economic activity. Diminished direct and indirect expenditures in community would negatively impact the induced effects in the community. These impacts will be highlighted throughout this section where appropriate.

3.6.4.1 Impacts to vessel owners, crew, quota shareholders and associated communities

Looking first at EBS snow crab vessel owners and the communities of historical ownership address, Table 3-15 shows the count of EBS snow crab vessels by community from 2006-2021, vessel ownership among states is concentrated in Washington at 56 percent and Alaska at 32 percent. The largest share of Washington ownership address is in the Seattle MSA at 51 percent, while the largest share of Alaska vessel ownership is concentrated in the communities of Kodiak (12 percent), Anchorage (12 percent), Homer (5 percent) and Seldovia (1 percent). From the perspective vessel owner information, these communities represent those most likely directly impacted by the action alternatives due to diminished or loss of EBS snow crab ex-vessel gross revenue. For example, Table 3-16 shows the overall pattern of distribution of EBS snow crab average annual ex-vessel gross revenue from 2006-2021 of \$122 million by vessel ownership address is Washington at 57 percent, Alaska at 30 percent, and Oregon/other states at 12 percent. Of the Washington communities, Seattle MSA has the highest portion of the average annual ex-vessel revenue from the EBS snow crab at 51 percent, while for Alaska communities, the annual average ex-vessel gross revenue from the EBS snow crab fishery is concentrated in Kodiak at 12 percent followed by Homer/Seldovia at 4 percent. All other Alaska communities combined captured the remaining 15 percent of annual average ex-vessel gross revenue.

Table 3-17 shows the dependency on the EBS snow crab fishery for vessel owners by community. For Alaska, the annual 2006-2021 average ex-vessel gross revenues from EBS snow crab fishery was \$33.4 million, while the annual average total ex-vessel gross revenue from all areas, gears, and fisheries for these same vessels and same period was \$73.5 million. As a percent of the total ex-vessel gross revenue, the EBS snow crab fishery on average contributed 45.4 percent of the ex-vessel gross revenue for vessel owners with an Alaska address. Of the Alaska communities with EBS snow crab ownership address, the eight vessels with a Kodiak address appear to be highly dependent on the EBS snow crab fishery with 50.6 percent of the total ex-vessel revenue coming from the EBS snow crab fishery. Vessel owners with Homer/Seldovia and other Alaska addresses also appear to be highly dependent on the EBS snow crab fishery with 50.4 percent and 41.2 percent of the total ex-vessel total ex-vessel gross revenue coming from the EBS snow crab fishery, respectively. For EBS snow crab vessel owners with Washington and Oregon/other states address, dependency on the EBS snow crab fishery is also substantial. For vessel owners with Washington address, 45.3 percent of their annual average total ex-vessel gross revenues from all areas, gears, and fisheries was from the EBS snow crab fishery, while vessel owners with Oregon/other states address the EBS snow crab contributed 45.3 percent of the annual average total ex-vessel gross revenue from all other fisheries (Table 3-17).

Looking at crew license address (Table 3-21) for crew of EBS snow crab vessels, the largest number of crew licenses for these EBS snow crab vessels were from Washington with an annual average of 242 licenses, while Alaska reported 211 annual average licenses. Of the Alaska communities, Kodiak had the largest number of crew licenses at an annual average of 211 followed by Anchorage/Palmer/Wasilla at 43 licenses, Homer/Seldovia at 28 licenses, and Dutch Harbor/Unalaska at 19 licenses.

From the perspective of EBS snow crab quota shareholders by address (Table 3-22) and their associated quota share revenues (Table 3-23) in communities of address, Washington followed by Alaska reported the highest annual average shares at 569 million shares and 321 million shares, respectively. As a community, Seattle MSA had the highest annual average shares at 427 million, while of the Alaska communities, Anchorage/Palmer/Wasilla had an annual average of 145 million shares, Kodiak as an annual average of 81 million shares, and Dillingham had 51 million shares.

Shifting to the impacts of the action alternatives from the perspective of vessel owners, crew, and quota shareholders, the loss of the EBS snow crab fishery under Alternative 2/option 1 relative to the Alternative 2/option 2 (the PPA) will likely be substantial. Clearly, the closure of the EBS snow crab fishery for 6 years under Alternative 2, option 1 would mean that the exvessel revenue for vessel owners

and quota shareholder, crew positions and their associated payments to crew would be foregone during the six-year rebuilding period. When compared to the average annual exvessel revenue from the EBS snow crab reported in the above tables, this loss represents a substantial economic impact for vessel owners, crew, quota shareholders, and the communities where these participants reside. As noted in Table 3-20, in 2021, 32 of the 60 vessels that participated in the EBS snow crab received 90 to 100 percent of their ex-vessel gross revenue from this fishery. The high recent high dependence on the EBS snow crab fishery for vessels is likely due to the closure of the directed BBRKC fishery last year. With the closure of EBS snow crab under Alternative 2/option 1 and roughly 50 percent of exvessel revenue originating from the EBS snow crab fishery (Table 3-17) and with little fishing opportunities from other crab fisheries with the closure of the BBRKC and groundfish fisheries (Table 3-19), many of the EBS snow crab vessel owners, crew, and quota shareholders will likely see substantial declines in payments to vessel owners, crew, and quota shareholders which could lead to consolidation of the snow crab fleet and result in substantial losses in crew opportunities in a future directed EBS snow crab fishery.

Most businesses that must invest in highly valued capital operate under credit instruments that provided needed cash flow and may have substantial loan financed debt. A complete closure of the EBS snow crab fishery for six-years would eliminate operating revenue from snow crab. As noted in Table 3-20, many harvesting operations are highly dependent on snow crab revenue and thus these operations may have difficulty maintaining their credit and debt instruments and could be forced to refinance, which may not be possible for some entities and such a situation could lead to business sale and/or bankruptcy. The extent to which such impacts on business operations may be realized is impossible to evaluate given the proprietary nature of business finance information.

Communities with historical EBS snow crab vessel ownership address, crew license address, and quota shareholder address, will likely experience negative impacts from Alternative 2/option 1 relative to Alternative 1 and Alternative 2/option 2 (the PPA). In general, the impacts to the communities under Alternative 2/option 1 would likely reduce indirect and induced expenditures by vessel owners, crew, and quota shareholders in the communities of residence. The severity of these impacts will likely depend on the economic diversity of the community. Communities that are highly dependent on the expenditures of EBS snow crab vessel owners, crew, and quota shareholders will be substantially impacted under Alternative 2/option 1. As indicated in Table 3-18, Table 3-19, Table 3-21, and Table 3-22, the Washington and Alaska communities most affected by the loss of expenditures in communities from payments to crew, and expenditures of snow crab fishing revenue from vessel owners and quota shareholders during the rebuilding period are likely Seattle MSA, Kodiak, Homer/Seldovia, and Anchorage/Palmer/Wasilla. For Seattle MSA and Anchorage/Palmer/Wasilla communities, these large economies with substantial diversity in a wide array of economic activities will help absorb the loss of EBS snow crab revenue expenditures due to the closure of the EBS snow crab fishery during rebuilding. For Kodiak, only 10 percent of the total annual average ex-vessel gross revenue was from EBS snow crab, while four percent of the total annual average ex-vessel gross revenue for Homer/Seldovia was from EBS snow crab fishery (Table 3-18). These communities, which have some crew, vessel owner, and quota shareholder dependency on the EBS snow crab fishery, do have economic diversity in other fisheries or other industries that may enable them to weather the loss of EBS snow crab expenditures under Alternative 2/option 1.

In comparing the impacts of Alternative 1 and Alternative 2/option 2 (the PPA), which both alternatives would allow for a directed fishery based on the state harvest strategy, the socioeconomic impacts could be improved for vessel owners, crew, and quota shareholders since a directed EBS snow crab fishery would be allowed during the rebuilding period. Under Alternative 1, the Council would not develop a rebuilding plan for EBS snow crab in accordance with MSA 304(e)(4)(A) guidelines and therefore selecting Alternative 1 would be in violation of these MSA guidelines. Relative to Alternative 2/option 2 (the PPA), there is the potential that the early years of the rebuilding plan could be closed to directed fishing.

As noted in Table 3-6, projected removals of EBS snow crab in the directed fishery plus bycatch simulation show very low removals of EBS snow crab during the first four years of the proposed rebuilding plan. These projected low removals could result the fishery being closed to directed fishing. There is substantial uncertainty surrounding the projected removal values and whether that results in a directed fishery since the TAC setting process is determined by the state of Alaska and varies annually based on numerous factors which will influence the TAC for EBS snow crab. The values presented in Table 3-6 are in no way a definitive prediction of a directed fishery throughout the rebuilding period but do provide better evidence of potential fishing activity throughout the rebuilding period under Alternative 2/option 2 (the PPA). If the stock is insufficient to support a directed fishery based on the State's harvest control rule during one of the rebuilding years, the fishery would be closed to directed fishing for that year. If under Alternative 2/option 2 (the PPA), the stock cannot support a directed fishery based on State of Alaska harvest strategy for several years as noted in the projected removals in Table 3-6, then the socioeconomics impacts under Alternative 2/option 2 (the PPA) would be similar to the impacts under Alternative 2/option 1.

Nevertheless, assuming there would be a directed fishery every year during the rebuilding program, the addition of a directed fishery under Alternative 2/option 2 (the PPA) for vessel owners, crew, and quota shareholders would provide critical ex-vessel payments from the harvesting of EBS snow crab that would keep the vessel and its crew active in the EBS snow crab fishery which would also provide benefits downstream for vessel owners, crew, quota shareholders, and the communities these participants reside.

Using the projected removals from Table 3-6 during the first four years of the rebuilding period to provide an estimate of the potential socioeconomic activity associated these projected removals and their associated TACs, these projected removals best match the retained catch associated with the 2021-2022 season. As noted in Table 3-13, during the 2021/2022 season, there were 42 active vessels in the EBS snow crab fishery associated with 5.5 million lbs. of retained catch. This retained catch generated an estimated \$34 million in gross ex-vessel revenue and an estimated ex-vessel price of \$6.09. Utilizing this ex-vessel price in conjunction with projected removals from Table 3-6, after accounting for average groundfish bycatch removals of .1317 percent (calculated using Table 3-8), the estimated ex-vessel gross revenue generated during the first four years of the rebuilding plan would range from \$23 million to \$36 million. However, since the ex-vessel price in 2021/2022 season was historically high driven by consumer demand during the covid-19 pandemic (see section 3.6.2.1), the 2005/2006-2021/2022 average ex-vessel price of \$3.03 was also utilized to estimate economic activity. Using the average ex-vessel price of \$3.03 combined with the forecasted removals of EBS snow crab (Table 3-13), after accounting for average groundfish bycatch removals of .1317 percent (calculated using Table 3-8), the ex-vessel gross revenue would range from \$11 million and \$18 million annually during the first four years of the rebuilding plan. This estimated gross ex-vessel revenue from the projected EBS snow crab removals will likely not be uniform in its distribution across all EBS snow crab participants. Harvesting operations that are highly dependent on the EBS snow crab fishery and are not diversified in other crab or groundfish fisheries may have greater difficulty maintaining their credit and debt instruments and could be forced to refinance or in some cases lead to business sale and/or bankruptcy during the years of projected low EBS snow crab removals. Harvest operators that exist the fishery would likely result in lost crew positions and would likely negatively impact communities. In contrast, harvest operators that are more diversified and less dependent on the EBS snow crab fishery would likely be better able to maintain their credit and debt instruments due to multiple sources of ex-vessel revenue during this period of projected low EBS snow crab TACs. These harvest operators would likely continue to provide crew positions and expenditures in the communities.

As for projected removals during the remaining two years of the rebuilding plan (see Table 3-6), these projected removals best matched the retained catch associated with the 2017/2018-2018/2019 seasons. As noted in Table 3-13, the retained catch of EBS snow crab was 19 million pounds in 2017/2018 and 28

million pounds in 2018/2019 and the ex-vessel price for these two seasons were \$4.02 and \$3.98 which generated \$76 million and \$109 million in ex-vessel gross revenue. during the last two years of the rebuilding plan, the economic activity associated with these harvest levels would likely provide greater economic activity in the last two years of the rebuilding plan. Using the average ex-vessel price of \$3.98 to \$4.02 combined with the forecasted removals of EBS snow crab (Table 3-13), after accounting for an estimated groundfish bycatch of .1317 percent, the estimated ex-vessel gross revenue would be approximately \$48 million for year five of the rebuilding plan and \$110 million for year six of the rebuilding period. Relevant to the first four years of the rebuilding plan, this estimated ex-vessel gross revenue will likely have less of a negative impact on the participates of the snow crab fishery. As noted above, the resulting distribution of ex-vessel gross revenue will likely not be uniform across all vessel owners, crew, quota shareholders, and communities.

Additionally, the importance of a fishery to the operations of vessels is not just a function of percentage contribution to overall gross revenues as, for example, a fishery may contribute revenue during what would otherwise be a slow time of year, which could be important for covering fixed cost, maintaining existing snow crab markets, helping to make or keep the vessel ready for future fishery operations, assist in employment/retention of crew, and/or maintaining favorable business relationships with processors, among other factors.

3.6.4.2 Impacts to processors and the communities where they are located

Overall, since 2006 the number of processors that have received EBS snow crab deliveries has ranged from a high of 15 in 2007 to a low of 10 in 2006 and 2018 (Table 3-25). In 2021, there were 11 processors that received EBS snow crab deliveries. The total first wholesale gross revenue for these 11 processors from processing EBS snow crab was \$274 million in 2021 (Table 3-26). Of that \$274 million first wholesale gross revenue from EBS snow crab, processors located in Akutan/King Cove/Kodiak/Saint Paul captured \$134 million or 46 percent of the total, processors located in Dutch Harbor garnered \$66 million or 29 percent of the total, and the remaining \$44 million or 26 percent was received by other processors.

In general, most of the processors that received EBS snow crab deliveries also process other crab, groundfish, as well as IFQ halibut and sablefish. As noted in Table 3-27, for processors located in Unalaska/Dutch Harbor, EBS snow crab contributed 12 percent of the total annual average first wholesale gross revenue, while for the combined Akutan/King Cove/Kodiak/Saint Paul processors the EBS snow crab fishery contributed 11 percent of the total annual average first wholesale gross revenue. Other revenue sources for Unalaska/Dutch Harbor processors and Akutan/King Cove/Kodiak/Saint Paul processors included other crab fisheries and groundfish fisheries. As noted in Table 3-29, annual average first wholesale gross revenue sources included BBRKC at \$34 million and groundfish at \$332 million for Akutan/King Cove/Kodiak/Saint Paul processors, while for Unalaska/Dutch Harbor processors, the BBRKC fishery contributed \$38 million and the groundfish fishery contributed \$258 million. Note that the BBRKC fishery was closed during the 2021/2022 season and will be closed during the 2022/2023 season, so first wholesale gross revenue from this fishery will be zero.

The importance of the EBS snow crab fishery for processors relative to other fisheries revenue sources is reflected in Table 3-30, which shows the annual average EBS snow crab first wholesale gross revenue as a percentage of annual average total first wholesale gross revenue from 2006 through 2021. Nearly half of the processors received less than 20 percent of their annual average total first wholesale gross revenue from the EBS snow crab fishery which could represent a substantial impact for these processors, but the diversity of other processing activity by these processors helps reduces the impact of a closed EBS snow crab fishery for six-years under Alternative 2/option 1. On the other end of the processor dependency spectrum, in 2020 and 2021, two processors were 90 percent to 100 percent dependent on the EBS snow

crab fishery. Of these two processors, one is a shorebased processor that may be severely impacted due to the loss of EBS snow crab processing.

Changes in the quantity of EBS snow crab delivered for processing can have both direct and indirect effect. The numbers of processor workers hired during the first several months of the year with associated implications for processor worker wages, local expenditures by processor workers in the community where the plant is located, as well as remittances of wages to primary residences supported by processing workers who are working away from home are examples of direct and indirect effects. Impacts to processors and the communities where the processors are based likely depend on the revenue diversity of the processor and/or the community. As noted above, for many of the processors, the EBS snow crab fishery is one of several fisheries the processor receives deliveries.

From the perspective of direct effects, Alternative 2/option 1, relative to the Alternative 1 and Alternative 2/option 2 (the PPA), would result in no EBS snow crab deliveries during the six-year rebuilding period which could range from minor impacts on total operating revenue to substantial impacts on operating revenue. The impacts to a processor from the closure of the EBS snow crab fishery during the six-year rebuilding period depends on how dependent the processor is on EBS snow crab deliveries, as well as its ability to adapt or diversify. As noted in Table 3-30, dependency is varied from little reliance on the EBS snow crab fishery to at least two processors in recent years that are highly dependent on EBS snow crab deliveries. Given the varied dependency on the EBS snow crab, impacts to the processors and the communities where the processors reside will also likely vary. For processors with little reliance on the EBS snow crab fishery, it would be expected that processors in this category would see reduced operating revenue from the loss of EBS snow crab deliveries which could result in reduced processor workers and/or reduced worker wages and the subsequent expenditures in the local communities where the processors reside and reduced remittance to worker's resident communities. Loss of processor operating revenue due to the closure of the EBS snow crab fishery could also reduce processor expenditures of goods and services in the community where the processor resides. In addition, like harvesters, processors likely operate under credit instruments that provided required cash flow and may also have substantial loan financed debt.

For at least one shorebased processor, Trident Seafoods, and the community the plant resides in, Saint Paul, the magnitude of impacts under Alternative 2/option 1 are likely severe since the processor and the community are highly dependent on EBS snow crab deliveries. As noted on Trident's website, the plant is the largest crab processing plant in the world. The plant can process and freeze more than 500,000 pounds of snow crab per day. The plant has processed snow crab, king crab, and Tanner crab in the past. During the peak of the snow crab season in February, the plant employs as many as 400 workers. Given the processor's focus on crab processing and the loss of EBS snow crab operating revenue and the potential continued loss of BBRKC operating revenue due to the continued closure of the fishery, it is likely the processor will be severely impacted by this loss of operating revenue under Alternative 2/option 1, which could have severe impacts on the community of Saint Paul as well. A substantial reduction or temporary loss of processing operations during part or all of the six-year rebuilding period could indirectly reduce or eliminate processor wages earned in the community, reduce or eliminate processor operating purchases of local goods and services, and reduce or eliminate processor worker expenditures in the community.

The loss of the processor would also likely impact processing of local halibut IFQ landings by the local community residents since they would no longer have a local processor for their IFQ halibut landings. Additionally, the community's CDQ fishing group (CBSFA) would also be impacted by the closure of the EBS snow crab fishery due to the loss of EBS snow crab revenue through CDQ snow crab quota and investments in CR Program harvesting and processing quota.

Saint Paul also receives tax revenue from a local sales tax on the sale of seafood that would be at risk if the processor reduced processing activity or shut down during the rebuilding period. As an example of the

risk to the community of Saint Paul, the City of Saint Paul's October 2022 Disaster Request indicates that the community is expecting a 100 percent drop in local sales tax on the sale of seafood and a 90 percent drop in state shared fish tax revenue due to the closure of the EBS snow crab fishery and the BBRKC fishery for the 2022/2023 season. The loss of revenue from seafood sales tax due to fishery related purchases and the loss of moorage, wharfage, and tariff collections for the 2022/2023 season would also impact provision of port services. Thus, by extension, an EBS snow crab fishery closure under Alternative 2/option 1 for six years, would place a considerable burden on the community Saint Paul and could likely result in severe reductions of services in the community. Based on information in the City of Saint Paul's October 2022 Disaster Request, the City of Saint Paul would likely be evaluating plans to reduce expenditures, reduce services provided by the community, seek new sources of revenue, lift current tax exemptions, and increase utility rates, adjust dockage, and wharfage fees due to a closure of the EBS snow crab fishery for six years under Alternative 2/option 1. In addition, Tanadgusix Corporation, the Alaska Native Village Corporation for Saint Paul, which also supports the crab fishery on Saint Paul Island would also be impacted by the loss of the directed fishing opportunity for EBS snow crab under Alternative 2/option 1. Examples of services and sales impacted would be moving and storage of crab pots, fuel sales to Trident Seafoods, fishing boats, and electricity generation specific to fisheries, and lodging of itinerant workers. As noted in the City of Saint Paul's October 2022 Disaster Request, Tanadgusix Corporation estimates for the 2022/2023 season as a complete loss from its 2020 to 2022 average revenue of \$2,369,792. This loss of revenue associated with the current economic conditions is likely an example of the economic conditions during the six-year rebuilding period under Alternative 2/option 1, which would likely impact city sales taxes, revenue for the corporation, and likely result in staffing changes that would result in diminished economic activity.

Alternative 1 and Alternative 2/option 2 (the PPA) would provide an opportunity for a directed EBS snow crab fishery under the State harvest strategy. Under Alternative 1, the Council would not develop a rebuilding plan for EBS snow crab in accordance with MSA 304(e)(4)(A) guidelines and therefore selecting Alternative 1 would be in violation of these MSA guidelines. Under Alternative 2/option 2 (the PPA), a rebuilding plan would be developed which would allow a directed snow crab fishery under the current State of Alaska harvest strategy. Nevertheless, there is the potential that the directed fishing could be closed even under Alternative 2/option 2 (the PPA). As noted in Table 3-6, projected removals of EBS snow crab in the directed fishery plus bycatch model show very low removals of EBS snow crab during the first four years of the proposed rebuilding plan. These projected low removals indicate a potential for EBS snow crab fishery being closed to directed fishing during the early years of the rebuilding plan. There is substantial uncertainty surrounding the projected removal values and whether these projections will result in a directed fishery since the TAC setting process is determined by the state of Alaska and varies annually based on numerous factors which will influence the TAC for EBS snow crab. The values presented in Table 3-6 are in no way a definitive prediction of a directed fishery throughout the rebuilding period but do provide better evidence of potential fishing activity throughout the rebuilding period under Alternative 2/option 2 (the PPA). If the stock is insufficient to support a directed fishery based on the State's harvest control rule during one of the rebuilding years, the fishery would be closed to directed fishing for that year. If under Alternative 2/option 2 (the PPA), the stock cannot support a directed fishery based on State of Alaska harvest strategy for several years as noted in the projected removals in Table 3-6, then the socioeconomics impacts under Alternative 2/option 2 (the PPA) would be similar to the impacts under Alternative 2/option 1.

Assuming there will be a directed fishery every year during the rebuilding program, the addition of a directed fishery under Alternative 2/option 2 (the PPA) for processors and communities would provide valuable operating revenue to allow for paying processor wages, processor expenditures of goods and services in the local community, expenditures by processor workers in the communities, as well as remittances of wages to primary residences supported by processing workers who are working away from home. In addition, the operating revenue from processing EBS snow crab would allow processors to make

payments toward financed credit and/or debt instruments. In general, given nearly all the EBS snow crab processors have a diversified revenue stream, it is likely most EBS snow crab processors would continue to process EBS snow crab despite the likelihood for economic inefficiency associated with processing small amounts of snow crab during the early years of the rebuilding plan. As noted in Table 3-6, the projected low EBS snow crab removals during these early years could be on par with EBS snow crab removals during the 2021/2022 season. During the 2021/2022 season, 5.5 million pounds of harvested EBS snow crab generated \$41 million in estimated first wholesale gross revenue (see Table 3-13).

Assuming similar first wholesale gross prices for EBS snow crab during the early years of the rebuilding plan, gross first wholesale revenue during these early years of the rebuilding plan would be similar to the 2021/2022 season. If first wholesale gross prices are lower given that 2021/2022 prices may be an outlier, then estimated first wholesale gross revenue could be significantly lower for the first four years of the rebuilding period. This projected decline in first wholesale gross revenue could result in some reduced processor workers and/or worker wages and the subsequent expenditures in the local communities where the processors reside and reduced remittance to worker's resident communities. Based on removal projects in Table 3-6, first wholesale gross revenue projects would likely increase in the later years of the rebuilding plan.

As noted in Table 3-30, there is at least one shorebased processor that is highly dependent on the EBS snow crab fishery. Processors that are highly dependent on the EBS snow crab with very limited revenue diversification are less likely to process EBS snow crab during the early years of the rebuilding plan. The small amount of projected EBS snow crab removals during the early years of the rebuilding plan may result in too small amount of snow crab deliveries to pay for the cost of the processing those deliveries. Based on the projected removals of EBS snow crab during the first four years of the rebuilding plan, the Trident processing plant based in Saint Paul could choose to not operate during the early years of the rebuilding plan since their revenue stream at the plant is less diversified than other processing plants. As noted in a personal communication with a Trident representative, in general, it is more costly to operate at low TAC levels than to have the shoreplants shuttered and pay for annual maintenance. However, long-term implications of doing so, for the community and regional delivery requirements, make the cost of not operating in St. Paul severe. If the plant is shuttered for a prolonged period, there would be additional costs related to the replacement non-stainless steel parts (e.g., drive changes and some bearings). However, as the projected removals of the EBS snow crab increase in the later years of the rebuilding plan (Table 3-6), there is greater potential for the Saint Paul processing plant to process EBS snow crab since the amount of deliveries of snow crab to the processor could pay for the cost of processing those deliveries. If the Saint Paul processor does process EBS snow crab, there would be a reciprocal improvement in the socioeconomic outlook for the community of Saint Paul given the importance of the facility to the community.

3.6.4.3 Impacts to CDQ groups

With regards to the CDQ groups, Alternative 2/option 1 relative to Alternative 1 and Alternative 2/option 2 (the PPA) will likely result in negative impacts for all of CDQ groups with some bearing a higher burden than others due to their level of investment in EBS snow crab quota, but all of the CDQ groups benefit from a diversified revenue stream which helps weather the rebuilding period. The absence of a directed EBS snow crab fishery for six years under Alternative 2/option 1 from the perspective of the CDQ allocation will likely have a negative impact on each of the CDQ groups due to loss of revenue from those allocations. From the perspective of EBS snow crab CDQ allocation, the impact will be slightly greater for BBEDC and CBSFA CDQ groups since they receive 20 percent of the 10 percent CDQ group allocation of EBS snow crab. NSEDC, YDFA, and CVRF groups would likely experience slightly less of an impact given their slightly smaller allocation of the 10 percent of the EBS snow crab allocation. APICDA, receiving 8 percent of the 10 percent EBS snow crab CDQ allocation would receive slightly less revenue from the loss of this fishery. CDQ groups would also be impacted based on their investments in the EBS snow crab harvester and processor quota. For those CDQ groups that are heavily invested in

the EBS snow crab quota, for example CBSFA, APICDA, CVRF, NSEDC, and BBEDC, would likely be severely impacted due to this loss of revenue due the closure of the directed EBS snow crab fishery. The loss of EBS snow crab CDQ earnings due to no directed fishing during the rebuilding period could reduce funding for CDQ programs. For example, CBSFA's Elders Residential Assistance Program, Unangam Tunuu Revitalization, and Bycatch Reduction Engineering Program, or APIDCA's Community Development Grant Program, large-scale multi-year infrastructure projects, and post-secondary and vocational education training scholarships are just a few of the programs that could be at risk. Reduced funding for these and other CDQ programs due to the loss of the directed EBS snow crab fishery could also diminish indirect and induced expenditures in the community. Money spent in the community by recipients of these CDQ programs helps support additional economic activity in the community.

From the perspective of Alternative 1 and Alternative 2/option 2 (the PPA), the socioeconomic impacts to the CDQ groups would likely improve. However, as noted above, under Alternative 1, the Council would not develop a rebuilding plan for EBS snow crab in accordance with MSA 304(e)(4)(A) guidelines and therefore selecting Alternative 1 would be in violation of these MSA guidelines. Under Alternative 2/option 2 (the PPA), a rebuilding plan would be developed which would allow a directed snow crab fishery under the current State of Alaska harvest strategy which given the potential of a directed EBS snow crab fishery as noted in projected annual EBS snow crab removals in Table 3-6, the socioeconomic impacts to the CDQ groups would likely improve. Projected removals from Table 3-6 are in no way a definitive prediction of a directed fishery throughout the rebuilding period but do provide better evidence of potential fishing activity throughout the rebuilding period under Alternative 2/option 2 (the PPA). However, if the stock is insufficient to support a directed fishery based on the State's harvest control rule, the fishery would be closed to directed fishing for that year. If under Alternative 2/option 2 (the PPA), the stock cannot support a directed fishery based on State of Alaska harvest strategy for several years as noted in the projected removals in Table 3-6, then the socioeconomic impacts under Alternative 2/option 2 (the PPA) would be similar to the impacts under Alternative 2/option 1.

Assuming there is a directed fishery every year during the rebuilding period, CDQ groups would receive earnings from their CDQ allocations of EBS snow crab in addition to earnings from their EBS snow crab investments. CDQ earnings under this rebuilding option could allow for continued funding of CDQ programs which in turn would continue the socioeconomic benefits to the communities impacted. The revenue from the EBS snow crab fishery during the rebuilding plan would allow the continue funding for CDQ programs like the CBSFA's Elders Residential Assistance Program or APIDCA's Community Development Grant Program. The continued funding of these CDQ program would also allow for continued indirect and induced expenditures in the CDQ group communities which would provide greater socioeconomic benefits to the communities.

3.6.4.4 Summary of Impacts

Given that Alternative 1 is the status quo alternative and would not implement a rebuilding plan for EBS snow crab, the alternative is a violation of MSA 304(e)(4)(A) guidelines and should not be selected as the preferred alternative.

Relative to the Alternative 2/option 2 (the PPA), which would provide the possibility of a directed EBS snow crab fishery, the guaranteed loss of the directed EBS snow crab fishery during the six-year rebuilding period under Alternative 2/option 1 could result in severe impacts for those associated with the fishery. The assured loss of directed EBS snow crab for six years would represent substantial losses for vessel owners, crew, harvesting and processing quota shareholders, processors, CDQ groups, as well as the associated communities in addition to the communities where the shorebased processors are located. When combined with the potential of annual directed fishery closures of the BBRKC fishery, the socioeconomic impacts to the participants of the EBS snow crab fishery could be accentuated even further under Alternative 2/option 1. Under Alternative 2/option 2 (the PPA), which allows for a directed fishing

during the six-year rebuilding period, the socioeconomic impacts of the rebuilding plan could be somewhat less than under Alternative 2/option 1, but still substantial. If, however, under Alternative 2/option 2 (the PPA), the stock cannot support a directed fishery based on the State of Alaska harvest strategy, then the socioeconomic impacts under Alternative 2/option 2 (the PPA) could be similar or identical to the impacts under Alternative 2/option 1.

3.7 Monitoring Progress of the Rebuilding Plan

As required under NS1 Guidelines, the Secretary must ensure that progress made under a rebuilding plan is adequate. Throughout the rebuilding plan for EBS snow crab (Alternative 2), assessment surveys and biological data collection would be continued and help to facilitate the determination of adequate progress.

The NMFS eastern Bering Sea bottom-trawl survey provides data for annual assessment of the status of crab stocks in the BSAI, including EBS snow crab, and would continue throughout rebuilding. The BSAI Crab Plan Team would annually report stock status and progress towards the rebuilt level in the Stock Assessment and Fishery Evaluation (SAFE) Report for the king and Tanner crab fisheries of the BSAI.

Additionally, ADF&G and NMFS will monitor directed fishery catch (allowable only under Alternative 2/option 2, the PPA) and bycatch of snow crab in other fisheries (under any alternative). ADF&G requires full observer coverage (100%) for catcher processors and partial coverage (30%) for catcher vessels participating in the fishery. Observers monitor harvest at sea and landings by catcher vessels shoreside processors. ADF&G reports the total harvest from the commercial fishery and that report will be included annually in the SAFE.

State and federal observer programs monitor bycatch with State coverage of the crab fisheries and federal monitoring of the groundfish trawl, pot and longline fisheries. Estimates of crab bycatch from all commercial fisheries will be reported annually in the SAFE and the CPT will assess that bycatch relative to the expectations and assumptions of the rebuilding plan.

3.7.1 Expectations for Recruitment during Rebuilding

One of the primary drivers for future stock health for most shellfish stocks is recruitment, which can be highly variable and driven by environmental conditions. As noted in section 2.2.2.1, varying assumptions about future recruitment and natural mortality have heavily influence projected rebuilding trajectories and associated time to rebuilding.

The average estimated recruitments of the period 1982-2017 and 2005-2019 are fairly similar, but when any mortality events are allowed in the projections, the ability of the stock to rebuild is hampered (compare Figure 2-2 to Figure 2-3). Under no mortality events in the projection period (i.e., the 1982-2017 scenario) the median projected stock rebuilt by 2029 with no fishing mortality. The rebuilding time was extended to 2034 under no fishing if mortality events were imposed during 1 in 7 years (on average).

It is the analysts goal in restating this to emphasize the expectation that uncertainty surrounding recruitment and mortality under the current ecosystem conditions may heavily influence the rate at which the stock is able to rebuild under the proposed projection parameters, regardless of the fishing mortality.

3.8 Draft Amendment 53 to the Bering Sea and Aleutian Islands Fishery Management Plan for King and Tanner Crabs

1. Remove content from section 6.2.1, 6.2.2, 6.2.3.
2. Modify section 6.2.4 to 6.2.1.

3. Modify section 6.2.5 to 6.2.2.
4. Add section 6.2.3 Bering Sea snow (*Chionoecetes opilio*) crab

NMFS declared EBS snow crab overfished on October 19, 2021, because the spawning biomass estimated from the NMFS trawl survey was below the minimum stock size threshold. The Council developed a rebuilding plan for the EBS snow crab to satisfy the requirements of section 304(e)(4)(A) of the Magnuson-Stevens Act and comply with the national standard guidelines at 50 CFR 600.310(e). The minimum time for rebuilding (T_{MIN}), taking into account the biology of the species and current environmental conditions, is 6 years, which under the National Standard 1 guidelines dictates a maximum time for rebuilding (T_{MAX}) of 10 years. Through this FMP, the State of Alaska is delegated management of the EBS snow crab fishery the State sets preseason TACs and GHGs, and season or area closures when the TAC or GHG is reached. The State of Alaska's EBS snow crab harvest strategy is provided in the Alaska Administrative Code at 5 AAC 35.517 and that strategy applies during rebuilding. The stock will be considered "rebuilt" when the stock reaches B_{MSY} in one year.

As required under NS1 Guidelines, the Secretary must ensure that progress made under a rebuilding plan is adequate. Throughout the rebuilding plan for EBS snow crab, assessment surveys and biological data collection would be continued and help to facilitate the determination of adequate progress. The NMFS eastern Bering Sea bottom-trawl survey provides data for annual assessment of the status of crab stocks in the BSAI, including EBS snow crab, and would continue throughout rebuilding. The BSAI Crab Plan Team would annually report stock status and progress towards the rebuilt level in the Stock Assessment and Fishery Evaluation (SAFE) Report for the king and Tanner crab fisheries of the BSAI.

Additionally, ADF&G and NMFS will monitor directed fishery catch and bycatch of snow crab in other fisheries. ADF&G requires full observer coverage (100%) for catcher processors and partial coverage (30%) for catcher vessels participating in the fishery. Observers monitor harvest at sea and landings by catcher vessels shoreside processors. ADF&G reports the total harvest from the commercial fishery and that report will be included annually in the SAFE.

State and federal observer programs monitor bycatch with State coverage of the crab fisheries and federal monitoring of the groundfish trawl, pot and longline fisheries. Estimates of crab bycatch from all commercial fisheries will be reported annually in the SAFE and the CPT will assess that bycatch relative to the expectations and assumptions of the rebuilding plan.

Under this rebuilding plan, changes to the components of the plan must; (1) comply with the existing criteria in the FMP and the national standard guidelines at 50 CFR 600.310(e), (2) be sufficient to rebuild the stock to the B_{MSY} level within a rebuilding time period that satisfies the requirements of section 304(e)(4)(A) of the Magnuson-Stevens Act, and (3) be consistent with applicable.

4 Magnuson-Stevens Act and FMP Considerations

4.1 Magnuson-Stevens Act National Standards

Below are the 10 National Standards as contained in the MSA and a brief discussion of how each alternative is consistent with the National Standards, where applicable. In recommending a preferred alternative, the Council must consider how to balance the national standards.

National Standard 1 — Conservation and management measures shall prevent overfishing while achieving, on a continuing basis, the optimum yield from each fishery for the United States fishing industry.

In terms of achieving “optimum yield” from the fishery, the Act defines “optimum,” with respect to yield from the fishery, as the amount of fish which—

(A) will provide the greatest overall benefit to the Nation, particularly with respect to food production and recreational opportunities, and taking into account the protection of marine ecosystems;

(B) is prescribed as such on the basis of the maximum sustainable yield from the fishery, as reduced by any relevant economic, social, or ecological factor; and

(C) in the case of an overfished fishery, provides for rebuilding to a level consistent with producing the maximum sustainable yield in such fishery.

The action alternative prevents overfishing of EBS snow crab by establishing a rebuilding plan that identifies catch levels that will rebuild the stock. Alternative 2 would implement a rebuilding plan under the parameters $T_{\min}=6$ and $T_{\max}=10$. The Council has yet to select a target time for rebuilding (T_{target}), but it will likely fall between 6 and 10 years. There are two rebuilding options. Alternative 2/option 1 would designate no directed EBS snow crab fishery, with bycatch removals only, and implications to the stock will be similar to those seen as a result of the 2022/2023 fishery closure. Alternative 2/option 2 (the PPA) would allow for directed harvest if an opening is triggered by threshold survey catches under the State harvest strategy.

National Standard 2 — Conservation and management measures shall be based upon the best scientific information available.

Information in this analysis represents the most current, comprehensive set of information available to the North Pacific Fishery Management Council (Council).

National Standard 3 — To the extent practicable, an individual stock of fish shall be managed as a unit throughout its range, and interrelated stocks of fish shall be managed as a unit or in close coordination.

This stock is managed as a single unit within the area defined as the stock boundary, and the rebuilding plan would be in effect throughout the management unit.

National Standard 4 — Conservation and management measures shall not discriminate between residents of different states. If it becomes necessary to allocate or assign fishing privileges among various United States fishermen, such allocation shall be: (A) fair and equitable to all such fishermen, (B) reasonably calculated to promote conservation, and (C) carried out in such a manner that no particular individual, corporation, or other entity acquires an excessive share of such privileges.

The proposed action considers closing the directed EBS snow crab fishery or allow a directed fishery while the stock is rebuilding. Thus, considerations under National Standard 4 pertain to the EBS snow crab vessel owners, quota shareholders, crew, and processors as directly affected by the proposed action. Nothing in the proposed alternatives considers residency as a criterion for the Council's decision. Residents of various states, including Alaska and the states of the Pacific Northwest, participate in the EBS snow crab fishery are directly or indirectly affected by the proposed action. A description of participants in the EBS snow crab fishery, including vessel owners, quota shareholders, and crew by ownership address and processors by location of shoreside processor is presented in section 3.6.3.

Nothing in the alternatives considers residency as a criterion for the Council's decision. Residents of the various states, including Alaska and states of the Pacific Northwest, participate in the fishery affected by this decision. No discriminations are made based on residency or any other criteria. Section 3.6.3 also contains detailed information for EBS snow crab fisheries on community engagement, dependency, and federally recognized tribal status and encompasses all state in which those communities are located.

National Standard 5 — Conservation and management measures shall, where practicable, consider efficiency in the utilization of fishery resources, except that no such measure shall have economic allocation as its sole purpose.

Efficiency in the context of the proposed action refers to economic efficiency. The analysis presents information, where appropriate, on economic efficiency under the proposed EBS snow crab rebuilding alternatives and provides information on the economic risks associated with these alternatives. Economic allocation is not the sole purpose of the proposed action.

National Standard 6 — Conservation and management measures shall take into account and allow for variations among, and contingencies in, fisheries, fishery resources, and catches.

The preliminary preferred action is consistent with this standard. Annual variability in EBS snow crab resource and catch are described in section 3.2.

National Standard 7 — Conservation and management measures shall, where practicable, minimize costs and avoid unnecessary duplication.

The preliminary preferred action is consistent with this standard.

National Standard 8 — Conservation and management measures shall, consistent with the conservation requirements of this Act (including the prevention of overfishing and rebuilding of overfished stocks), take into account the importance of fishery resources to fishing communities by utilizing economic and social data that meet the requirements of National Standard 2, in order to (A) provide for the sustained participation of such communities, and (B) to the extent practicable, minimize adverse economic impacts on such communities.

Pursuant to the MSA section 304(e)(4)(A) and the National Standard Guidelines, the purpose of this proposed action is to develop a rebuilding plan to prevent overfishing and to rebuild the EBS snow crab stock. A summary of the level of fishery engagement and dependence in communities with regard to EBS snow crab fishing is provided in section 3.6. Through 2006 and 2021, the overall community processing dependence as measured by the amount of first wholesale gross revenue processed in each community that is from the EBS snow crab fishery is 7 percent for the combined communities of Akutan/King Cove/Kodiak/Saint Paul, 8 percent for Unalaska/Dutch Harbor, and 6 percent for all other communities (see Table 3-28). The information provided throughout section 3.6.3 masks the impacts of the alternatives being considered for some communities due to the need to protect confidential data. The action alternatives, although likely substantial for Unalaska, Kodiak and some other Alaska communities and

likely severe for Saint Paul, are projected to provide for the sustained participation of the community once the EBS snow crab are rebuilt.

Additionally, Alternative 2/option 2 (the PPA) would allow for a directed fishery based the Alaska harvest control rule, which if open for directed fishing and the local community processor is operating would provide processor operating revenue to pay for local processor wages and processor expenditures of goods and services in the local community. Expenditures by the processor and processor workers in the communities would also generate additional economic activity through induced expenditures. There is also the potential for harvesting and processing operations that are highly dependent on the EBS snow crab fishery and are less economically diversified could choose to not operate or even exit the fishery all together during the early years of the rebuilding period based on projected removals (see Table 3-6). The loss of these harvesters and processor operations during the early years of the rebuilding plan could negatively impact communities that rely on EBS snow crab fishery through loss of expenditures by harvesters in the community, loss of expenditures by processors and their workers in the community, and loss of community tax revenue from the local sales tax on the sale of seafood.

National Standard 9 — Conservation and management measures shall, to the extent practicable, (A) minimize bycatch, and (B) to the extent bycatch cannot be avoided, minimize the mortality of such bycatch.

No alternatives propose new management measures to reduce PSC of EBS snow crab in the groundfish fisheries. Bycatch rates in non-target fisheries showed no substantial effect on rebuilding time. Projections using average bycatch shows that the median time for stock recovery was not differentiable from no fishing scenario. Additionally, the time for stock recovery was minimally affected in projections that used 5x and 100x level of average observed bycatch as seen in the rebuilding projections. Therefore, analyst concluded that recovery of the EBS snow crab stock is likely to not be affected by current or predicted bycatch levels, based on average historical bycatch. Additionally, Alternative 2/option 1 and Alternative 2/option 2 (the PPA) would maintain all of the existing measures that minimize fishery impacts on EBS snow crab throughout the rebuilding period. Bycatch in the groundfish fisheries or other-directed crab fisheries are not expected to directly impact the success in the rebuilding under any of the proposed alternatives.

National Standard 10 — Conservation and management measures shall, to the extent practicable, promote the safety of human life at sea.

The alternatives under consideration appear to be consistent with this standard. None of the alternatives or options proposed would change safety requirements for fishing vessels. No safety issues have been identified relative to the proposed action.

4.2 Section 303(a)(9) Fisheries Impact Statement

Section 303(a)(9) of the MSA requires that a fishery impact statement be prepared for each FMP or FMP amendment. A fishery impact statement is required to assess, specify, and analyze the likely effects, if any, including the cumulative conservation, economic, and social impacts, of the conservation and management measures on, and possible mitigation measures for (a) participants in the fisheries and fishing communities affected by the plan amendment; (b) participants in the fisheries conducted in adjacent areas under the authority of another Council; and (c) the safety of human life at sea, including whether and to what extent such measures may affect the safety of participants in the fishery.

The EA prepared for this plan amendment constitutes the fishery impact statement. The likely effects of the proposed action are analyzed and described throughout the EA. The effects on participants in the fisheries and fishing communities are analyzed in section 3.6.2. No effects of the proposed action on safety of human life at sea are anticipated since no changes in fishery management measures are

proposed. Based on the information reported in this section, there is no need to update the Fishery Impact Statement included in the FMP.

The proposed action affects the EBS snow crab fishery in the EEZ off Alaska, which is under the jurisdiction of the North Pacific Fishery Management Council and the State of Alaska. Impacts on participants in fisheries conducted in adjacent areas under the jurisdiction of other Councils are not anticipated as a result of this action.

4.3 Council's Ecosystem Vision Statement

In February 2014, the Council adopted, as Council policy, the following:

Ecosystem Approach for the North Pacific Fishery Management Council

Value Statement

The Gulf of Alaska, Bering Sea, and Aleutian Islands are some of the most biologically productive and unique marine ecosystems in the world, supporting globally significant populations of marine mammals, seabirds, fish, and shellfish. This region produces over half the nation's seafood and supports robust fishing communities, recreational fisheries, and a subsistence way of life. The Arctic ecosystem is a dynamic environment that is experiencing an unprecedented rate of loss of sea ice and other effects of climate change, resulting in elevated levels of risk and uncertainty. The North Pacific Fishery Management Council has an important stewardship responsibility for these resources, their productivity, and their sustainability for future generations.

Vision Statement

The Council envisions sustainable fisheries that provide benefits for harvesters, processors, recreational and subsistence users, and fishing communities, which (1) are maintained by healthy, productive, biodiverse, resilient marine ecosystems that support a range of services; (2) support robust populations of marine species at all trophic levels, including marine mammals and seabirds; and (3) are managed using a precautionary, transparent, and inclusive process that allows for analyses of tradeoffs, accounts for changing conditions, and mitigates threats.

Implementation Strategy

The Council intends that fishery management explicitly take into account environmental variability and uncertainty, changes and trends in climate and oceanographic conditions, fluctuations in productivity for managed species and associated ecosystem components, such as habitats and non-managed species, and relationships between marine species. Implementation will be responsive to changes in the ecosystem and our understanding of those dynamics, incorporate the best available science (including local and traditional knowledge), and engage scientists, managers, and the public.

The vision statement shall be given effect through all of the Council's work, including long-term planning initiatives, fishery management actions, and science planning to support ecosystem-based fishery management.

In considering this action, the Council is being consistent with its ecosystem approach policy. This action establishes a plan for rebuilding the EBS snow crab stock, taking into account the biology of the stock and prevailing ecosystem conditions. This is supportive of the Council's intention to prevent overfishing and to adjust catch levels as necessary to account for uncertainty and ecosystem factors.

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RKCSA Closure Emergency Action

At this meeting, the Council considered and recommend against emergency rulemaking for a proposed action that would have closed the Red King Crab Savings Area and Savings Subarea to fishing with all gear types for January 1 – June 30, 2023. The Council did initiate a separate action to consider crab abundance-triggered area closures through normal rulemaking.

After reviewing the emergency rule analysis, the Council determined the proposed action did not meet the requirements established under NMFS' Policy Guidelines for the Use of Emergency Rules, specifically criterion 1 and 3. For criterion 1, the Council determined the situation did not result from recent, unforeseen events or recently discovered circumstances. While Bristol Bay red king crab abundance is low, the stock is not overfished and overfishing is not occurring, and the stock is regularly assessed so this the decline in abundance is not unforeseen or recently discovered. For criterion 3, Council members believe that available information does not support a finding that the immediate benefits of

emergency rulemaking outweigh the value of advance notice, public comment, and deliberative consideration of the impacts on participants under the normal rulemaking process. Although the analysis indicates the proposed action could provide habitat benefits through reduced bottom contact by trawl gear and potentially reduced Bristol Bay red king crab mortality, these impacts are identified as uncertain and highly dependent on assumptions that the closure will move trawl and fixed gear fleets into areas with lower crab impacts. The analysis also indicates potentially negative impacts to other prohibited species like salmon, halibut and herring and fleet operations.

In response to discussion, testimony and analysis on this topic, the Council initiated a separate action to consider crab abundance-triggered area closures through normal rulemaking. The Council adopted a purpose and need statement and provided two alternatives (which are not mutually exclusive) for an initial review analysis.

The first alternative considers implementing an annual closure of the Red King Crab Savings Area and Red King Crab Savings Subarea to all commercial groundfish fishing gears. The existing closure for non-pelagic trawl gear is not changed. The closure would be triggered either by the closure of the BBRKC fishery (Option 1) or by a threshold for the area-swept biomass of BBRKC (50,000t; Option 2). This alternative also considers exempting hook-and-line gear (Sub-option 1) and/or pot gear (Sub-option 2).

The second alternative considers implementing a closure of Area 512 for fishing for Pacific cod with pot gear. The closures would again be triggered by either the closure of the BBRKC fishery (Option 1) or by a threshold for the area-swept biomass of BBRKC (50,000t; Option 2).

The analysis will also evaluate the potential trade-offs and challenges of establishing dynamic closure areas to promote BBRKC. In addition, the analysis will provide an expanded discussion of the performance standard applicable to vessels in the directed pollock fishery and the regulatory definition of pelagic trawl gear. The expanded discussion will include background on the rationale for and information used to establish the performance standard and gear definition to help evaluate whether the performance standard and gear definition are meeting Council objectives.

Staff contact is Sam Cunningham

Snow Crab Rebuilding

At this meeting, the Council made an initial review analysis of a draft environmental assessment to rebuild the Eastern Bering Sea snow crab stock in compliance with section 304(e)(3) of the Magnuson-Stevens Fishery Conservation and Management Act. The impacts

of the alternatives considered on EBS snow crab stock, fishery participants, habitat, and other components of the human environment are described in the analysis.

After reviewing the draft, the Council recommended scheduling the analysis for final action and selected the following as their preliminary preferred alternative:

Alternative 2: Set a target rebuilding time frame for the number of years necessary to rebuild the stock to the BMSY level at a probability $\geq 50\%$. The stock will be considered “rebuilt” once it reaches BMSY.

Option 2: Allow the directed fishery to open based on the state harvest strategy while the stock is rebuilding.

In speaking to the motion, the Council noted that the preliminary preferred alternative is consistent with the mandate under the Magnuson Stevens Act to rebuild stocks and is also consistent with National Standard 1 in that it would establish a plan estimated to rebuild the stock in less than 10 years but allows directed fishing to occur under the state harvest strategy while the stock rebuilds to a level that achieves optimum yield. Maintaining the economic opportunity for a directed commercial fishery under the state harvest strategy is essential for harvesters, processors, and communities. Selecting this option as the preliminary preferred alternative does not extend the proposed rebuilding timeline.

The Council also requested, based on the SSC recommendations, if possible, the analysts should add:

1. the projected catch during the rebuilding period to inform the evaluation of the potential economic effects; and
2. information on the status of fishing-dependent communities through the 2021/22 season and CDQ ownership/participation.

The analysis will also evaluate Alternative 1 in the expected effects section of the alternatives.

Staff contacts are Jon McCracken and Sarah Rheinsmith.

***NATIONAL MARINE FISHERIES SERVICE INSTRUCTION 01-111-05
MARCH 20, 2007***

***National Marine Fisheries Procedures for RFA
and E.O. 12866 Review Process***

***GUIDELINES FOR ECONOMIC REVIEW OF
NATIONAL MARINE FISHERIES SERVICE REGULATORY ACTIONS***

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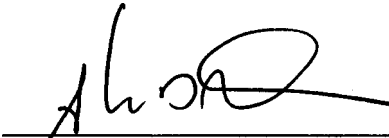
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SUMMARY OF REVISIONS: *This document revises the “Guidelines for Economic Analysis of Fishery Management Actions (August 16, 2000) as follows:*

- *Revises instructions regarding OMB Circular #A-4;*
- *Exempts NMFS from the peer review requirements of OMB’s bulletin on that topic for both the RIR and RFAA under most circumstances;*
- *Clarifies and distinguishes the scope of analyses required under E.O. 12866 and RFA with regard to small entities;*
- *Provides recent SBA modifications of monetary-based definitions of small entities;*
- *Clarifies the scope of analyses required for certification of no significant impact under RFA;*
- *Advises procedure on the development and distribution of the Small Entity Compliance Guide concomitant with the rulemaking process. Suggested language for the FRFA is also provided;*
- *Clarifies the requirements of periodic reviews of significant rules, under Section 610 of the RFA;*
- *Clarifies means of presentation of the RFAA, whether it be a stand alone document or as part of a NEPA document or proposed rule;*

- *Provides guidance on compliance with E.O. 13272;*
- *Provides guidance on development of RFAA for Emergency Regulation or interim measures to reduce overfishing, under Section 305(c) of the Magnuson-Stevens Act;*
- *Broadens the application of regulatory analysis to include those implemented by the Offices of Protected Resources and Habitat Conservation and other NMFS line offices in addition to those conducted by the Office of Sustainable Fisheries; and*
- *Provides a new size standard for the definition of a small entity for meeting the requirements of the RFA regarding vessels that both harvest and process fish (referred to as catcher-processors).*

Signed

 3/6/07

Alan Risenhoover Date
Director
Office of Sustainable Fisheries

**GUIDELINES FOR ECONOMIC REVIEWS OF
NATIONAL MARINE FISHERIES SERVICE REGULATORY ACTIONS**

NATIONAL MARINE FISHERIES SERVICE

SILVER SPRING, MARYLAND 20910

March, 2007

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I. INTRODUCTION

The purpose of this document is to provide guidance on understanding and meeting the procedural and analytical requirements of Executive Order (E.O.) 12866 and the Regulatory Flexibility Act (RFA) (5 U.S.C. 601 et seq.) for regulatory actions promulgated by the National Marine Fisheries Service (NMFS).

NMFS prepares a Regulatory Impact Review (RIR), consistent with E.O. 12866, which includes an analysis of the economic effects of the preferred and alternative actions, in contrast to taking “no action”. The RIR is intended to assist the Regional Fishery Management Councils (Council) and NMFS in selecting the regulatory approach that maximizes net benefits to the Nation (including potential economic, environmental, public health and safety, and other advantages; distributive impacts; and equity), unless a statute requires another regulatory approach.

Regulatory Flexibility Act Analysis (RFAA) is necessary to satisfy the requirements of the RFA. The RFAA assesses the impacts of the proposed/final rule on small entities and describes steps the agency has taken to minimize any significant economic impact on small entities while achieving regulatory goals. The general intent of the RIR and RFAA requirements is to make the decision process open and transparent so that all can understand the what, where, and why of regulatory decision-making and can agree that the required steps of the process were followed. The analyses provide decision-makers and the public with the Agency's reasonable estimates of the economic impacts of proposed actions and of their alternatives.

These Guidelines were developed by a team of NMFS economists, Council staff, attorneys from the Office of General Counsel/Department of Commerce (OGC/DOC), and attorneys from the National Oceanic & Atmospheric Administration General Counsel for Fisheries (NOAA/GCF). In comparison to the previous RIR/RFAA guidelines, revised in August, 2000, these guidelines would --

- Revise instructions regarding OMB Circular #A-4;
- Explain that RIRs and RFAAs are, for the most part, not subject to the requirements of OMB's peer review bulletin;
- Clarify and distinguish the scope of analyses required under E.O. 12866 and RFA with regard to small entities;
- Provide recent SBA modifications of monetary-based definitions of small entities;
- Clarify the scope of analyses required for certification of no significant impact under RFA;
- Advise procedure on the development and distribution of the Small Entity Compliance Guide concomitant with the rulemaking process. Suggested language for the FRFA is also provided;

- Clarify the requirements of periodic reviews of significant rules, under Section 610 of the RFA;
- Clarify means of presentation of the RFAA, whether it be a stand alone document or as part of a NEPA document or proposed rule;
- Provide guidance on compliance with E.O. 13272;
- Provide guidance on development of RFAA for Emergency Regulation or interim measures to reduce overfishing, under Section 305(c) of the Magnuson-Stevens Act;
- Broaden the application of regulatory analysis to include those implemented by the Offices of Protected Resources and Habitat Conservation and other NMFS line offices in addition to those conducted by the Office of Sustainable Fisheries; and
- Provide a new size standard for the definition of a small entity for meeting the requirements of the RFA regarding vessels that both harvest and process fish (referred to as catcher-processors).

Although the RIR and the RFAA should be undertaken by those with economic expertise, these guidelines are written to ensure that non-economists understand what should be in the RIR and the RFAA. When developing the RIR and the RFAA, the analyst is expected to make a reasonable effort to organize the relevant information and supporting analyses, given the significance of the issue, projected time tables, and available resources. At a minimum, the RIR and the RFAA should include a good qualitative discussion of the economic effects of the selected alternatives. Quantification of these effects is desirable, but the analyst needs to weigh such quantification against the significance of the issue and available studies and resources. Generally, a good qualitative discussion of the expected effects would be better than poor quantitative analyses.

Economic analyses are also required, to varying degrees, under the Magnuson-Stevens Fishery Conservation and Management Act (Magnuson-Stevens Act), the National Environmental Policy Act (NEPA), the Endangered Species Act (ESA), the Marine Mammal Protection Act (MMPA), and other applicable laws. The analyst should be aware of these other laws, as he/she will often be working with other analysts conducting analyses to meet the requirements of these laws. For example, section 303(a)(9) of the Magnuson-Stevens Act requires a Fishery Impact Statement (FIS). This includes an analysis of the effects of a proposed action on participants in the fishery and on fishing communities.

Regulatory analyses for NMFS management actions should strive to simultaneously meet the requirements of E.O. 12866, the RFA, and other laws, including Magnuson-Stevens Act, NEPA, MMPA, and ESA. Appendix A contains brief descriptions of the requirements of these laws. Economic analyses done early in the development of the regulatory process, along with biological, environmental, and social information, allow decision-makers to identify and explore the full range of management alternatives. Integrating the analytical requirements of the RIR,

the RFAA, and other economic analyses into the Council and NMFS decision-making process will ensure that the best scientific information available can be used. Decision-makers will have a full suite of information available to make informed decisions for the resources and for all those who benefit from them.

The level of analysis may be constrained by a lack of available data and quantitative models. However, the analysis conducted should be sufficient to allow the Council and NMFS, on behalf of the Secretary of Commerce, to make informed decisions and to present, quantitatively and/or qualitatively, the expected economic effects for the management alternatives under consideration (i.e., the selected management alternatives).

These guidelines identify a step-wise approach that will aid in identifying data requirements and conducting economic analyses for regulatory actions. Although these guidelines focus on economic analyses for meeting the requirements of E.O. 12866 and the RFA, it should be emphasized that the first step in the development of a fishery management plan or a regulatory action for a federally managed fishery or any other NMFS regulatory action (other than identifying the goals and objectives) is a description of the biological, economic, social, and cultural characteristics of the fishery, affected entities, and communities. This integrated multi-disciplinary approach provides information on the data available and enables the analyst to identify data required for the analyses.

The following sections present details on the process and analyses necessary to satisfy the requirements of E.O. 12866 and the RFA. Section II provides a general framework that could be utilized for economic analyses of regulatory actions. In Section III, the recommendation is made to have a preliminary analysis of the economic effects of the selected alternatives available prior to the determination of the preferred alternative. Sections IV and V describe the specific analytical and procedural requirements for NMFS regulatory actions, as established by E.O. 12866 and the RFA, respectively.

This document is also likely to serve as a reference document. In this regard, it makes explicit use of cross referencing and indicates to the reader where to turn for greater details. The repetition of key ideas or concepts is intended to aid the reader.

II. GENERAL FRAMEWORK FOR ECONOMIC ANALYSES OF REGULATORY ACTIONS

These guidelines do not prescribe methods. Rather, they identify analytical elements that should be addressed and the scope of analysis required under applicable law. Embodied in these guidelines is the principle that a well developed qualitative analysis may be preferable to a poorly specified complex analytical model. Economists may use several analytical options to meet the spirit and requirements of E.O. 12866, the RFA, and other applicable laws. The appropriate options depend on the circumstances to be analyzed, available data, the accumulated knowledge of the fishery and of other potentially affected entities, and on the nature of the regulatory action. The options may include, but are not limited to, complex multi-sector bioeconomic models, sparser static breakeven analysis, theory-informed qualitative descriptions, and other accepted forms of economic analysis.

Analysis of NMFS regulatory actions, both qualitative and quantitative, requires considering the relevant sociological, economic, and biological aspects of the rulemaking. The economic analyses of the effects of alternative management actions are expected to include quantifiable measures to the fullest extent that they can be usefully estimated, as well as qualitative measures of the effects that are difficult to quantify but, nevertheless, essential to consider.

Analytical requirements for E.O. 12866 and the RFA overlap substantially. Although a benefit-cost analytical framework is prominent in meeting the intent of E.O. 12866, it also requires broad consideration of the distributive effects and economic burden that may be imposed on individuals, businesses of differing sizes, as well as small communities and governmental entities. Thus, the level and types of analyses required under E.O. 12866 mirror, to some extent, those required under the RFA.

Meeting the broad analytical requirements of E.O. 12866 requires consideration of both benefits and costs of regulatory alternatives from a National perspective, as well as from that of the private individual or firm.

But, even though the analytical requirements are similar, the RFA has specific process and content requirements that are not contained in E.O. 12866. Likewise, E.O. 12866 has specific requirements not contained in the RFA. Nevertheless, a carefully designed analysis could satisfy both requirements.

The economic framework below examines how a regulatory action affects demand for fishery products, recreational fishing opportunities, the supply of such products, and market interactions, which, in turn, affect fishing decisions, dependent communities, and the conditions of living marine resources and their habitats. Analysis of these considerations may be subdivided into the following four basic components:

(1) The first component is an analysis of potential changes in prices, timing/quantity/quality/forms produced or consumed, fishing or observational trips, etc., as a result of changing supply and demand conditions in the marketplace. This information can be used to determine consumer surplus for various fishery products or activities and provides a partial measure of net benefits from the fishery. Expected price changes may be characterized by using a graphical analysis accounting for levels of imports, exports, domestic landings of substitute and complementary fishery products and other consumer goods, disposable income, and other effects.

(2) The second component is an examination of the change in revenues and operating costs for firms or individuals in the fishery, in response to changes in market, biological conditions, and fishery management regulations. Analysis of firm-level changes provides an indication of how producer surplus may change and, for small entities, the impact of regulatory actions. This firm-level analysis characterizes changes in harvesting costs and outputs in the fishery, and may also be used to assess changes in potential industry output levels and fishing season length. Similar analyses can also be developed for the recreational sector and for non-consumptive users of the resource.

(3) The third component is an analysis of how the regulation is expected to affect fishing fleets, and the fishery dependent communities they support. Fleet size and composition

may change in response to market prices, biological conditions, and/or the regulatory environment. Consideration of price and operating cost changes will permit an evaluation of how aggregate fleet size and composition may change. In the absence of either reliable cost or price data, a qualitative discussion of changes in fleet size and composition may be presented. Projected changes in size, composition, and geographic distribution of fishing fleets may permit extrapolation of fleet level impacts to the communities (or regions) from which the fleets operate. Participation rates within recreational fishing modes, and for non-consumptive user groups, should be addressed in a similar manner, where relevant.

(4) The fourth component of this economic framework makes use of the biological analysis that explains the response of the stock or stocks of living marine resources to the proposed regulation. Fishing mortality is a function of effort levels that are determined by market and biological conditions and by fishery regulations. By treating the change in stock size as a factor in the economic objectives of individual fishermen, or of the fleet as a whole, anticipated changes in fishing effort and their impact on the subsequent size of fish stocks and other living marine resources can be evaluated. It should be remembered that non-consumptive user groups assign values to the resource, as well (e.g., avoidance of bycatch of non-target fish, protected species of turtles, and marine mammals). These non-consumptive values may affect optimal stock sizes and influence the net benefit to the Nation associated with a proposed action.

By melding these four components into an overall economic framework, a reasoned assessment of the expected direction of change in net benefits to the Nation, as well as the specific effects on individual entities of a proposed regulatory action, may be evaluated. For each of the four components discussed above, a quantitative analysis should be substituted for a qualitative analysis whenever feasible (i.e., when adequate data, resources, and defensible analytical models are available). It should be noted that a complex empirical model is not necessarily needed to analyze proposed changes for all regulatory actions. In many cases, the analysis will consist of a mix of qualitative and quantitative information. The resulting estimates of the changes in the consumer surplus associated with use and non-use values, producer surplus, management and enforcement costs, fleet size, employment, community¹ effects, and stock abundance are examples of the types of information that may be used by fishery managers to determine whether their objectives and goals are achievable and to compare regulatory alternatives.

III. PRELIMINARY REGULATORY ECONOMIC EVALUATION

Although not required by law, NMFS recommends that a preliminary evaluation describing the expected economic effects of the selected alternatives be undertaken when the

¹ Fishing community, as defined for purposes of this document, refers to the definition in the Regulatory Flexibility Act, not the National Standard 8 fishing community definition in the Magnuson-Stevens Act.

alternatives are developed but before a preferred alternative is identified. The primary intent of this recommended analysis is to provide early consideration of all relevant economic effects of a proposed regulatory action, not to delay or put up roadblocks to an action.

In addition, such preliminary economic analyses could be used to solicit early public comments on the expected economic effects of the alternatives proposed, and as a platform from which information could be obtained to address the requirements of various applicable laws (e.g., E.O. 12866 and the RFA). The preliminary evaluation may be included in the document that goes out to public hearing or for public comment.

For purposes of these guidelines, this preliminary analysis will be labeled a “Preliminary Regulatory Economic Evaluation” (PREE). The PREE should describe, to the fullest extent practicable at this stage of development, the general economic effects that may be reasonably anticipated to occur upon implementation of each alternative management action under consideration. In keeping with applicable law (E.O. 12866 and the RFA), these effects may include effects on net benefits, distributive impacts, and small and large entities.

Depending on the specificity of the alternatives and the number and complexity of proposed alternatives, the PREE may be largely qualitative or may provide quantitative estimates of economic impact. At a minimum, a qualitative discussion of the expected economic impacts of the proposed alternatives should be provided. A quantitative analysis should be substituted for qualitative assessments when available data and resources are available. However, given the preliminary nature of the analysis, and its purpose in soliciting information and data through public comment, the analyst should use reasoned judgment in determining the level of analysis necessary for a particular issue. Regardless of which approach is used (qualitative, quantitative, or a combination of both), the PREE should provide the reader with an overall framework for assessing economic impacts.

IV. REGULATORY IMPACT REVIEW

The objective of Executive Order (E.O.) 12866 (58 FR 51735, October 4, 1993) is to improve the Federal regulatory system. One of the purposes of the RIR is to comply with the requirements of E.O. 12866. The regulatory philosophy of E.O. 12866 is reflected in the following statements:

Federal agencies should promulgate only such regulations as are required by law, are necessary to interpret the law, or are made necessary by compelling public need, such as material failures of private markets to protect or improve the health and safety of the public, the environment, or the well-being of the American people. In deciding whether and how to regulate, agencies should assess all costs and benefits of available regulatory alternatives, including the alternative of not regulating. Costs and benefits shall be understood to include both quantifiable measures (to the fullest extent that these can be usefully estimated) and qualitative measures of costs and benefits that are difficult to quantify, but nevertheless essential to consider. Further, in choosing among alternative regulatory approaches, agencies should select those approaches that maximize net benefits (including potential economic, environmental, public health and safety, and other

advantages, distributive impacts; and equity), unless a statute requires another regulatory approach.

To ensure that the agencies' regulatory programs are consistent with this philosophy, agencies should adhere to the following principles, to the extent permitted by law and where applicable:

- (1) Each agency shall identify the problem that it intends to address (including, where applicable, the failures of private markets or public institutions that warrant new agency action) as well as assess the significance of that problem.
- (2) Each agency shall examine whether existing regulations (or other law) have created, or contributed to, the problem that a new regulation is intended to correct and whether those regulations (or other law) should be modified to achieve the intended goal of regulation more effectively.
- (3) Each agency shall identify and assess available alternatives to direct regulation, including providing economic incentives to encourage the desired behavior, such as user fees or marketable permits, or providing information upon which choices can be made by the public.
- (4) In setting regulatory priorities, each agency shall consider, to the extent reasonable, the degree and nature of the risks posed by various substances or activities within its jurisdiction.
- (5) When an agency determines that a regulation is the best available method of achieving the regulatory objective, it shall design its regulations in the most cost-effective manner to achieve the regulatory objective. In doing so, each agency shall consider incentives for innovation, consistency, predictability, the costs of enforcement and compliance (to the government, regulated entities, and the public), flexibility, distributive impacts, and equity.
- (6) Each agency shall assess both the costs and the benefits of the intended regulation and, recognizing that some costs and benefits are difficult to quantify, propose or adopt a regulation only upon a reasoned determination that the benefits of the intended regulation justify its costs.
- (7) Each agency shall base its decisions on the best reasonably obtainable scientific, technical, economic, and other information concerning the need for, and consequences of, the intended regulation.
- (8) Each agency shall identify and assess alternative forms of regulation and shall, to the extent feasible, specify performance objectives, rather than the behavior or manner of compliance that regulated entities must adopt.

(9) Wherever feasible, agencies shall seek views of appropriate State, local, and tribal officials before imposing regulatory requirements that might significantly or uniquely affect those governmental entities. Each agency shall assess the effects of Federal regulations on State, local, and tribal governments, including specifically the availability of resources to carry out those mandates, and seek to minimize those burdens that uniquely or significantly affect such governmental entities, consistent with achieving regulatory objectives. In addition, as appropriate, agencies shall seek to harmonize Federal regulatory actions with related State, local, and tribal regulatory and other governmental functions.

(10) Each agency shall avoid regulations that are inconsistent, incompatible, or duplicative with its other regulations or those of other Federal agencies.

(11) Each agency shall tailor its regulations to impose the least burden on society, including individuals, businesses of differing sizes, and other entities (including small communities and governmental entities), consistent with obtaining the regulatory objectives, taking into account, among other things, and to the extent practicable, the costs of cumulative regulations.

(12) Each agency shall draft its regulations to be simple and easy to understand, with the goal of minimizing the potential for uncertainty and litigation arising from such uncertainty.

IV.1. Key Elements of the Regulatory Impact Review

The key elements of the Regulatory Impact Review (RIR) for NMFS management actions include --

- A description of the management goals and objectives;
- A description of the fishery and/or other affected entities;
- A statement of the problem;
- A description of each selected alternative,² including the “no action” alternative; and,
- An economic analysis of the expected effects of each selected alternative relative to the baseline.³

² Throughout this document, the term “selected alternatives” refers to the alternatives a Council or NMFS determines will be analyzed in the RIR and RFAA (Regulatory Flexibility Act Analysis).

³ The baseline is what is likely to occur in the absence of the proposed action, i.e., the status quo. See discussion of alternatives on page 12.

If these elements are already included in another section of the document, the appropriate section must be referred to under the RIR.

If a proposed action is determined to be significant under E.O. 12866, the analysis undergoes further scrutiny by the Office of Management and Budget (OMB) to ensure that it meets the requirements of E.O. 12866. A “significant regulatory action” means any regulatory action that is likely to result in a rule that may --

- Have an annual effect on the economy of \$100 million or more or adversely affect in a material way the economy, a sector of the economy, productivity, competition, jobs, the environment, public health or safety, or State, local, or tribal governments or communities;
- Create a serious inconsistency or otherwise interfere with an action taken or planned by another agency;
- Materially alter the budgetary impact of entitlements, grants, user fees, or loan programs or the rights and obligations of recipients thereof; or
- Raise novel legal or policy issues arising out of legal mandates, the President's priorities, or the principles set forth in the Executive Order.

OMB's Guidelines for the Conduct of Regulatory Analyses prepared under E.O. 12866 was issued as OMB Circular No. A-4 on September 17, 2003. The Circular provides guidance to Federal agencies on the development of regulatory analysis, as required under Section 6(a)(3)(c) of Executive Order 12866, “Regulatory Planning and Review,” the Regulatory Right-to-Know Act, and a variety of related authorities. The Circular also provides guidance to agencies on the regulatory accounting statements that are required under the Regulatory Right-to-Know Act. These Guidelines became effective for economically significant proposed rules submitted to OMB on or after January 1, 2004, and for economically significant final rules submitted on or after January 1, 2005. Circular A-4 can be obtained at:

<http://www.whitehouse.gov/omb/circulars>

Economically significant rules are those expected to have an annual effect on the economy of \$100 million or more or adversely affect in a material way the economy, a sector of the economy, productivity, competition, jobs, the environment, public health or safety, or State, local, or tribal governments or communities. Both benefit-cost analysis and cost-effectiveness analysis provide a systematic framework for identifying and evaluating the likely outcomes of alternative regulatory choices. An economically significant rulemaking should be supported by both types of analysis wherever possible. OMB specifically suggests that cost effectiveness analysis be prepared for all economically significant rulemakings for which the primary benefits are improved public health and safety to the extent that a valid effectiveness measure can be developed to represent expected health and safety outcomes. While Circular A-4 has replaced former guidance on regulatory analysis of economically significant rules (specifically, Section 1 of memorandum m00-08, March 22, 2000), NMFS suggests that the regulatory analysis

performed for significant rules be referred to as a Regulatory Impact Assessment (RIA) instead of RIR to clarify that the rule is economically significant (Please see Presidential Review of Agency Rulemaking by OIRA, September 20, 2001, http://www.whitehouse.gov/OMB/infoereg/oira_review-process.html). However, the content and scope of the RIA has to be sufficient to meet the requirements of E.O. 12866 using the guidelines found in Circular A-4.

RIR “Objectives” Section. The management objectives should be discussed or referred to so that they can be used as criteria in the evaluation of the potential success or failure of each alternative management measure.

Objectives must often take into account the requirements of multiple laws and mandates, such as the Magnuson-Stevens Act, E.O. 12866, RFA, NEPA, Unfunded Mandates Reform Act (UMRA), ESA, and MMPA. These laws and mandates should be referenced, as appropriate. A summary of the requirements of these other applicable laws is included in Appendix A of these NMFS Guidelines.

RIR “Description” Section. This section of the RIR provides a description of affected entities. For example, the RIR for fishery management actions should include a depiction of how the fishery is conducted, the utilization pattern, trends, observed deviations, and the current status. This description should provide managers with insight into who is fishing, when and where fishing occurs, what species are targeted and caught, what is retained and what is discarded, the numbers and sizes of businesses involved in the fishery and supporting activities, the number, size, and location of dependent communities, and the relationship of various segments of the affected industry.

RIR “Problem Statement” Section. The problem statement should identify the problem that the proposed action intends to address (including, where applicable, the failures of private markets or public institutions that warrant new action by the agency), as well as assess the significance of that problem. It should also examine whether existing regulations (or other law) have created, or contributed to, the problem that a new regulation intends to correct and whether those regulations (or other law) should be modified to achieve, more effectively, the intended goal of the new regulation.

RIR “Alternatives” Section. NMFS, not the analyst, is required to ensure that a range of feasible alternatives is included in the regulatory document. Although no minimum number of alternatives must be analyzed, NMFS should consider the “no action” alternative and the most significant other alternatives. The “no action” alternative should be the basis of comparison for other alternatives. However, the “no action” alternative does not necessarily mean a continuation of the present situation, but instead is the most likely scenario for the future, in the absence of other alternative actions.

Many times, alternatives are eliminated from further consideration early in the regulatory process. Examples include alternatives that are determined to be infeasible for various reasons, or which are precluded by law. To enhance the administrative record, these eliminated alternatives should be included, perhaps in an appendix to the final document, with a brief explanation of why they were eliminated from further consideration. However, these “considered and rejected alternatives” should not be analyzed in the RIR.

Each selected alternative should be described completely. Information should be presented in detail on the regulatory measures to be imposed, the process of implementing the measures, and the timing for implementation.

IV.2. Analysis of Alternatives

The objective of the analysis is to describe clearly and concisely the economic effects of the various alternatives. This will enable the agency to determine the regulatory alternative that maximizes net benefits to the Nation, including potential economic, environmental, public health and safety, and other advantages, distributive impacts, and equity. Economic analysis can provide a quantitative or qualitative estimate of changes in net benefits, expressed in both monetary and non-monetary terms. Economic analysis can also provide the basis for describing the distributive impacts of regulatory actions. For distributive analyses, characterization of the magnitude and the direction of change in the distribution of benefits and costs of regulatory actions are of principal concern. If substantial differences in distributive impacts among the selected alternatives exist, the analysis can provide an estimate of the differences in net economic benefits among the selected alternatives, as well as an estimate of the distributional differences among those alternatives.

As a basis for estimating the economic impacts of the management measures, the analyst should compare, in a straightforward manner, how each alternative would impact the Nation when compared to the no-action alternative. The cumulative impact of each selected alternative within a management measure should be analyzed to the extent practicable. The analyst should also consider each element of a proposed measure separately when a selected alternative contains a number of distinct measures. For example, if a Council proposes three separate alternatives for a fishery, and each includes minimum fish size, possession limit / trip limit, and closed season, the individual and combined effects of each management measure should be analyzed in context (i.e., by alternative). The components of the analysis are described below.

IV.3. Identification of Expected Effects

The types and direction of expected effects on the living marine resources, their habitats, and those who benefit from these resources should be discussed. The types of effects to consider include the following:

- Changes in net benefits within a benefit-cost framework.
- Changes in benefits and costs of groups of individuals, businesses of differing sizes, and other entities (including small communities and governmental entities).
- Changes in income and employment.
- Cumulative impacts of regulations.
- Changes in other social concerns.

Such effects may be the product of regulatory action-induced changes to the following:

- The goods and services that are available from the use and existence of living marine resources and the benefits they provide.

- The factors of production (e.g., capital, labor, and living marine resources) used to provide those goods and services, the cost of and returns from using the factors of production, and the payments made for their use.

IV.3.a. Changes in Net Benefits within a Benefit-Cost Framework

Benefit-cost analysis is conducted to evaluate the net social benefit arising from changes in consumer and producer surpluses that are expected to occur upon implementation of a regulatory action. The proper comparison is 'with the action' to 'without the action,' rather than to 'before and after the action,' since certain changes may occur even without action and should not be attributed to the regulation. In general, benefits are measured by willingness to pay (WTP), and costs, by opportunity costs. Opportunity costs reflect the foregone benefits from the use of a resource in one activity as compared to the best alternative use.⁴

Benefits may accrue as surpluses to consumers and/or producers. Total Consumer Surplus (CS) is the difference between the amounts consumers are willing to pay for products or services, and the amounts they actually pay. Thus, CS represents net benefits to consumers. Net benefit to producers is producer surplus (PS). Total PS is the difference between the amounts producers actually receive for providing goods and services, and the economic cost producers bear to do so. Economic costs are measured by the opportunity cost of all resources, including the raw materials, physical capital and human capital used in the process of supplying these goods and services to consumers.

Benefits and costs may accrue to consumers or producers not only through their own direct activity, but also through changes in public expenditures or receipts that may redirect resources for use elsewhere in the economy. From a social perspective, many public expenditures represent transfer payments, in that they do not require consumption of any additional resources. However, public expenditures provide a variety of services that do have value. For example, enforcement of fishery regulations provides economic benefits to the Nation by improving compliance behavior. That greater compliance provides enhanced assurance that the regulations will achieve their intended purpose. From a budgetary perspective, the cost of enforcement is equivalent to the total public expenditure devoted to enforcement. The economic cost of enforcement is measured by the opportunity cost of devoting resources to enforcement vis à vis some other public or private use and/or by the opportunity cost of diverting enforcement resources from one fishery to another.

Benefits and costs are measured from the perspective of the Nation, rather than from that of private firms or individuals. Benefits enjoyed by other nations are not included, although tax payments by foreign owners, and export revenues, are benefits to the Nation. Because of the national accounting stance, opportunity costs (whenever possible), rather than accounting costs, are employed. Forgone interest, depreciation, some taxes, and subsidies are considered transfer

⁴ Alternatively, willingness to accept (WTA) may be used instead of WTP. The choice of benefit measure will depend upon the policy context of the regulatory change. WTP will be appropriate when measuring benefits for increment in market or non-market goods. Although, under certain circumstances, WTP and WTA could yield theoretically equivalent surplus measures, WTA may be preferable when valuing decrements in market or non-market goods.

payments, from the perspective of society, rather than expenditures of real resources and, hence, are considered private rather than economic costs. Secondary costs and benefits are generally excluded when opportunity cost or WTP is used to measure costs and benefits, since their inclusion would be double counting. For example, the benefits of a stock-rebuilding program may be reflected in increased values of participating vessels. Provided economic costs and benefits are measured as opportunity costs and WTP, respectively, the capital gains associated with increased vessel value would already be reflected in the benefit-cost analysis. If the analyst were to also add the increased value of capital assets, this would count the program benefits twice.

If there are no market distortions and all goods are traded in markets, CS and PS can, at least theoretically, be approximated by market demand and supply curves. PS can alternatively be calculated from revenue and cost data, using opportunity rather than accounting costs.

When there is market failure or relevant market distortions, such as those that result from, say, non-competitive markets or regulated open-access management, market supply and demand curves and market prices are biased. The extent of the bias depends on many factors. It will often not be possible to measure the effect of these distortions, but their possible existence and direction of bias should be noted where applicable. A sensitivity analysis may help shed light on the importance of the bias.

Not all goods and services important to people are exchanged through markets, nor receive market prices. Including non-market values may be particularly important when considering amenities, such as habitat, ecosystem, recreational experiences, and protected resources, or issues affecting cultural heritage, historical and/or archeological assets, or other unique community resources. Non-market goods can be further categorized by whether they provide consumptive or non-consumptive use value (sometimes called direct and indirect use values), or non-use value. Non-use value includes existence value, which is the WTP for the continued existence of a good or service, over and above the WTP for potential or actual use of that asset, and bequest value, which is the WTP to conserve the good so that future generations can enjoy it. Whenever practicable, non-market values should be monetized (e.g., consumers' WTP) using appropriate valuation techniques, such as travel cost, stated preference, contingent valuation, or hedonic methods. In cases where estimation of non-market values may not be practicable, they must be treated qualitatively, in order to provide a complete accounting of costs and benefits attributable to an action.

For economic analysis of regulatory actions, changes in net benefits are measured by the difference in the present value of the discounted stream of net benefits of regulatory action, as compared to the status quo. In this context, a positive result means that the net present value of the regulatory action exceeds that of the status quo. Conversely, a negative result indicates that the status quo yields higher net present value than the regulatory action. Given that the primary purpose for the analysis is an assessment of how net benefits may be expected to change relative to the status quo, the analyst may choose to focus only on those economic costs and benefits that are expected to change. If, for example, fixed costs for fishing firms are expected to be unaffected, any change in costs may be fully captured by changes in operating costs, thereby obviating the need to estimate fixed costs. Similarly, if retail market supplies are not expected to change, due to ready availability of imports, a given regulatory action may have little or no

impact on consumers. In this instance, changes in net benefits will be fully captured by factors other than consumer surplus.

In instances where benefits are identical, regardless of the regulatory choice, or where a specific action is mandated by statute or some other binding ruling, a cost-effectiveness analysis may be used to make comparisons across alternatives. A cost-effectiveness analysis does not seek to determine whether or not regulatory action is warranted. Rather, a cost-effectiveness analysis seeks to find the regulatory design that minimizes costs. Typically, a cost-effectiveness analysis cannot be used to rank regulatory alternatives as compared to the status quo.

IV.3.b. Changes in the Distribution of Benefits and Costs

Changes in the distribution of benefits and costs reflect changes in the benefits and costs of groups of individuals, businesses of differing sizes, and other entities (including small communities and governmental entities). For businesses, the change in accounting profit can be used as a measure of the change in net benefits. Profit is a widely used term, but is generally understood to be the result of subtracting costs from gross receipts over a period of time. Defined in this manner, calculation of profit will be affected by differences in both cost-accounting conventions and accounting conventions applied to gross receipts. Similarly, the change in net benefits to governmental entities can be approximated by changes in revenues and costs using normal accounting practices.

The change in net benefits to consumers can be approximated by the change in consumer surplus, just as it would be in a benefit-cost analysis.

IV.3.c. Changes in Income and Employment

Regional economic models, including input-output models, can be used to estimate the regional income and employment impacts of alternative regulatory actions. These models provide measures of the changes in economic activity by region, not measures of net benefits. In the absence of these models, which can take substantial time and effort to develop and update, base sector models can be used, or qualitative assessments can be made.

IV.3.d. Cumulative Impact of Regulations

Imposition of more restrictive or multiple regulations to control symptoms of the common property externality may result in ever-increasing costs of management borne by fishermen and other regulated entities. Although the marginal economic effect of each regulatory action may be small, the cumulative impacts of several such actions over time may be large. Accounting for cumulative impacts may be of particular concern when multiple framework adjustments are made between full amendments to an FMP. Analyses that focus on the limited effects of a given management measure could miss important economic effects (both positive and negative) of the management program as a whole.

Accounting for cumulative impacts is difficult conceptually, and even more difficult to analyze. One of the difficulties is determining the benchmark to use in assessing the cumulative impacts. Conceptual or empirical analyses may need to explicitly account for the management history in a fishery, by including previous regulatory instruments in the analyses. By capturing these effects within the analytical framework, cumulative impacts can be made explicit for fishery managers.

IV.3.e. Changes in Other Social Concerns

The changes with respect to social concerns that are not captured in the preceding categories of effects should be addressed. Such concerns may be explicitly or implicitly identified in the problem statement, or they may arise during the development and review of alternative management actions.

IV.3.f. Qualitative Analysis of Expected Economic Effects

At a minimum, a qualitative analysis of the expected economic effects of each selected alternative to the status quo is required. In developing this section, the analyst first defines the baseline or "no action" condition, which provides the standard against which all other alternative actions are compared. The baseline is what is likely to occur in the absence of any of the proposed actions. Once the baseline condition is established, the incremental economic effects of each alternative, relative to the baseline, can be assessed. The specific economic effects to be analyzed should fall under the general areas of concern identified in Section IV.3.(a - e).

When quantifiable measures of the effects cannot be usefully estimated, because of the nature of the data and other resources available for the analyses, the types and models that would be required to usefully estimate such measures should be identified.

IV.3.g. Quantitative Analysis of Expected Economic Effects

If adequate data and models are available to provide useful estimates of quantifiable measures of the expected economic effects, a quantitative analysis of the effects of the selected alternatives should be substituted for the qualitative analysis described in the previous section, when this is the appropriate option. The quantitative analysis should use generally accepted methods to provide an understanding of the economic consequences of the selected alternatives. In many cases, only a small amount of quantification will be practicable. This could include, if appropriate, presenting empirical analysis from previously published sources, focus group input, expert opinion groups, as well as the analyst's own economic analysis. Any such information should be used in accordance with applicable statutes, such as the Federal Advisory Committee Act (5 U.S.C. App. 2). Good management requires that an effort be made to provide reasonably precise comparisons of the selected alternatives.

IV.4. Summary of Expected Economic Effects

E.O. 12866 defines net benefits in terms of potential economic, environmental, public health and safety, and other advantages; distributive impacts; and equity. With this very broad definition of net benefits in mind, the incremental benefits and costs of all alternatives, relative to the "no action" alternative as a baseline, should be summarized in the schedule. The schedule should (1) list all benefits and costs of each alternative, either monetized or non-monetized; (2) identify when the benefits and costs would occur; and (3) identify to whom the benefits and costs would accrue. All monetized benefits and costs should be in terms of their present value and should be presented as incremental changes relative to the baseline. Plausible ranges of estimates of benefits and costs should be provided where the estimate is sensitive to uncertain parameters, such as the rate of compliance, the rate of biological recovery, or other relevant variables.

IV.5. General Considerations in Developing the Analysis

IV.5.a. Forecasting

Evaluation of alternative actions should be based on the most likely conditions expected to exist in the future, with and without the proposed management actions. The forecast uses analysis of conditions expected to prevail without the proposed rule. The expected conditions may well differ from the existing conditions.

Forecasts should be made for selected years over the period of the analysis (see section c below) to indicate impacts of changes in economic and other conditions. During the period of analysis, if national or regional economic conditions are expected to change significantly, the changes should be factored into the analysis, if possible. For example, in the analysis of short-term effects, such factors as resource availability, utilization, and mobility may be considered in the analytical framework.

IV.5.b. Discount Rates

The costs and benefits that result from regulations usually occur at different times. Capital investments and some costs required by regulations tend to be concentrated at the outset, whereas benefits often occur at later dates. Some method must be used to permit comparisons between costs and benefits that have different time profiles. Discounting, which transforms future benefits and costs into "present values," should be utilized where appropriate. Direct comparisons between costs incurred and benefits realized at different time periods can then be made. The social discount rate used in an economic analysis may differ from the interest rate used in a private accounting analysis.

The Office of Management and Budget (OMB) has provided "Guidelines and Discount Rates for Benefit-Cost Analysis of Federal Programs" in Circular Number A-94 distributed by Transmittal Memorandum Number 64 (October 29, 1992). This Circular specifies certain discount rates that will be updated annually when the interest rate and inflation assumptions in the budget are changed. The goal of this circular is to promote efficient resource allocation through well-informed decision-making by the Federal Government. It provides general guidance for conducting benefit-cost and cost-effectiveness analyses. It also provides specific guidance on the discount rates to be used in evaluating Federal programs whose benefits and costs are distributed over time. Copies of the Circular may be obtained from the OMB Publications Office (202-395-7332). This information can also be obtained from the OMB web site: <http://www.whitehouse.gov/OMB/circulars/a094/a094.html>

Section 8.b.1. of the Circular specifies a real discount rate of 7 percent for computing net present value (NPV) when doing constant-dollar, benefit-cost analyses of proposed investments and regulations. Please note that the rates presented in Appendix C to OMB Circular No. A-94 do not apply to regulatory analysis or benefit-cost analysis of public investment. They are to be used for lease-purchase and cost-effectiveness analysis, as specified in the Circular. Circular A-4 encourages the analyst to present analyses using both 3 and 7 percent. The analyst may also use other discount rates, if the use of such alternative rates can be justified. Circular A-4 cautions that if you are uncertain about the nature of the opportunity cost, then you should present a higher discount rate as a further sensitivity analysis as well as using 3 and 7 percent.

An alternative that is often used is the social rate of time preference.⁵ Special approaches may also be appropriate when comparing benefits and costs across generations. One approach is to follow the discounting method discussed above, but to address the inter-generational equity, fairness, and uncertainty issues explicitly, rather than by modifying the discount rate.

IV.5.c. Period of Analysis

A general guideline for the period of analysis cannot be established for all fishery management actions, since there is such a wide diversity of possible situations and measures to be dealt with. The analyst should determine the appropriate period over which the analysis will be conducted, but, in all cases, he or she should provide an explanation of the specific period chosen that conforms to accepted benefit-cost analysis practices. For example, the period of analysis could reflect the time it takes for the fishery to move from its initial equilibrium, along the expansion path, to the final equilibrium point (including the time needed for the present value of costs and benefits to approximate zero) due to the adoption of the proposed regulation, holding all other influence constant. In some cases, the lack of necessary data will limit the period of analysis. However, a reasonable attempt should be made to conduct the analysis over a sufficient period of time to allow a consideration of all expected effects. Choosing a period of analysis that is too short may bias the analysis toward costs, where costs are incurred in the short-term and benefits are realized later. The period of the analysis should be the same for each alternative, including the "no action" alternative (i.e., all alternatives should be analyzed over the time frame that is appropriate for the alternative having the longest stream of costs and/or benefits).

IV.5.d. Risk and Uncertainty

The results of economic impact analyses should be examined to evaluate the uncertainty inherent in the data or in various assumptions. Areas of sensitivity should be described clearly so that decisions can be made with knowledge of the degree of reliability of the information presented. Situations of risk are defined as those in which the potential outcomes can be described in reasonably well-known distributions of benefits and costs. Situations of uncertainty are defined as those in which potential outcomes cannot be described with known probabilities. Reducing risk and uncertainty may involve increased costs or loss of benefits. The benefits and costs of reducing risk and uncertainty should be considered in the analytical and decision-making process.

Three fundamental types of analyses are possible. First, areas of risk and uncertainty can be qualitatively described. These qualitative descriptions are especially appropriate when reliable economic data or analytical models are unavailable. Second, a formal sensitivity analysis can be conducted in which important parameters are systematically varied and the

⁵ The social rate of time preference reflects the discount rate at which society is indifferent between a payment now and a correspondingly larger payment in a future year. It may be lower than the average real return on investment because, as a result of taxes and other distortions, individuals do not receive the full return on their investments. Most analysts use the average real rate on long-term Treasury bonds to represent the social rate of time preference. For the last 15 years, this rate has been in the range of 3 to 5 percent.

impact on expected economic effects evaluated. Sensitivity analysis most frequently varies key variables one at a time. Third, a formal risk analysis can be conducted through Monte Carlo simulation. A formal risk analysis provides expected values and distributions for a given probability distribution. A key consideration is the possible correlation among variables and the appropriate level of aggregation of variables. The use of conservative or best estimates or the use of a risk premium added to the social discount rate is not recommended.

IV.6. Information Requirements

Given the analytical requirements of the Magnuson-Stevens Act, ESA, MMPA, and other applicable laws, an economic analysis related to the performance of the relevant commercial and recreational users, non-consumptive users, processing sector, and retail or other market sectors is needed for the same period of time as the biological estimates. At a minimum, a qualitative analysis should discuss the relative magnitude of changes in performance. The qualitative components of the analysis should be replaced with quantitative components when this is the appropriate option. Information should be tailored to the sector(s) being analyzed, including commercial fishing and processing, recreational and subsistence fishing, and non-consumptive uses of fishery or other living marine resources. Examples of the information that should be provided in an RIR, if relevant to the analysis, may include the following:

- Expected levels or changes in participation (number of fishing vessels and/or anglers, etc.) and activity (number of fishing trips, days at sea, etc.).
- Expected levels or changes in harvests (commercial, recreational, and subsistence) and their distribution by sector.
- Expected levels or changes in non-consumptive use of the resource.
- Expected changes in prices (commercial ex-vessel prices and recreational access prices (e.g., charter fees)).
- Expected changes in harvesting costs (fixed and variable costs, including capital and labor costs), as well as equivalent costs for non-consumptive use activities.
- Expected levels and costs of processing.
- Expected changes in benefits or costs incurred by specific user groups, including effects on small entities.
- Expected effects on employment.
- Expected effects on profits, competitive position, productivity or efficiency of individual fishermen, user groups, or fishing communities.
- Expected effects on the reporting burden.
- Expected impacts on recreational and subsistence use, including changes in participation and catch rates and, to the extent practicable, their consumer surplus; for subsistence fishing, food and cultural availability.
- Expected management and implementation costs attributable to the action, including enforcement costs.
- Expected effects on non-use values.
- Expected effects on fishing capacity.

In addition, the MMPA and the ESA related economic analyses may need to include information in addition to those listed above. By way of example, Section 4(b)(2) of the ESA,

requires NMFS to designate critical habitat for listed species, based on the best available scientific data and after taking consideration the economic and other impacts. A critical habitat designation does not apply to citizens engaged in activities on private lands that do not involve a Federal agency, and only affects activities where Federal funding, permits, or projects are involved.⁶ The economic analysis serves to estimate impacts from designating an area as critical habitat. Upon consideration of economic impacts and other impacts such as those to national security, an area may be excluded from critical habitat designation by NMFS or the Secretary of Commerce when the benefits of such exclusion outweigh the benefits of inclusion, unless the failure to designate the area as critical habitat will result in the extinction of the listed species. Restrictions within areas designated as critical habitat only impact those projects in which there is a Federal nexus, triggering a Section 7 consultation under ESA. Meetings with relevant stakeholder agencies, members of the business community and general public would aid the process of gathering this information. While the information needed for the economic analyses may vary based on regional and species-specific habitat requirements (for instance, land-based habitat requirements may be prescribed as well), some information that may need to be considered in the relevant economic analyses in addition to those items listed above include, but are not limited, to the following:

- Federal permits (e.g. section 404 and National Pollutant Discharge Elimination System (NPDES) storm water permit under the Clean Water Act, requirements under the Federal Power Act)
- Federal funding for projects (such as funding from the Federal Highway Administration for transportation projects)
- Expected impacts due to potential changes in business practices (including farming, timber sales) and administrative costs from having to enter into Section 7 consultation, if there was no need to consult in the absence of critical habitat designation (e.g. project delays), and
- Expected impacts of implementing mitigation measures.

Other MMPA and ESA related actions may also require consideration of “non-fisheries” impacts to coastal industries such as those related to travel and tourism, equipment rental, or commercial shipping, to name a few examples. These impacts will vary regionally and by design of particular actions.

IV.7. Analytical Procedures

In general, the complexity of the analytical framework that should be used depends on the scope and magnitude of the problem, the number of regulatory alternatives, and the ability to measure the economic effects.

⁶Under Section 7 of the ESA, all Federal agencies must ensure that any actions they authorize, fund, or carry out are not likely to jeopardize the continued existence of a listed species, or destroy or adversely modify its designated critical habitat.

Generally accepted methodologies should be used in determining the economic effects of each selected alternative. Specific methodologies for examining the economic effects of alternative management actions are not detailed here because such methodologies are well documented elsewhere (see bibliographic references and suggested readings for literature on methodologies).

IV.8. Analysis of Framework Management Measures

The purpose of a framework measure is to "build in" flexibility to provide the opportunity to adjust to problems caused by the natural variability of a fishery and/or the lack of complete information early in the decision-making process.

NMFS Operational Guidelines for the Fishery Management Plan Process require that every framework measure be analyzed and that the analysis be available to the public for comment at some time prior to implementation. The analysis may be provided at the same time the framework is added to the FMP, or it may be provided subsequently when the framework action is actually taken. The extent of analysis, notification, and comment required will depend on the specificity and analysis provided when the framework was established.

The critical decision points where flexibility is required must be identified in framework measures. Also, the exact manner in which the framework will allow decisions to be made at those points must be described. It is necessary to show how this framework and its decision process will affect expected or average values of the important variables under various management alternatives.

When no further analysis is provided for proposed measures under a framework action, the analyst should clearly show that the current situation in the fishery has not changed from the time the analyses were done and that the specific regulatory action to be taken under the framework was analyzed adequately.

IV.9. Exemption from the Peer Review Bulletin

The Final Information Quality Bulletin for Peer Review excludes Regulatory Impact Analyses (RIAs) including RFAAs from coverage under the Bulletin. However, original data and formal analytical models used by agencies in economic analyses to support RIAs will be subject to peer review. The general benefit-cost model, as suggested by Circular A-4, is an accepted method of estimating impacts under E.O. 12866 and would be exempt from peer review. RIA documents themselves are already reviewed through an interagency review process under E.O. 12866 that involves application of the principles and methods defined in OMB Circular A-4.

V. REGULATORY FLEXIBILITY ACT PROCESS AND ANALYSIS

The purpose of the RFA (5 U.S.C. 601 et seq.) is to establish a principle of regulatory issuance that agencies shall endeavor, consistent with the objectives of the rule and of applicable statutes, to fit regulatory and informational requirements to the scale of businesses, organizations, and governmental jurisdictions subject to regulation. To achieve this principle, agencies are required to solicit and consider flexible regulatory proposals and to explain the rationale for their actions to assure that such proposals are given serious consideration.

With the exception discussed below, the RFA requires agencies to conduct an Initial Regulatory Flexibility Analysis (IRFA) and Final Regulatory Flexibility Analysis (FRFA) for each proposed and final rule, respectively. The IRFA and FRFA are designed to assess the impacts that various regulatory alternatives would have on small entities, including small businesses, and to determine ways to minimize adverse impacts. Under the RFA, an agency does not need to conduct an IRFA or FRFA if a certification can be made that the proposed rule, if adopted, will not have a significant adverse economic impact on a substantial number of small entities.

It should be emphasized that the RFA does not require that the alternative with the least cost or with the least impact on small entities be selected as the preferred alternative. The RFA does not contain any decision criteria; instead the purpose of the RFA is to inform the agency, as well as the public, of the expected economic impacts of the various alternatives contained in the FMP, FMP amendment, or other regulatory document and to ensure that the agency considers alternatives that minimize the expected impacts while meeting the goals and objectives of the regulatory documents and applicable statutes. Note that, when an FRFA is prepared, it must include a statement of the factual, policy, and legal reasons for selecting the alternative adopted and explain why each of the other significant alternatives to the rule considered by the agency which offset the impact on small entities was rejected.

A good regulatory flexibility analysis will ensure that --

- Reasonable alternatives to the preferred alternative/proposed action are identified.
- Managers understand and are able to address public comment on the more controversial aspects of a regulatory flexibility analysis.
- The proposal competes well against other social goals, regardless of legislative mandates, in light of other administration priorities.
- The proposal will move rapidly through the regulatory process at OMB and SBA's Office of Advocacy.
- The proposal is likely to withstand legal challenges.

The RFAA must address the impacts of a proposed rule only on small entities subject to the regulation (i.e., small entities to which the rule will directly apply) and not on all small entities that are affected by the regulation (i.e., small entities to which the rule will indirectly apply). This guidance provides for examining subsets of entities to which the rule will apply if the rule is likely to affect some of those entities differently than others. (See section V.1. "Steps for Fulfilling the RFA Requirements" for guidance on tiering.) It is important to note that under the E.O. 12866 requirement for analysis of burdens of a regulation on small entities, as contained in the eleventh principle, the economic impact on all small entities that are affected by a regulation, regardless of whether they are directly or indirectly impacted, is analyzed. However, impacts under E.O. 12866 need not be identified at the vessel or firm level in the RIR, whereas, these levels remains the focus of the RFAA.

V.1. Steps for Fulfilling the RFA Requirements

The steps for conducting the RFAA may be done in a number of ways, with responsibilities shared among the Councils, NMFS' Regions and Centers, and Headquarter's offices, but NMFS retains the ultimate responsibility to ensure that an adequate RFAA is completed.

Part 121 of Title 13, Code of Federal Regulations (CFR), sets forth, by North American Industry Classification System (NAICS) categories, the maximum number of employees or average annual gross receipts a business may have to be considered a small entity for RFAA purposes. See 13 C.F.R. § 121.201. Under this provision, the U.S. Small Business Administration established criteria for businesses in the fishery sector to qualify as small entities. As of August, 2006, the size standards for commercial fishing and other fishing related businesses are noted below. In addition to those size standards, there may be non-fishery related entities that are directly impacted by NMFS regulations, especially in regard to economic analysis of protected species or critical habitat under the ESA and MMPA. All size standards are subject to change based on price inflation or other economic indicators. For this reason, the analyst is advised to check the SBA Office of Size Standards website prior to conducting any RFAA: <http://www.sba.gov/size/index.html>

Provision is made under SBA rules, for an agency to develop its own industry-specific definitions. Under this provision, for example, NMFS (after consultation with the Office of Advocacy at the U.S. Small Business Administration and an opportunity for public comment) may establish size criteria that differ from those established by the SBA Office of Size Standards, but only for use by NMFS, and further, only for purposes of conducting an analysis of economic impacts, in fulfillment of the agency's obligations under the RFA. To utilize this provision, NMFS must publish such size standards in the Federal Register. An example of this is the catcher-processor definition, presently under discussion, found below.

The RFA recognizes and defines three kinds of small entities: small businesses, small organizations, and small governmental jurisdictions. The current size standards for Magnuson-Stevens Act related rules are as follows:

Any fish-harvesting or hatchery business is a small business if it is independently owned and operated and not dominant in its field of operation and if it has total annual gross receipts not in excess of \$4.0 million. Total annual gross receipts should include those of affiliates when practicable and appropriate to do so.

Any vessel which both harvests and processes fish (also referred to as a catcher-processor) is currently considered a small business if its combined total annual gross receipts (including all affiliates, worldwide, where practicable and appropriate) are not in excess of \$4.0 million. However, NMFS is currently proposing a new size standard for catcher-processors operating in the Pacific and North Pacific to be combined total annual gross receipts not in excess of \$20.0 million. A final determination on this size standard has not yet been made.

For related industries involved in seafood product preparation and processing (NAICS 31171) including canned and cured fish and seafood (NAICS 31711) and fresh and

frozen seafood processing (NAICS 31172), a small business is one that employs 500 or fewer employees.

For the wholesale industry, a small business is one that employs 100 or fewer employees.

For marinas and charter/party boats, a small business is one with total annual gross receipts not in excess of \$6.5 million.

A small organization is any not-for-profit enterprise that is independently owned and operated and not dominant in its field.

A small government jurisdiction is any government or district with a population of 50,000 or fewer persons.

Although, at a minimum, the RFA requires a bifurcation between small and large entities, the analyst may choose to create classes or tiers from among the identified universe of small entities, when appropriate. The creation of separate classes of small entities may be appropriate when a regulatory action is expected to have differential impacts on firms, based on their sizes or other characteristics. For example, even among a directly regulated class of vessel operators all of whom meet the SBA size criterion for inclusion as “small,” for RFAA purposes, the smallest vessels among these may be less able to adapt to a regulatory action than the relatively larger vessels, due to their limited range. In such a case, it may be appropriate to identify a “universe” of entities, from among those already defined as “small” under SBA criteria, and evaluate any unique, disproportionately burdensome aspects of the proposed action accruing to these operators.

V.1.a. Certification Process

The Regulatory Flexibility Act requires Federal agencies to conduct a full RFAA unless the agency can certify that the proposed and/or final rule would not have a “significant economic impact on a substantial number of small entities.” This determination can be made at either the proposed or final rule stage. If the agency can certify, it need not prepare an IRFA, a FRFA, or a “Small Entity Compliance Guide” (Guide), or undertake a subsequent periodic review of such rules.

The information from the PREE, or from other relevant economic analyses, may indicate whether there is or is not a factual basis upon which to certify that the preferred alternative would not have a “significant adverse economic impact on a substantial number of small entities.” When the factual basis to certify exists, the agency has the option of certifying. The decision on whether or not to apply certification criteria or to certify should be made after the final decision on the preferred alternative. This will ensure that this process is done only once for a particular regulatory action. The certification process is the only time NMFS must make a determination that a rule will not have a significant economic impact on a substantial number of small entities. For IRFAs and FRFAs, it is not required to perform an analysis for significance or substantial number provided that impacts of the selected alternatives are provided to the public.

The NMFS Regional Administrator/Office Director, using analyses and rationale provided by the Council or NMFS, prepares a memorandum from the Chief Counsel for Regulation (CC/Regs) of the Department of Commerce to the Chief Counsel for Advocacy of the Office of Advocacy at the U.S. Small Business Administration (SBA) certifying and setting forth the factual basis for the certification. Generally, the body of the letter is quoted in the classification section of the proposed rulemaking. The CC/Regs will sign and transmit the certification to SBA at the time the notice of proposed rulemaking or final rulemaking is published in the Federal Register, along with a statement providing the factual basis for such certification.

“Boilerplate” notice language should not be used by the agency in its statement on the factual basis for a certification or in the equally important ancillary requests for public comment. If the agency has conducted the appropriate analysis, it can offer clear, concise, declarative statements that address each of the six points below and reflect the specifics of the proposed rule.

The Office of Advocacy at the SBA recommends that the certification statement include the following:

1. A statement of basis and purpose of the rule. This should include the statutory basis for the regulation, and the objectives of the rule including a brief description of the context.
2. A description and estimate of the number of small entities to which the rule applies. This should describe how the universe of regulated entities was determined (and segmented) and details on the relevant economic and functional characteristics of those entities. This element should provide clear information on the range and scope of the regulation and the analysis which supports the certification.
3. Description and estimate of economic impacts on small entities, by entity size and industry. This should include the rationale for the certification decision, based on the criteria specified in the next element, as well as a summary of the basic analysis supporting that determination. Unlike the analysis for economic impact performed under the IRFA where all alternatives are analyzed (see below), for certification of no significant impact only the proposed alternative is analyzed.
4. An explanation of the criteria used to evaluate whether the rule would impose “significant economic impacts”. These guidelines suggest two criteria to consider in determining the significance of regulatory impacts, namely, disproportionality and profitability.⁷ These criteria relate to the basic purpose of the RFA, i.e., to consider the effect of regulations on small businesses and other small entities, recognizing that regulations frequently do not provide for short-term cash reserves to finance operations through several months or years until the positive

⁷ The concept of profitability may not be appropriate for a non-profit small organization or a small government jurisdiction. For these groups, disproportionality may be the appropriate standard.

effects of the regulation start paying off. If either criterion is met for a substantial number of small entities, the rule should not be certified.

Disproportionality. Do the regulations place a substantial number of small entities at a significant competitive disadvantage to large entities? If the answer is “Yes,” the rule should not be certified.

Whenever a disproportional effect on profits, costs, or net revenues is expected to occur for a substantial number of small entities, the test is adjudged to be met, and the rule should not be certified.

This criterion compares the effect of the regulatory action between small and large entities (using the SBA approved size definition of “small” entity), not the difference between segments of small entities.⁸ However, if an appreciable segment of small entities is disproportionately affected relative to large entities, even if the average small entity is not affected, the test would be adjudged to be met, and the rule should not be certified.

Profitability. Does the regulation significantly reduce profit for a substantial number of small entities? If the answer is “Yes,” the rule should not be certified. The thrust of the analysis should be short- and medium-term in nature. While 1 year may be considered short-term, the analyst may consider shorter periods, e.g., six months for which the fishery is open, or longer periods, e.g., two years after which the regulation sunsets. Whichever period is selected, the analyst must provide a rationale for that choice as well as a discussion of how the findings may be affected by the choice. Profit is a widely used term and is generally understood to be the result of subtracting costs from gross receipts over a period of time. Defined in this manner, calculation of profit will be affected by differences in both cost accounting conventions and accounting conventions applied to gross receipts. In general, the analysis should focus on the ability of the firm to meet both short-term (operating costs plus payments on other short-term obligations) and long-term debt (principal and interest payments on plant and equipment) obligations using generally accepted accounting practices for the regulated industry. The selected accounting practices will depend upon available data. Whichever accounting rules are selected, the analyst must describe the assumptions and should discuss how the findings may be affected by these assumptions.

Ultimately, the question the RFA analysis needs to answer is whether in the short- and medium-term, the costs (or reduction in revenues) imposed by the regulation can be absorbed by the firm (due to higher than average profitability) or passed on to its customers. If these costs (or reductions in revenues) cannot be absorbed so

⁸ Impacts within segments of small entities can be evaluated by the second criterion.

that either profits are reduced significantly or the solvency (ability to meet long term debt payments) of a substantial number of small entities is clearly threatened, then the impact of the rule is significant and the agency should not certify.

5. An explanation of the criteria used to evaluate whether the rule would impose impacts on “a substantial number” of small entities.

The term “substantial number” has no specific statutory definition and the criterion does not lend itself to objective standards applicable across all regulatory actions. Rather, “substantial number” depends upon the context of the action, the problem to be addressed, and the structure of the regulated industry. The SBA casts “substantial” within the context of “more than just a few” or De Minimis (“too few to care about”) criteria.⁹ In some cases consideration of “substantial number” may go beyond merely counting the number of regulated small entities that are impacted significantly. For example, a fishery may have a large number of participants, but only a few of them may account for the majority of landings. In such cases, a substantial number of small entities may be adjudged to be significantly impacted, even though there may be a large number of insignificantly impacted small entities.

Generally, a rule is determined to affect a substantial number of entities if it impacts more than just a few small entities. In a borderline case, the rule’s effect on the structure of the regulated industry or the controversiality of the rule might tip the balance in favor of determining that a substantial number of entities would be affected.

6. A description of, and an explanation of the basis for, assumptions used. This should describe the data sources and analytical methods used in the analyses, variability, and uncertainty in the cost and revenue estimates, explain the assumptions used, and indicate the extent to which the results were affected by those assumptions.

V.1.b. Initial Regulatory Flexibility Analysis

In addition to the economic impact analysis, Section 603 (b) of the RFA identifies the elements that should be included in the IRFA. These are as follows:

A description of the reasons why action by the agency is being considered.

A succinct statement of the objectives of, and legal basis for, the proposed rule.

A description and, where feasible, an estimate of the number of small entities to which the proposed rule will apply.

⁹ See page 18, U. S. Small Business Administration, Office of Advocacy, “A Guide for Government Agencies: How to Comply with the Regulatory Flexibility Act,” May, 2003.

A description of the projected reporting, record-keeping, and other compliance requirements of the proposed rule, including an estimate of the classes of small entities which will be subject to the requirements of the report or record.

An identification, to the extent practicable, of all relevant Federal rules, which may duplicate, overlap, or conflict with the proposed rule.

Each IRFA shall also contain a description of any significant alternatives to the proposed rule which accomplish the stated objectives of applicable statutes and which minimize any significant economic impact of the proposed rule on small entities. Consistent with the stated objectives of the applicable statutes, the analysis shall discuss significant alternatives such as --

The establishment of differing compliance or reporting requirements or timetables that take into account the resources available to small entities.

The clarification, consolidation, or simplification of compliance and reporting requirements under the rule for such small entities.

The use of performance rather than design standards.

An exemption from coverage of the rule, or any part thereof, for such small entities.

The IRFA should estimate the costs associated with each of the selected alternatives and identify the classes of small entities that will be subjected to the costs. The relevant costs include both direct compliance costs, reporting, record-keeping, and other administrative costs. Note that compliance costs are broadly defined to include the value of forgone fishing opportunities, increased operating costs, and costs associated with higher levels of debt servicing. The IRFA should compare the costs of compliance for small and large entities to determine whether any small entities are disproportionately affected. If all entities in the industry are small entities, the costs imposed on the typical, representative, median, or average entity in a particular segment of the industry should be analyzed. The resulting effects of business closures on production and employment should be estimated. Other sectors of the economy indirectly affected by the proposed rule should be considered in the RIR.

The discussions in the following two paragraphs refer to process, rather than analysis. Since the regulatory development process varies by region, in some cases the analyst would not be involved in the process described below. In cases in which the analyst is a Council staff member, he or she would likely be involved in this process.

As indicated above, the RFA requires consideration of alternatives that accomplish the stated objectives of the applicable statutes and that minimize any significant economic impacts on small entities. The IRFA should identify any significant alternatives to the proposed rule that would minimize economic impacts on small entities, if such alternatives exist. The RFA requires that the alternatives be part of the IRFA to ensure that the public will have adequate opportunity to comment on them and to suggest other alternatives. If there is an alternative with

less of an impact on small entities that meets the stated objectives, the IRFA should explain why the preferred alternative was selected over the alternative with lower impact.

The certification process is the only time NMFS must make a determination that a rule will not have a significant economic impact on a substantial number of small entities. For IRFAs and FRFAs, the analyst is not required to perform an analysis for significant impact or substantial number if impacts of the selected alternatives¹⁰ are provided to the public.

A rationale should be provided to explain any unavoidable adverse effects on small entities that are necessary to achieve the objectives. For documents that are prepared by the Councils, if a Council fails to fully comply with the RFA requirements for an IRFA, NMFS may elect to return a Council's recommendation as incomplete or may supplement a Council's IRFA submission by adding language to the preamble of the proposed rule. In such an instance, the IRFA will be considered to consist of a Council's submission as supplemented by the preamble.

V.1.c. Final Regulatory Flexibility Analysis

An agency must prepare a Final Regulatory Flexibility Analysis (FRFA) if it has published in the Federal Register a general notice of proposed rulemaking, unless the agency certifies that the rule, if adopted, will not have a significant adverse economic impact on a substantial number of small entities. NMFS prepares the FRFA at the end of the public comment period. The FRFA or a summary should normally be published in the Federal Register with the final rule.

Section 604(a) of the RFA identifies the elements that should be in the FRFA in addition to the analysis of impacts:

- A succinct statement of the need for, and objectives of, the rule.
- A summary of the significant issues raised by the public comments in response to the IRFA, a summary of the assessment of the agency of such issues, and a statement of any changes made in the proposed rule as a result of such comments.
- A description and an estimate of the number of small entities to which the rule will apply or an explanation why no such estimate is available.
- A description of the projected reporting, record-keeping, and other compliance requirements of the rule, including an estimate of the classes of small entities which will be subject to the requirement and the type of professional skills necessary for the preparation of the report or record.
- A description of the steps the agency has taken to minimize the significant adverse economic impact on small entities, consistent with the stated objectives of applicable statutes, including a statement of the factual, policy, and legal reasons for selecting the alternative adopted in the final rule and the reason that each one of the other significant alternatives to the rule considered by the agency was rejected.

¹⁰ See footnote 1

The FRFA may be based on the IRFA but should reflect new data developed during the comment period and comply with the above requirements. Often, in order to comply, the FRFA will consist of the IRFA and of portions of the preamble to the final rule.

The remaining sections deal with process, rather than analysis. Since the regulatory development process varies by region, in some cases the analyst would not be involved in the process described in the remaining sections. In other cases where the analyst is a council staff member, he or she would likely be involved in this process.

V.2. Small Entity Compliance Guides

Pursuant to the Small Business Regulatory Enforcement Fairness Act (SBREFA) of March 29, 1996 (P.L. 104-121), NMFS is required to publish one or more guides to assist small entities in complying with a rule or group of related rules when a final regulatory flexibility analysis is prepared under section 604 of title 5, United States Code.

SBREFA requires that Federal agencies explain the actions a small entity is required to take to comply with a rule or group of rules. The agency shall, in its sole discretion, taking into account the subject matter of the rule and the language of relevant statutes, ensure that the guide is written using sufficiently plain language likely to be understood by affected small entities. Agencies may prepare separate guides covering groups or classes of similarly affected small entities, and may cooperate with associations of small entities to develop and distribute such guides.

As part of the rulemaking process, whenever a final regulatory flexibility analysis is prepared, a small entity compliance guide(s) must be made available through various avenues to small entities to which the rule will apply. Such avenues could include direct mailing of compliance guides to affected entities and various organizations, along with appropriate access to the same via publication in the Federal Register, and distribution of brochures or leaflets made available at in areas or locations frequented by the affected entities. Compliance guides, must be included on NMFS's websites and distributed to affected parties upon publication of the final rule in the Federal Register, or shortly thereafter.

Every FRFA or FRFA summary should contain the following paragraph, or a variation thereof.

Section 212 of the Small Business Regulatory Enforcement Fairness Act of 1996 states that, for each rule or group of related rules for which an agency is required to prepare a FRFA, the agency shall publish one or more guides to assist small entities in complying with the rule, and shall designate such publications as "small entity compliance guides." The agency shall explain the actions a small entity is required to take to comply with a rule or group of rules. As part of this rulemaking process, a small entity compliance guide was prepared. [Insert recipients of the small entity compliance guide, e.g., "The guide was sent to all vessel owners issued a (name of species) permit, and to all Federal dealers issued a (name of species) permit."] In addition, copies of this final rule and guide (e.g., a permit holder letter) are available from the Regional Administrator (see ADDRESSES) and are also available at the following web site: <insert primary website for lead office>.

V.3. Waiving or Delaying the Preparation of an RFAA

The requirement to prepare some or all of an IRFA may be waived or delayed by an agency head when an emergency makes compliance impracticable. To effectuate such a delay or waiver, a notice must be published in the Federal Register, no later than the date of publication of the final rule. That publication must include a written finding, with reasons therefore, that the final rule is being promulgated in response to an emergency that makes timely compliance with the requirements to prepare an IRFA impracticable.

An agency head may delay completion of a FRFA up to 180 days after the final rule is published in the Federal Register, by publishing in the Federal Register, no later than the date of publication of the final rule, a written finding, with reasons that the final rule is being promulgated in response to an emergency that makes compliance with the requirements to prepare a FRFA impracticable. Note that preparation of an FRFA may not be waived. The rule will lapse and have no effect if a FRFA is not prepared within this time period. Further, the rule may not be re-promulgated until a FRFA has been prepared.

V.4. Relationship of the Regulatory Flexibility Act to Other Applicable Law

The RFA requires that the agency identify and consider alternatives that minimize the adverse impacts of a regulation on small entities subject to the regulation, but it does not require that the agency select any particular alternative, such as the alternative with the least cost or with the least adverse impact on small entities. However, if there is an alternative (other than the preferred) with less of an adverse impact on small entities, rationale must be provided for selecting the preferred over that alternative. Section 606 of the RFA (5 U.S.C 606) states that the requirements to prepare an IRFA and a FRFA do not alter standards otherwise applicable by law to agency action. Regardless of the requirement to conduct an RFAA (or, for that matter, an RIR), the regulatory action taken must be consistent with the Magnuson-Stevens Act and with other applicable law.

V.5. Involvement of Small Entities in the Rulemaking

The RFA mandates that, if a rule will have a significant economic impact on small entities, the agency involved will take steps to assure that small entities will have an opportunity to participate in the rulemaking. Possible steps suggested by the RFA include the following:

- Providing a statement accompanying an advance notice of rulemaking that the proposed rule might have a significant economic impact on a substantial number of small entities.
- Publishing a notice in publications likely to be obtained by small entities.
- Directly notifying affected parties, including representatives of participants in adjacent areas.
- Conducting open conferences or public hearings, intending to include representatives of fisheries that might be affected by possible regulatory changes. The chances of public acceptance of regulations are improved by involving all concerned/affected groups in all phases of the process, including data collection.

The Magnuson-Stevens Act process provides for public participation in FMP and amendment development. Public input (including small entities) is provided throughout the regulatory development process through Council members who represent coastal states, industry, and environmental groups; Advisory Panels; Scientific and Statistical Committees; Social and Economic Panels; Plan Development Teams; and ad hoc committees that Councils or the Secretary appoint when necessary. Public notification of each of these meetings is required, and public testimony is routinely taken. Further, some public meetings are recorded, and meeting summaries may be prepared. A record of the number of opportunities for small entity input may be constructed by listing the dates and locations of each public meeting held in which the proposed regulation was discussed. This record may be enhanced by including meeting summaries, attendance lists, and key issues identified by small entities. In many cases, this will satisfy the RFA requirements for public input (which must be documented in the FRFA).

V.6. Periodic Review of Significant Rules

The current Guidelines contain a requirement for periodic reviews of significant rules under Section 610 of the RFA. For NMFS purposes, significant rules are considered to be those for which a FRFA was prepared.

All rules containing a FRFA or FRFA summary should be reviewed except: (1) annual or multi-year specifications requiring proposed and final rulemaking, (2) in-season actions requiring proposed and final rulemaking, and (3) temporary or emergency rules requiring proposed and final rulemaking.

For each FMP or ongoing/long-term environmental plan the analyst should determine the final rules to be reviewed and compare each final rule with current regulations to ascertain current status of the final rule such as continuing, revised, or rescinded. If the rule has been rescinded cease the review and report the date, if available, of the rescission.

For revised and continuing rules the analyst should:

- (1) describe in qualitative terms the management measures contained in each final rule;
- (2) describe in qualitative terms the economic impacts of regulations presented in the FRFA and the public comment received on specific regulations;
- (3) determine if the rule overlaps, duplicates or conflicts with other Federal rules, and to the extent feasible with State and local government rules;
- (4) describe the length of time since the rule has been evaluated and the degree to which technology, economic conditions, or other factors have changed in the area affected by the rule;
- (5) assess the complexity of the rule;
- (6) recommend the continuation or rescission of the rule; and

- (7) explain the need for the rule to be continued or rescinded (for rules that are continued, explain the rationale for the conclusion that there is no available alternative that is consistent with stated objectives of applicable statutes and that would reduce significant impacts).

V.7. Exemption from the Peer Review Bulletin

The Final Information Quality Bulletin for Peer Review excludes Regulatory Impact Analyses (RIAs) including RFAAs from coverage under the Bulletin. However, original data and formal analytical models used by agencies in economic analyses to support RIAs will be subject to peer review. The general benefit-cost model, as suggested by Circular A-4, is an accepted method of estimating impacts under E.O. 12866 and would be exempt from peer review. RIA documents themselves are already reviewed through an interagency review process under E.O. 12866 that involves application of the principles and methods defined in OMB Circular A-4.

V.8. Preparation of “Stand-Alone” RFAA

The RFA does not require preparation of an individual, “stand-alone” RFAA. Federal agencies are afforded a great degree of discretion in terms of drafting their RFAAs, as long as the analytical and process requirements are met. According to the First Circuit Court of Appeals, “section 604 prescribes the content of a FRFA only - it does not demand a particular mode of presentation.” See Associated Fisheries of Maine, Inc. v. Daley, 127 F.3d 104(1st Cir. 1997). The RFA’s legislative history demonstrates that an agency “may incorporate in a regulatory flexibility analysis any data or analysis contained in any other impact statement or analysis required by law.” Therefore, the RFAA may be included in a NEPA document, as a stand-alone document, in the classification section of the rule in its entirety, or any combination.

V.9. Compliance with Executive Order 13272

Executive Order 13272 “Proper Consideration of Small Entities in Agency Rulemaking” (67 Fed. Reg. 53,461, August 16, 2002) requires Federal agencies to undertake additional consultation with the Chief Counsel for Advocacy of the United States Small Business Administration (SBA) in certain circumstances. Pursuant to the Executive Order, NMFS is required to notify SBA of any draft rule that may have a significant adverse economic impact on a substantial number of small entities under the RFA. The draft rule should be transmitted to SBA when NMFS submits a draft rule to the OIRA at the Office of Management and Budget under Executive Order 12866. In such a case, CC/Reg will submit the rule to SBA. If no submission to OIRA is required, NMFS shall transmit the draft rule to SBA within a reasonable time prior to publication (i.e., 3-5 business days in advance of publication). NMFS should consult with CC/Reg on the appropriate time for submitting a rule to SBA when submission to OIRA is not required. Furthermore, NMFS is required to consider any comments provided by SBA regarding a draft rule and respond to such comments in its final rule.

V.10. Preparation of an RFAA when NMFS Develops an Emergency Regulation or Interim Measure to Reduce Overfishing Under Section 305(c) of the Magnuson-Stevens Act

The analytical requirements of the RFA apply whenever an agency is required by law to issue a proposed rule. As such, when NMFS, under section 305(c) of the Magnuson-Stevens

Act, prepares an emergency regulation or interim measure to reduce overfishing, and publishes only a final rule pursuant to a good cause waiver or other APA exception, the analytical requirements of the RFA would not be applicable. In addition, if NMFS extends an emergency regulation or interim measure for an additional 180 days, consistent with section 305(c), there is no requirement to prepare an FRFA for the extension because the triggering event (i.e., notice of proposed rulemaking) did not occur.

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APPENDIX A: Summaries of Legislative Requirements of Other Applicable Law

Appendix A provides summaries of the legislative requirements for the National Environmental Policy Act (NEPA), Magnuson-Stevens Fishery Conservation and Management Act (Magnuson-Stevens Act), Unfunded Mandates Reform Act (UMRA), Congressional Review of Agency Rulemaking, Endangered Species Act (ESA), and Marine Mammal Protection Act (MMPA).

National Environmental Policy Act

The NEPA (42 U.S.C. 4371 *et seq.*) requires a report on any proposed major Federal actions significantly affecting the quality of the human environment. The National Oceanic & Atmospheric Administration (NOAA) policy requires NEPA analysis for significant fishery actions. The required analysis includes evaluation of the following: (1) fishery impact on species protected under the Endangered Species Act and the Marine Mammal Protection Act, and (2) impacts on non-target fish species (e.g., bycatch or other incidental fishing mortality), and on fishery habitats.

Often, the first step in complying with NEPA is to conduct an Environmental Assessment (EA), which is a brief analysis of the environmental impacts of the proposed action and its alternatives, including sufficient evidence to determine whether the action may have a significant impact on the human environment. Alternatively, if it is clear that the proposed action will have significant impacts, the agency may prepare an Environmental Impact Statement (EIS) without first preparing an EA. If the EA indicates the action will have no significant impact, including economic impacts, on the human environment, a Finding of No Significant Impact (FONSI) is prepared. If the proposed action may result in significant impact on the human environment, an EIS is required. An EIS is a detailed report that describes the proposed action, the need for action, alternatives considered, the environment affected by the action, and the environmental consequences of the proposed action and reasonable alternatives (NOAA Administrative Order 216-6).

Magnuson-Stevens Fishery Conservation and Management Act and the National Standards

The Magnuson-Stevens Act (16 USC 1801 *et seq.*) requires the development and implementation of conservation and management measures to prevent overfishing, rebuild stocks, and promote the long-term health and sustainability of fisheries. Under section 303(a)(9) of the Magnuson-Stevens Act, any fishery management plan must include a Fishery Impact Statement (FIS), which assesses, specifies, and describes the likely effects, if any, of the conservation and management measures on participants in the fishery or fisheries being managed, fishing communities, and participants in fisheries in adjacent areas. Analyses for FIS requirements should include assessments and descriptions of the economic and social impacts of the proposed action on various components of the fishery being managed, over the entire range of the regulated species, on participants in the fishery and in other fisheries, and on fishing communities.

Eight of the ten national standards for fishery conservation and management have implications for economic analysis:

1. National standard 1 requires that “Conservation and management measures shall prevent overfishing while achieving, on a continuing basis, the optimum yield from each fishery for the United States fishing industry;” where “optimum yield” is defined in terms of the amount of fish which will provide the greatest overall benefit to the Nation.

2. National standard 2 requires that “conservation and management measures shall be based upon the best scientific information available.”

3. National standard 4 requires that “Conservation and management measures shall not discriminate between residents of different states. If it becomes necessary to allocate or assign fishing privileges among various United States fishermen, such allocation shall be (A) fair and equitable to all such fishermen; (B) reasonably calculated to promote conservation; and (C) carried out in such manner that no particular individual, corporation, or other entity acquires an excessive share of such privileges.”

4. National standard 5 requires that “Conservation and management measures shall, where practicable, consider efficiency in the utilization of fishery resources; except that no such measure shall have economic allocation as its sole purpose.”

5. National standard 7 requires that “Conservation and management measures shall, where practicable, minimize costs and avoid unnecessary duplication.”

6. National standard 8 states that “Conservation and management measures shall, consistent with the conservation requirements of this Act (including the prevention of overfishing and rebuilding of overfished stocks) take into account the importance of fishery resources to fishing communities in order to (A) provide for the sustained participation of such communities, and (B) to the extent practicable, minimize adverse economic impacts on such communities.”

7. National standard 9 requires that “Conservation and management measures shall, to the extent practicable, (A) minimize bycatch; and (B) to the extent bycatch cannot be avoided, minimize the mortality of such bycatch.” NMFS has defined the term “to the extent practicable” to include a consideration of the effects of reducing bycatch and bycatch mortality on the overall benefit to the Nation.

8. National standard 10 requires that “conservation and management measures shall, to the extent practicable, promote the safety of human life at sea.”

Unfunded Mandates Reform Act of 1995

The UMRA (Public Law 104-4) requires agencies to prepare a report if a Federal rule that includes a Federal mandate may result in the expenditure by state, local, and tribal governments in the aggregate, or the private sector, of \$100 million or more (adjusted for inflation) in any one year. The report must --

1. Identify the Federal law under which the rule is being promulgated.

2. Provide a qualitative or quantitative assessment of the anticipated costs and benefits of the mandate, including an analysis of the extent to which costs may be paid with Federal financial assistance and to which there are available Federal resources to carry out the mandate.

3. Provide estimates of the future compliance costs of the mandate, any disproportionate budgetary effects on particular regions, state, local, or tribal governments, urban or rural or other types of communities, or segments of the private sector.

4. Provide estimates of the effect on the national economy (e.g., on productivity, economic growth, full employment, creation of productive jobs, international competitiveness).

5. Describe the agency's consultation with elected representatives of the state, local, or tribal governments.

6. Summarize comments received.

7. Summarize the agency's evaluation of the comments.

8. Identify and consider a reasonable number of alternatives.

It should be noted that UMRA has a decisional criterion that the RFA does not have. A Federal agency is required to select the alternative with the least cost or with the least impact. Specifically, if a report is required, the agency must select the least cost, most cost effective, or least burdensome alternative that achieves the objectives of the rule.

Congressional Review of Agency Rulemaking

Under the Congressional Review of Agency Rulemaking Act (5 U.S.C. 801 et seq.), prior to promulgating a rule, agencies are required to submit to each house of Congress and to the Comptroller General a copy of a proposed rule, a statement as to whether it is a major rule, and the proposed effective date of the rule. If the rule is a major rule, the Comptroller General is required to report to Congress on whether the agency has complied with benefit-cost analyses required by E.O. 12866, the UMRF, the RFA, and any other applicable law or Executive Order.

Endangered Species Act

Section 7 of the ESA (16 U.S.C. 1531 et seq.) requires Federal agencies to use their authorities to conserve endangered and threatened species. The ESA does not allow consideration of economic impacts in making species listing decisions. Public comments are solicited before a final decision is made on the listing. Critical habitat necessary for the continued survival of a species should be designated at the time a species is listed. Section 4(b)(2) of the ESA requires that designation of critical habitat be based on the best scientific data available and after considering economic impacts, the impact on national security, and any other relevant impact, of specifying any particular area as critical habitat. The Secretary of Commerce

may exclude an area from critical habitat if the benefits of exclusion outweigh the benefits of designation, unless excluding the area will result in the extinction of the species concerned. A critical habitat designation only affects activities where Federal funding, permits, or projects are involved.¹¹ These requirements do not apply to citizens engaged in activities on private land that do not involve a Federal agency.

In terms of fishery management actions, if a proposed action may affect an endangered or threatened species, a Section 7 consultation must be conducted to ensure that the action is not likely to jeopardize the continued existence of the species or adversely modify critical habitat.

The ESA also requires the development of recovery plans. The recovery planning process must include site-specific management measures as well as estimates of the time and costs required to carry out those measures. The recovery planning process may include economic considerations.

Marine Mammal Protection Act

The MMPA (16 U.S.C. 1361 *et seq.*) recognizes that certain species of marine mammals are in danger of extinction or depletion. All [marine mammals](#) are protected under the MMPA. The MMPA prohibits, with certain exceptions, the take of marine mammals in U.S. waters and by U.S. citizens on the high seas, and the importation of marine mammals and marine mammal products into the U.S.

Congress passed the Marine Mammal Protection Act of 1972, to address, among others, the following concerns: (1) some marine mammal species or stocks may be in danger of extinction or depletion due to human activities, (2) these species or stocks must not be allowed to fall below their optimum sustainable population level, and (3) measures should be taken to replenish these species or stock. The MMPA was amended in 1994 to provide for the following: (1) certain exceptions to the take prohibitions, such as for Alaska Native subsistence and permits/authorizations for scientific research (2) A program to authorize and control the taking of marine mammals incidental to commercial fishing operations, and (3) stock assessment reports for all marine mammal stocks under U.S. jurisdiction.

¹¹ Under Section 7 of the ESA, all Federal agencies must ensure that any actions they authorize, fund, or carry out are not likely to jeopardize the continued existence of an endangered or threatened species, or destroy or adversely modify its designated critical habitat.

APPENDIX B: Typical Regulatory Process

FISHERY MANAGEMENT COUNCIL ACTIONS

The following provides a description of how the Council regulatory process typically works. This is not a recommendation on how the process should work. There are regional variations regarding timing and content of required supporting documents in the approval process. Each Council and NMFS Regional Office have agreed upon operational plan under which Council actions are implemented.

Problem Identification, Options Paper, Scoping, & Public Input:

These steps are initiated when a problem is identified through the annual report to Congress on the status of fish stocks, results from stock assessments, environmental concerns, public comments, etc. The Council prepares a scoping document explaining the problem and provides a number of options for dealing with the problem. This document goes through the public scoping process to obtain public input on the options that would be considered for solving the problem.

Council Reviews Public Input, Approves Alternatives:

This process enables the Council to select alternatives (during Council meetings) to be included in the Fishery Management Plan (FMP) or amendment. Usually the options are narrowed down, but, sometimes, new options are added.

Prepare Public Hearing Document, Preliminary Analysis, Public Hearing:

The alternatives are well defined. Sometimes a preferred alternative is selected. A preliminary analysis is prepared to indicate expected impacts of alternatives. A draft environmental assessment (EA) or a draft environmental impact statement (DEIS) is prepared at this time. If a DEIS is prepared, it is published in the Federal Register (FR) with a 45-day comment period. Comments on the DEIS are sent to the Council.

Note that the Preliminary Regulatory Economic Evaluation (PREE) or similar analysis recommended in the guidelines should be done at this time.

The document is taken to a series of public hearings. The number of hearings depends on the nature of the problem and the geographic extent of the fishery under consideration. The Council's Scientific & Statistical Committee (SSC), Social & Economic Panel (SEP), Advisory Panel (AP), and similar groups review and provide comments on the draft document.

Council Reviews Public Comments, Selects Preferred Alternative, Approves Document for Secretarial Review:

This process could go through more than one Council meeting. Detailed economic analysis of the likely impacts of the alternatives should be available at this time. The economic analysis should be done to meet the requirements of the National Environmental Protection Act, E.O. 12866, and the Regulatory Flexibility Act. SSC, SEP, AP, and similar bodies meet to review measures in the draft document. The Council takes public comment at the meeting where the final vote is taken for the submission of the document.

Formal Regulatory Flexibility Analysis Process:

Once the regulatory document is approved for Secretarial review, the analyst uses information provided in the economic analysis to determine whether there is a factual basis to recommend certifying that the preferred alternative would not result in “significant economic impact on a substantial number of small entities.” If this is possible, the analyst does not prepare an IRFA but provides the factual basis in the regulatory document. The regulatory package is then transmitted to the Regional Administrator. However, if there is no factual basis to recommend certification, the analyst prepares an IRFA which is included in the package before transmittal.

Council Submits Regulatory Package for Secretarial Review:

This package could include the transmittal letter, final FMP, FMP amendment, regulatory amendment, with FEIS (if an DEIS was prepared), an EA (if an DEIS was not prepared), an RIR, an RFAA with an IRFA (if one was prepared), or an annual specifications document; proposed rule, including the codified section of the rule. In some cases, the NMFS Region prepares the proposed rule with the concurrence of the Council.

Regional Administrator (RA) Reviews Regulatory Package:

During this review process, any serious deficiencies in the analytical supporting documents (i.e., lack of supporting analysis for PRA requirements or deficient RIR/IRFA analyses) for the regulatory package are resolved with the Council. Sometimes supplementary analysis or documents are prepared.

RA Transmits Issues Advisory (IA) to Assistant Administrator for Fisheries (F):

When the regulatory package is complete, the RA sends an IA to F providing a summary of major features of the management measures, outlining any controversial issues, and stating the controversy. If need be, HQ and RA discuss and resolve issues. Once resolved, F signs off on the IA and requests that RA formally transmit regulatory package to HQ.

Formal Transmittal of Regulatory Package, Formal Review Process Begins:

The day the package is formally transmitted is the day the regulatory clock starts ticking. At this stage, the FMP/amendment begins tracking through one process and the regulations track through another process.

**Notice of Availability Published in the Federal Register (FR):
FMP/Amendment**

Within 5 days, a notice of availability (NOA) is published in the FR. A comment period is open for 60 days from the date of publication of the NOA. Public comments are received. Comments could affect approvability of measures proposed in the FMP/amendment. At the end of the 60-day comment period, NMFS HQ has 30 days to approve, partially approve, or disapprove the FMP/Amendment.

Regulations

During an initial 15-day period, NMFS (HQ and the Region) and NOAA General Counsel for Fisheries (NOAA - GCF) evaluate the proposed rule to make sure the measures it contains are consistent with the FMP/Amendment, the Magnuson-Stevens Act and other applicable law. If the determination is affirmative, the proposed rule and its measures continue to be reviewed and processed by the Office of Sustainable Fisheries (F/SF) and NOAA - GCF for publication in the Federal Register. If the determination is negative, NMFS on behalf of the Secretary of Commerce notifies the Council in writing of inconsistencies and provide recommendations on revisions to make the proposed measures consistent with the FMP/amendment, the Magnuson-Stevens Act, and other applicable law (see section 304 of the Magnuson-Stevens Act).

There is normally a 45-day comment period during which comments are solicited from the public. At the end of the comment period, the NMFS Region compiles all comments received and prepares responses to those comments. NMFS responds to all comments received on the FMP/amendment and the rule. These comments could pertain to the Magnuson-Stevens Act, including national standards, E.O. 12866, IRFA, NEPA, and to other applicable law. These comments and responses are included in the final rule. If an IRFA has been already prepared, an FRFA is now being prepared, which should address comments pertaining to the IRFA and to any changes in the analysis contained in the IRFA as a result of the comments received. The final rule is published 30 days after the end of the comment period. The final rule becomes effective 30 days after it is published in the Federal Register unless there is a waiver or an extension of the 30-day period.

Initial OMB Clearance:

OMB reviews those rules that it determines significant under E.O. 12866. NMFS prepares a listing document for OMB which indicates whether NMFS considers the rule to be significant or not. This is sent to OMB during the initial evaluation of the proposed rule (15-day period). If OMB concurs with NMFS that the rule is not significant, the OMB review process ends at this point. However, if OMB overrules a NMFS determination of not significant and they determine the rule is significant under E.O. 12866, OMB advises the Office of General Counsel /Department of Commerce (OGC/DOC) and OGC/DOC informs NMFS. OMB has to give clearance before any proposed rule that is determined to be significant is published. When the rule is determined to be significant, the analysis goes through more scrutiny by OMB to ensure that the requirements of E.O. 12866 are met. If any part of the required analysis is missing, OMB requests additional analysis to correct this deficiency. If OMB determines that the rulemaking is significant under E.O. 12866, it also reviews and clears the final rule before it is published in the Federal Register. OMB usually reviews the rule only, but occasionally, it reviews also the FMP or amendment.

Small Business Administration (SBA):

The proposed rule, along with the IRFA (if one was prepared) or the certification letter (if the agency decides to recommend certification to OGC/DOC), is sent to the Chief Counsel for Advocacy of the Small Business Administration at the same time the proposed rule is sent to the Office of Federal Register for publication in the Federal Register. SBA has 45 days to comment.



UNITED STATES DEPARTMENT OF COMMERCE
National Oceanic and Atmospheric Administration
NATIONAL MARINE FISHERIES SERVICE
Southeast Regional Office
263 13th Avenue South
St. Petersburg, Florida 33701-5505
<https://www.fisheries.noaa.gov/region/southeast>

F/SER25:FH

Mr. Mel Bell, Chair
South Atlantic Fishery Management Council
4055 Faber Place Drive, Suite 201
North Charleston, South Carolina 29405

Dear Mr. Bell:

By this letter, I am disapproving Amendment 10 to the Fishery Management Plan (FMP) for the Coral, Coral Reefs, and Live/Hard Bottom Habitats of the South Atlantic Region (Coral FMP). As you know, Amendment 10 to the Coral FMP (Coral Amendment 10) would establish an approximately 22 mi² shrimp fishery access area (SFAA) along the eastern boundary of the northern extension of the Oculina Bank Habitat Area of Particular Concern (OHAPC) where fishing for and possessing rock shrimp are currently prohibited. The South Atlantic Fishery Management Council (Council) developed Coral Amendment 10 to help achieve optimum yield in the South Atlantic rock shrimp portion of the shrimp fishery and increase economic and social benefits to rock shrimp fishermen by increasing access to historic rock shrimp fishing grounds.

The Council approved Coral Amendment 10 for Secretarial review and implementation at its September 2021 meeting. In undertaking this review, section 304(a)(1) of the Magnuson-Stevens Fishery Conservation and Management Act (Magnuson-Stevens Act) requires the Secretary of Commerce (Secretary) to make a determination as to whether Coral Amendment 10 is consistent with the Magnuson-Stevens Act and other applicable laws and publish a notice of availability for the amendment in the *Federal Register*. Section 304(a)(2) of the Magnuson-Stevens Act requires this review to take into account the information, views, and comments received on the amendment from interested persons.

NOAA Fisheries noticed the availability of Coral Amendment 10 on April 29, 2022, and accepted public comments on the amendment through June 28, 2022 (87 FR 25438) [RIN 0648-BL09]. NOAA Fisheries received 353 distinct comments during that time period; however, several were form letters signed by many individuals, bringing the total number of people who commented on the Amendment to 32,200. Commenters included the Florida Fish and Wildlife Commission, commercial and recreational fishermen, fishing organizations, environmental groups, and the general public. The vast majority of comments opposed the proposed action stating that it is not based on the best scientific information available; fails to minimize adverse fishing impacts on essential fish habitat (EFH) to the extent practicable; is inconsistent with the goals and objectives of the Coral FMP; and is inconsistent with the goals and recommendations of the Biden-Harris Administration's *America the Beautiful* initiative, including the goal to conserve at least 30% of U.S. lands and waters by 2030, per Executive Order 14008 on *Tackling the Climate Crisis at Home and Abroad*.

NOAA Fisheries' review of Coral Amendment 10 determined the amendment and supporting analyses do not adequately demonstrate how the Council's proposed action is consistent with (1) Section 303(a)(7) of the Magnuson-Stevens Act, which requires FMPs to minimize to the extent practicable the adverse effects of fishing on EFH, and the goals and objectives of the Coral FMP, specifically in regards to protection of EFH; and (2) Section 301(a)(9) of the Magnuson-Stevens Act, which requires fishery conservation and management measures to minimize bycatch to the extent practicable and, to the extent bycatch cannot be avoided, minimize the mortality of such bycatch.



Essential Fish Habitat

Section 303(a)(7) of the Magnuson-Stevens Act and its implementing guidelines require that amendments to FMPs or the regulations implementing the amendments ensure the FMP continues to minimize to the extent practicable adverse effects on EFH caused by fishing. FMPs must explain the reasons for the Council's conclusions regarding new actions that minimize to the extent practicable the adverse effects of fishing on EFH (50 CFR 600.815(a)(2)(ii)).

The action proposed in Coral Amendment 10 has the potential to adversely affect habitat within the proposed SFAA and OHAPC. While *Oculina* coral, including coral rubble and high relief mounds, have not been reported inside the proposed SFAA, rock shrimp trawling could impact any low-relief hard bottom and coral rubble within the proposed SFAA that are providing substrate for corals to recruit and recover from previous trawling events. Such impacts could be direct impacts from shrimp trawl gear interactions with coral habitat or indirect impacts from sediment plumes created by shrimp trawl gear. However, the likelihood and degree of any such impacts are unknown due to a paucity of habitat mapping and characterization data for this area. By reducing between 42%-54% of the existing buffer between fishing grounds and known high relief habitat occurring west of the OHAPC boundary, the proposed action also could indirectly impact coral habitat within the OHAPC through sedimentation.

The Council established the OHAPC in the original 1984 Coral FMP. The Coral FMP restricted the harvest of corals, with one of the FMP Objectives being to “minimize, as appropriate, adverse human impacts on coral and coral reefs.” To further this goal, the Council prohibited bottom trawling for snapper-grouper species through Amendment 1 to the Snapper-Grouper FMP in 1988. In 1991, Amendment 1 to the Coral FMP defined overfishing for prohibited corals (including *Oculina* coral) as an annual harvest that exceeds optimum yield, with the optimum yield equal to zero.

When the Council expanded the size of the OHAPC via Amendment 4 to the Coral FMP (effective in 2000), they added more fishing gear restrictions, including prohibiting bottom trawling and fishing for rock shrimp. Coral Amendment 4 was part of the Council’s Comprehensive EFH Amendment, and in that amendment, the Council also established the OHAPC as an EFH-habitat area of particular concern (EFH-HAPC) for coral and snapper-grouper species. An EFH-HAPC for these species is a portion of EFH that is “particularly vulnerable to fishing activities.” (50 CFR 600.815(a)(2)(i)). The Council stated that expanding the OHAPC would provide additional protection for EFH.

In 2015, the Council’s Amendment 8 to the Coral FMP (Coral Amendment 8) further expanded the OHAPC to include the area at issue in Coral Amendment 10 based on new scientific information showing an extensive deep water coral ecosystem. From Amendment 8, page 86: “Bottom trawling is considered the most ecologically destructive fishing method (Chuenpagdee et al. 2003; Morgan and Chuenpagdee 2003). This gear type, used to target shrimp species in the South Atlantic living on or just above the seafloor, has been shown to severely impact deep-water coral ecosystems (Fosså et al. 2002; Puglise et al. 2005).”

In determining whether it is practicable to minimize an adverse effect from fishing, Councils should consider the nature and extent of the adverse effect on EFH and the long and short-term costs and benefits of potential management measures to EFH, associated fisheries, and the Nation, consistent with National Standard 7. In approving Coral Amendment 10, the Council considered that future trawling activity would likely occur where rock shrimp have been previously harvested, in low relief bottom areas already impacted by past fishing activities. However, the Council also recognized that the risk of directly impacting such habitat cannot be completely eliminated, as the rock shrimp portion of the fishery is prosecuted along the boundary of a closed area on a regular basis. To address the concerns that a portion of the proposed SFAA was identified through modeling as suitable deep-water coral habitat, NOAA

Fisheries proposed conducting scientific surveys (video, remotely operated vehicle, and possibly multi-beam sonar) to better characterize the area. The agency was able to survey a portion of the SFAA from May 30 - June 3, 2022 (30.11% of the North-South range of the SFAA; <1% of the total SFAA), and was unable to verify the presence of rock shrimp, *Oculina* coral, coral rubble, or suitable coral habitat in the area. Coral Amendment 10 concludes the net economic benefits of the SFAA cannot be quantified given the likely variability in usage of the area, as well as the exhibited variability in overall participation in the regional rock shrimp portion of the shrimp fishery. More robust analysis is needed to demonstrate these benefits support a determination that the proposed action in Coral Amendment 10 minimizes adverse impacts to EFH to the extent practicable.

Bycatch

The national standard guidelines for complying with Section 301(a)(9) of the Magnuson-Stevens Act require Councils to assess the effects of each management measure on the amount and type of bycatch and bycatch mortality in the fishery and to select measures that, to the extent practicable, will minimize bycatch and bycatch mortality. A determination of whether a conservation and management measure minimizes bycatch or bycatch mortality to the extent practicable, consistent with other national standards and maximization of net benefits to the Nation, should consider the following factors: (A) Population effects for the bycatch species; (B) Ecological effects due to changes in the bycatch of that species (effects on other species in the ecosystem); (C) Changes in the bycatch of other species of fish and the resulting population and ecosystem effects; (D) Effects on marine mammals and birds; (E) Changes in fishing, processing, disposal, and marketing costs; (F) Changes in fishing practices and behavior of fishermen; (G) Changes in research, administration, and enforcement costs and management effectiveness; (H) Changes in the economic, social, or cultural value of fishing activities and nonconsumptive uses of fishery resources; (I) Changes in the distribution of benefits and costs; and (J) Social effects. The Councils should adhere to the precautionary approach found in the Food and Agriculture Organization of the United Nations Code of Conduct for Responsible Fisheries (Article 6.5), when faced with uncertainty concerning any of these factors (50 CFR 600.350(d)(3)).

The comprehensive bycatch practicability analysis supporting Coral Amendment 8 concluded the action to expand the OHAPC to include the area at issue in Coral Amendment 10 would benefit coral communities and ecosystems. From Coral Amendment 8, page G-3: “These proposed actions would prevent fisheries from expanding into the proposed areas along with associated bycatch. Therefore, the actions in Coral Amendment 8 would likely result in long-term, positive ecological benefits and prevent disruptive changes that could occur in the community structure of coral reef ecosystems.” The cursory discussion of bycatch in section 3.2.2 of Coral Amendment 10 concludes the proposed action is not expected to result in significant changes in bycatch within the proposed areas since there is only a small amount of predicted fishing effort. More robust analysis through a Bycatch Practicability Analysis is needed to demonstrate that the proposed action in Coral Amendment 10 minimizes bycatch and bycatch mortality to the extent practicable.

Conclusion

If a Council FMP or amendment is disapproved based on inconsistencies with the Magnuson-Stevens Act or other applicable law, section 304(a)(3) of the Magnuson-Stevens Act requires the Secretary to recommend actions the Council could take to conform the amendment to the relevant legal requirements. Section 304(a)(4) of the Magnuson-Stevens Act provides Councils the opportunity to revise and submit revised amendments for Secretarial review after addressing the relevant legal requirements. As discussed above, to conform Coral Amendment 10 to the requirements of applicable law, the Council must adequately address through more robust analyses (1) how the proposed action minimizes to the extent practicable the adverse effects of fishing on EFH, as required by Section 303(a)(7) of the Magnuson-Stevens Act, and is consistent with the goals and objectives of the Coral FMP,

specifically in regards to protection of EFH; and (2) how the proposed action minimizes bycatch to the extent practicable and, to the extent bycatch cannot be avoided, minimizes the mortality of such bycatch, as required by National Standard 9 of the Magnuson-Stevens Act. In addition, the Council should consider input received during the public comment period when revising the amendment, including comments related to best scientific information available, minimizing impacts to EFH, inconsistencies with goals and objectives of the Coral FMP and the *America the Beautiful* initiative.

Thank you for your work on Coral Amendment 10. I appreciate and share your desire to increase economic and social benefits in the rock shrimp fishery, along with the interest many have expressed in ensuring the continued protection of vulnerable coral habitat. Our nation's valuable ecosystems, like the OHAPC, play an important role in combating the climate crisis. I trust you will agree that it is in the best interest of all stakeholders and resources potentially impacted by this action to ensure we sufficiently analyze and balance the tradeoffs and practicability of this important decision.

Sincerely,

Andrew J. Strelcheck
Regional Administrator

cc: John Carmichael
Myra Brower
John McGovern, Ph.D.
Rick DeVictor
Frank Helies
Monica Smit-Brunello



UNITED STATES DEPARTMENT OF COMMERCE
National Oceanic and Atmospheric Administration
NATIONAL MARINE FISHERIES SERVICE
Northwest Region
7600 Sand Point Way N.E., Bldg. 1
Seattle, WA 98115

December 27, 2010

Mr. Mark Cedergreen, Chair
Pacific Fishery Management Council
7700 NE Ambassador Place
Portland, Oregon 97220

Dear Mr. Cedergreen:

By this letter, I am partially approving Amendment 23 and disapproving Amendment 16-5 to the Pacific Coast Groundfish Fishery Management Plan (FMP). Amendment 23 revises relevant sections of the Groundfish FMP to ensure they are consistent with the Magnuson-Stevens Fishery Conservation and Management Act of 2006 (MSA) and National Standard 1 Guidelines. The guidelines describe fishery management approaches to meet the objectives of MSA National Standard 1 (MSA Section 301). Amendment 16-5 revises rebuilding plans for the following species: Bocaccio south of 40°10' north latitude; canary rockfish; cowcod south of 40°10' north latitude; darkblotched rockfish, Pacific Ocean Perch, widow rockfish, and yelloweye rockfish. This amendment also created a rebuilding plan for petrale sole, which was declared overfished on February 9, 2010. In addition, Amendment 16-5 modifies status determination criteria for flatfish and establishes a new precautionary harvest control rule for flatfish.

The National Marine Fisheries Service (NMFS) strongly supports the Council's efforts with these amendments to increase the accounting for scientific and management uncertainty, prevent overfishing, and rebuild overfished species.

NMFS has determined that, except for the removal of dusky and dwarf-red rockfish from the FMP, Amendment 23 is consistent with the national standards and other provisions of the MSA and other applicable laws. A further discussion of the basis for NMFS' disapproval of the removal of the two species from the FMP is provided in the Issues Attachment 1 below.

As you are aware, on April 29, 2010, the District Court for the Northern District of California ruled in part against NMFS in a case on the 2009-2010 harvest specifications (*Natural Resources Defense Council v. Locke*). Specifically, the court found that NMFS violated:

- National Standard 2 of the MSA by failing to use the best scientific information available on the economic status of fishing communities;



- Section 304(e)(4)(A)(i) of the MSA by establishing rebuilding plans for darkblotched rockfish, cowcod and yelloweye rockfish that do not rebuild those species in time periods that are “as short as possible.”

As thoroughly discussed at the November meeting, NMFS concluded that we would not be able to make necessary analytical refinements to the Draft Environmental Impact Statement (EIS) for Amendment 16-5 in time for NMFS to make a final decision by the deadlines imposed in Section 304(a) of the Magnuson-Stevens Act. Therefore, Amendment 16-5 is being disapproved. Because of the disapproval, NMFS’ implementation of the 2011 specifications and management measures will be accomplished, in part, under the emergency authority of the MSA. NMFS requests that the Council revise as appropriate and resubmit Amendment 16-5 for implementation of the 2012 specifications and management measures. NMFS understands that the Council has already preliminarily scheduled time in the spring/summer 2011 meeting agendas to accomplish this task.

Items for Disapproval

Amendment 16-5 is being disapproved because there is not currently an adequate EIS to support decision-making. The Magnuson-Stevens Act (16 U.S.C. 1854(a)(3)) requires that before approving an FMP or amendment, NMFS must review the FMP for consistency with the measures of the MSA itself, as well as other applicable law. One of the primary tools that NMFS uses to accomplish this review is an adequate Final EIS, drafted in a manner consistent with the guidance contained within NAO 216-6 (Environmental Review Procedures For Implementing The National Environmental Policy Act). At this time, there is not any Final EIS, let alone an adequate Final EIS, to serve as a basis for either approving or disapproving Amendment 16-5 to the FMP. The Region will continue to work with Council staff to reorganize and refine the Draft EIS analyses to publish a Final EIS in time to support a final decision by the court-ordered April 29, 2011 deadline.

Amendment 23 as recommended by the Council also removes dusky and dwarf-red rockfish from the list of species in the FMP, and from the minor rockfish complex as a management group. The FMP currently specifies that it covers “rockfish,” which includes all genera and species of the family Scorpaenidae¹, *even if not listed*, that occur in the Washington, Oregon, and California area.” NMFS believes that removing any species of “rockfish” from the FMP at this time is a fundamental change to one of the foundations of groundfish management, and as such, should be accompanied by a thorough analysis to support such a change. As described below, a thorough analysis was not provided. Therefore, NMFS is partially approving Amendment 23, which would leave these species in the FMP.

¹ *Scorpaenidae* genera (*Sebastes*, *Scorpaena*, *Scorpaenodes*, and *Sebastolobus*)

The rationale given by the Council for removing dusky and dwarf red rockfish was that there are very few historical landings of these species. However, these two species are not the only species with little or no landing history that are currently in the FMP. Further, when NMFS looked at Pacific whiting data from the NORPAC database, which includes observer sampling information, there were historical catches of dusky rockfish in most years, including over 2 metric tons in 2003. Although the catches were relatively small in most years, they show that this species is regularly subject to mortality in this fishery. In addition, there are other rockfish species currently in the FMP that have very small landings; our review revealed at least 15 species with less than 1,000 pounds of landings in 2009.

Although the FMP includes all rockfish species, individual management of all species of rockfish is not feasible because some species are rarely caught in fisheries, and in many cases there is insufficient information on certain species on which to base management decisions. In response to this circumstance of rare catches and limited scientific information, many species in the FMP are managed within a stock complex. This is the case for both dusky and dwarf-red rockfish, which are members of the minor shelf rockfish complex. Stock complexes are an important management tool for managing many rockfish species on the West Coast.

Removal of a species from the FMP would reflect a determination that conservation and management measures are not necessary. NMFS is not prepared to make that determination at this time without a more comprehensive discussion and review of how the Council and NMFS manage rockfish genera and species within the family of scorpaenidae. NMFS understands that the Council is planning to refine the existing stock complexes for the 2013-14 specifications and management measures. NMFS believes it is prudent, from a management perspective, to take a comprehensive look at all species and complexes in the FMP before deciding to remove or add any. The option to designate species as ecosystem component species is also another option that may be explored through the Council process.

As discussed above, NMFS is approving the remainder of Amendment 23. However, we note that public comments highlighted a lack of clarity in the amendment with respect to the connection between Annual Catch Limits (ACLs) and Accountability Measures (AMs). While NMFS does not believe this is a basis on which to disapprove the amendment, we agree that this lack of clarity needs to be addressed through the development and submission of an additional amendment to the FMP. The National Standard 1 Guidelines require that the FMP describe AMs to prevent ACLs from being exceeded, and to correct or mitigate overages of the ACLs if they occur. Amendment 23 contains an extensive suite of inseason and other management measures, some of which are referred to as accountability measures, but the amendment does not clearly articulate the connection between ACLs and AMs, and it is not entirely clear which management

measures constitute AMs. NMFS requests that the Council address this issue in conjunction with its resubmission of Amendment 16-5, and we are committed to working with Council staff to develop the appropriate language.

Amendments 23 and 16-5 were developed over several Council meetings and required significant efforts by the Council, its advisory bodies, and its staff. NMFS will continue to work with the Council to make the changes necessary to Amendment 16-5 for approval for implementation for 2012, as well as the refinements to ACL provisions discussed above. The deliberations by the Council and its advisory bodies, and the work of members of the public at Council meetings and in their home ports, demonstrate an impressive commitment to the principles of the Magnuson-Stevens Conservation and Management Act. Thank you for your time and effort.

Sincerely,

A handwritten signature in black ink, appearing to read "Frank Stelle", written over a horizontal line.

For William W. Stelle, Jr.
Regional Administrator

Issues Attachment
Rationale for partial disapproval of Amendment 23

The Pacific Coast Groundfish Fishery Management Plan was established to manage all “rockfish” genera and species within the family of scorpaenidae, even if not listed individually in the FMP, that occur in Washington, Oregon and California. Although the FMP includes all rockfish species, individual management of all species of rockfish is not feasible because some species are rarely caught in fisheries, and in many cases there is insufficient information to manage a species specifically. In response to this circumstance of rare catches and limited scientific information, many species in the FMP are managed within a stock complex. This is the case for both dusky and dwarf-red rockfish, which are members of the minor shelf rockfish complex.

Stock complexes are an important management tool for managing many rockfish species on the West Coast. NMFS believes it is prudent, from a management perspective, to take a comprehensive look at all species and complexes in the FMP before deciding to remove or add any. The option to designate species as ecosystem component species is also another option NMFS will explore through the Council process.

The rationale given by the Council for removing dusky and dwarf red rockfish was that there are very few historical landings of these species. However, these two species are not the only species with little or no landing history that are currently in the FMP. Further, when NMFS looked at data from the NORPAC database, observer sub-sampling data in the Pacific whiting fishery indicated that incidental catch of dusky rockfish has historically been reported (Table 1). Although the landings were relatively small, they show that this species has been landed. There are other rockfish species currently in the FMP that have very small landings (Table 2). Removal of a species from the FMP would reflect a determination that conservation and management measures are not necessary. NMFS is not prepared to make that determination at this time, without a more comprehensive discussion and review of how the Council and NMFS manage “rockfish” genera and species within the family of scorpaenidae.

The Northwest Fisheries Science Center indicated that dusky and dwarf-red rockfish most likely do not occur in Washington, Oregon or California in numbers sufficient enough to determine abundance or conduct a stock assessment. However, this does not lead NMFS to conclude that these species should be removed from the FMP at this time. NMFS believes that as the stock complexes are further refined, it may be appropriate to remove species from the FMP. However, this discussion and related policy decisions should be addressed in a comprehensive, as opposed to piecemeal fashion. Finally, with the implementation of the trawl rationalization program and full catch accounting of both catch and discards, more information will be available to inform any decisions on removal of species from the FMP.

Landings of dusky rockfish
NORPAC Database

Table 1. Landings of dusky rockfish in individual hauls from the NORPAC database.	
Haul Date	Metric Tons
12/12/1990	0.212
7/23/1991	0.001
9/10/1992	0.036
9/12/1992	0.233
9/12/1992	0.524
9/12/1992	1.220
10/6/1992	0.015
10/6/1992	0.074
4/15/1994	0.817
4/16/1994	0.132
4/16/1994	0.273
5/2/1995	0.027
5/15/1996	0.009
5/18/1998	0.016
5/19/1998	0.005
5/16/1999	0.067
5/23/1999	0.005
5/15/2000	0.014
5/15/2000	0.017
7/17/2000	0.013
7/17/2000	0.017
7/17/2000	0.071
7/17/2000	0.079
7/17/2000	0.093
7/17/2000	0.164
7/22/2000	0.083
7/22/2000	0.479
7/23/2000	0.086
7/23/2000	0.137
7/23/2000	0.250
7/23/2000	2.074
7/24/2000	0.041
9/4/2000	0.002
10/14/2001	0.003
10/24/2001	0.010
10/24/2001	0.018
5/17/2002	0.003
5/15/2003	0.005

Table 2. Landings (in pounds) of rockfish species in the FMP in 2009
PACFIN Database

Table 2. Landings (in pounds) of rockfish species in the FMP in 2009	
Species	Pounds
UNSPECIFIED SHELF ROCKFISH	7
FLAG ROCKFISH	20
CALIFORNIA SCORPIONFISH	24
QUILLBACK	27
OLIVE	50
VERMILION	63
SHORTBELLY	102
BROWN	115
STRIPETAIL	144
UNSPECIFIED ROCKFISH	148
UNSPECIFIED SMALL REDS ROCKFISH	172
YELLOWEYE	189
ROSETHORN	228
NORTH UNSPECIFIED NEAR-SHORE ROCKFISH	304
UNSPECIFIED REDS ROCKFISH	309
BLACKSPOTTED	360
GREENSPOTTED	539
BLACK	593
SILVERGREY	690
GREENBLOTCHED	797



**NOAA
FISHERIES**

In April 2020, NOAA Fisheries prepared its first national report on the regional impacts of COVID-19 on the commercial, recreational and aquaculture sectors. This report updates that initial assessment, capturing economic changes experienced by the fishing industry as the country began its phased reopening along with infusion of Federal funding through the CARES Act. NOAA Fisheries will continue to use this information to identify economic hardship where it exists and identify pathways for enhancing the resilience of the U.S. seafood and fisheries industries.

Alaska Snapshot, January-August 2020



Alaska Fisheries Impacts from COVID-19

Overview

This report is intended to supplement the prior Alaska Fisheries Impacts from COVID-19 documents, which include a more thorough review of COVID-19 actions taken by the state and local governments in Alaska and our finalized revenue trends from 2014-2018 (the most recent data currently finalized and completed). Herein, we provide a summary of the current harvest amounts and estimate projected revenues through August 2020.

Harvest volumes are down in Alaska approximately 15% compared with the baseline through August, and prices are generally expected to be lower in 2020 than prior years, which is confirmed by export data through June, and estimated revenues have dropped by over one third over the same period. Industry has reported that they have spent over \$50 million (McKinley Group September 2020)¹ to reduce the risk of COVID-19 transmission among harvesters, processors, and the local communities while still providing important seafood for the U.S. and international markets as well as providing food security for many Alaskans. The seafood industry has been fairly successful in Alaska at limiting virus spread, but they had to deal with a substantial reduction in transportation options in many Western Alaska and Aleutian Islands communities and limited ability to switch crews throughout the fishing seasons to date. NOAA Fisheries Alaska Regional Office has been instrumental in devising solutions with industry to allow the continuation of fishing operations and limit the need for fisheries closures which would otherwise lead to vessel downtime and higher crew turnover, increasing the risk of COVID-19 transmission.

Alaska remains the region of the United States with the largest export values over the baseline January-June period (about \$1 billion) but experienced a 7% decline in value in 2020 relative to 2019. There was also a large (48%) decline in imports to Alaska, but those only represent a very small amount (approximately \$5 million). China is the U.S.'s top export country for January-June seafood exports, and is a major importer of Alaska seafood. Exports to China (including cod, crab, and pollock) decreased from the \$515 million baseline average to \$386 million in 2020, a 25% decline. However, the U.S. dollar weakened from January to August 2020 against the Euro (-6%)² and Yen (-3%)³ which should have helped U.S. and Alaska seafood export competitiveness.

Commercial Fisheries Landings Trends and Impacts through August 2020

The figures below represent landings through August 2020 from in-season Alaska Department of Fish and Game (ADF&G) fish tickets (which are still considered preliminary). Revenues in 2020 are estimated using in-season ADF&G fish ticket prices for groundfish and the at-sea prices are estimated as their shoreside equivalents or imputed based on best available data. ADF&G in-season prices and historic Commercial Fisheries Entry Commission (CFEC) prices are used to estimate salmon and herring revenues for 2020. All values are deflated to \$2019 using the GDP deflator.

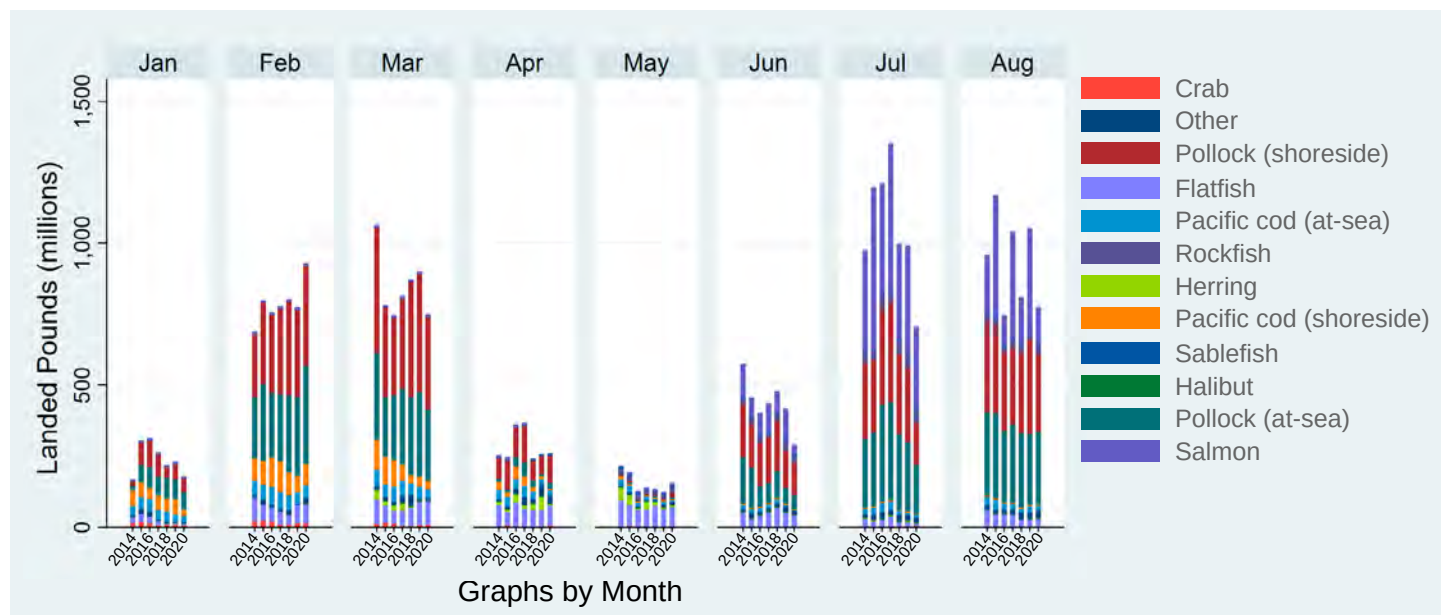


Figure 1. Landings by month and year from January-August.

Total landings from January through August 2020 are 15% below 2019 levels (Figure 1), a reduction of 695 million pounds from 4.74 billion pounds to 4.03 billion pounds (Figure 2). These trends are largely due to a 71% and 45% decline in harvest volume for herring and salmon, respectively, and a decline of 18% for halibut, 16% for Pacific cod at-sea and 13% for Pacific cod shoreside compared with 2019 levels. Landings in 2020 fell by 17% or 815 million pounds compared to the 2015-2019 baseline period as a result of a 74%, 44%, 36%, 39%, and 17% reduction in herring, salmon, Pacific cod at-sea, Pacific cod shoreside, and halibut landings, respectively.⁴ In contrast, crab, flatfish and rockfish harvests are up 3%, 4%, and 11%, respectively, compared with the 2015-2019 period.

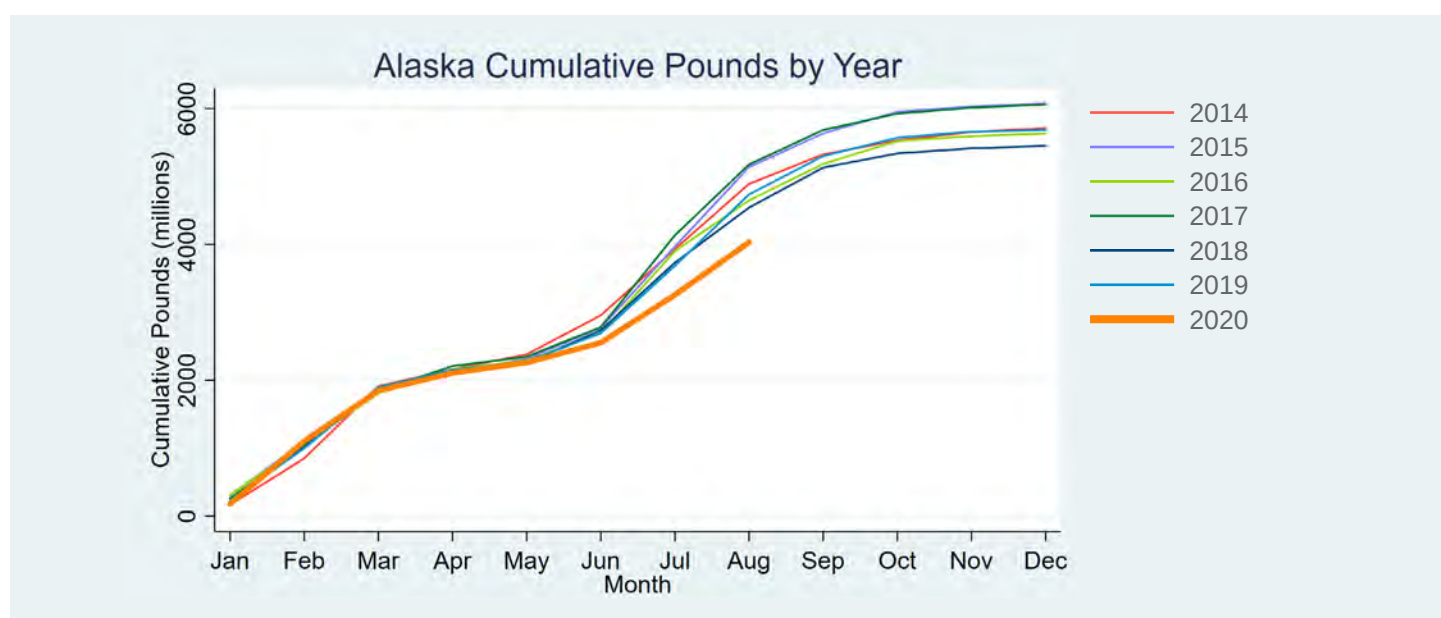


Figure 2. Alaska cumulative landed weight by month and year.

Estimated landings revenue from January through August 2020 are 30% below 2019 levels (a decline of \$436 million from \$1.48 billion to \$1.04 billion) and 35% below the 2015-2019 baseline period (a decline of \$549 million from \$1.59 billion; Figures 3 and 4). This is a result of the combination of lower volumes and lower prices across many species. The largest components of the decrease in value over this period include a 67% decline in the harvest value for herring, a 61% reduction in salmon, a 37% reduction in halibut revenues, a 30% decline in shoreside Pacific cod, and a 17% decrease in the value of flatfish compared with 2019 levels. Compared with the 2015-2019 baseline, salmon revenues are estimated to have declined by 58%, herring revenues decreased by 55%, halibut revenues dropped 48%, Pacific cod at-sea was reduced by 45%, shoreside Pacific cod decreased by 41%, sablefish declined by 26%, while pollock at-sea and shoreside fell 16% and 15%, respectively, and flatfish declined by 4%. The two bright spots compared with the baseline were a 17% increase in crab revenues and a 6% increase in rockfish revenues.

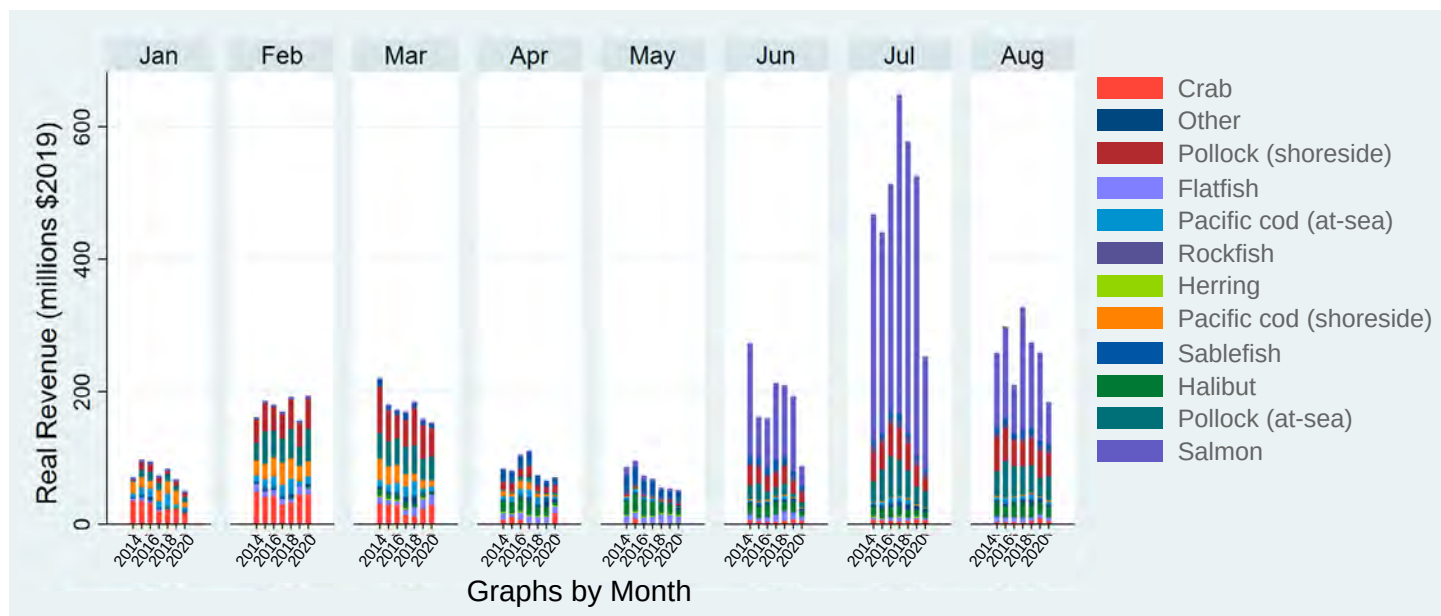


Figure 3. Estimated landings revenue (\$2019) by month and year from January-August.

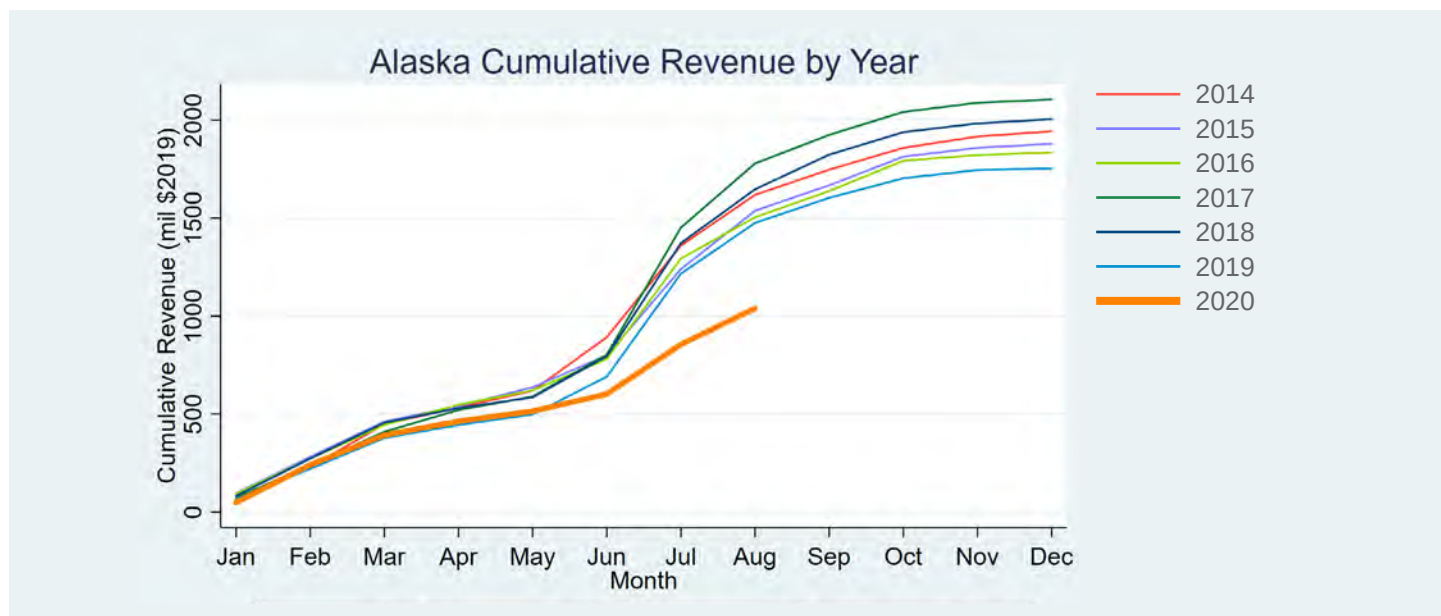


Figure 4. Total cumulative revenue by month and year.

While the novel coronavirus initially stoked fears of closing portions of the Alaska salmon fishery over the summer, the fisheries were prosecuted but with limited processing capacity, resulting in less value-added products and ultimately substantially lower ex-vessel prices. This is seen in Figure 4 as a further deviation from the recent past in cumulative revenues throughout the year, as the salmon fishery typically starts in late June and runs through August.

Figure 5 presents cumulative revenues by month and year for several major species groupings in Alaska in 2020 (the thick orange line) compared with each of the years 2014-2019 (line colors correspond to the legend in Figures 2 and 4). The following section provides a brief summary of cumulative harvest and revenue trends of several key species in Alaska from January through August.

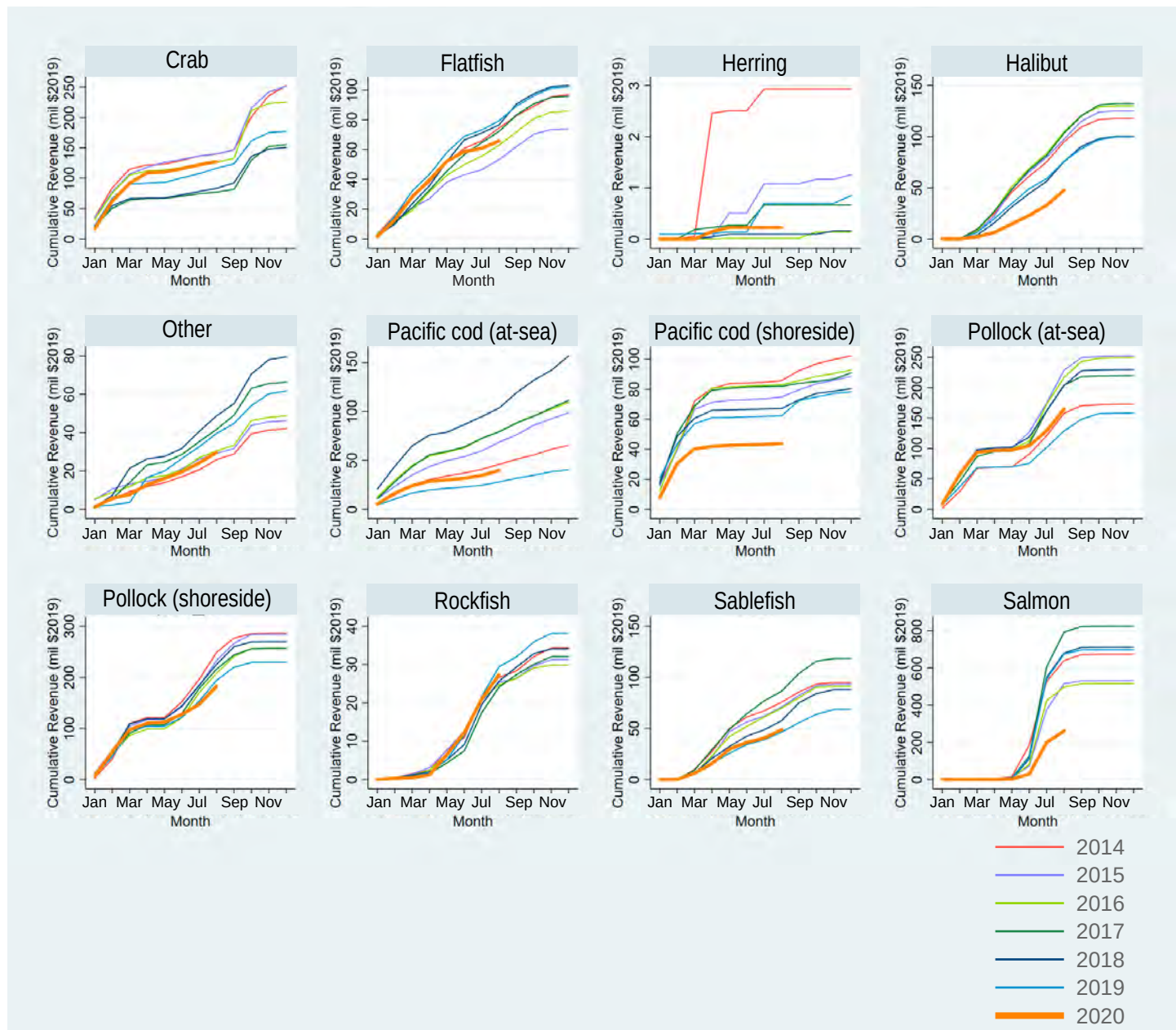


Figure 5. Cumulative revenue by fishery, month, and year.

CRAB

There was nearly full (99-100%) utilization of the total allowable catch (TAC) for all opened Bering Sea and Aleutian Islands (BSAI) crab fisheries: Bristol Bay red king crab, Bering Sea snow crab, and Aleutian Islands golden king crab. The TAC in the 2019-2020 Bristol Bay king crab fishery was a historic low, while the TAC for Bering Sea snow crab increased for the third straight season, but remains at historically low levels. For all crab species in Alaska, total landings from January-August of 2020 were 18% above the same period in 2019 but only 3% above the 2015-2019 average. There continues to be some indications of strong demand for crab throughout the COVID-19 pandemic, as crab revenues are up 9% through August of 2020 relative to the same period in 2019 and 17% compared with the 2015-2019 period (Figure 5).

FLATFISH

January to August flatfish catch in 2020 is up 1% from the 2019 level and is 4% higher than the 2015-2019 average by volume, but revenues are down 17% compared with 2019 and 4% compared with 2015-2019 (Figure 5).

HALIBUT

Prices have gone up for halibut relative to the beginning of the season but are still down over last year (estimated 10-15% for fresh and 25% for frozen). Many fishermen switched over to salmon during the summer and anticipate harvesting their remaining halibut quota in the fall. January to August 2020 has seen a large reduction in halibut landings of approximately 18% from the 2019 level and 2015-2019 average. Combined with lower prices, this has resulted in a decline in revenues of 37% in January to August 2020 relative to the same period in 2019 and a 48% decline relative to the average from January to August from 2015-2019 (Figure 5). However, many still expect that the full TAC will be harvested given the flexibility in leasing individual fishing quotas (IFQs) recently provided for by emergency transfer provisions instituted by the North Pacific Fishery Management Council (NPFMC) in response to the pandemic.

PACIFIC COD

Ex-vessel prices in 2020 have decreased slightly from 2019 but remain above 2014-2018 levels. Consistent with TAC declines in Pacific cod in the BSAI and Gulf of Alaska (GOA), shoreside catch of Pacific cod have declined by approximately 14% from January to August of 2020 relative to the same period in 2019, while shoreside catches are actually down 39% from the average from January to August during the 2015-2019 period. In combination with a slight decrease in ex-vessel prices in 2020, this has resulted in a reduction in shoreside Pacific cod revenue from January to August of approximately 30% compared with 2019 and 41% for the first half of 2020 relative to the average of the first half of the year from 2015-2019 (Figure 5).

For the at-sea sector, fishing conditions were mixed this past summer after a good A season. Prices in 2019 and 2020 are down relative to 2018 but are within 2014-2017 levels. Prices were okay this past winter but have softened as we approach the fall. There is a lot of inventory and many people are holding, hoping that prices will rebound. There is a lot of Russian cod on the market that is lowering prices in Japan and Europe. Consistent with TAC declines in Pacific cod in the BSAI and GOA, at-sea catch of Pacific cod have declined by approximately 16% from January to August 2020 relative to the same period in 2019, while at-sea catches are actually down 36% from the average January to August from 2015-2019. In combination with a slight increase in ex-vessel prices in 2020, this has resulted in a reduction in at-sea Pacific cod revenue of approximately 45% for the first half of 2020 relative to the average of first half of the year from 2015-2019, but a surprising 42% increase relative to the historically low revenue over this period in 2019 (Figure 5).

POLLOCK

For the shoreside sector, fishing was not very good (especially in the south where the shoreside sector predominantly harvests) this

past summer and therefore some vessels are waiting to fish later in the season; 2020 shoreside pollock harvests through August are down 12% compared with 2019 and the 2015-2019 average (Figure 5). Shoreside pollock prices are relatively stable in both GOA and BSAI relative to 2019 and are within historical range. Shoreside revenues are 6% lower than 2019 at this point in the year and over 15% below the average over 2015-2019.

For the at-sea sector, prices are down slightly now from the expectations for B Season. Pollock surimi exports to Japan are down significantly through May. However, surimi exports to the

European Union (EU) remain strong. Pollock fillet exports in March and April (the peak following A season) were below typical levels. 2020 at-sea pollock harvests are slightly below recent periods (7% less than the 2015-2019 average and 3% lower than 2019), but the revenues through August are lower than the 2015-2019 average by approximately 16% while the at-sea pollock revenues are up nearly 28% in 2020 compared with the same period in 2019, which was the lowest revenue year over this period for this sector (Figure 5).

ROCKFISH

Rockfish ex-vessel prices are within 2014-2019 norms. Rockfish harvests through August 2020 are down 4% from 2019 levels but are 11% higher than the average January to August period of 2015-2019. Similarly, revenues through August of 2020 are down 7% from the same period in 2019 but increased by 6% over the 2015-2019 period (Figure 5).

SABLEFISH

Fleet is reporting sablefish prices to be down by 55% compared to last year. This has a lot to do with the size of the fish, which are still on the smaller side, and the size-based pricing structure for sablefish. However, the large amount of small fish and market changes are bringing prices down across the board. There is an expectation that not all of the TAC is going to be landed for sablefish as a result. This was also true in 2019. Additionally, only 40% of the sablefish quota from the GOA Rockfish Program (secondary species) has been taken compared to 76% at this time last year, due to the very low price they are receiving for trawl caught sablefish.

Some are also reporting a shift in rents away from the permit holder and toward vessel operators as many individuals intend to take advantage of the special COVID-19 leasing provisions enacted by the NPFMC and International Pacific Halibut Commission (IPHC). Those transfers did not start clearing NOAA Fisheries Restricted Access Management (RAM) until early July, and may partially explain the large amount of IFQ remaining to catch this fall. Additionally, a number of individuals are also leaving their 10% rollover in the water hoping for better prices next year. Sablefish landings through August 2020 were 5% below 2019 levels but 4% above the 2015-2019 average, however revenue is down 26% and 2020 revenue is only 5% above the lowest revenue over this period, experienced in 2019 (Figure 5).

SALMON

Earlier in the year, it was unclear to what extent the wild Alaska salmon fisheries would be prosecuted this year, but the salmon fisheries were operating at slightly lower capacity on the harvesting and processing side as COVID-19 precautions are limiting

dense aggregations of individuals. Additionally, several runs did not meet their projected size outside of Bristol Bay. The somewhat compressed season and lack of processing throughput capacity has also led to a decrease in the amount of value-added products created in the Bristol Bay fishery, likely leading to substantially lower prices compared with previous years. For all salmon species in Alaska, total landings from January-August of 2020 were 46% below the same period in 2019 and 44% below the 2015-2019 average. In combination with a decrease in ex-vessel prices in 2020, this has resulted in an estimated reduction in salmon revenue of 61% through August of 2020 relative to 2019 and a 59% decrease relative to the average the same period from 2015-2019 (Figure 5).

Charter Sector Impacts

In response to COVID-19, the NPFMC and IPHC relaxed some halibut-related charter fishing restrictions for the remainder of the season effective June 15, which relaxed size limits in Areas 2C and 3A, and a relaxed annual limit and no day of the week restrictions for charter halibut fishing in Area 3A. However, reports from the field suggest that charter fishing is still well below normal levels for this time of year throughout Alaska, with some in industry estimating between 30-50% losses for the season. A more representative industry poll (May 2020) with n=66 respondents suggested “68% [of charter businesses] have indicated up to a 50% drop in bookings from this time last year.” There is another industry poll planned for October.

Historically, 31.5% of total trips take place in May (7.1%) and June (24.4%) for a first half total of approximately 32% of total annual trips, while nearly all of the remaining trips take place from July-September. There is some evidence of a marginal increase in fishing activity in July, as Pacific halibut (a primary target species on charter trips) harvest through July picked up relative to May and June, but remains lower than previous years. Preliminary estimates of Pacific halibut harvest in IPHC Area 2C (Southeast Alaska) through July is slightly less than half of the harvest at the same time in 2019. In IPHC Area 3A (Southcentral Alaska), the other main area for charter fishing in Alaska, Pacific halibut harvest through July was at about 71% of the harvest for the same time in 2019. The harvest trends likely do not correspond in a one-to-one fashion with numbers of trips, and updated trip information is unfortunately not available at this time.

Endnotes

- 1 <https://www.alaskaseafood.org/covid-19-impact-reports/>
- 2 <https://www.x-rates.com/average/?from=USD&to=EUR&amount=1&year=2020> (accessed October 9, 2020).
- 3 <https://www.x-rates.com/average/?from=USD&to=JPY&amount=1&year=2020> (accessed October 9, 2020).
- 4 Note that the federal Gulf of Alaska Pacific cod fishery is closed in 2020 and the federal Bering Sea and Aleutian Islands total allowable catch (TAC) was reduced by 15% in 2020 relative to 2019.



U.S. Secretary of Commerce
Wilbur Ross

Acting Under Secretary of Commerce
for Oceans and Atmosphere
Dr. Neil Jacobs

Assistant Administrator for Fisheries
Chris Oliver

January 2021

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**NOAA
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West Coast Snapshot, March-July 2020



West Coast Fisheries Impacts from COVID-19

Commercial Fisheries Landings Trends and Impacts¹ through July 2020

In 2019, approximately 3,300 non-tribal commercial fishing vessels operated in the West Coast fisheries and landed products valued at \$526 million (including aquaculture shellfish, where available). The top commercial fisheries by landings revenue in 2019 were crab, groundfish (including whiting), and shrimp. Many of the participants in West Coast commercial fisheries are active year-round, shifting from one fishery to another as seasons open and close. Broadly speaking, West Coast fishermen were initially affected by the cessation of exports caused by the closure of Asian seafood markets beginning in January 2020. Losses were compounded by the sharp decrease in demand for seafood, both domestically and globally, as countries imposed social distancing restrictions that closed restaurants where the majority of seafood is consumed. Some in the industry are concerned about whether markets will fully recover.

To assess potential effects of the pandemic on West Coast fisheries, we calculate the five-year median of landings (2015-2019) for each species group and region to provide a baseline comparison to 2020 landings. A five-year baseline is chosen because it corresponds to the time-frame used for fishery disaster declarations. Data for these comparisons are available from January through (and including) July 2020. Year-to-date comparisons include all landings from January through July. Comparisons restricted to March through July are also provided to emphasize impacts likely related to COVID-19, since the period coincides with the introduction of stay-at-home orders and restaurant closures on the West Coast. Ex-vessel revenue is adjusted for inflation. Ex-vessel revenue is adjusted for inflation. In order to protect confidentiality, data with too few observations are suppressed in the figures below.

Year-to-date, total West Coast shoreside commercial ex-vessel revenue was 9% lower than the baseline period. From March-July 2020, total ex-vessel revenue was 31% lower than the 2015-2019 baseline. At the start of the year, coast-wide ex-vessel revenue was higher than the baseline period, largely due to Dungeness crab landings, but fell below the baseline starting in March and continuing through July, which coincides with the start of stay-at-home orders and restaurant closures on the West Coast. Figure 1 illustrates ex-vessel revenue by species group generated during the months of March through July each year between 2015 and 2020. Total commercial fishing revenue in 2020 during these months was lower than any of the previous five years. The same is true of specific

fisheries including tuna, Individual Fishing Quota (IFQ) trawl, and fixed gear non-whiting groundfish. Revenue from market squid has been higher in 2020 thus far, and is reported to be experiencing an abundant season, particularly relative to last year.

Comparisons of 2020 landings to the baseline period vary across states. In California, the fisheries that contributed most to year-to-date ex-vessel revenue were crab, market squid, and salmon. Total ex-vessel revenue was 2% lower than the baseline period for the year-to-date and 16% lower from March through July. In Oregon, the fisheries that contributed most to year-to-date ex-vessel revenue were crab, shrimp, and shoreside Pacific whiting. Total ex-vessel revenue was 2% higher than the baseline period, and 21% lower from March through July. In Washington, the fisheries that contributed most to year-to-date ex-vessel revenue were crab and shrimp. Total ex-vessel revenue in Washington was 24% lower than the baseline period, and 40% lower from March through July.

Total Coast-Wide Commercial Ex-Vessel Revenue (March-July) by Species Group, 2015-2020

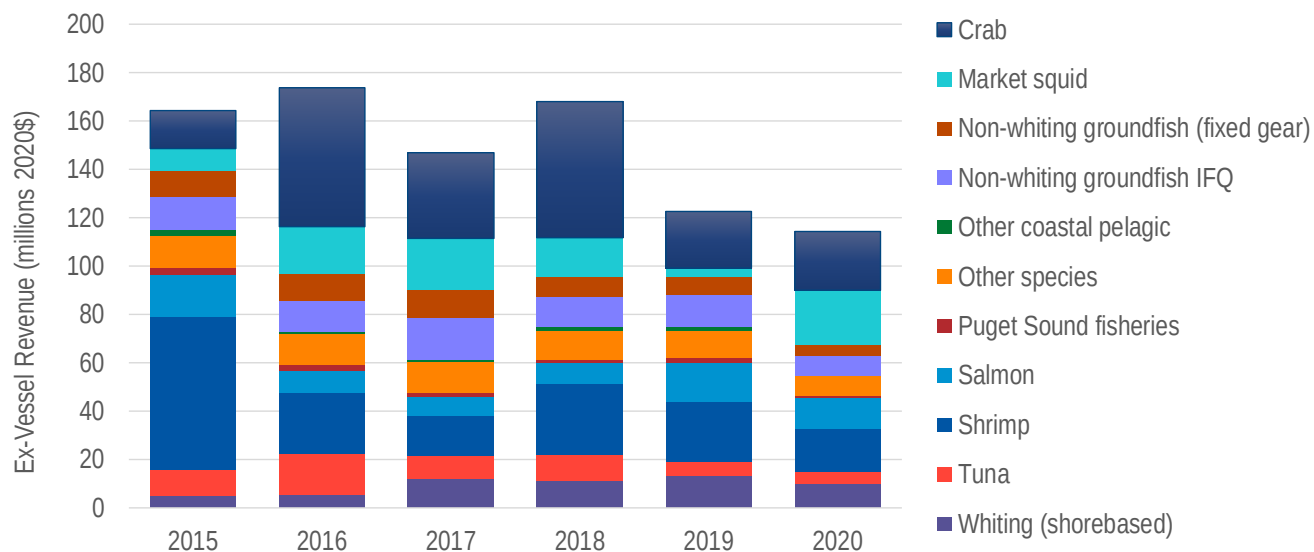


Figure 1. Total coast-wide commercial ex-vessel revenue generated during the months of March through July each year between 2015 and 2020 by species group.

NON-WHITING GROUND FISH FISHERY

From 2015-2019, the non-whiting groundfish fisheries contributed to about 11% of coast-wide shoreside ex-vessel revenue and approximately 970 commercial fishing vessels participated annually. Vessels target multiple species, including sablefish (also called black cod), rockfish, petrale sole, and Dover sole, and primarily use trawl nets and fixed gear (e.g., longlines, pots). Year-to-date, 2020 ex-vessel revenue was 44% lower than the baseline (2015-2019 median), and landed weight was 21% lower.

The non-whiting IFQ bottom trawl fishery operates year-round and accounts for approximately half of the non-whiting groundfish fishery ex-vessel revenue in 2020 thus far. Year-to-date, this fishery has experienced a 51% decrease in ex-vessel revenue and a 31% decrease in landed volume relative to the 5-year median. The

Coast-Wide Non-Whiting Groundfish Ex-Vessel Revenue

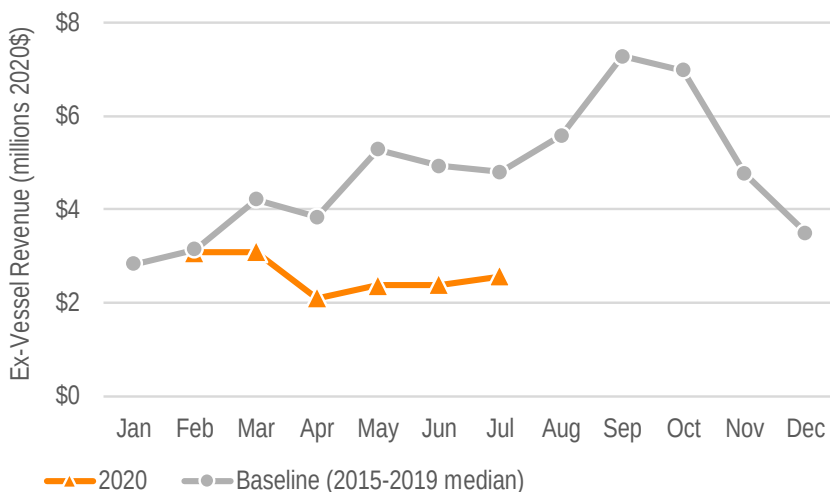


Figure 2. Coast-wide non-whiting groundfish ex-vessel revenue (millions of 2020 dollars).

fixed gear sablefish fishery is primarily executed between April and October. From March-July, ex-vessel revenue was 65% lower and volume landed was 50% lower than the 5-year median.

The non-whiting groundfish fisheries are reportedly experiencing challenging market situations for select species. For example, sablefish, a high-valued species typically exported or sold to restaurants, has been impacted by both the decline in export in Asian markets as well as the decline in domestic demand due to restaurant closures. The weighted average sablefish ex-vessel price between March and July 2020 was \$1.22 per pound, down from \$1.80 per pound in 2019.

SALMON FISHERY

The commercial, ocean troll salmon fishery on the West Coast typically contributes 3-4 percent of fisheries revenue with over 600 participating vessels. Participants target multiple, mixed stocks and the season typically runs from April through September with rotating open periods throughout. Year-to-date, total ex-vessel revenue and landed weight in the salmon fishery were higher than the 5-year median (2015-2019) by 39% and 76% respectively. These increases over the baseline period are due to conditions in California, which experienced average to above-average landings and ex-vessel prices. By contrast, landings were down considerably in Oregon and Washington.

Salmon volume and revenue in California from May, the first month the fishery was open, to July were substantially higher than the 5-year (2015-2019) median. Landed weight was 249% higher and revenue was 173% higher than the median value. This is likely because of a low baseline value due to poor stock conditions in previous years, average to above-average ex-vessel prices this year, and reasonably good stock conditions this year. Salmon fishermen in California report average to above-average prices this summer (approximately \$9 per pound). Demand for salmon has been surprisingly strong, as buyers have made up some of the lost revenue from sales to restaurants by increasing sales to grocery stores. Much higher sales to direct-to-consumer channels, though still a relatively small proportion of demand, and sales to markets in the Midwest and East Coast have helped support prices.

In Oregon, March-July ex-vessel revenue was 59% lower and volume landed was 58% lower than the 5-year median. Demand for Oregon salmon has been hurt by sharply reduced restaurant demand, which typically accounts for most of the market. Prices for salmon in Oregon have remained at or near historical levels, however, supported by increased direct-to-market sales, some increased demand from supermarkets, and very low landings. In Washington, March-July, ex-vessel revenue was 82% lower and volume landed was 80% lower than the 5-year median.

Portions of unused early season catch limits have been rolled over into late summer seasons in ocean troll fisheries in Washington and Oregon. The salmon fishery in Washington and northern Oregon in May and June caught only 15% of its early season allowable catch (2,141 out of 13,820 fish north of Cape Falcon; all of the unused portion was re-allocated to the July-September season). Late season reports indicate that it is unlikely that harvesters will take all of the quota. The southern Oregon ocean troll fishery harvested 24% of the June quota (165 out of 700 Chinook salmon), with a portion of the uncaught quota re-allocated to late summer (330 out of 535 uncaught fish). The full portion was not rolled over because the July and August fishery has a relatively high impact on the Klamath River fall-run Chinook stock, which is in poor condition.

SHRIMP FISHERY

From 2015-2019, the shrimp fisheries contributed to about 9% of coast-wide shoreside ex-vessel revenue and around 180 commercial fishing vessels participated annually. Shrimp revenue is largely comprised of the pink shrimp fishery, which runs from April through October, as well as spot prawns. Year-to-date, 2020 ex-vessel revenue was 28% lower than the baseline period, while landed weight was similar, indicating that the price per pound decreased. The weighted average pink shrimp ex-vessel price in 2020 thus far was \$0.50 per pound, down from \$0.72 per pound in 2019, and the spot prawn price in 2020 was \$12.36 per pound, down from \$12.90 per pound in 2019.

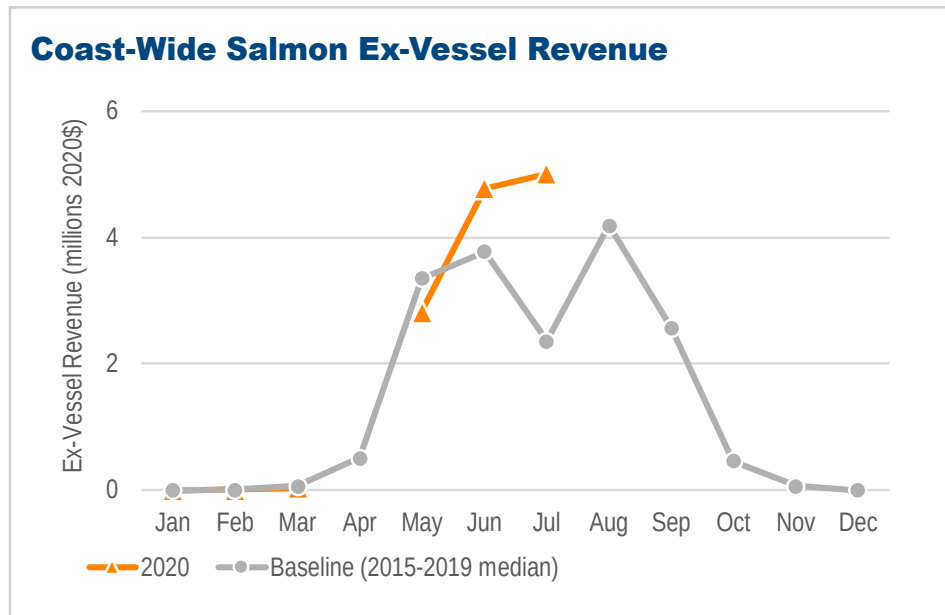


Figure 3. Coast-wide salmon ex-vessel revenue (millions of 2020 dollars).

Shrimp landings in 2020 thus far for Oregon and Washington were primarily composed of pink shrimp. In Oregon, landed weight was 5% lower and ex-vessel revenue was 38% lower than the baseline period. In Washington, landed weight was 23% higher than the baseline, while ex-vessel revenue was 6% lower. In California, ex-vessel revenue from shrimp landings since the start of the year was largely composed of prawn. Year-to-date, landed weight was 58% lower and ex-vessel revenue was 25% lower than the baseline. From March-July, landed weight was 63% lower and ex-vessel revenue was 30% lower. Industry feedback indicated that demand from restaurants for live products declined for shrimp in particular, due to their perishability and poor transportability once cooked, and they were endeavoring to expand direct-to-consumer sales.

WHITING FISHERY

Data that are available indicate that whiting harvest volume is higher than the baseline period but revenues are lower due to lower prices. The at-sea and shoreside Pacific whiting seasons open on May 15 each year, and whiting vessels in the at-sea sector typically depart the fishery to fish for Pollock in Alaska from July through September. Thus, the data available only reflect the first two and a half months of the at-sea season. Through July of 2020, volume was 34% higher than the 5-year median in the at-sea catcher-processor sector and 3% lower in the at-sea mothership sector. In the shoreside sector, landed weight was 5% higher than the 5-year median while ex-vessel revenue was 11% lower, indicating a lower average ex-vessel price. From May through July, ex-vessel prices in the shoreside sector were 25% lower than the baseline.

Vessels active in the at-sea whiting fishery have experienced a number of publicized outbreaks, which has likely increased the cost of operating in the fishery. Three Pacific whiting catcher-processor vessels (American Dynasty, American Triumph, and Northern Jaeger) experienced outbreaks of COVID-19 onboard, which required a 14-day quarantine before resuming fishing. One mothership processor decided to limit their activities to Alaska to mitigate the risk associated with COVID-19, which

Coast-Wide Shrimp Ex-Vessel Revenue

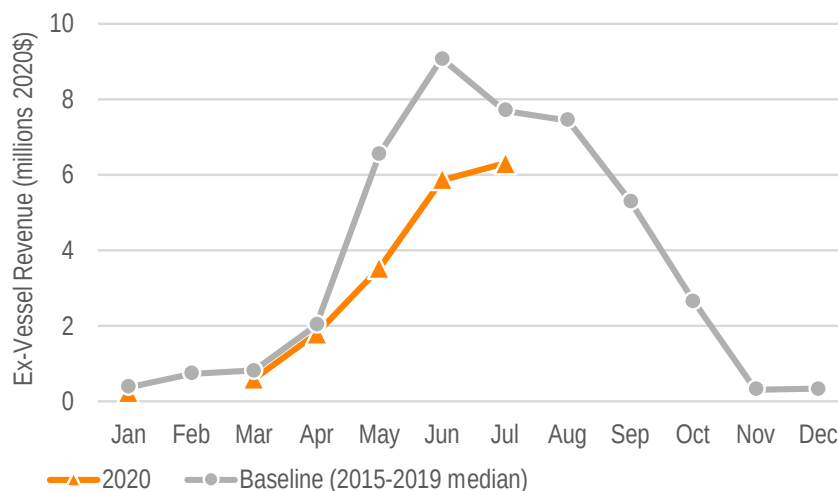
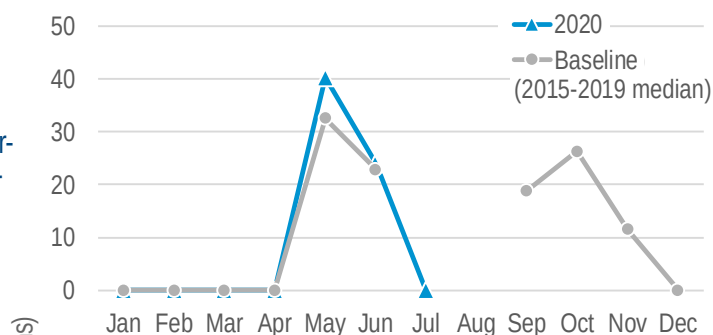
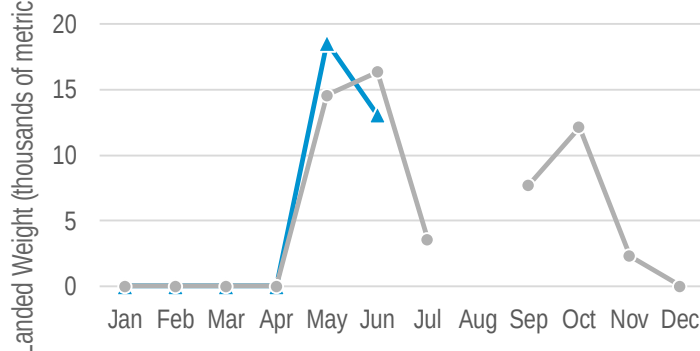


Figure 4. Coast-wide shrimp ex-vessel revenue (millions of 2020 dollars).

Pacific Whiting Volume landed in catcher-processor sector



Pacific Whiting Volume landed in mothership sector



Pacific Whiting Volume landed in shoreside sector

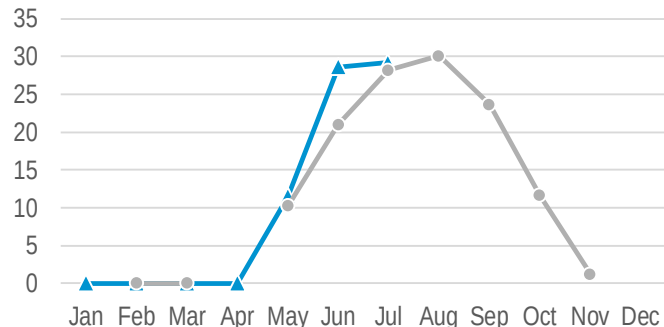


Figure 5. Total volume landed (metric tons) in at-sea and shoreside Pacific whiting fisheries.

left some catcher vessels without a processing platform. To provide operational flexibility, NOAA Fisheries issued an emergency rule to allow eligible mothership and catcher-processor vessels to operate as either type of processing platform during the 2020 Pacific whiting season.²

CRAB FISHERY

The Dungeness crab fishery, which comprises the majority of the West Coast crab landings revenue, was well into the 2019/2020 fishing season when COVID-19 began to affect markets for crab. Dungeness crab fisheries generally open in early December with most landings (approximately 80% of landed weight in the baseline) occurring between December and February. While much of the fishing season was over, there is some indication in the landings statistics that markets for crab experienced some weakness near the end of the fishing season. Industry feedback indicated that demand from Asian nations that import crab was down early in the year, and the high-value live crab market was effectively cancelled for the fishing season. This is exhibited in U.S. trade data, which indicates that all crab exports are down 25% from the baseline average. Trade data indicate that falling prices near the end of the fishing season potentially exhibit some market weakness: January-February prices were 8% higher than the baseline period, but March-July was 17% lower.

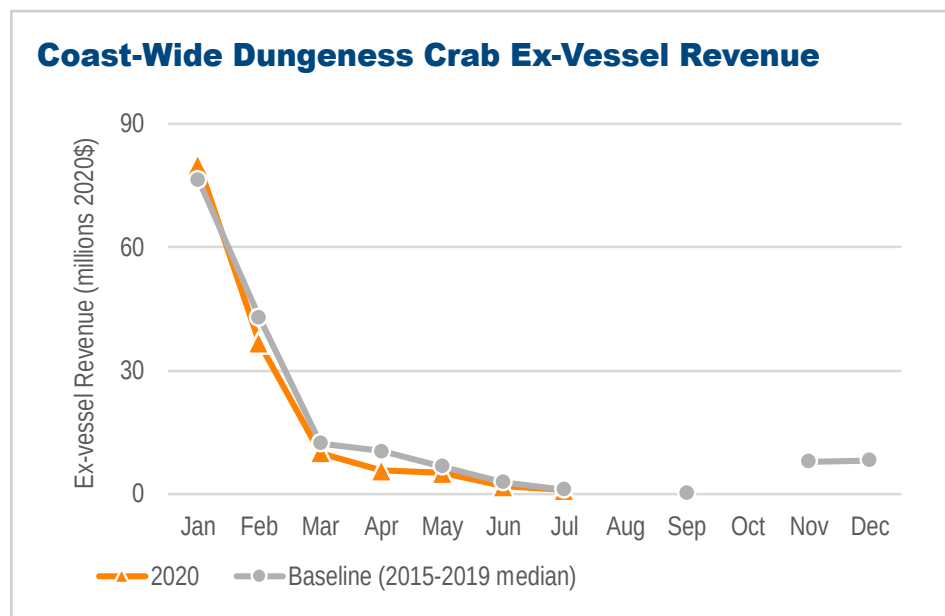


Figure 6. Coast-wide Dungeness crab ex-vessel revenue (millions of 2020 dollars).

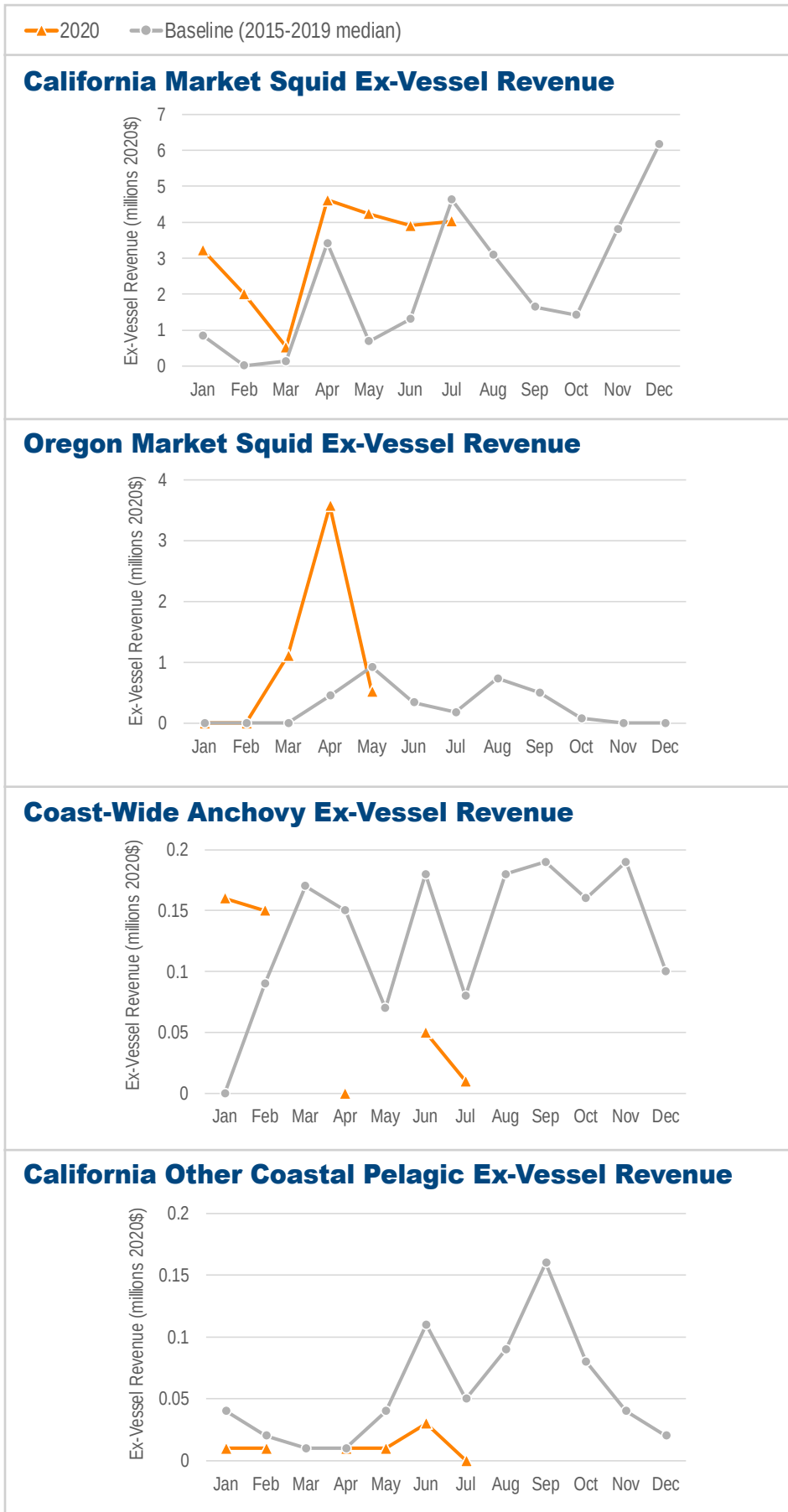
Region-wide, year-to-date ex-vessel revenue and landed weight in the Dungeness crab fisheries were lower than the 5-year median (2015-2019), and declines from the March-July time frame were greater, particularly for Washington and California.³ Region-wide year-to-date revenue was down 4% and weight was down 12%. From March-July, ex-vessel revenue was 32% lower and volume landed was 26% lower than the 5-year median. In Oregon year-to-date, Dungeness crab total volume and ex-vessel revenue were slightly higher than the 5-year median (2015-2019), by 4% and 6% respectively; and from March-July, ex-vessel revenue was similar to the 5-year median and volume landed was 9% higher. In California year-to-date, Dungeness crab total volume and ex-vessel revenue were 18% and 15% lower than the 5-year median; and from March-July, ex-vessel revenue was 59% lower and landed weight was 52% lower. In Washington year-to-date, Dungeness crab total volume and ex-vessel revenue were both 16% lower than the 5-year median (2015-2019); and from March-July, ex-vessel revenue was 29% lower and landed weight was 22% lower.

Concerns have been expressed that both quantity demanded and prices of crab will be depressed in the upcoming crab season. Many crab buyers typically freeze 4-5 months of inventory to sell it over the summer months, when the fishery is closed. Restaurant closures have nearly eliminated retail sales this summer and as a result, buyers report a large inventory of frozen crab still on hand.

COASTAL PELAGIC FISHERIES

The Coastal Pelagic Species (CPS) fishery targets the following: anchovy, market squid, sardine, and other coastal pelagics (which include mackerel). During the 2015 through 2018 period, over 94% of total CPS landings and revenues occurred in California. The share of revenues from the Pacific Northwest were higher earlier in the decade but declined due to the closure of the West Coast major directed commercial Pacific sardine fishery in 2015 (over 70% of Pacific sardine revenues occurred in the Pacific Northwest in the 2009-2015 period). The primary directed fishery for Pacific sardine has been closed since 2015. Beginning in 2018, there has been a major increase in market squid landings and revenues in the state of Oregon.

Coast-wide, the year-to-date total ex-vessel revenue and landed weight in coastal pelagic fisheries were higher than the 5-year median (2015-2019) by 36% and 5% respectively. From March-July, ex-vessel revenue was 28% higher and volume landed was 17% lower than the 5-year median. In California year-to-date, total ex-vessel revenue in coastal pelagic fisheries was higher than the 5-year median (2015-2019) by 20%, while landed weight was similar. From March-July, ex-vessel revenue was 20% higher and volume landed was 27% lower than the 5-year median. In Oregon, the year-to-date total ex-vessel revenue and landed weight in coastal pelagic fisheries were higher than the 5-year median (2015-2019) by 224% and 54% respectively.



For market squid, the year-to-date total ex-vessel revenue and landed weight were higher than the 5-year median (2015-2019) by 41% and 25% respectively. From March-July, ex-vessel revenue was 39% higher and volume landed was 13% higher than the 5-year median. In California, the year-to-date total ex-vessel revenue and landed weight in the market squid fisheries were higher than the 5-year median (2015-2019) by 22% and 9% respectively. From March-July, ex-vessel revenue was 32% higher and volume landed was 6% lower than the 5-year median. In Oregon, the year-to-date total ex-vessel revenue and landed weight in the market squid fisheries were higher than the 5-year median (2015-2019) by 228% and 201% respectively.

For anchovy, the year-to-date total ex-vessel revenue and landed weight were lower than the 5-year median (2015-2019) by 56% and 42% respectively. From March-July, ex-vessel revenue was 80% lower and volume landed was 71% lower than the 5-year median. A majority of the anchovy landings occurred in California.

Landing revenues for Pacific and other mackerel fisheries are captured in the other coastal pelagic fishery category. As can be seen by Figure 7, landing revenues for this species grouping are significantly down relative to the prior 5-year median.

HIGHLY MIGRATORY SPECIES FISHERIES

Year-to-date, total ex-vessel revenue and landed weight in the tuna fisheries were lower than the 5-year median (2015-2019) by 52% and 55%, respectively. From March-July, ex-vessel revenue was 56% lower and volume landed was 57% lower than the 5-year median. In Oregon, year-to-date, total ex-vessel revenue and landed weight in the tuna fisheries were lower than the 5-year median (2015-2019) by 77% and 80%, respectively. In California, year-to-date, total ex-vessel revenue and landed weight in the tuna fisheries were lower than the 5-year median (2015-2019) by 8% and 33%, respectively. From March-July, ex-vessel revenue was 16% lower and volume landed was 41% lower than the 5-year median. For Washington, year-to-date, total ex-vessel revenue and landed weight in the tuna fisheries were lower than the 5-year median (2015-2019) by 51% and 39%, respectively.

Figure 7. Coastal Pelagic Species ex-vessel revenue by type and location (millions of 2020 dollars).

Preliminary data suggest that August revenues for 2020 West Coast tuna are approaching the 5-year median (Figure 8). Anecdotally, these data are thought to be 85-90% complete, suggesting August 2020 revenues production may be comparable to levels of recent years.

Commercial albacore fishing industry members have reported extra costs due to protective measures such as grocery delivery services to vessels coming into port, to avoid requiring crew to leave the boat and face risk of contracting or spreading COVID-19. It is not clear to what degree these added costs are reducing profitability.

U.S. tropical tuna purse seine vessels in both the Eastern and the Western and Central Pacific Oceans face declining catch. Crew are not allowed to leave vessels to fly home, there are no observers, and drydocking is not possible. Vessel operators face increasing difficulties in obtaining parts and supplies. Landings prices are down, and profits are adversely impacted.

Coast-Wide Tuna Ex-Vessel Revenue

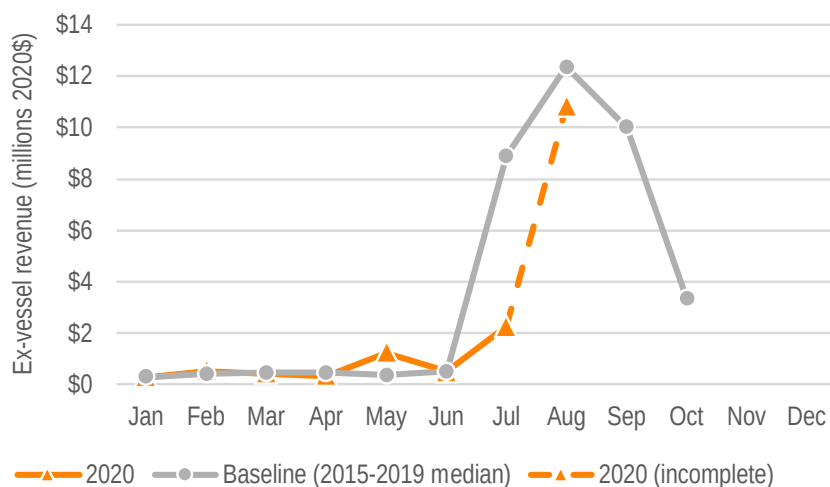


Figure 8. Coast-wide tuna ex-vessel revenue (millions of 2020 dollars). Dotted line indicates incomplete data.

Seafood Processors

Approximately 1,000 buyers purchased seafood from harvesters and producers in the West Coast Region in 2019. In 2020, these processors have faced increased costs and reduced demand coastwide, and there are compounding effects of wildfires and a water supply emergency in Newport, OR that halted processing operations. There have been a number of reported COVID-19 outbreaks at facilities. News sources indicate that some processing plants temporarily closed due to outbreaks at facilities in locations including Astoria/Warrenton, OR and Newport, OR. Closures reportedly ranged from several days to two weeks. In the last few months, there were several reports of employees testing positive for the virus; however, facilities did not close.

On the demand side, interviews with seafood processors conducted early during the COVID-19 outbreak indicate that the fresh seafood market was greatly affected by restaurant closures. There are reports that industry (at various levels of the supply chain) shifted towards other markets, including direct-to-consumer and community-supported fishery (CSF) sales. While some inroads have been made, there is still a substantial learning curve to make it successful. Interviews with large tuna processing companies in September indicate consumer retail demand for shelf-stable products remains strong but institutional demand for frozen products remains adversely impacted. Interviews with highly migratory seafood processors indicate that retail is strong but wholesale is moving smaller volumes than normal, but with a slight increase as restaurants reopen or expand operations. General uncertainty continues to negatively impact wholesale prices.

Processors are facing cost increases on several fronts. In addition to the cost of closures due to outbreaks, state agencies reportedly have been working on changes to infectious disease protocols, which, while not in place yet, has involved effort to ensure compliance. There have been changes to requirements for workers on processing lines. Distancing, installation of plexiglass shields between positions, and hiring on-site medical personnel are actions that some processors have taken.

Cold storage shortage is an issue that will likely continue to affect prices in the months ahead. Restaurant closures have pushed much seafood into cold storage, and most available freezer space is full. The U.S. Department of Agriculture (USDA) has tried to work with processors to find other outlets for currently frozen products in institutions such as prisons and schools. However, that program has been largely unsuccessful on the West Coast as the products in cold storage tend to be higher priced, luxury products such as Dungeness crab.

Recreational Fishing — For-Hire Sector

All modes of recreational fishing are now open after closures. In most jurisdictions, the relaxation of restrictions started with the opening of boat launches for private boats, followed by opening of public beaches, then for-hire operations. The majority of for-hire operations began operating as local guidelines allowed — with many operations starting in mid to late-May. Vessel launch and trip booking websites outline protocols in place for social distancing. A small number of charter vessels located on Indian reservations remain shut down as the reservations remain closed to the public. Vessels are carrying fewer passengers per trip, with some operations increasing ticket prices to make up for fewer passengers.

Retail tackle shops have reopened after the lifting of state and local mandated closures to non-essential business. Closures and openings varied significantly by jurisdiction. Some fishing tackle shops with online infrastructure were able to continue selling and shipping goods; others were able to take orders over the phone for pick up by appointment. Shops without an online presence were likely more impacted by a longer period of closure and/or no sales.

Many fishing trade shows were cancelled, or postponed only to be later cancelled, due to limits on public gatherings, closure of host locations, and logistics needed to change locations and dates of such large events. At least one fishing show for 2021 has already been cancelled. Fishing tournaments were also cancelled due to stay at home orders. As restrictions have been lifted, some tournaments have resumed.

For-hire effort update from RecFIN data

Across the West Coast, for-hire angler trips were considerably lower in 2020 relative to the 5-year median. However, in recent months, effort has returned to levels near the historical median. The availability of current data is a limiting factor in making a comparison in California due to COVID-related closure of the sampling program. For-hire effort data are available from January through July in both Washington and Oregon. In California, estimating effort relies on sampling that was not done from April through June, due to COVID-19 restrictions, so these data might remain missing. Based on all months for which RecFIN data were available for 2020, year-to-date reductions for the for-hire recreational fishing sector were as follows: 62% for Washington, 32% for Oregon, and 32% for California. Focusing on March through July only, reductions for for-hire trips were as follows: 62% for Washington, 31% for Oregon, and 36% for California.

Washington For-Hire Trips

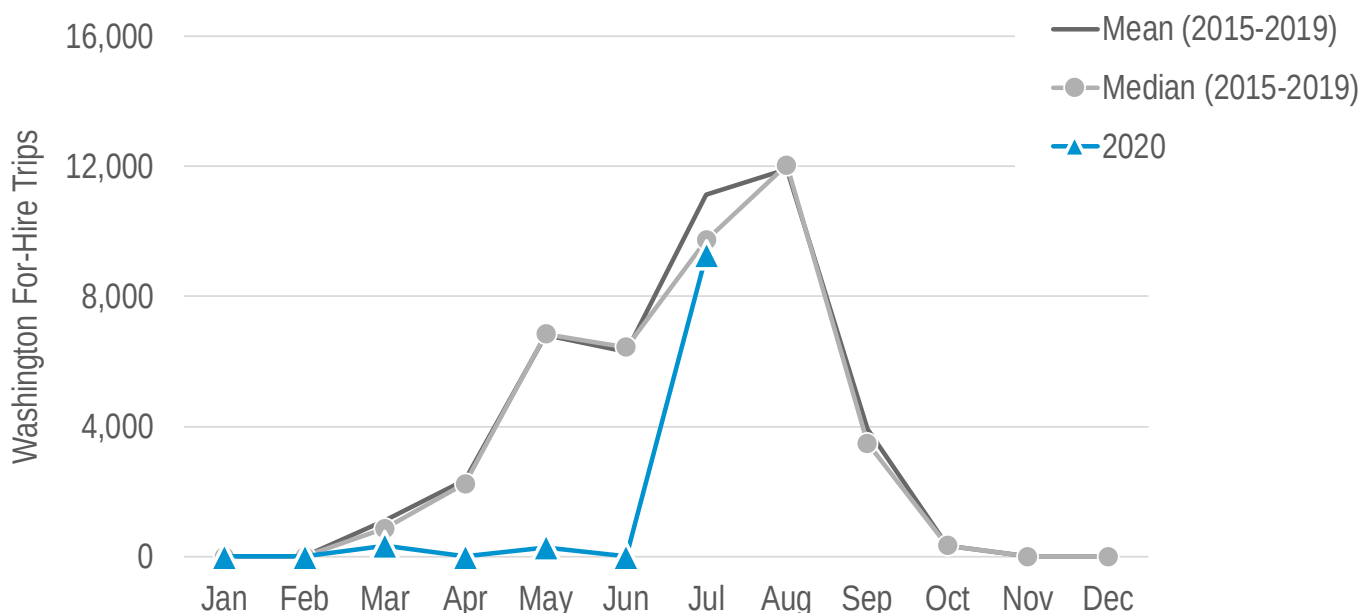


Figure 9. For-hire trips in Washington.

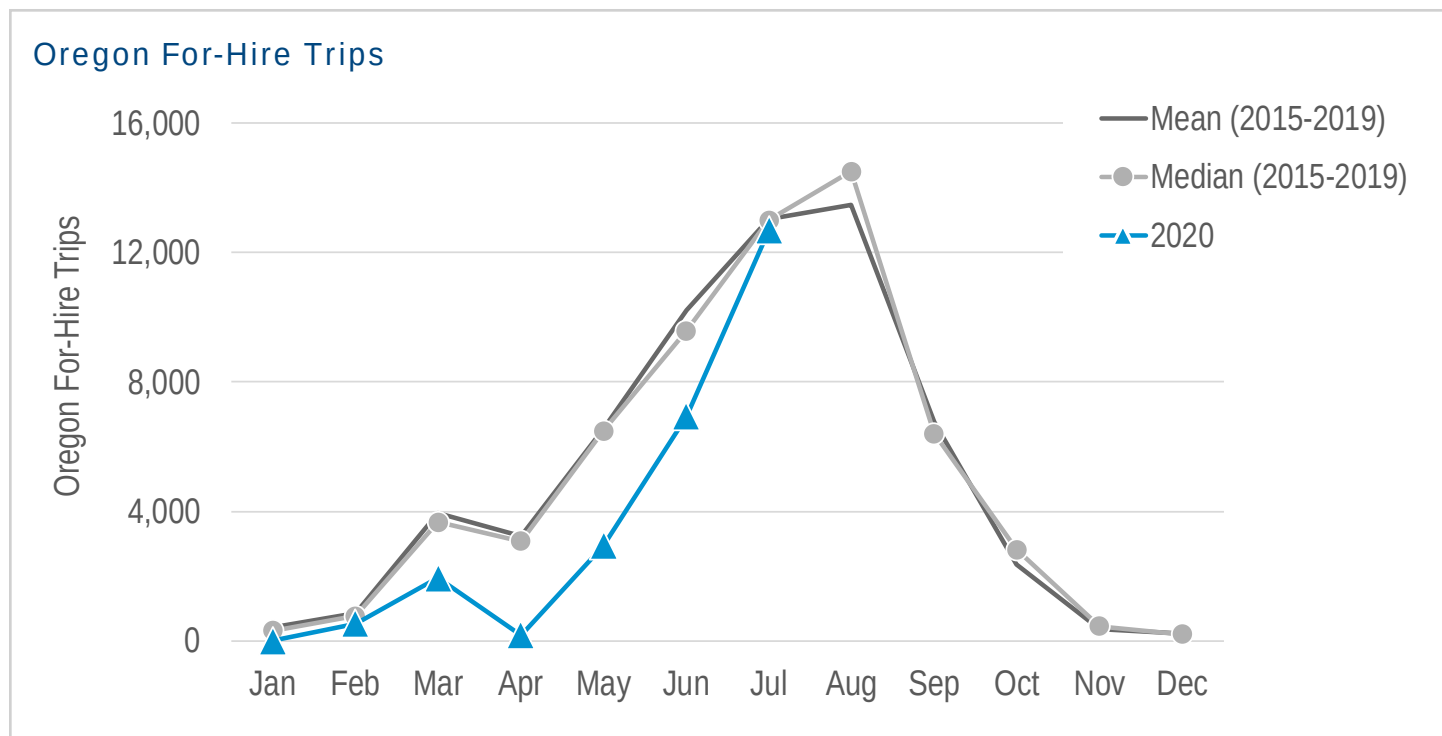


Figure 10. For-hire trips in Oregon.

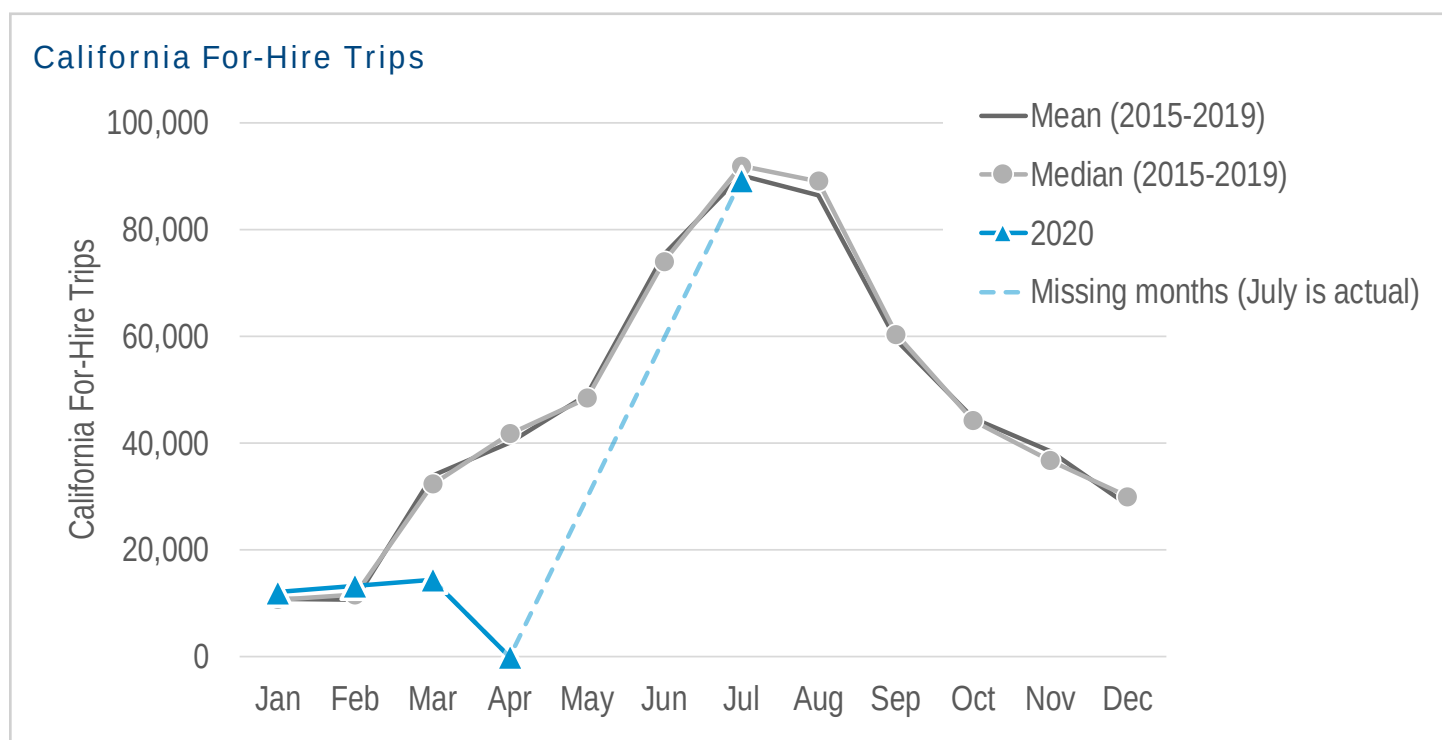


Figure 11. For-hire trips in California.

Endnotes

- 1 Data source: https://dataexplorer.northwestscience.fisheries.noaa.gov/fisheye/landings_tracker/
- 2 USOFR. 2020. 50 CFR Part 660: Fisheries Off West Coast States; Emergency Action To Temporarily Remove Seasonal Processing Limitations for Pacific Whiting Motherships and Catcher-Processors (RIN 0648-BJ83). Federal Register 85:119(19 June 2020):37027–37030.
- 3 For the Dungeness crab fishery, we remove 2015 and 2016 from this calculation because the fishery received a Federal fishery disaster designation in both years. Removing the disaster years puts the median within 4% of the average from 2011 to 2019.



U.S. Secretary of Commerce

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January 2021

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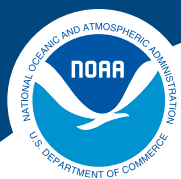
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NOAA
FISHERIES

In April 2020, NOAA Fisheries prepared its first national report on the regional impacts of COVID-19 on the commercial, recreational and aquaculture sectors. This report updates that initial assessment, capturing economic changes experienced by the fishing industry as the country began its phased reopening along with infusion of Federal funding through the CARES Act. NOAA Fisheries will continue to use this information to identify economic hardship where it exists and identify pathways for enhancing the resilience of the U.S. seafood and fisheries industries.

Pacific Islands Snapshot, March-July 2020



Pacific Islands Fisheries Impacts from COVID-19

Hawai'i

The State of Hawai'i implemented numerous protective measures to prevent the spread of the novel coronavirus beginning in mid-March, including social distancing (March 13) and cancellation of public gatherings (March 15), a state-wide stay-at-home work-at-home order (March 25), and all persons entering the State of Hawai'i (visitors and returning residents) were required to self-quarantine for 14 days or for the duration of their stay in Hawai'i, whichever is shorter (March 26).¹

Along with many other states, these restrictions were slowly relaxed between the months of May and July, as the islands staged an incremental reopening strategy. However, surges in domestic cases in June and July precluded the State of Hawai'i from relaxing travel quarantine restrictions. Initial efforts to roll out a program that would allow for travelers with pre-travel negative test results to bypass quarantine was planned to take effect on August 1, but this has since been pushed back to September 1, later to October 1, and is currently proposed to launch in mid-October.² While outside the time period for this analysis, it should be noted that local case counts spiked during August and September, and the islands returned to a strict lockdown with renewed state-wide stay-at-home orders for a period of 4 weeks (August 27-September 23).³

Hawai'i's largest industry, tourism, which provides high demand for Hawai'i seafood products, remains shuttered, creating significant economic hardship statewide. Cumulative visitor counts for the months of April-July 2020 (53 thousand visitors) are down 98.5% from this same April-July period in 2019 (3.6 million visitors).⁴ Unfortunately, the forecast is bleak with the State of Hawai'i Department of Business, Economic Development, and Tourism (DBEDT) predicting annual visitor counts in 2020 to be down 72% from 2019, with visitor spending to be approximately 68% less than 2019.⁵ While seasonally-adjusted unemployment rates for the State of Hawai'i have improved from 23.8% (April) to 13.1% (July), these levels have consistently exceeded national unemployment rates of 14.4% and 10.5% for the months of April and July, respectively. While fishing and seafood markets are classified as an "Essential Business", the Hawai'i fishing and seafood industry has experienced significant economic impacts as a result of global COVID-19 spread.

The Hawai'i fishing and seafood industry is an integrated food production and supply system that links fishermen to our nation's only fresh tuna auction, the fish auction buyers (mainly wholesalers), and ultimately retailers and restaurants in Hawai'i and across the United States. The COVID-19 pandemic has virtually eliminated market demand for Hawai'i seafood in local restaurants, which are heavily dependent on tourism, and severely restricted the mainland US retail market. What has

remained is the local retail market in Hawai'i. This significant reduction in market demand has cascaded through market channels to the fishing sector, which has faced significant reductions in fish prices, and the market has struggled to balance supply with reduced demand. The economic viability of fishermen, the fish auction, and fish processors continue to be threatened by the economic effects associated with pandemic restrictions and shifts in demand.

Despite these challenges the fishing community (commercial fishers, non-commercial fishers, seafood distributors) in the Pacific Islands Region plays a vital role in supporting local food systems, nutrition, food security, and social cohesion.⁶ This importance is amplified in the face of natural disasters and human health crises, and fishing communities across the Pacific Islands Region have adapted to continue these crucial functions in the face of this unprecedented disruption.

Commercial Fisheries Landings Trends and Impacts through July 2020

In 2018, Honolulu was the nation's #6 port in seafood value (\$106 million).⁷ On average, the Hawai'i longline fishery comprises approximately 97% of Honolulu fishery revenues and over 85% of fishery revenues for the State of Hawai'i. Other important commercial fisheries in Hawai'i include small boat fishers targeting tunas and other highly-migratory species, as well as bottomfish, nearshore, and reef fish species. State of Hawai'i commercial fisheries are year-round fisheries, with revenues peaking in the months from March through June, with a second seasonal peak during the holiday months of December and January. During 2018, there were 121 active seafood dealers, over 2,500 licensed fishers with commercial sales, and a world-renowned charter/for-hire industry.

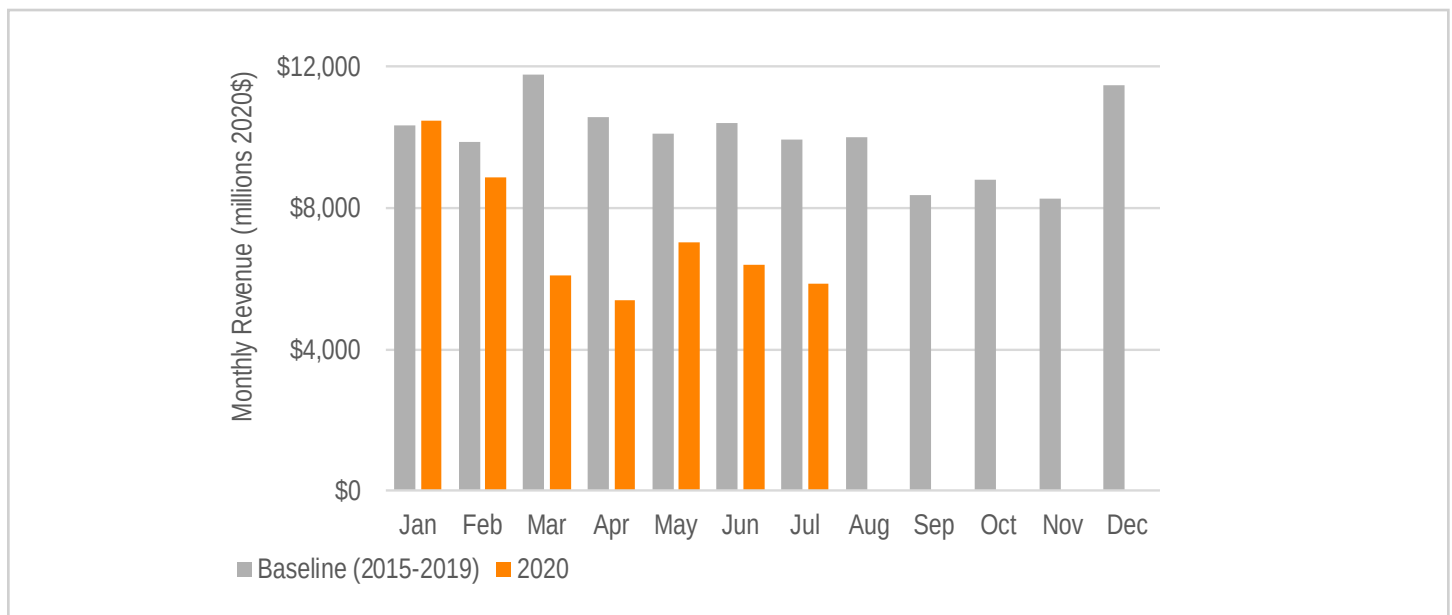


Figure 1. Monthly Hawai'i commercial landings revenue, 2020 relative to 2015-2019 (Inflation-adjusted, 2020 dollars).

Average monthly State of Hawai'i commercial fishing inflation-adjusted revenues over the baseline period of 2015-2019 was approximately \$10 million (adjusted to January 2020 dollars, using Bureau of Labor Statistics Hawai'i all-urban consumer price index). Preliminary monthly revenue estimates for 2020 show clear impacts from COVID-19. Market price declines of just under 75% that hit in mid-March⁸ and held over the last two weeks of the month, coupled with industry-imposed landing limits, resulted in industry revenue declines of approximately 48% for the month of March relative to baseline monthly averages. The expectation, at the time, of sustained industry struggles has been realized.

Throughout this report, where possible, 2020 data are presented. These data are preliminary and subject to change.

Hawai'i commercial fishery revenues (March-July 2020) have experienced an approximate 42% decline relative to the 5-year baseline (2015-2019) for the months of March-July. Cumulative preliminary 2020 commercial fishery revenues (March-July 2020) are approximately \$22 million below the 5-year baseline for this period (March-July 2015-2019). Similarly there have been reductions in active fishers and seafood dealers, on account of COVID-19 impacts and restrictions.

Table 1. Hawai'i Commercial Fisheries Performance: Percentage Change, 2020 compared to Baseline (2015-2019).*

Month	Revenues	Landings	Prices	Fishers**	Dealers**
March	-48.4	-24.3	-31.8	-28.6	-16.7
April	-49.1	-29.4	-27.8	-38.8	-36.7
May	-30.4	-25.0	-7.2	-36.7	-28.2
June	-38.7	-43.4	+8.3	-32.4	-33.1
July	-41.1	-34.4	-10.3	-37.4	-33.9
Average (March-July)	-41.5	-31.3	-13.8	-34.8	-29.7

* Source: State of Hawai'i Division of Aquatic Resources, 2020 data are preliminary (as of 9/11/20).

** Number of unique fishers submitting commercial reports, number of unique dealers submitting dealer reports.

Commercial Fishing Impacts

LOGLINE FISHERY

During 2019, there were 150 active longline fishing vessels, landing approximately 26.5 million pounds, valued at nearly \$95 million. Due to the recent sharp decline in landed fish prices, beginning on March 14, longline fishing operations have seen significant revenue losses. The Hawai'i longline fishery has taken nearly the full impact (approximately 94%) of cumulative preliminary 2020 Hawai'i commercial fishery revenue declines between March and July 2020 relative to the 5-year baseline for this period (March-July). Preliminary Hawai'i Longline commercial fishery revenues (March-July 2020) have experienced an approximate 45% (or \$20.7 million) decline relative to the 5-year baseline (2015-2019) for the months of March-July.

Between March 14 and July 31, the average weekly price at the United Fishing Agency (UFA) auction for the two primary target species (bigeye tuna and swordfish) were down approximately 20% compared to pre-COVID 2020 prices (January 4 – March 13). Price improvements were seen during mid-May through June. However, this was largely due to a nearly 50% reduction in landings levels as the market worked to balance supply with local demand (industry-imposed vessel-level and daily landings restrictions have been maintained since mid-March). It should be noted, during May and June there also was improved access to mainland markets as states began to open up, coupled with the relaxing of local restrictions which helped pick up prices for these key target species. However, as restrictions heightened in many states during July to match rising case counts, the industry saw these price gains disappear, highlighting the direct impacts of COVID-19 restrictions on the Hawai'i seafood market.

Counter to fishery target species, the average weekly price at the UFA auction for combined mix/whitefish species,⁹ primarily marketed to the foodservice sector were down approximately 54% compared to pre-COVID 2020 prices. These species have seen little to no rebound in recent months and comprise a large portion of fishery landings, significantly dampening industry revenues.

Table 2. Hawai'i Longline Fishery Performance: Percentage change, 2020 compared to Baseline (2015-2019).*

Month	Revenues	Landings	Prices	Fishers**
March	-48.9	-21.3	-35.0	-2.2
April	-51.5	-31.3	-29.4	-16.2
May	-33.6	-28.0	-7.7	-8.8
June	-47.5	-52.0	+9.4	-19.6
July	-45.3	-38.2	-11.4	-6.1
Average (March-July)	-45.3	-34.2	-14.8	-10.6

* Source: State of Hawai'i Division of Aquatic Resources, 2020 data are preliminary (as of 9/11/20).

** Number of unique fishers submitting commercial reports.

To support the local community and alleviate challenges in matching supply and demand, the Hawai'i seafood industry has established new partnerships with community organizations. In late April, the Hawai'i seafood industry donated approximately 2,000 pounds of fresh fish to the Hawai'i Foodbank, and through this partnership Hawai'i Foodbank purchased \$50,000 worth of seafood landed by Hawai'i longline vessels.¹⁰ In early July, the City of Honolulu has committed \$2.6 million in federal coronavirus relief funds to help the

fishing industry via a “fish-to-dish” program¹¹ which will work in partnership between the Hawai‘i Foodbank and the Hawai‘i Longline Association, the United Fishing Agency, which runs the fish auction, and the Hawai‘i Seafood Council to distribute fish to people in need in the community. An estimated 350,000 servings of fresh fish are expected to be distributed to the community over the next five months through the Hawai‘i Foodbank and its partner agencies. The long-term financial outlook for the Hawai‘i longline fishery remains highly uncertain and depends on both local and national recovery efforts.

The top COVID-19 related factors affecting business for the Hawai‘i longline fishing sector continue to be:

- Reduction of market prices and landed value (particularly for mix/whitefish species).
- Reduced market demand from foodservice sector.
- Market competition with cheaper foreign imported frozen products.
- Reduced opportunities for credit offered by supply companies (e.g. fuel).

OTHER COMMERCIAL FISHERIES

Other important commercial fisheries in Hawai‘i include small boat, spear, and nearshore fishers targeting tunas and other highly-migratory species, as well as bottomfish, nearshore, and reef fish species. During 2018, there were just under 1,700 commercially-licensed fishers in the State of Hawai‘i, landing nearly 5 million pounds valued at approximately \$16 million. These fishers have faced similar negative pricing impacts as the longline fishery on account of COVID-19 as they market their fish through the UFA auction, dealers/processors, restaurants, retail storefronts, and within their community. Historically low prices and state-wide stay-at-home orders severely limited commercial small boat fishing effort during March. However, as local restrictions relaxed, while there were fewer active fishers, difficult economic conditions likely drove increased fishing activity (preliminary 2020 revenues in June exceeded baseline revenues). Many commercial small boat fishers have been forced to and/or chose to shift to marketing their fish via social media, within community networks, and developing value-added products (e.g., smoking) with their catch. Pursuing these marketing channels, coupled with significant reductions in longline fishery landings, likely helped prices during this time period.

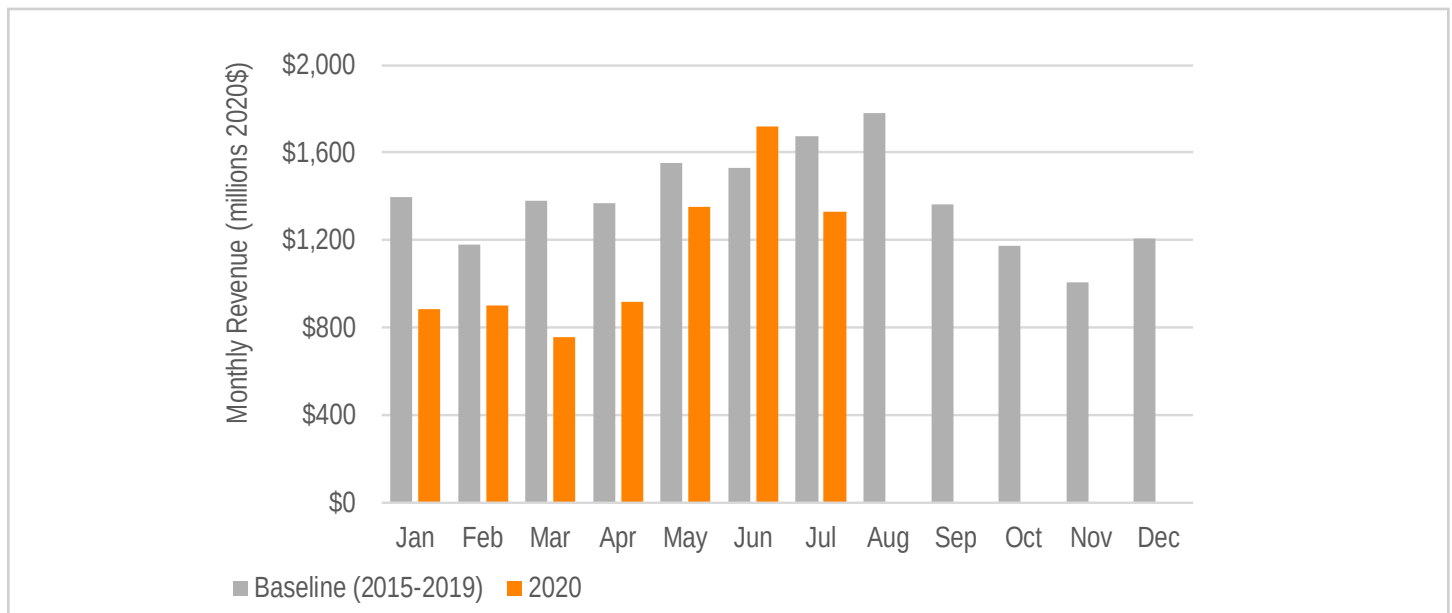


Figure 2. Monthly Hawai‘i other (non-longline) commercial fisheries landings revenue, 2020 relative to 2015-2019 (Inflation-adjusted, 2020 dollars).

Preliminary Hawai‘i other (non-longline) commercial fishery revenues (March-July 2020) experienced an approximate 20% decline relative to the 5-year baseline (2015-2019) for the months of March-July. Preliminary cumulative 2020 fishery revenues March through

July 2020 are approximately \$1.4 million below the 5-year baseline for this period (March-July). There have been notable reductions in active fishers and seafood dealers, on account of COVID-19 impacts and restrictions.

Table 3. Hawai'i Other (Non-Longline) Commercial Fisheries Performance: Percentage Change, 2020 compared to Baseline (2015-2019).*

Month	Revenues	Landings	Prices	Fishers**	Dealers**
March	-45.0	-46.0	+2.0	-33.4	-16.9
April	-32.8	-16.8	-19.2	-42.7	-37.4
May	-12.9	-6.8	-6.5	-41.0	-28.7
June	+12.3	+2.9	+9.2	-34.2	-33.6
July	-20.5	-17.3	-3.9	-41.9	-34.7
Average (March-July)	-19.7	-16.8	-3.7	-38.7	-30.2

* Source: State of Hawai'i Division of Aquatic Resources, 2020 data are preliminary (as of 9/11/20).

** Number of unique fishers submitting commercial reports, number of unique dealers submitting dealer reports.

These fishers (along with thousands of non-commercial fishers) play vital roles in supporting local food systems, nutrition, food security, and community social cohesion.¹² This importance is amplified in the face of natural disasters and human health crises. A public Facebook group *Hawai'i Fishers Feeding Families* was established in mid-April to promote fisher contributions to local food security. As of July, over 1,200 individual fishers have posted a cumulative estimate of over 10,500 pounds of fish that have helped to feed over 11,500 people across the State of Hawai'i.

Seafood Dealers/Processors

The Hawai'i longline fishery supplies ice-chilled, high quality fresh fish that seafood dealers and processors distribute to Hawai'i and U.S. mainland foodservice and retail markets. The Hawai'i seafood industry produces over 80% of U.S. domestic landings of bigeye and yellowfin tuna and 55% of the Nation's domestic supply of swordfish. There are around 1,000 workers in the State of Hawai'i (directly) employed in wholesale seafood and supply businesses and thousands (indirectly) employed in local restaurants supplied by Hawai'i commercial fisheries. As shown in previous tables, between March and July 2020 there were approximately 30% fewer active dealers in the State of Hawai'i relative to the 5-year baseline (2015-2019) for the months of March-July.

The Hawai'i seafood market was hit extremely hard beginning in mid-March.¹³ Beginning March 15, vessel landing limits and daily target total volume restrictions have been imposed by the Hawai'i Longline Association (HLA) and the United Fishing Agency (UFA) auction, and these restrictions hold to date. The Hawai'i longline fishery is Hawai'i's largest food producing industry and severely reduced supply has important food security considerations.

Current challenges for the industry are similar to those outlined in the early days of the COVID-19 shock and involve matching fishery supply with local consumer retail demand as fresh air freight for all seafood products is limited, reducing US mainland market access. In an effort to mitigate low prices, more than one processor has begun direct marketing in an effort to generate cash flow and move product. An additional concern is competition with cheaper frozen import product inventories, and with no disruption

to ocean shipping channels, this poses a significant short term challenge due to price competition in local retail markets, as local communities endure harsh economic conditions and dramatic increases in statewide unemployment stressing food budgets.

The Western Pacific Fishery Management Council (WPFMC) undertook communications¹⁴ with Hawai'i seafood dealers and processors during the months of May and June. Approximately 28 businesses engaged in these communications. A majority (86%) experienced negative revenue impacts in the first quarter of 2020, ranging from -5% to -95%, with an average of -43%. Over 72% of businesses reduced their workforce in the past year, with an average of 35% staff reduction (47 workers, with a minimum of two and maximum of 140). A combined 80% indicated that the COVID-19 pandemic has been somewhat, substantially, or completely responsible for these losses. Seafood businesses whose most important customers were restaurants experienced the greatest average loss of employees, followed by businesses whose most important customers were retail establishments, and then businesses whose most important customers were distributors. A majority expect negative impacts for the remaining quarters of 2020, in the -25% to -75% range.

As stated by one representative, "COVID 19, combined with the effects of the Hawai'i travel quarantine, has almost destroyed our business, it's been extremely impactful and making it hard for all of us [in the seafood industry] to survive."

The top COVID-19 related factors affecting business for seafood dealers/processors continue to be:

- Reduced demand across all markets (mainland and Hawai'i; retail and particularly restaurants).
- Managing inventory (decreasing storage capacity for fresh local product).
- Shipping/distribution constraints - reduction in air cargo capacity as airlines have limited flights.

Charter/For-Hire Impacts

As the only state in the U.S. where marlin and other trophy billfish can be reliably caught year-round, Hawai'i is well known among serious anglers as a destination for big game fishing trips. Although Kailua-Kona (Big Island) draws in many return and regular patrons with its world renowned tournaments, historical harbor and promises of trophy fish, for the most part, Hawai'i's charter operations host first-time patrons visiting the state. The Hawai'i charter/for-hire industry has been effectively closed for business as of mid-March on account of social distancing mandates, stay-at-home orders, drastic reduction in visitor numbers,¹⁵ visitor quarantine mandates, and suspension of harbor operations and commercial ocean activities, including tournaments.¹⁶ Charter/for-hire permit restrictions were relaxed in late-May/early June, but social distancing and tourism restrictions have precluded any industry rebound to date.

Preliminary estimates of reported charter fishing trips for the month of March were down 66% relative to the baseline of 2015-2019. Approximately 39 charter/for-hire fishing trips were reported statewide between April and July 2020, a 99.3% decline from the cumulative average of 3,200 trips taken during this time period, using the baseline of 2015-2019. The 2020 Hawai'i International Billfish Tournament (HIBT) was cancelled due to COVID-19, and several local tournaments scheduled between March and July were also cancelled. The 2020 Hawai'i Marlin Tournament series has been held as scheduled (it began in early July), but with significant reductions in participation from previous years (about one-third of traditional participation levels).¹⁷

To date, COVID-19 and the restrictions in place to mitigate spread have imposed catastrophic financial burden on charter operators in Hawai'i and deprive the state of significant economic contributions through supporting industries,¹⁸ the scientific community of valuable tagging data, and many operators indicate the viability of the operations in the near future is highly uncertain.

American Samoa

American Samoa implemented strict protective measures to prevent the spread of the novel coronavirus including social distancing and cancellation of public gatherings associated with a public emergency declaration (March 18), and island-wide stay-at-home work-at-home order (March 24). Businesses have been subject to a curfew, and restaurants, bars and nightclubs, which are primary destinations for island seafood can serve only 10 or fewer customers, leading many to shut down entirely, unable to cover payroll, rent, utilities, and other fixed costs. Perhaps most significantly, on March 30, all flights between Hawai'i and American Samoa were suspended for 30 days,¹⁹ and these flight restrictions have been extended and broadened, resulting in no commercial flights, so the island is effectively closed. As of July, there are no reported positive cases of COVID-19 in American Samoa.

The American Samoa Government imposed a 15-day quarantine on foreign vessels that have visited other ports, met up with another vessel, or fueled via a tanker. These vessels supply the StarKist Samoa cannery. Cannery staff and stevedores are allowed onboard to offload product, but the crew are not allowed to depart vessels. The Department of Health has maintained diligence to ensure crew are safe and limiting exposure to local populations. Given ongoing travel restrictions many skippers and crews have not been able to return home for over 6 months.

Commercial Fishing Impacts

PURSE SEINE FISHERY

In 2020, there have been 24 active purse seine fishing vessels delivering product to the StarKist Samoa cannery in American Samoa. Due to ongoing travel restrictions, these operations are under continued strain maintaining morale for skippers and crew facing long onboard confinement. While observer requirements continue to be waived on a case-by-case basis, a significant operational cost borne by industry was the repatriation of fishery observers. Concerns expressed in March by Bill Gibbons-Fly, executive director of the American Tunaboat Association (ATA) which represents the U.S. Pacific tuna purse-seine fleet, continue to date, "The increasing travel constraints throughout the Pacific are complicating efforts to get crew, repair parts, technicians and supplies to boats in a timely fashion". Operational difficulties associated with COVID-19 restrictions such as limited at-sea transshipment, inability to change

crews and conduct routine maintenance, quarantine periods associated with port calls, on top of other factors (low prices [particularly March-June]²⁰ and fully utilized high seas allocation) will likely negatively impact fishery effort and performance.

LONGLINE FISHERY

The American Samoa longline fishery operates out of Pago Pago, American Samoa. In 2019, there were 17 active vessels that took approximately 100 trips, landing approximately 3 million pounds valued at about \$4 million. The primary target is albacore tuna and the fishery delivers primarily to StarKist Samoa.

The months of May through July are typically the most productive season for this fishery. The fishery has faced significant economic struggles in recent years,²¹ but preliminary 2020 estimates would suggest that despite some reductions in the number of active vessels in 2020, fishery performance appears to mirror 2019. However, local fisheries experts observed that the longline fleet has struggled to recruit fishing crew. Many of the fishing crew originate from Apia in Western Samoa, and travel restrictions have prevented international workers from returning to American Samoa. Some longline boats have adapted by sharing crew members.

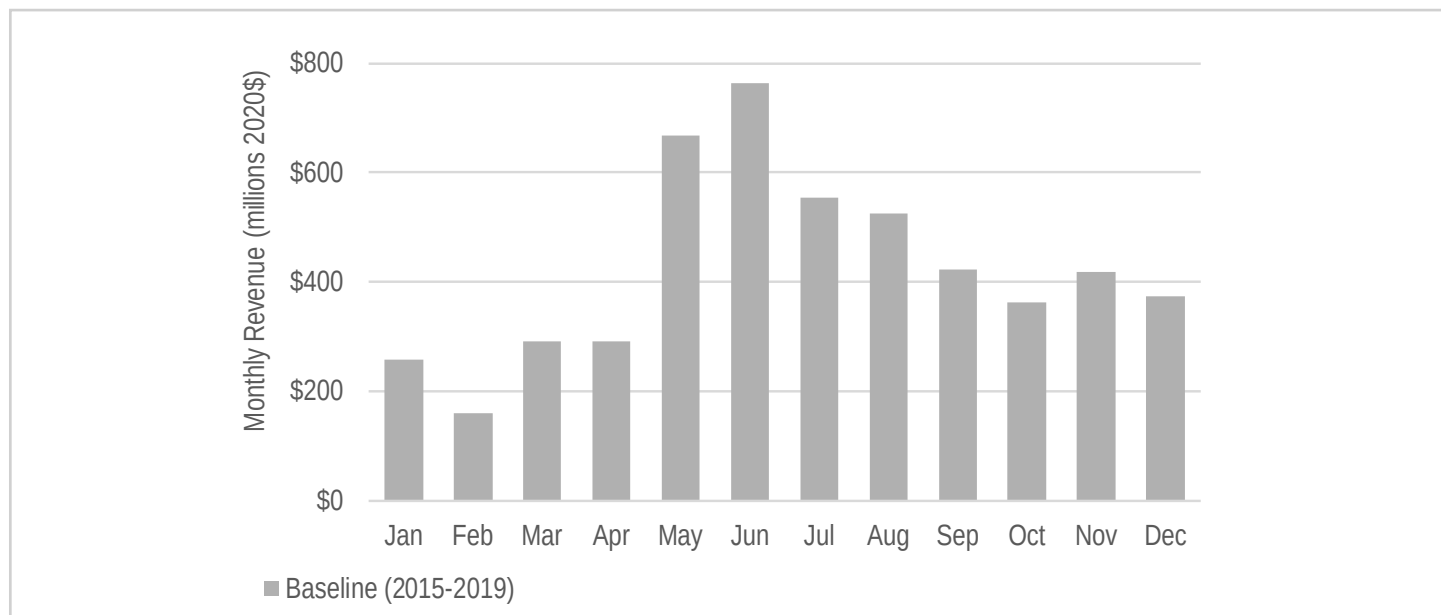


Figure 3. American Samoa longline average monthly fishery revenues: 2015 - 2019 (Inflation adjusted to 2018 dollars).

OTHER COMMERCIAL FISHERIES

American Samoa alia and small boat fisheries are a mix of subsistence, cultural, recreational, and quasi-commercial fishers. Fish and fishing are an integral part of the culture and important component of the social fabric in American Samoa. In addition to social importance, most fishers consider the fish they catch to be an important source of food for their families. Fishing is critically important in terms of building and maintaining social and community networks, perpetuating fishing traditions, and providing fish to local communities as a source of food security. During the baseline period of 2015-2019, average annual commercial fishery revenues for American Samoa other commercial fisheries was approximately \$339 thousand.²²

A group of local fishing experts reported that, in spite of COVID-19, part-time alia and small boat fishing has continued as normal, including sports fishing events. They also confirmed that strong cultural traditions of sharing and resource distribution as a source of community resilience have mitigated any effects of private sector job loss.

Seafood Dealers/Processors

StarKist Samoa, the largest local private employer on island with about 2,000 workers, has received exempt status from the American Samoa Governor's emergency declarations, allowing it to maintain operations, given it operates evening and sometimes weekend shifts.²³ While fish supply has remained steady, given contributions from U.S. and foreign purse seine vessels, and the plant has been operating at full capacity, keeping up with demand has been challenging and the cannery has faced numerous challenges due to COVID-19 restrictions. Flight restrictions to and from American Samoa have increased the cost of airfreight for the cannery. Additionally, flight restrictions have hampered plant maintenance projects, constrained professional service contracts, and disrupted new recruitment for cannery workers. Despite these challenges, StarKist Samoa continues to play a vital role in the U.S. food supply chain with average annual canned tuna exports to the United States of approximately \$400 million per year, in recent years.²⁴ The risk of COVID-19

to cannery operations cannot be overstated as any positive cases in American Samoa would likely put cannery operations at significant risk, jeopardizing the American Samoa economy and U.S. seafood supply chain.

Commonwealth of the Northern Mariana Islands

The Commonwealth of the Northern Mariana Islands (CNMI) implemented strict protective measures to prevent the spread of the novel coronavirus including, social distancing and cancellation of public gatherings associated with a public emergency declaration coupled with a stay-at-home work-at-home order (March 17),²⁵ and all inbound travelers, including returning residents are required to undergo a 14-day quarantine (March 23).²⁶

Tourism is by far the largest industry in the CNMI and COVID-19 impacts began in February, with 11 major hotels collectively reporting the lowest occupancy rates ever recorded at less than 20%. Hotels started the planning stage of laying off employees, closing entire wings, closing restaurants, and suspending contracts for outsourced services.²⁷ March 2020 visitor arrivals were down 85% from last year and tourism was effectively shut down in recent months due to flight suspensions for all non-residents. This loss of livelihood, coupled with federal immigration work visa policies, has led to significant out-migration of international workers from the Philippines, the Marshall Islands, and the Federated States of Micronesia (FSM). As of July 1, the Marianas Visitors Authority has outlined procedures²⁸ that would allow non-resident visitors to avoid a 14-day quarantine, a first step to rebuilding tourism in the islands.

A recent survey conducted by the Saipan Chamber of Commerce²⁹ found that 55% of responding businesses have either made large operational reductions or are temporarily closed; 72% have either reduced staff or decreased hours; 40% have had to reduce over a quarter of their total staff; 79% have seen a reduction in revenue for more than four weeks; and 47% have potentially lost more than \$100,000 in revenue due to COVID-19 and the economic downturn.

In response to this drastic and dramatic decline in tourists, the CNMI government implemented austerity measures to balance projected budget shortfalls, which among other measures include 16-hour cuts (shift to a 64 hour bi-weekly work schedule) for government employees.³⁰ The first two confirmed positive cases for COVID-19 in the CNMI occurred on March 29. Since then there have been 59 COVID-19 cases, half of which were detected in people arriving from outside CNMI.

CNMI small boat fisheries are a mix of subsistence, cultural, recreational, and quasi-commercial fishers. Fish and fishing are an integral part of the culture and important component of the social fabric in the CNMI. In addition to social importance, most fishers consider the fish they catch to be an important source of food for their families. Fishing is critically important in terms of building and maintaining social and community networks, perpetuating fishing traditions, and providing fish to local communities as a source of food security.³¹ During the baseline period of 2015-2019, average annual commercial fishery revenues for CNMI fisheries were approximately \$686 thousand.³²

In considering COVID-19 impacts to the CNMI fishing community, the Saipan Fishermen's Association cancelled their annual Mahimahi Fishing Derby scheduled for March 28. In response to the first two positive cases, one of which had travelled to Tinian, on March 30 the Tinian government implemented a "sunset-to-sunrise" curfew and closed the harbor to recreational and commercial fishing.³³ This restriction was also enforced in Saipan and effectively shut down night-time spear and bottom fishing. The following day, fishing outside the reef was also banned, and all but one boat ramp was closed so that fishing activities could be closely monitored. This shut down the small boat pelagic fisheries. This also effectively shut down the fresh fish market. The community petitioned to relax the restrictions on fishing, and in April and May changes were made to the curfew, and boat ramp access which has allowed for fishing again.

COVID-19 regulations have had a sudden and severe impact on fishing in CNMI. While access to fishing has returned, the market for fresh fish, which relied heavily on a tourism economy, has been greatly reduced with low demand. The majority of fresh fish available to the population now come from people who fish for themselves or their family.

Guam

Guam implemented strict protective measures to prevent the spread of the novel coronavirus, including social distancing, cancellation of public gatherings, public emergency declaration (March 14), an island-wide stay-at-home work-at-home order, and closure of non-essential businesses (March 19).³⁴ Tourism impacts began in February, with major hotels experiencing roughly 30% reductions in occupancy rates.³⁵ March arrivals were down nearly 77% relative to 2019, creating significant economic hardships. Reopening of travel to Guam from the countries of Japan, South Korea, and Taiwan (originally scheduled for July 1) has been postponed until further notice. Military personnel movement and patronage of restaurants and bars has also been restricted intermittently.³⁶

Guam small boat fisheries are a mix of subsistence, cultural, recreational, and quasi-commercial fishers. Fish and fishing are an integral part of the culture and important component of the social fabric on Guam. In addition to social importance, most fishers consider the fish they catch to be an important source of food for their families. Fishing is critically important in terms of building and

maintaining social and community networks, perpetuating fishing traditions, and providing fish to local communities as a source of food security.³⁷ During the baseline period of 2015-2019, average annual commercial fishery revenues for Guam were approximately \$550 thousand.³⁸

The Guam Fishermen's Cooperative Association (GFCA) is a central component of Guam's contemporary fishing industry that continues to pursue and broaden its original mission of providing marketing services, fuel, and ice for its small-boat fishermen members. Organized in 1976, GFCA's influence has become pervasive, providing a variety of benefits not just to its members, but for fisheries conservation, marine education, and the greater Guam community.³⁹ Amid coronavirus concerns, the GFCA remains open to support fishers and provide seafood to the local community, whose restaurant options for fresh fish have dwindled. Stay-at-home measures, however, have had negative impacts on GFCA operations due to a limited customer base.

The GFCA has reported a 70-80% decline in sales due to declines in tourism, tough economic conditions, and a shift in local consumers' preferences away from fresh product toward emergency supplies. Additionally, the timing of Guam's SCUBA spearfishing ban coincided with the start of COVID-19 impacts, limiting reef fish availability for the GFCA until the start of the bottomfish season. The GFCA is currently selling more non-traditional products, including tilapia and shrimp. Concerns about NOAA's discontinued bio-sampling and commercial receipt data collection arose during discussions with the GFCA, given their provision of key insights into COVID-19 impacts to Guam's fisheries and fishing community.

Discussions with members of the local fishing community indicate that the charter fishing industry has been mostly shut down since March. However, charter businesses and their employees have maintained their credentials and boat basin infrastructure at personal expense (e.g. drug tests, licenses, registration, harbor slips).

Observations from local fishers and business owners indicate that older fishers' (50-60+ years old) participation has declined during the pandemic given health concerns. This has implications for local seafood supply, given this age group's association with the regular provision of fresh seafood to Guam and more frequent fishing trips. Discussions suggest that younger fishers have sustained participation better than their older counterparts, but tend to fish less frequently. Logistical considerations for fishing activity have also shifted given COVID-19 concerns. One fisher described creating a "bubble" with 1-2 consistent fishing partners, gathering fishing supplies on fewer, more intentional trips to local shops, and embarking on fewer fishing trips lasting all day to maximize catch.

Overall, however, discussants from Guam's fisheries described an increase in fishing participation, given that many other activities have been classified non-essential. One discussant described inconsistencies between local government and enforcement officers' understanding of the essential classification of subsistence fishing. Given gathering restrictions and health concerns, the sharing of seafood has become less tied to community socializing. Instead, discussants described fishers prioritizing the food security of close family and friends, with remaining catch used for bartering or sale, for example, at the GFCA. Fishing has thus provided a mechanism for resilience during the pandemic.

A discussion with a Hagåtña restaurant owner revealed challenges in acquiring local fresh fish as quasi-commercial fishers prioritize fish for their close family and friends, and commercial fishers seek alternative sources of income. The discussant noted high hotel vacancy rates and closing restaurants which have made it difficult for commercial fishers to sell their catch. The discussant described her restaurant's revenues declining by 50-75%, with less than half of the original restaurant staff still actively working. The type of fish available has also changed. Where before tuna, mahimahi, marlin, and wahoo were commonly purchased from local fishers, now many fishers are focusing on bottomfish, perhaps for their high comparative market value. The restaurant owner estimated that fishing revenues are at an all-time low, despite that fishing activity has increased overall.

Social Impacts

In addition to the concern facing the rest of the United States over the strain COVID-19 will bring to the health care system, the Pacific Islands Region experiences a number of unique concerns. Residents tend to live in large, crowded, multi-generational households⁴⁰ with a higher proportion of older residents than most other states,⁴¹ placing the population in the region at higher risk. In addition, extended families are often spread throughout the region, which includes many remote island-based communities where medical facilities are limited. In American Samoa, arriving passengers from one flight on 3/26/2020 filled the capacity of their health care center and resulted in a halt on passenger travel between Honolulu and Pago Pago.⁴² Airlines have significantly limited flights across the Pacific Islands Region. As time progresses, increasing travel restrictions are forcing individuals into indefinite periods of separation from loved ones causing significant added anxiety. The fishing communities (both commercial and non-commercial) in the Pacific Islands Region play a vital role in supporting local food systems, nutrition, food security, and supporting community social cohesion. This importance is amplified in the face of natural disasters and human health crises, and fishing communities are adapting their operations to be able to continue these crucial functions in the face of unprecedented disruptions and increasing isolation.

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- 37 <https://repository.library.noaa.gov/view/noaa/4420>
- 38 <https://apps-pifsc.fisheries.noaa.gov/wpacfin/home.php>
- 39 <https://repository.library.noaa.gov/view/noaa/3583>
- 40 <https://www2.census.gov/library/publications/2012/acs/acsbr11-03.pdf>
- 41 <https://www.prb.org/which-us-states-are-the-oldest/>
- 42 <https://samoanews.com/update-hawaiian-air-service-american-samoa-suspended>

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Northeast Snapshot, January-June 2020



Northeast Fisheries Impacts from COVID-19

Commercial Fisheries Landings Trends and Impacts through June 2020

From 2015 to 2019 an average of approximately 13,500 commercial fishing vessels operated in the Northeast region, accounting for an average of \$1.82 billion in ex-vessel revenue. Of these vessels, about 3,400 held permits issued by the Greater Atlantic Regional Fisheries Office (GARFO), which accounted for 69% of region-wide harvest revenue while the remaining 31% of regional fishing revenue was landed by 10,100 vessels that either fished in state waters or fished in the EEZ for species that are not regulated by a Federal Fishery Management Plan (FMP) or held federal permits for Highly Migratory Species (HMS) (e.g. tunas, swordfish, and sharks) that are not issued by GARFO. The Northeast has a variety of commercial fisheries, with lobsters and scallops alone averaging just over \$1 billion (64%) of total landings revenue from 2015 to 2019 (see Figure 1). Important fisheries for blue crab, squids, groundfish, menhaden, surfclams, ocean quahogs, summer flounder, black sea bass, scup, monkfish, and Jonah crab accounted for an additional \$392 million. These fisheries combined with lobster and scallops accounted for an average of 86% of landings revenue.

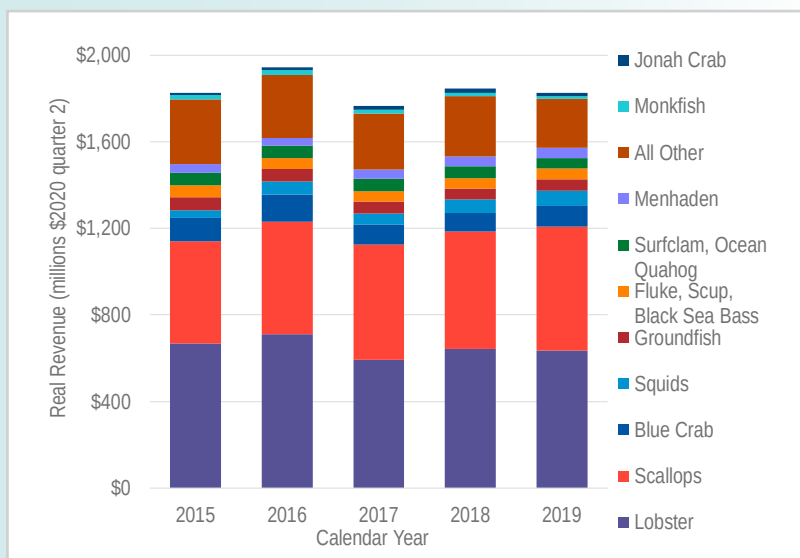


Figure 1. Inflation adjusted ex-vessel revenue for the top 10 Northeast region fisheries.

Although most Northeast fisheries are prosecuted year-round, there are seasonal differences among fisheries that have implications for the timing and magnitude of potential market disruptions resulting from COVID-19. For many fisheries, 40 to 60% of annual revenue is earned during the months from January to June. Figure 2 plots the 2015-2019 January-Dec baseline monthly average revenue from lobster and species that are regulated by the Regional Councils (New England and Mid-Atlantic) and 2020 Jan-Jun revenues.¹ Calendar year 2020 revenues exceeded baseline revenue during January and February but have been below baseline March to June for a cumulative difference of -\$126 million. The majority of the cumulative reduction in 2020 revenues occurred in April and May (70%) of which \$79 million was associated with reduced revenue from American lobster (\$21 million) and sea scallops (\$58 million) (see Figure 3). The reduction in sea scallop revenue was partly due to a 17% reduction in the scallop quota from 2019 levels that took effect at the start of the scallop fishing year on April 1, 2020. Based on 2015-2019 average share of landings for April (10.7%) and May (15.4%) expected 2020 landings for a 51.6 million pound quota would be 5.5 and 7.9 million pounds in April and May respectively. Actual landings were 4.4 million pounds in April and 5.3 million pounds in May for an aggregate reduction of 3.7 million pounds. Note that June scallop landings were 7.3 million pounds, which is nearly equal to what would be expected based on 14% of a 51.6 million pound quota and is similar to the baseline average landings for June. However, June scallop prices were nearly 13% below the 2015-2019 average resulting in 2020 June revenues \$8.7 million below baseline. In fact, lower prices have been a general trend for many species throughout the Northeast region.

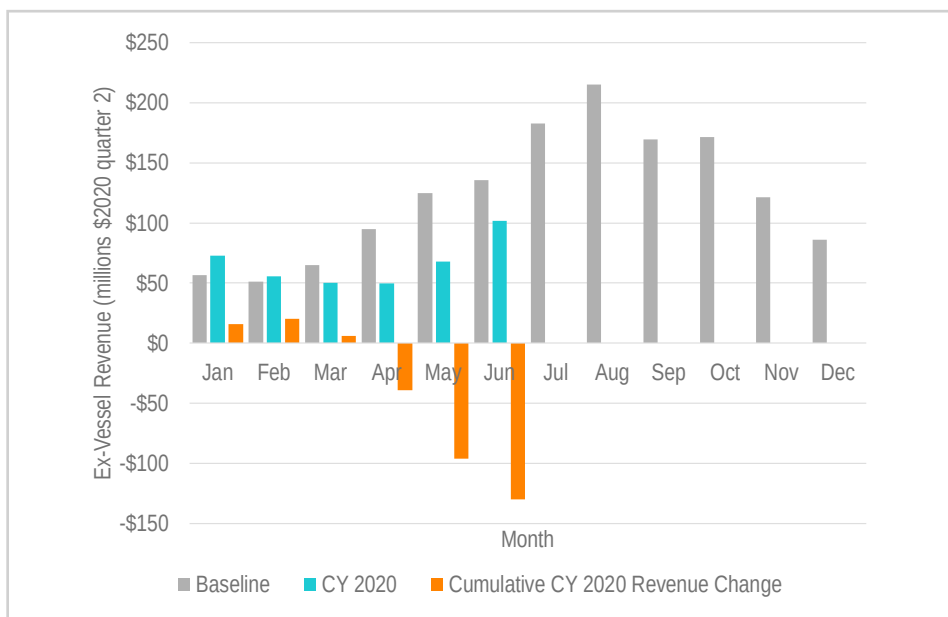


Figure 2. Ex-vessel monthly revenue for the baseline (2015-2019), CY2020, and the cumulative CY2020 revenue change.

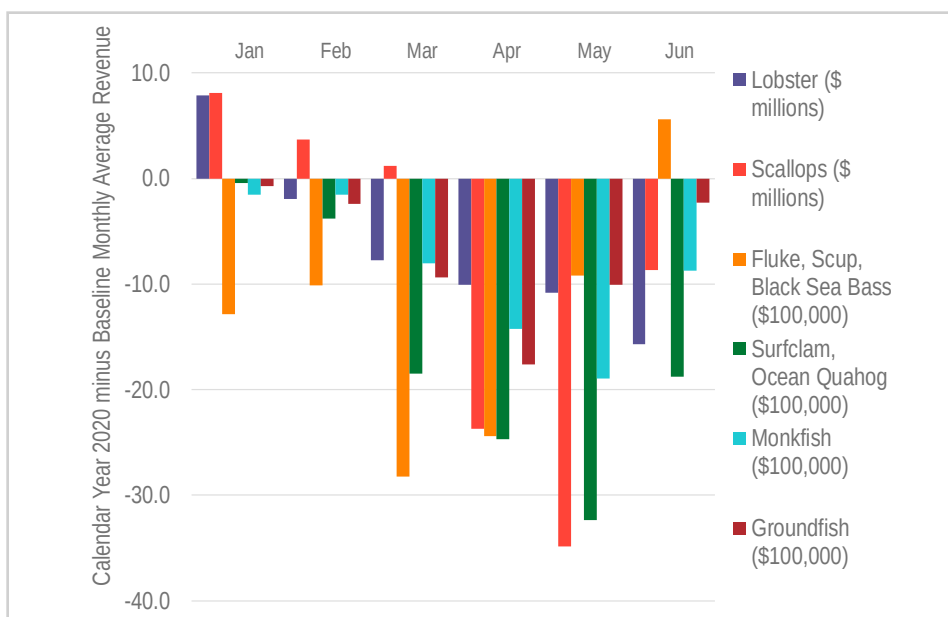


Figure 3. CY2020 monthly revenue minus the baseline monthly average monthly revenue.

The general decline in landings and prices, hence revenues has affected the number of federally permitted vessels that have landed fish with a federally permitted dealer in the Northeast region. During 2015-2019, the number of vessels was at its lowest during February (1,314) then increased through July to 2,657 vessels before tapering off throughout the rest of the year. The number of vessels reporting sales through a Northeast region dealer was above the 2015-2019 baseline average in both January and February but began to fall to 15.3% below the baseline in March and continued to run about 25% below the baseline in both April and May (see Figure 5). A survey was conducted to ascertain how commercial harvesters may have been affected by COVID-19. Key findings from that survey are as follows:

83% OF COMMERCIAL HARVESTERS WERE AFFECTED BY THE COVID-19 PANDEMIC

- Fishing was the primary source of income for 63% of commercial harvesters. On average responding commercial harvester operators had owned a vessel for 32 years.
- Compared to business operations from January to June of 2019:
 - 17% had reduced the number of trips.
 - 14% had difficulty finding supplies.
 - 60% experienced lack of markets; low prices; limited access to marinas.
- 78% of commercial harvest operators stopped fishing for some period of time.
 - 21% stopped fishing for less than 1 month.
 - 48% stopped fishing for 1 to 3 months.
 - 11% stopped fishing for more than 3 months.
 - 15% had stopped fishing indefinitely with plans to resume.
 - Less than 1% had gone out of business.
- On average, responding commercial harvester operators are operating at 50% of trips compared to June/July of 2019.
- 76% of responding commercial harvester operators had not reduced the number of employees.
- 91% of responding commercial harvester operators had reduced revenue, 3% had increased revenue.
 - Average reduction in revenue was 51%.
 - Average increase in revenue was 73%.

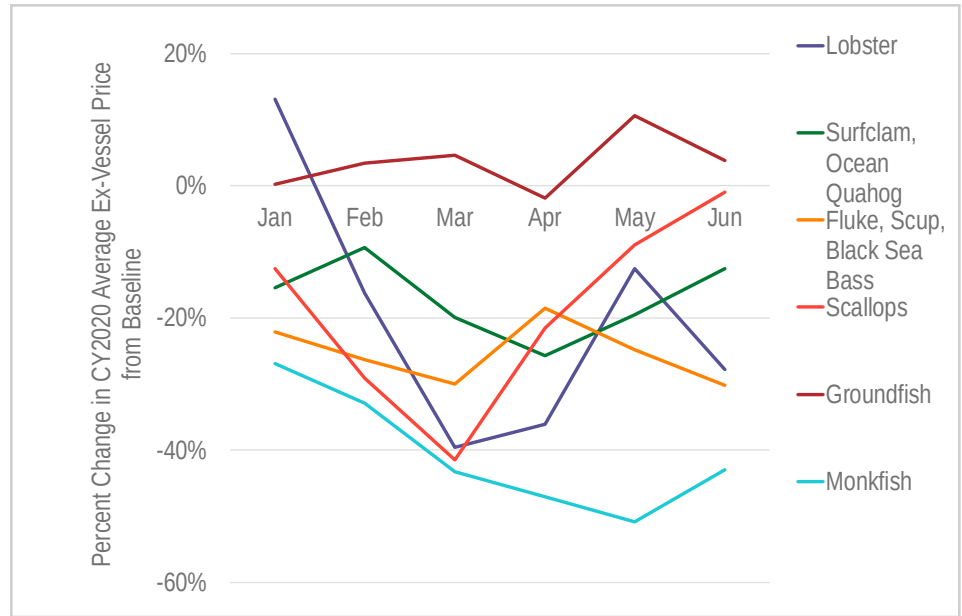


Figure 4. Percent change in CY2020 average ex-vessel price from baseline (2015-2019).

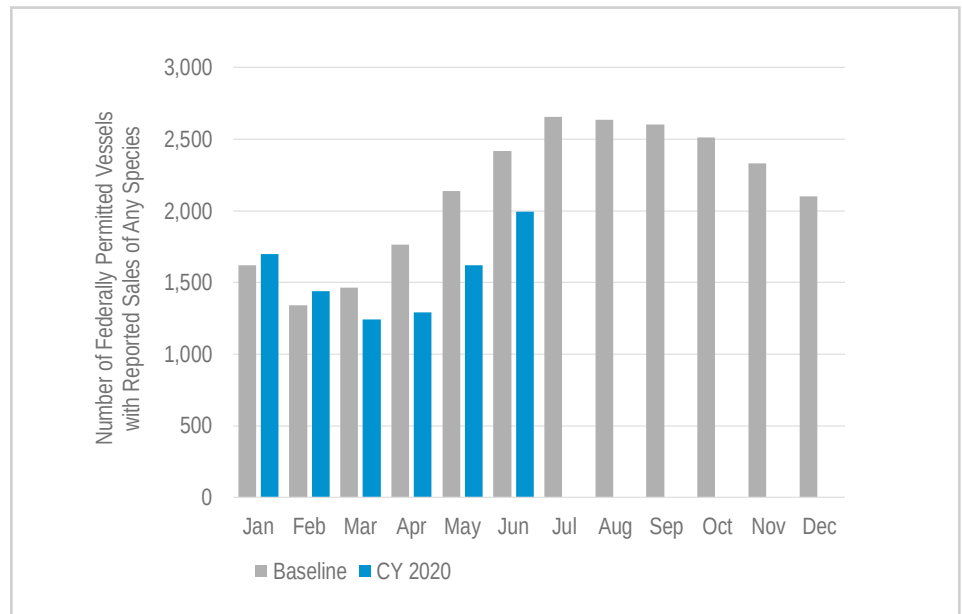


Figure 5. Number of federally permitted vessels with reported sales of any species.

The top three COVID-19 pandemic factors having the largest impact on the business were:

- Instructed not to fish by dealer/processor (30%).
- Low prices for fish (30%).
- Lack of markets or buyers (8%).

Seafood Dealers/Processors

During 2015-2019 an average 640 federally permitted seafood dealers reported purchase of fish and/or shellfish from a fishing vessel in the Northeast Region. Although some seafood products that are primarily exported have a high volume of exports (e.g., lobster and monkfish), much of the Northeast region's seafood product is consumed domestically. There are a mix of species that are primarily sold fresh to restaurants with limited processing, and these had sizable price and revenue declines due to closures of restaurants. Changes in seafood consumption habits have shifted to frozen shelf-stable products and value added processing to meet increased demand for preparation at home, which has been noted as a significant change in seafood consumption from away-from-home to at-home consumption. The changes in seafood wholesale and retail markets have resulted in a decline in the number of federally permitted dealers that have purchased seafood by a month-over-month average of nearly 11% from January to June 2020 compared to 2015-2019 baseline January to June monthly average (see Figure 6).

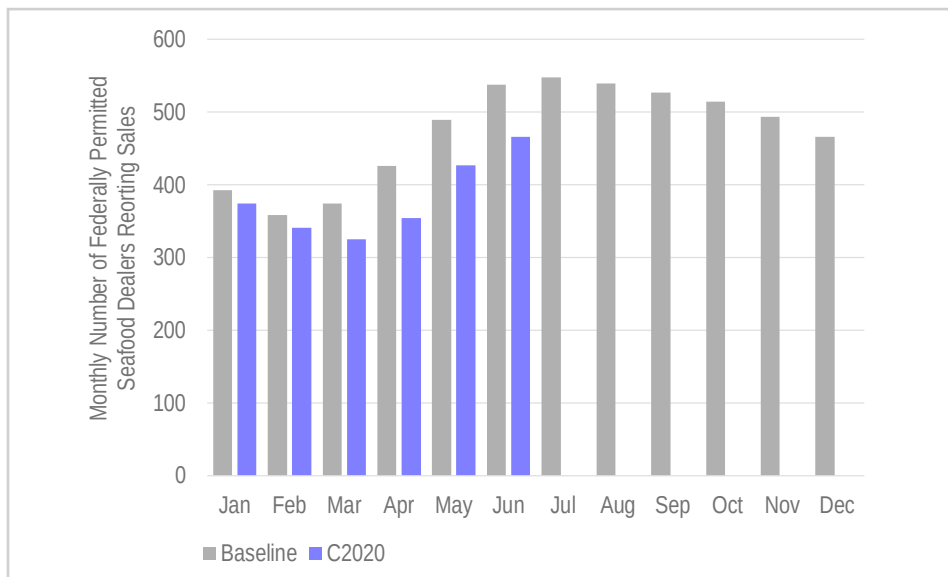


Figure 6. Monthly number of federally permitted seafood dealers reporting sales for January to June 2020, as compared to baseline monthly averages (2015-2019).

NOAA Fisheries conducted a survey of seafood dealers to ascertain the impacts of the COVID-19 pandemic on business operations over the January to June, 2020 period. Key findings from that survey are as follows:

SEAFOOD DEALER/PROCESSORS — 91% OF RESPONDENTS WERE IMPACTED

- Most common impacts were:
 - Reduced operations or business hours (35%).
 - Reduced sales to restaurants, retail, or grocery stores (35%).
- 35% of dealer/processors closed their business operations for some period of time.
 - 29% were closed for less than 1 month.
 - 48% were closed for 1 to 3 months.
 - 6% were closed for more than 3 months.
 - 16% have closed indefinitely with plans to reopen.
- On average, responding dealer/processors were operating at 58% of business activity compared to June/July of 2019.
- Average number of on-site employees was 15. 37% of businesses had reduced the number of on-site employees by an average of 3 people. By contrast, 10% of businesses increased the number of employees by an average of 10 people.
- 85% of dealer/processors reported reduced sales since January 2020 while 8% reported increased sales.
 - Of those having increased sales, revenues increased an average of 15%.
 - Of those with reduced sales, revenues decreased by an average of 44%.

The top three COVID-19 pandemic factors having the largest impact on the business were:

- Dealer/processors listed low seafood prices (29%).
- Loss of markets or buyers (19%).
- Loss of employees (12%).

Recreational Fishing — For-Hire

The recreational for-hire sector in the Northeast region includes a range of services from trips that carry six or fewer passengers that may focus on large game (e.g., tunas or sharks) or small game (e.g., bluefish or striped bass), to operations that carry a large number of anglers that focus on bottom fishing for species such as groundfish, black sea bass, scup, and summer flounder. During 2015 to 2019 for-hire operators in the New England and Mid-Atlantic regions combined collected passenger fees averaging \$123.5 million (2020 \$) providing recreational fishing services on an average of 1.2 million angler trips. Of these angler trips 888 thousand (73%) were taken in Mid-Atlantic states from NY to VA and 315 thousand angler trips were taken in the New England region.² Demand for party/charter services is seasonal with the majority of trips taken from May to August in both New England (84%) and Mid-Atlantic MRIP regions (74%) although the season is longer in the Mid-Atlantic (see Figure 7).

Given the close proximity within which for-hire recreational fishing occurs and guidance for social distancing had a large effect on angler trips as restrictions on gathering and closures of non-essential business were implemented in New England and Mid-Atlantic states particularly during March and lasting through much of May and into June for some states. The timing and relative severity of the restrictions on gatherings is measured as an index rating from 0 to 4 where 0 is no restrictions at all and 4 would be limitations of 5 to 10 or fewer (see Figure 8).³ With the exception of VA with an index value of 3, all other states were at a 2; allowing gatherings ranging from 50 to 250. By the end of March Maine

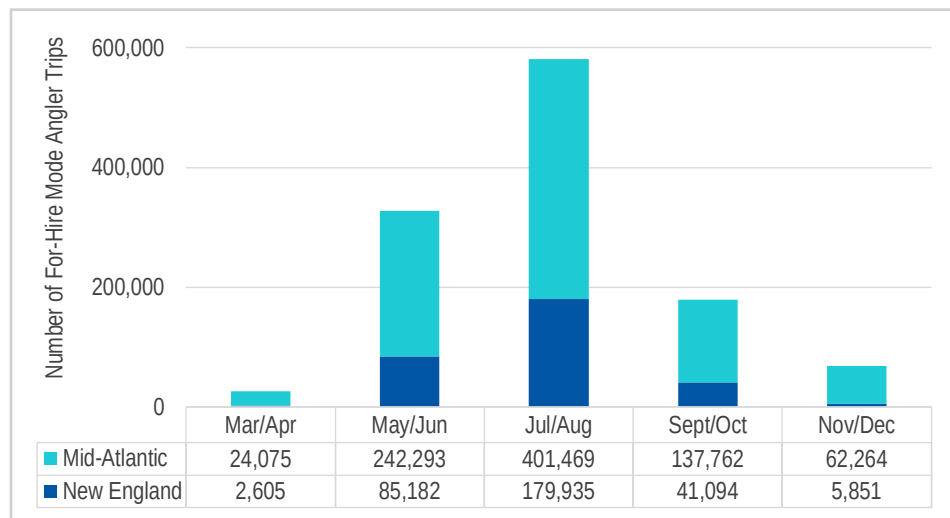


Figure 7. Average number of for-hire angler trips by wave for New England and Mid-Atlantic MRIP regions for a 2015 to 2019 baseline.

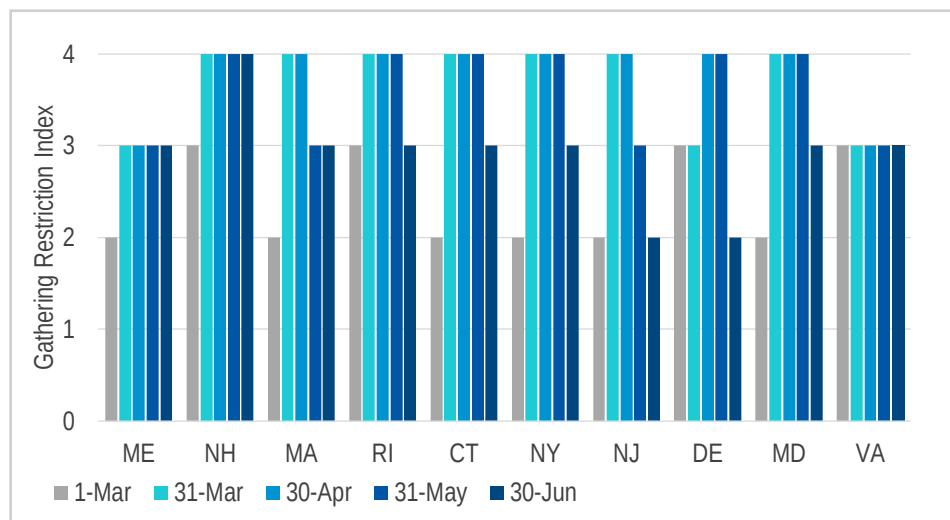


Figure 8. Gathering Restriction Index for March 1, 2020 and Index Value on the last day of the month for March to June, 2020.³

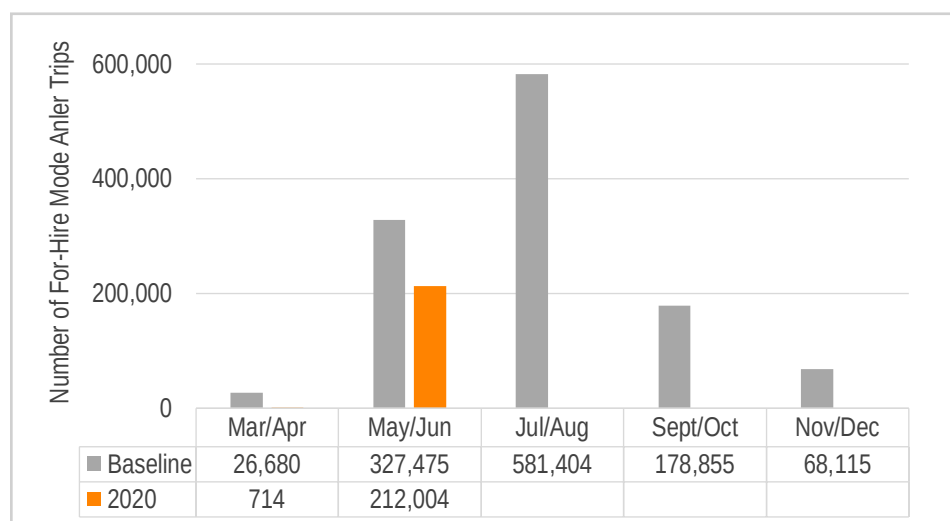


Figure 9. Number of for-hire angler trips taken during 2020 wave 2 and wave 3 compared to the baseline average (2015-2019).

and Virginia were at a 3 but all other states had implemented much more restrictive limits on gathering. In most states these limits remained in place during April and May with easing of restrictions in many states by the end of June.

Even though statewide restrictions may not necessarily apply to for-hire businesses specifically they may have a dampening impact on the demand for party/charter fishing trips. During wave 2 (March and April) when limits on gatherings were most restrictive the number of for-hire angler trips in the New England and Mid-Atlantic regions combined fell from a 2015-2019 baseline average of about 26.7 thousand to 714. (see Figure 9). With some easing of the restrictions on gatherings the number of for-hire angler trips during wave 3 (May and June) increased to 212 thousand but was still 35% lower than the baseline average of 327.5 thousand angler trips.

To obtain more information on impacts on the for-hire sector NOAA Fisheries conducted a survey of for-hire operators in the New England and Mid-Atlantic regions to get their perspectives on the impact of the response to COVID-19 has had on their business over the January to June, 2020 period compared to their business over the same period from 2019. Key for-hire interview findings include:

91% OF PARTY/CHARTER OPERATORS WERE AFFECTED BY THE COVID-19 PANDEMIC

- For 35% of responding fishing was their primary source of income.
- Compared to business operations from January to June of 2019:
 - 45% had reduced the number of trips.

13% had difficulty finding supplies.

- 6% had difficulty obtaining bait.
- 87% of responding party/charter operators stopped fishing for some period of time.
 - 9% stopped fishing for less than 1 month.
 - 63% stopped fishing for 1 to 3 months.
 - 11% stopped fishing for more than 3 months.
 - 16% had stopped fishing indefinitely with plans to resume.
 - 2% went out of business.
- On average party/charter operators are operating at 42% compared to June/July of 2019.
- 75% of responding party/charter operators had not reduced the number of employees.
- 87% of responding party/charter operators had reduced revenue.
 - Average reduction in revenue was 58%.

The top three COVID-19 pandemic factors having the largest impact on the business were:

- Restrictions by state and local governments (56%).
- Lack of passengers (7%).
- Loss of crew (6%).



U.S. Secretary of Commerce
Wilbur Ross

Acting Under Secretary of Commerce
for Oceans and Atmosphere
Dr. Neil Jacobs

Assistant Administrator for Fisheries
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January 2021

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Endnotes

- 1 Fisheries that take place exclusively in State waters, or nearly so, were excluded from Figure 2 because available data were incomplete.
- 2 All for-hire data were based on Marine Recreational Information Program (MRIP) regions, which includes North Carolina in the South Atlantic region. For this reason, trends and impacts on the for-hire sector in North Carolina are reported in the Southeast Region section of this report.
- 3 Source: Oxford Corona Virus Government Response Tracker (<https://www.bsg.ox.ac.uk/research/research-projects/coronavirus-government-response-tracker>).

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Atlantic HMS Snapshot, January-July 2020

Atlantic Highly Migratory Species Fisheries Impacts from COVID-19

The Atlantic Highly Migratory Species (HMS) fishery is a high-value fishery comprised of seven key species and species groups: bluefin tuna, BAYS tunas (bigeye, albacore, yellowfin, and skipjack), swordfish and sharks. All of the HMS species/species groups are harvested year-round (within open seasons) but most species typically have lower average monthly landings revenue (\$1.8 million per month) from February through April as compared to other months (\$3.2 million per month) throughout the year. All high-end products, particularly bluefin, bigeye and yellowfin tunas, that are typically sold to restaurants suffered major price declines in March and April as restaurant restrictions were implemented domestically and air travel restrictions slowed global shipping of exports.

Commercial Fisheries Landings Trends and Impacts through July 2020

Overall, Atlantic HMS ex-vessel revenue was 11.4% lower in the first half of 2020 as compared to 2019. COVID-19 impacted landings starting in the second half of March and resulted in second quarter landings decreasing 36.3% as compared to 2019. To date, April experienced the steepest monthly decline in Atlantic HMS landings value with a 66% decrease as compared to April of 2019. Since the April low of \$439,000 in ex-vessel landings of Atlantic HMS, commercial landings have improved significantly. July landings exceeded \$3.1 million, but they were still 25% lower than landings for July of 2019.

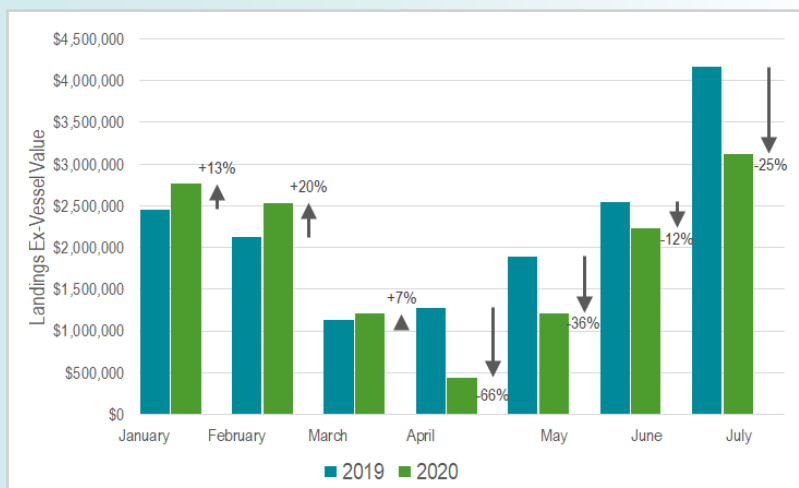


Figure 1. Atlantic HMS commercial landings (ex-vessel value).

In terms of species, bluefin tuna landings have been most impacted. April landings value declined by 69.2%, May landings value declined by 87.5%, June landings value declined by 49.1%, and July landings value declined by 57.4% as compared to the same months in 2019. This decline is primarily due to a decrease in demand for high end sushi and a decrease in the availability of shipping to the international market. The average ex-vessel price per pound dressed weight for bluefin tuna declined by 19.5% (\$5.30/lb compared to \$6.58/lb in 2019) in the second quarter of 2020 as compared to the second quarter of 2019. The average ex-vessel price per pound dressed weight for yellowfin tuna declined by 24.8% (\$3.17/lb compared to \$4.22/lb in 2019) in the second quarter of 2020 as compared to the second quarter of 2019. The average ex-vessel price per pound dressed weight for bigeye tuna declined by 2.8% (\$5.80/lb compared to \$5.96/lb in 2019) in the second quarter of 2020 as compared to the second quarter of 2019. However, the average ex-vessel price per pound dressed weight for swordfish increased by 20.2% (\$5.31/lb compared to \$4.41/lb in 2019) in the second quarter of 2020 as compared to the second quarter of 2019. There have been some indications that there has been an increase in demand for swordfish by the domestic retail market and some decreases in swordfish imports, which could have contributed to the price increase. Regardless of the price increase, the overall value of swordfish landings decreased 65.9% in April, 36.9% in May, and 12.0% in June as compared to those months in 2019. In July, the value of swordfish landings recovered substantially and far exceeded the weak July 2019 harvest.

Atlantic HMS Commercial Fleet

One high frequency indicator of fleet activity available to NOAA Fisheries is the number of hail outs reported by Vessel Monitoring Systems (VMS). VMS is required by vessels equipped with pelagic longline onboard; bottom longline vessels fishing off South Carolina, North Carolina, and Virginia from January through July 31st; gillnet vessels issued a directed shark limited access permit operating near the Southeast U.S. Monitoring Area from December to March 31; and purse seine vessels. The number of trips taken by the Atlantic HMS vessels equipped with VMS decreased by 54% in April. However, by May, vessels resumed taking trips that were comparable to 2019 levels and continued doing so for the remainder of the summer. However, the distance, length, and level of fishing effort were likely reduced as a result of the market shifts associated with COVID-19.

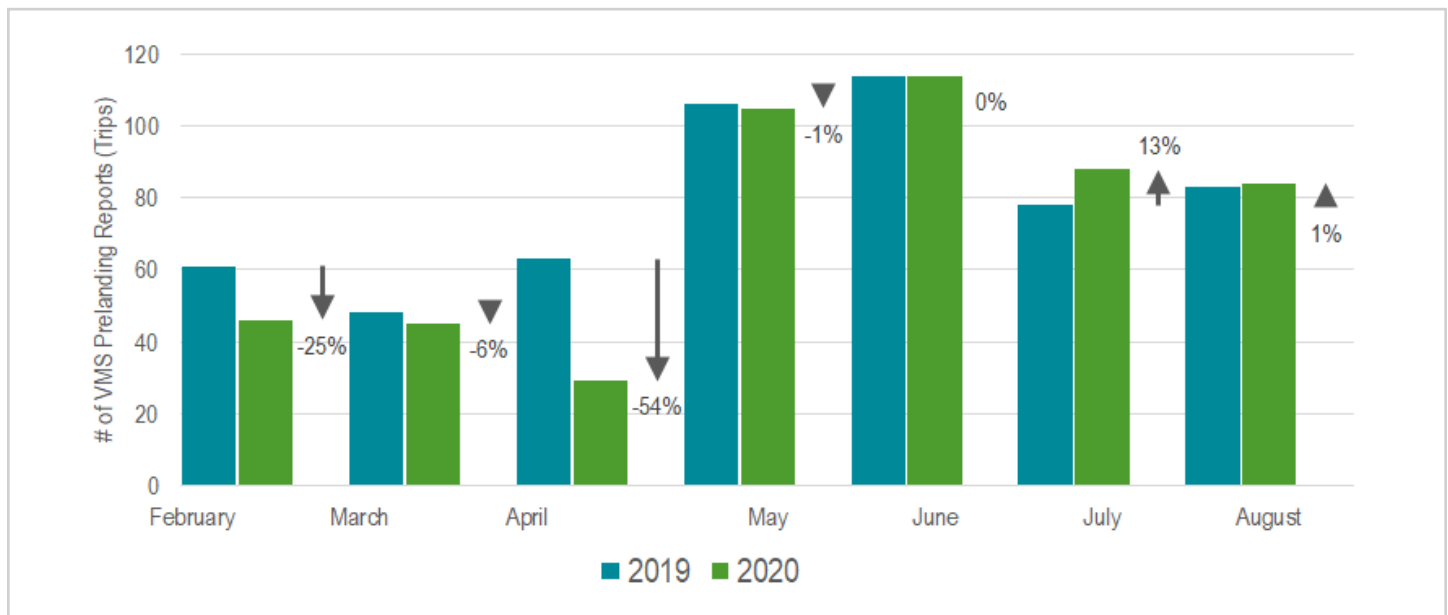


Figure 2. HMS commercial vessel activity (trips).*

*Based on VMS hail outs for Atlantic HMS vessels.

Atlantic Pelagic Longline Fleet

Pelagic longline vessels are also required to report on each gear set via their VMS unit. That data can be used to measure the fishing effort of this fleet in real time. Effort in the Atlantic pelagic longline fleet decreased by 20% in March and by 63% in April as compared to the same months in 2019. By June, effort had recovered substantially and actually exceeded June 2019 effort by 4%. July effort was just 5% below 2019 and August was just 5% below 2019 effort levels.

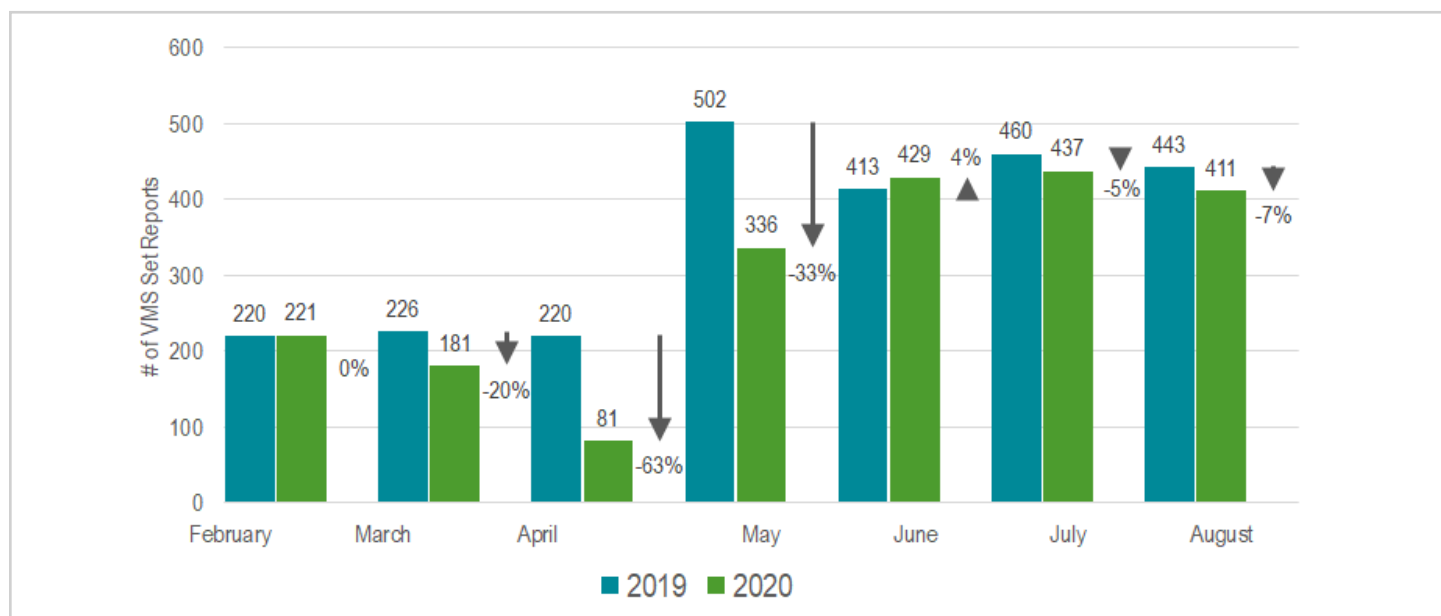


Figure 3. Pelagic longline fleet effort for 2020.

Seafood Dealers

While there were 153 seafood dealers that handled Atlantic HMS product in 2019, to date in 2020, there have been 150 dealers that handled Atlantic HMS product. There have only been two or three Atlantic HMS dealers that contacted the agency stating that they shut down due to COVID-19.

Recreational Fishing

CHARTER SECTOR

During the spring months when the COVID-19 pandemic first arose, the active HMS for-hire fishery was relegated to the South Atlantic and Gulf of Mexico, as is typical for that time of the year. The primary targets of the HMS for-hire fishery at this time of year include tuna and sailfish. Atlantic HMS staff contacted 24 HMS charter/headboat permit holders in the South Atlantic ($n = 13$) and Gulf of Mexico ($n = 11$) states in the month of April. Of those contacted, 100 percent reported that their business had been impacted by the COVID-19 pandemic and the associated closures and social-distancing measures put in place by their states, with 19 vessels reporting that all of their April bookings had been cancelled. On average, vessel captains reported that 97 percent of their April bookings had been cancelled. Most captains reported additional cancellations extending into May with four reporting cancellations into June. NOAA Port Agents also canvassed the Southeast for-hire industry in late March, and reported reductions in effort ranging from 80-100 percent by state. Approximately 63 percent of those captains interviewed reported having to lay off or reduce the hours of their staff, with reductions in hours averaging 95 percent. (Most employees of for-hire operations are fishing mates that are technically classified as 1099 employees or independent contractors.) Those captains that did not report laying off staff either did not have any, or kept what little staff they had on to assist with boat maintenance projects. For-hire operators cited hotel and restaurant closures as among the primary factors impacting their business.

TOURNAMENTS

A 2016 study conducted by NOAA Fisheries estimated that HMS tournaments conducted in the Atlantic, Gulf of Mexico, and the Caribbean contributed approximately \$129 million in annual economic impacts to the United States economy. Since 2009, an average of 259 HMS tournaments have registered with NOAA Fisheries each year. Atlantic HMS tournament registrations were almost unchanged in the first quarter of 2020 (24 versus 23 in 2019). Second quarter HMS tournament registrations were down 39.8% compared to 2019 (56 versus 93 in 2019). Overall, there are 24.9% fewer registered tournaments occurring through August 2020 compared to 2019 (151 versus 201). These numbers reflect the adjustments made for 8 tournaments that registered in 2020 but later reported cancelling their events and 5 postponements; however, it is likely additional events in the second quarter made such changes without reporting them to NOAA Fisheries. Compared to 2019, 50 fewer tournaments have been registered in 2020 through the month of August.

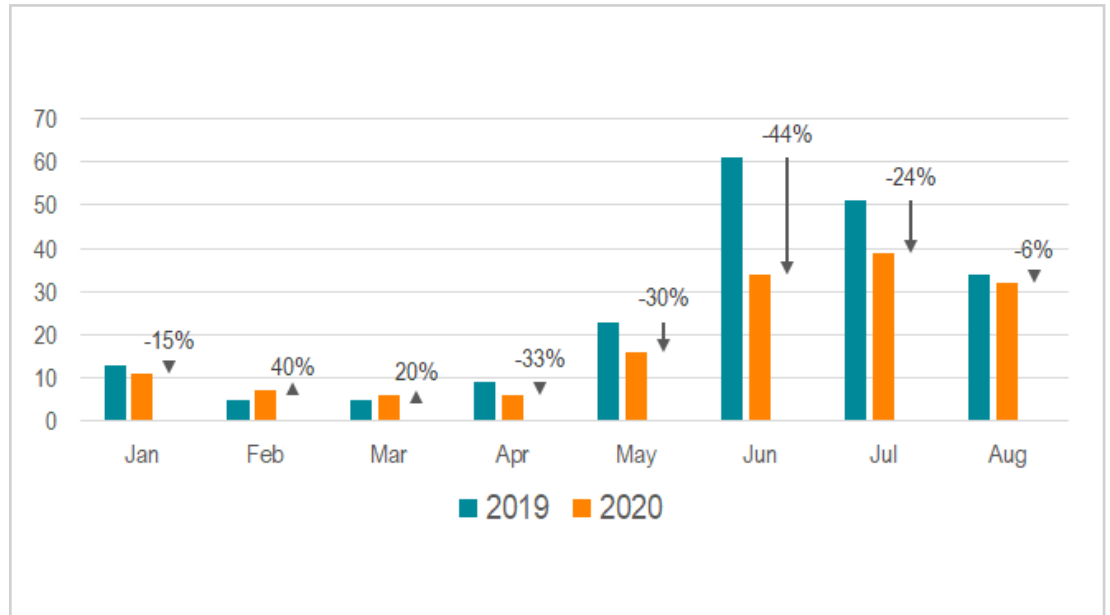


Figure 4. Number of Atlantic HMS tournament registrations.



U.S. Secretary of Commerce

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Southeast Snapshot, January-June 2020



Southeast Fisheries Impacts from COVID-19

Commercial Fisheries Landings Trends and Impacts through June 2020

From 2014 to 2018, approximately 10,000 commercial fishing vessels operated in the Southeast fisheries and landed products valued on average at \$1.1 billion annually. The top three commercial fisheries by landings revenue in the Gulf of Mexico are shrimp, oysters, and menhaden; the top fisheries by landings revenue in the South Atlantic Region are blue crab, shrimp, and snapper-grouper. Many of the participants in Southeast commercial fisheries are active year-round, reflecting the seasonality of these fisheries (see Figure 1). For the period March to May, which corresponds to the period in 2020 during which social distancing measures were first implemented and most restrictive, the shrimp fishery is somewhat less active than it is during other parts of the year, with 16% of annual shrimp landings revenue earned during this 3-month period. In contrast, 25% to 30% of black sea bass, blue crab, Eastern oysters, Gulf of Mexico reef fish and South Atlantic snapper-grouper landings revenue is earned during this period while 33% and 66% of deep-sea golden crab and dolphinfish-wahoo, respectively, is earned during this period.

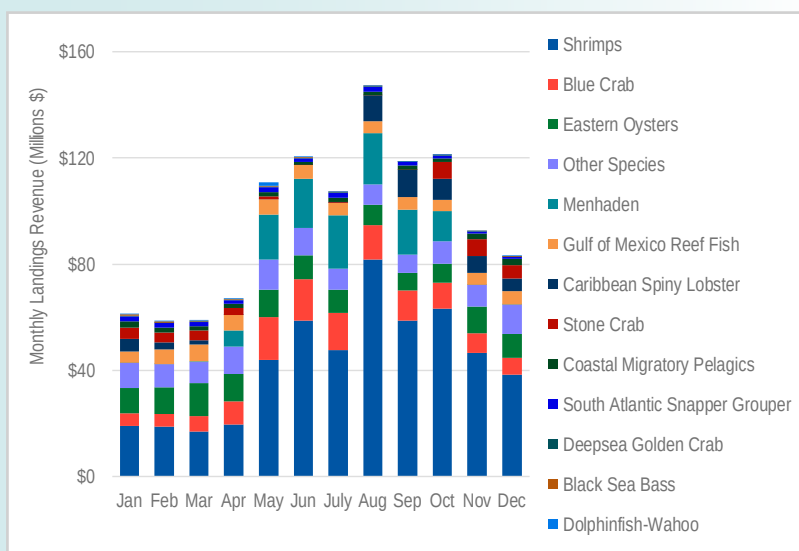


Figure 1. Southeast Average Monthly Landings Revenue for Species, 2014 to 2018.

To assess the effects of COVID-19 on Southeast commercial fisheries, 2019 and 2020 data were obtained from the SEFSC Commercial Landings Monitoring System. Note that these data are preliminary and representative of only selected federally-managed species. Due to incomplete data, some large Southeast fisheries, such as Eastern oysters, blue crab, shrimp, and menhaden, as well as landings reported to states are excluded from this analysis. In addition, a multitude of minor species are harvested by commercial operations in southeast waters which are not regulated by federal quotas and are also not included in this analysis.

Overall, the analysis of monthly landings revenue data for selected federally-managed species (excludes shrimp and menhaden) for January to June 2020 compared with 2019 for the same period indicates that landings revenue is down 23% for the first six months of 2020 relative to the same time period in 2019 (Figure 2). Landings revenue declined 11% in January and February relative to 2019 but then dropped sharply (36%) in March through May, with April experiencing the largest monthly decline (-48%) (Figure 3).

Table 1 summarizes impacts to ex-vessel landings revenues and average ex-vessel prices for the selected components of southeast commercial fisheries included in Figures 2 and 3 above. In the Gulf of Mexico, landings revenue of king mackerel increased by 12% in the first six months of 2020 relative to 2019 while landings revenue of king mackerel in the South Atlantic decreased 13%. Landings revenue of Spanish mackerel decreased significantly in both regions during March and April while average prices increased by 32% in the Gulf of Mexico and 15% in the South Atlantic. While landings of Spanish mackerel have not recovered in the Gulf, landings revenue in the South Atlantic sector increased by 50% in May and June 2020 relative to 2019, signaling that this sector is improving.

Some fisheries, however, incurred significant reductions in landings revenue in 2020. Historically, most landings of dolphinfish in the South Atlantic are recorded in May and June. Landings revenue of dolphinfish declined by 53% in May and June 2020 relative to 2019. It is not clear whether the fleet will be able to make up this shortfall in the remaining six months of 2020. In the South Atlantic Region, landings revenue in the black sea bass fishery and deep-sea golden crab fishery declined by 65% and 55%, respectively. Landings revenue in the commercially important yellowtail snapper fishery declined by 43% in the South Atlantic Region and 55% in the Gulf of Mexico. Landings revenue also significantly declined for several important Gulf of Mexico reef fish species not managed under an individual fishing quota (IFQ) program, such as greater amberjack (42%) and vermilion snapper (34%).

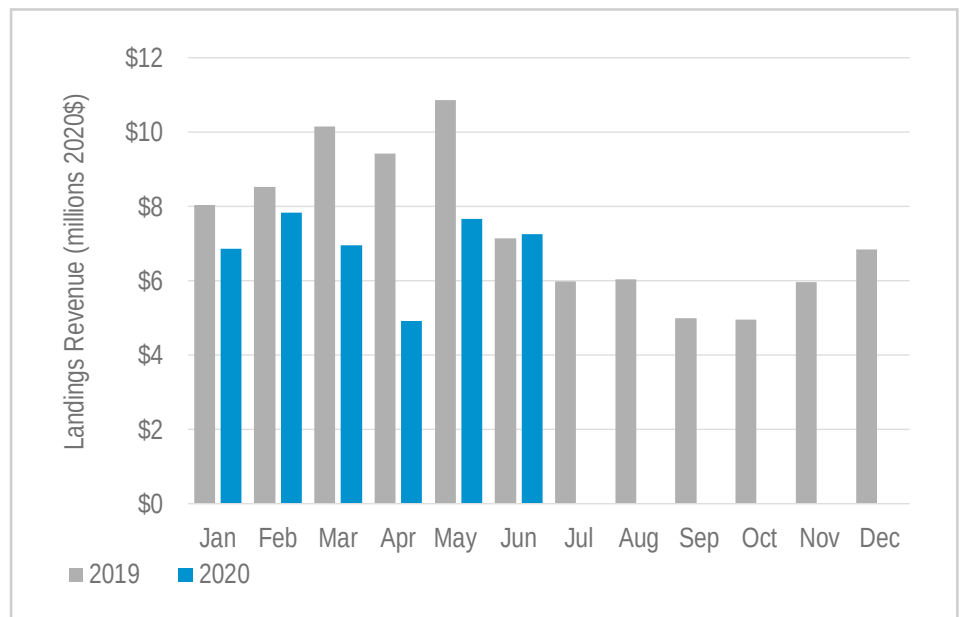


Figure 2. Southeast monthly landings revenue in 2019 and 2020.

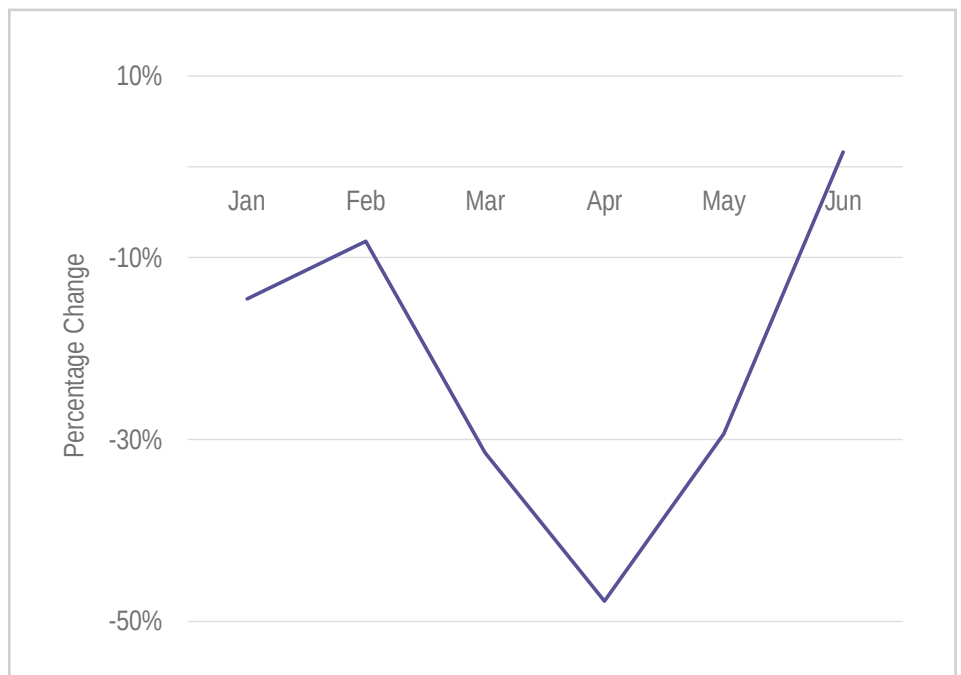


Figure 3. Percentage change in 2020 Southeast monthly landings revenue relative to 2019 monthly landings revenue.

Table 1. Percentage changes in ex-vessel landings revenues and average ex-vessel prices for major southeast commercial finfish fisheries for January-June 2020 relative to January-June 2019.

Aggregate Fishery	Key Species Sectors	Percent Change in Ex-Vessel Landings Revenue	Percent Change in Average Ex-Vessel Price
Gulf of Mexico Coastal Migratory Pelagics	Gulf of Mexico Coastal Migratory Pelagics (all sectors)	-9%	-26%
	King mackerel	12%	0%
	Spanish mackerel	-73%	32%
Gulf of Mexico Reef Fish	Gulf of Mexico Reef Fish (all sectors)	-20%	-2%
	IFQ species*	-17%	N/A
	Greater amberjack	-42%	8%
	Vermilion snapper	-34%	-2%
	Yellowtail snapper	-55%	-8%
South Atlantic Black Sea Bass	South Atlantic Black Sea Bass (all sectors)	-65%	21%
South Atlantic Coastal Migratory Pelagics	South Atlantic Coastal Migratory Pelagics (all sectors)	-12%	7%
	King mackerel	-13%	-1%
	Spanish mackerel	-9%	15%
South Atlantic Deepsea Golden Crab	South Atlantic Deepsea Golden Crab (all sectors)	-55%	19%
South Atlantic Dolphin-Wahoo	South Atlantic Dolphin-Wahoo (all sectors)	-54%	3%
	Dolphinfish	-53%	4%
South Atlantic Snapper Grouper	South Atlantic Snapper Grouper (all sectors)	-35%	-5%
	Deep water complex	-18%	-4%
	Gray triggerfish	-52%	-40%
	Greater amberjack	-32%	2%
	Vermilion snapper	-37%	0%
	Yellowtail snapper	-43%	-9%

***The Red Snapper IFQ program was implemented in 2007. The Grouper-Tilefish IFQ program was established in 2010 and is comprised of five share categories (gag, red grouper, shallow water groupers, deep water groupers and tilefishes). Due to multiple species categories, an aggregate average IFQ species price is not available.**

Several commercially important reef fish species are managed under the Gulf of Mexico Red Snapper and Grouper-Tilefish individual fishing quota (IFQ) programs. IFQ programs are management regulations that set species-specific total allowable catch (i.e. quota) for a fishing season and distribute shares of a dedicated portion of the quota through shares to individual accounts. These shares result in annual allocations of the yearly dedicated quota which are transferable among approved entities and also enable shareholders to postpone harvesting their quota if market conditions are unfavorable, as occurred this past spring as social distancing

measures implemented to reduce the spread of COVID-19 closed restaurants both domestically and globally.

Figure 4 reveals underlying trends in the economics of the Gulf of Mexico IFQ fisheries. A steep decline in landings revenue from Gulf of Mexico IFQ species was reported in March and April 2020 but beginning in May, landings revenue started to rebound. Cumulative landings revenue of all IFQ species through June 2020 were 17% less than cumulative landings in the first six months of 2019. Furthermore, cumulative landings of IFQ species through

June 2020 represent 41% of the total annual quota compared to 44% during the same time period in 2019. However, this is higher than the average of 34% during the first six months of 2017-2019.

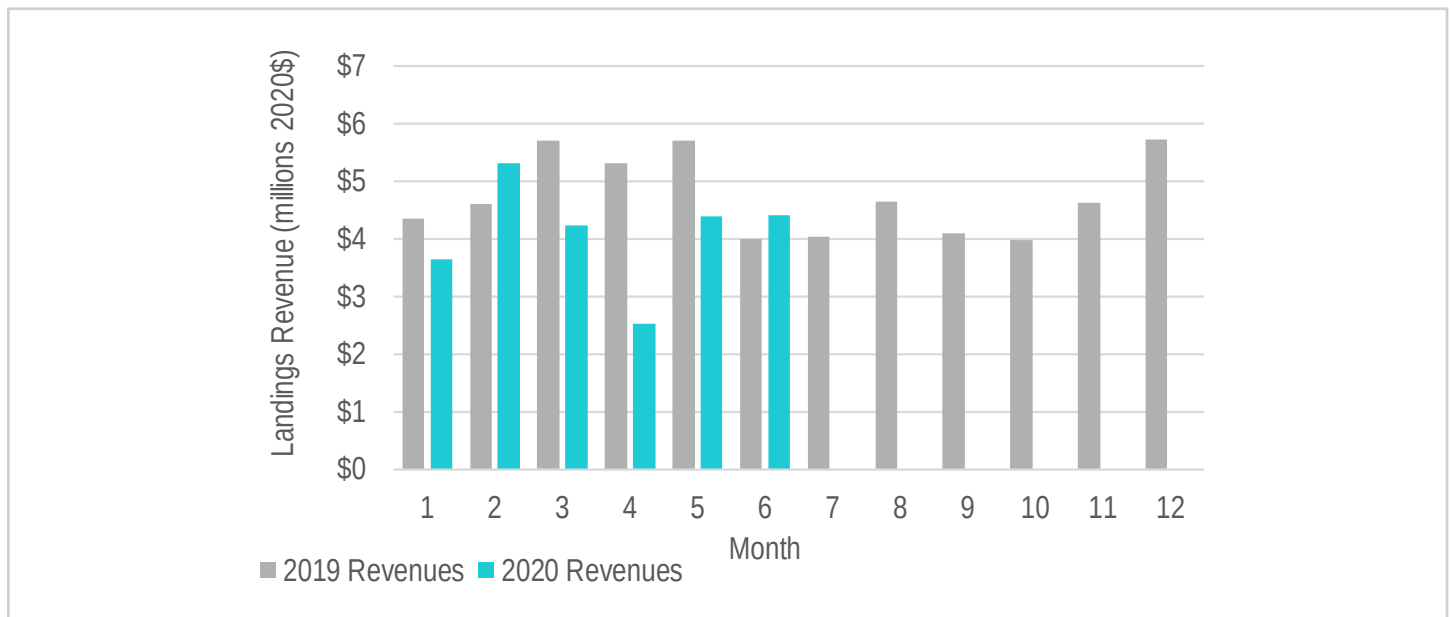


Figure 4. Landings revenues for Gulf of Mexico IFQ species by month for 2019 and 2020.

SPINY LOBSTER FISHERY

Data from the state of Florida shows that over the first 3 months of 2020 in comparison to 2019, spiny lobster landings declined by 42.5%, average price declined by 25.4% and dockside value declined by 57.1%. The lobster season is closed from April-August. With the available data we cannot clearly determine the cause of the decline. However, because the spiny lobster fishery relies heavily upon the Chinese market, the impact of tariffs and the closure of the Chinese market in January contributed significantly to the very steep declines the fishery experienced in the first quarter of 2020.

COMMERCIAL FISHING PERMIT HOLDER SURVEY

A total of 327 commercial fishing permit holders were contacted in the eight states in the Gulf of Mexico and South Atlantic Regions between July 5 and September 15, 2020 regarding the impacts of COVID-19 related factors on their businesses through the first half of the year (January - June/July 2020). Of those contacted, 88% reported that COVID-19 related factors had affected their businesses. For 80% of respondents affected by COVID-19 related factors, commercial fishing represented their primary source of income. Some 94% of affected commercial fishing permit holders reported reduced revenues in the first half of 2020 in comparison to the same period in 2019, with revenue declining on average 56%. Only 1% reported increased revenue in comparison to the same period in 2019. Twenty-eight percent (28%) of commercial permit holders reported a reduction of employees with an average decrease of 2 employees. Less than 1% of commercial permit holders reported an increase in numbers of employees.

Comparison of 2020 with first half of 2019

- 94% of commercial permit holders reported revenue losses.
- Commercial permit holders reported between <1% and 100% revenue losses, for an average 56% decrease in revenue among those reporting losses.
- 28% of commercial permit holders reported a reduction in employees. <1% reported an increase and 66% reported no change.
- On average, commercial fishing permit holders reported operating at 44% of normal fishing activity compared to the same period in 2019.

Top three COVID-19 related factors affecting commercial permit holders:

- Instructed not to fish by dealer/processor (36%).
- Low prices for seafood (22%).
- No crew available (9%).

Eighty-six percent (86%) of commercial permit holders that responded to the survey stopped fishing for some period of time in the first half of 2020. Of these, 12% stopped fishing for less than a month, 46% stopped fishing from 1 to 3 months, and 16% stopped fishing for more than three months. Ten percent (10%) had ceased operations indefinitely but had plans to eventually resume taking trips. Less than 1% of survey respondents reported permanently shutting down their commercial fishing operations. The most common ways that COVID-19 related factors impacted commercial fishing permit holders included reduced number of trips (21%) and difficulty obtaining supplies other than bait (19%), although 53% percent responded “Other” which includes a wide range of impacts.

Seafood Dealers

There were 3,240 seafood dealers operating in the Southeast Region in 2018. Notwithstanding a few seafood products that are exported (e.g., spiny lobster to China), much of the Southeast Region’s product is consumed domestically, with a high proportion dependent upon restaurant sales. King mackerel is a notable exception; it is sold primarily in grocery stores and there remains a strong market for this product, although especially in the Gulf of Mexico, it is a seasonal fishery with the majority of sales in January/February and April/May.

NOAA Fisheries interviewed 270 seafood dealers and processors between July 5 and September 15, 2020 regarding the impacts of COVID-19 related factors on their businesses through the first half of the year (through June/July 2020). Some 89% of seafood dealers/processors reported that they had been impacted by COVID-19 related factors in the first half of 2020. Of those affected, 89% reported reduced revenue in comparison to the same period in 2019, with revenue declining on average 55%. Only 6% of those affected reported sales improved in comparison to the same period in 2019 with an average 20% increase in revenue. Some 46% of affected businesses reported a reduction of on-site employees with an average decrease of 8 employees. Only 2.5% of affected businesses reported an increase in numbers of employees (average of 4 employees).

Comparison of 2020 with first half of 2019

- 89% of dealers reported revenue losses.
- Dealers reported between <1% and 100% revenue losses, for an average 55% decrease in revenue among those reporting losses.
- 46% of dealers reported a reduction in employees. 6% reported an increase and 47% reported no change.
- On average, dealer processors were operating at 48% of business activity compared to the same period in 2019.

Forty-eight percent (48%) of affected dealer/processors closed their business operations for some period of time in the first half of 2020. Of these, 23% were closed for less than a month, 40% were closed from 1 to 3 months, and 11% were closed for more than three months. Twenty-two percent (22%) were closed indefinitely but had plans to eventually reopen. Only 3% of survey respondents reported permanently shutting down their operations. The most common impacts of COVID-19 related factors on seafood dealer processors included reduced operations or business hours (43%), reduced sales to restaurants, retail or grocery stores (20%), and lack of local product (11%).

Top three COVID-19 related factors affecting seafood dealer/processors:

- Low seafood prices (33%).
- Implementing health safety measures (19%).
- Loss of employees (17%).

Recreational Fishing — For-Hire Sector

There were approximately 4,000 and 1,800 charter operations in the Gulf of Mexico and South Atlantic Regions, respectively, in 2019. There were also an estimated 63 headboat vessels in the South Atlantic and 69 in the Gulf of Mexico in 2019. From 2016 to 2019, anglers in the Southeast took an average of 1,523,770 charter trips annually.

In 2020, charter revenues from January through the end of June from North Carolina to Mississippi were \$141.7 million. This represented an overall 24% decrease in average revenues for this same time period for the years 2017 to 2019. The decrease in revenues is largely attributable to COVID-19 closures that took place in March to May 2020. For example, from January-February 2020, total charter trips and revenues in the southeast were up 13% and 14% respectively over the 2017-2019 average. In comparison, from March-April 2020, charter trips and revenue declined by 72% in comparison to the 2017-2019 period. Finally, from May-June 2020, trips and revenues were only down by 4.5% in comparison to the 2017-2019 average for those months. During this same period (May-June), South Carolina had a 29% increase in revenue and Alabama had a 10% increase in revenue; Georgia had the largest decline at 47%, but both East and West Florida showed minor declines of 2% and 5%, respectively.¹

Headboat revenue across the Southeast decreased by 34% overall across states in the Southeast (NC to TX) from January to June 2020 compared to the average revenues for the same time period between 2017 and 2019. For May and June, 2020 headboat revenues were \$10 million compared to the average \$13 million, representing a 21% decrease in comparison to the same time period in 2017-2019. Texas was the only state with an increase in headboat revenues in May/June (7%); Louisiana had the largest decrease in comparison to May/June of 2017-19 (56%).

FOR-HIRE OPERATORS SURVEY

A total of 486 for-hire businesses were contacted in the eight states in the Gulf of Mexico and South Atlantic Regions between July 5 and September 15, 2020 regarding the impacts of COVID-19 related factors on their businesses through the first half of the year (January through June/July 2020). Of those contacted, 95% reported that COVID-19 related factors had affected their businesses. For 69% of affected respondents, charter/for-hire fishing represented their primary source of income. Some 94% of affected for-hire operators reported reduced revenues in the first half of 2020 in comparison to the same period in 2019, with revenue declining on average 58%. Only 2% of those affected reported that sales improved in comparison to the same period in 2019, with an average 30% increase in revenue. Some 12% of affected for-hire businesses reported a reduction of employees with an

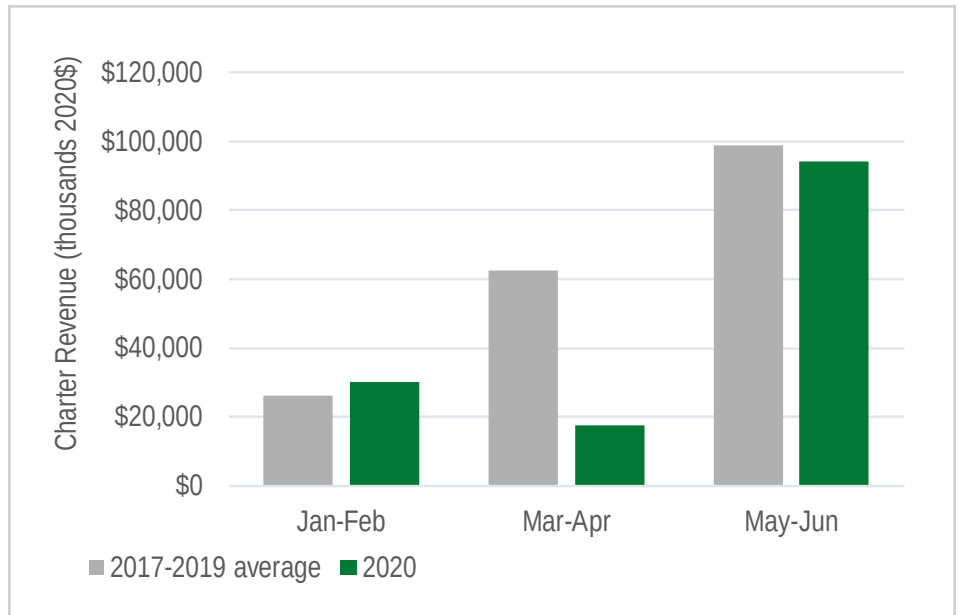


Figure 5. Southeast charter revenues for the baseline (2017-2019) and 2020.*

*January-February excludes South Carolina and Georgia. Texas and Louisiana are not included in these figures due to lack of 2020 data for those states.

average decrease of 2 employees. Less than 1% of affected for-hire operations reported an increase in numbers of employees.

Comparison of 2020 with first half of 2019

- 94% of affected for-hire operators reported revenue losses.
- Affected for-hire businesses reported between 5% and 100% revenue losses, for an average 58% decrease in revenue among those reporting losses.
- 12% of affected for-hire operators reported a reduction in employees. <1% reported an increase and 85% reported no change.
- On average, affected for-hire operations were operating at 53% of normal fishing activity compared to the same period in 2019.

Top three COVID-19 related factors affecting the for-hire sector:

- Restrictions by state and local governments (57%).
- Lack of charter/party/for-hire clients (10%).
- Loss of crew (3%).

Eighty-eight percent (88%) of affected party/charter/for-hire businesses stopped taking fishing trips for some period of time in the first half of 2020. Of these, 5% stopped taking trips for less than a month, 64% stopped taking trips from 1 to 3 months, and 11% stopped taking trips for more than three months. Eight percent (8%) had ceased operations indefinitely but had plans to eventually resume taking trips. Only 1% of survey respondents reported permanently shutting down their charter operations. The most common ways that COVID-19 related factors affected charter/party/for-hire operations included reduced number of trips (49%) and difficulty obtaining supplies other than bait (22%).

Puerto Rico

Between 2013 and 2018, dockside revenues in Puerto Rico averaged about \$5 million for the first six months of the year (January-June). On March 15, 2020, the Governor instituted a two-week closure (curfew) for the majority of businesses on the island of Puerto Rico. However, commercial fishermen were exempted from the curfew and fish cooperatives (*villas pesqueras*) and fish stores could also sell seafood products.

COMMERCIAL FISHING SURVEY

A total of 318 small-scale commercial fishermen were contacted in Puerto Rico in July and August of 2020 regarding the impacts of COVID-19 related factors on their fishing operations through the first half of the year (January-June/July 2020). Of those contacted, 96% reported that COVID-19 related factors had affected their fishing operations. For 73% of the respondents affected by COVID-19 related factors, commercial fishing represented their primary source of income. Some 87% of affected commercial fishing permit holders reported reduced revenues in the first half of 2020 in comparison to the same period in 2019, with revenue declining on average 65%. Only one fisherman reported increased revenue in comparison to the same period in 2019. Twenty-five percent (25%) of commercial fishermen reported a reduction of employees/crew.

Puerto Rico: Comparison of 2020 with first half of 2019

- 87% of small-scale fishermen reported revenue losses.
- Commercial permit holders reported between <1% and 100% revenue losses, for an average 65% decrease in revenue among those reporting losses.
- 28% of commercial permit holders reported a reduction in employees. <1% reported an increase and 66% reported no change.
- On average, commercial fishing permit holders reported operating at 44% of normal fishing activity compared to the same period in 2019.

Top three COVID-19 related factors affecting Puerto Rican commercial harvesters:

- Lack of markets or buyers (79%).
- State and local government restrictions (71%).
- Implementing health safety measures (48%).

Ninety-four percent (94%) of commercial fishermen stopped fishing for some period of time in the first half of 2020. Of these, 13% stopped fishing for less than a month, 43% stopped fishing from 1 to 3 months, and 33% stopped fishing for more than three months. Nine percent (9%) had ceased operations indefinitely but had plans to eventually resume taking trips. Less than 1% of survey respondents went out of business. The most common ways that COVID-19 related factors impacted commercial fishing permit holders included reduced number of trips (83%), lack of markets (30%), and difficulty obtaining supplies other than bait (21%). Six percent (6%) of Puerto Rican fishermen reported switching to new fisheries.

SEAFOOD DEALER SURVEY

In Puerto Rico, there are approximately 70 active landing centers and independent dealers that market most of the production locally. Most of Puerto Rico's product is consumed domestically, with a high proportion of sales going to restaurants and hotels.

NOAA Fisheries surveyed 47 seafood dealers and processors in Puerto Rico in July/August 2020 regarding the impacts of COVID-19 related factors on their businesses through the first half of the year (through June/July 2020). 98% of seafood dealers/processors reported that they had been impacted by COVID-19 related factors. Of those affected, ninety-three percent (93%) reported reduced revenues, with an average decrease of 56%. Only 2% (one dealer) reported increased revenues. Some 43% of affected businesses reported a reduction of on-site employees, while 56% did not change the number of employees.

Puerto Rico: Comparison of 2020 with first half of 2019

- 93% of affected dealers reported reduction in sales.
- Dealers reported an average 56% decrease in revenue for those reporting losses.
- 43% of dealers reported a reduction in employees and 57% reported no change.
- On average, affected dealer processors were operating at 33% of business activity compared to the same period in 2019.

Top three COVID-19 related factors affecting seafood dealer/processors:

- State and local market restrictions (87%).
- Loss of markets or buyers (77%).
- Implementing health safety measures (70%).

Eighty-seven percent (87%) of affected dealer/processors closed their business operations for some period of time in the first half of 2020. Of these, 30% were closed for less than a month,

45% were closed from 1 to 3 months, and 15% were closed for more than three months. Ten percent (10%) were closed indefinitely but had plans to eventually reopen. No dealer/processors in Puerto Rico reported having gone out of business. The most common impacts of COVID-19 related factors on seafood dealer processors included reduced sales to restaurants, retail, or grocery stores (94%), lack of product (81%), and reduced operations or business hours (60%).

CHARTER OPERATORS SURVEY PUERTO RICO

There are currently around 30 charter operators in Puerto Rico, although the number can increase to 50 on a seasonal basis when demand is high through the addition of mostly informal, unlicensed operators. A total of 16 charter operators were contacted in Puerto Rico in July/August 2020 regarding the impacts of COVID-19 related factors on their businesses through the first half of the year (January-June/July 2020). Of those contacted, 100% reported that COVID-19 related factors had affected their businesses. For 88% of affected respondents, charter/for-hire fishing represented their primary source of income. Some 100% of charter operators reported reduced revenues in the first half of 2020 in comparison to the same period in 2019, with revenue declining on average 85%. Some 50% of affected for-hire businesses reported a reduction of employees.

Eighty-eight percent (88%) of affected party/charter/for-hire businesses stopped taking fishing trips for some period of time in the first half of 2020. Of these, 7% stopped taking trips for more than three months, 67% had ceased operations indefinitely with plans to resume, and 20% went out of business.

Comparison of 2020 with first half of 2019, Puerto Rico Charters

- 100% of affected for-hire operators reported revenue losses.
- Affected for-hire businesses reported an average 85% decrease in revenue.
- 50% of affected for-hire operators reported a reduction in employees and 44% reported no change in crew/employees.
- On average, affected for-hire operations were operating at 1.3% of normal fishing activity (most reported no activity).

Top three COVID-19 related factors affecting the charter sector in Puerto Rico:

- Lack of charter clients (94%).
- State and local government restrictions (94%).
- Implementing health safety measures (38%).

U.S. Virgin Islands (USVI)

In early 2020, many commercial and charter fishermen in USVI were still struggling to recover from the devastating impacts of hurricanes Irma and Maria in late 2017. Some commercial fishermen were fishing on others boats trying to accumulate funds for boat repairs and to purchase new gear. Charter fishermen were just starting to recover from the decline in tourism related to hotel closures and infrastructure damage related to the storms.

In mid-March 2020, the Governor announced the closure of USVI to all tourists after 17 individuals on the islands tested positive for COVID-19. The first closure lasted until mid-July. After a brief reopening to tourism, the USVI was once again closed down as the COVID-19 threshold was exceeded.

Between August 24 and September 9, 2020, NOAA contacted 87 commercial and charter fishermen on the islands of St. Thomas, St. Croix, and St. John by phone to administer a survey regarding the impacts of COVID-19 related factors on their fishing operations during the first half of the year (January-June 2020). Although there may be some differences between islands, the following sections cover all of USVI. Due to the lack of a significant dealer/processor sector in USVI (since most catch is sold directly to consumers either in the market or directly to restaurants or stores) this sector was not included in the survey.

COMMERCIAL FISHING SURVEY

A total of 63 small-scale commercial fishermen responded to the survey in USVI. Of those contacted, 86% reported that COVID-19 related factors had affected their fishing operations. For 63% of the respondents affected by COVID-19 related factors, commercial fishing represented their primary source of income. Some 87% of affected commercial fishing permit holders reported reduced revenues in the first half of 2020 in comparison to the same period in 2019, with revenue declining on average 53%. Thirty-one percent (31%) of commercial fishermen reported having experienced a reduction in employees/crew with an average of 1.3 fewer crew members; 69% had not reduced the number of crew.

USVI: Comparison of 2020 with first half of 2019

- 87% of USVI commercial fishermen reported revenue losses.
- Affected commercial fishermen reported an average decrease in revenue of 53%
- 31% of commercial fishermen reported a reduction in crew. 69% reported no change.
- On average, commercial fishermen reported operating at 48% of normal fishing activity compared to the same period in 2019.

Top three COVID-19 related factors affecting commercial harvesters in USVI:

- Implementing health safety measures (63%).
- State and local government restrictions (61%).
- Lack of markets or buyers (56%).

Eighty-one percent (81%) of commercial fishermen in USVI stopped fishing for some period of time in the first half of 2020. Of these, 30% stopped fishing for less than a month, 50% stopped fishing from 1 to 3 months, and 1% stopped fishing for more than three months. Two percent (2%) had ceased operations indefinitely but had plans to eventually resume fishing.

FOR-HIRE OPERATORS SURVEY

A total of 18 charter businesses responded to the survey in USVI. Of those contacted, 89% reported that COVID-19 related factors had affected their businesses. For 72% of affected respondents, charter/for-hire fishing represented their primary source of income. One hundred percent (100%) of affected for-hire operators reported reduced revenues in the first half of 2020 in comparison to the same period in 2019, with revenue declining on average 77%. Some 31% of affected for-hire businesses reported that they reduced their number of employees and another 67% did not change the number of employees

Comparison of 2020 with first half of 2019

- 100% of affected for-hire operators reported revenue losses.
- Affected for-hire businesses reported an average revenue decrease of 58%.
- 31% of affected for-hire operators reported a reduction in crew/employees.
- On average, affected for-hire operations were operating at 39% of normal fishing activity compared to the same period in 2019.

Top three COVID-19 related factors affecting the for-hire sector:

- Lack of charter/party/for-hire clients (79%).
- Restrictions by state and local governments (74%).
- Implementing health safety measures (42%).

Ninety-four percent (94%) of affected party/charter/for-hire businesses stopped taking fishing trips for some period of time in the first half of 2020. Of these, 20% stopped taking trips for less than a month, 27% stopped taking trips from 1 to 3 months, 33% stopped taking trips for more than three months, and 20% stopped fishing indefinitely with plans to resume. We received no reports of charters going out of business.

Endnotes

1 Revenues are in 2020 dollars. Texas and Louisiana are not included in these figures as we don't have monthly or wave estimates yet for 2020.



U.S. Secretary of Commerce

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January 2021

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Status of FMP Amendments
February 05, 2023

FMP Amendment Status	Lead Council and SF Staff	Date of Council Final Action	Start Regional Review	Transmittal Date of Action to NMFS HQ for Review	Proposed FMP Amendment Notice of Availability Published in <i>Federal Register</i>	Proposed Rule Published in <i>Federal Register</i>	Final Rule or Notice of Approval Published in <i>Federal Register</i>
Amendment 52 to the FMP for BSAI King and Tanner Crabs (KTC); EDR Revisions Approved: January 05, 2023	Marrinan Mackey	February 2022	PR: 08/05/22 FR: 12/14/22	PR: 10/05/22 FR: 12/23/22	Oct 6, 2022 87 FR 60638 EOC: 12/05/22	Nov 1, 2022 87 FR 65724 EOC: 12/01/22	Feb 6, 2023 Effective: 03/08/23
Amendment 113 to the GOA FMP; Rockfish Program Adjustments	McCracken	June 2022					
Amendment 122 (BSAI) PCTC Program Decision date: March 30, 2023	McCracken Warpinski	October 2021	PR: 08/23/22	PR: 12/22/22	Dec 30, 2022 87 FR 80519 EOC: 02/28/23	Feb 9, 2023 EOC: 03/13/23	
Amendment 123 to the Fishery Management Plan for Groundfish of the Bering Sea and Aleutian Islands Management Area; Halibut Abundance-Based Management of A80 PSC Limit Decision date: March 09, 2023	Stram Mansfield	December 2021	PR: 06/24/22	PR: 11/02/22	Nov 9, 2022 87 FR 67665 EOC: 02/07/23	Dec 9, 2022 87 FR 75570 EOC: 01/23/23	
Amendment 125 to the BSAI FMP; Pacific Cod Small Boat Access	Happala	October 2022					
Amendment 126 to the BSAI FMP and Amendment 114 to the GOA FMP; Trawl Electronic Monitoring	Henry	October 2022					

"EOC" means end of the comment period.

Status of FMP Amendments
February 05, 2023

FMP Amendment Status	Lead Council and SF Staff	Date of Council Final Action	Start Regional Review	Transmittal Date of Action to NMFS HQ for Review	Proposed FMP Amendment Notice of Availability Published in <i>Federal Register</i>	Proposed Rule Published in <i>Federal Register</i>	Final Rule or Notice of Approval Published in <i>Federal Register</i>
Amendment 124 to the BSAI FMP for Groundfish and Amendment 112 to the GOA FMP for Groundfish to Revise IFQ Program Regulations Approved: January 27, 2023	Cleaver Jahn	April 2022	PR: 08/29/22 FR: 01/09/23	PR: 10/28/22 FR: 01/24/23	Nov 2, 2022 87 FR 66125 EOC: 01/03/23	Nov 23, 2022 87 FR 71559 EOC: 12/23/22	

"EOC" means end of the comment period.

Status of Regulatory Amendments
February 05, 2023

Regulatory Amendment Status	Lead Council and SF Staff	Date of Council Final Action	Start Regional Review	Transmittal Date of Action to NMFS HQ for Review	Proposed Rule Published in <i>Federal Register</i>	Final Rule Published in <i>Federal Register</i>
Observer Provider and Small C/P Reporting Changes						
Regulatory Amendment to Clarify Timing of Haul Designation for Trawl Catcher Processors in the GOA and BSAI Groundfish Fisheries	NA Cates					
RQE Fee Collection Program	Marrinan	April 2022				
BSAI Pot CP Monitoring Requirements	Evans Smith		PR: 10/03/22	PR: 12/22/22		
Bering Sea and Aleutian Islands: Proposed 2023 and 2024 Groundfish Harvest Specifications	Stram Whitney	December 2022	PR: 10/14/22	PR: 11/28/22	Dec 14, 2022 87 FR 76435 EOC: 01/13/23	
Gulf of Alaska; Proposed 2023 and 2024 Harvest Specifications for Groundfish	Cleaver Davis	December 2022	PR: 10/13/22 FR: 12/22/22	PR: 11/15/22	Dec 2, 2022 87 FR 74102 EOC: 01/03/23	

"EOC" means end of the comment period.

Status of Other Federal Register Notices
February 05, 2023

Title	Lead SF Staff	Start Regional Review	Transmittal Date of Action to NMFS HQ for Review	Notice Published in <i>Federal Register</i>
Cost Recovery Fee Notice for IFQ Program for 2022	Smith	12/19/22	12/21/22	Dec 28, 2022 87 FR 79869 Effective: 12/28/22
Cost Recovery Fee Notice for Rockfish Program for 2022	Hadfield	01/13/23	01/19/23	Jan 27, 2023 88 FR 5311 Effective: 01/27/23
Observer Program Fee Notice of Standard Ex-Vessel Prices for 2023	Hadfield	12/16/22	12/21/22	Dec 29, 2022 87 FR 80164 Effective: 01/01/23

"EOC" means end of the comment period.